Compilated Notes on Real Analysis

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The Origin of Measure Theory

One of the most fundamental concepts in classical geometry is the measure $m(E)$ of an object E in one-dimensional . In one, two, and three dimensions, we refer to this measure as the length, area, or volume of E, respectively. In classical geometric approaches, the measure of an object is typically computed by dividing it into a finite number of components, applying rigid motions (such as translations or rotations) to each part, and then reassembling these parts to form a simpler object with the same area. One may also obtain lower and upper bounds for the measure of a shape by calculating the measures of certain inscribed or circumscribed figures. This ancient ideas can be traced back at least to the work of Archimedes. The reasonableness of this method is often justified by geometric intuition, or simply by assuming the existence of a measure m that can be applied to all objects E, and that this measure adheres to a set of geometrically plausible axioms.

However, with the development of modern mathematics, our study of geometry has been placed within the context of the set of real numbers $\mathbf{R}$ or $\mathbf{R}^n$, where the measure of general subsets is not intuitively obvious. In this regard, we must carefully reexamine our earlier intuition: Can we simply assume the existence of a measure m that applies to all subsets and satisfies a set of reasonable axioms? The axioms we first consider are those presented by Lebesgue in his doctoral thesis (Under the assumption that the measure of a nonempty set is nonnegative and, without loss of generality, setting m([0,1]) = 1):

- Nonnegativity,
- Translation invariance,
- Addivity.

In this chapter, I will introduce the work of mathematicians Jordan and Lebesgue on the aforementioned issues, providing readers with a broad understanding of the motivations behind measure theory research. The content references mathematician Terence Tao's "An Introduction to Measure Theory, mathematician H. Lebesgue's doctoral thesis "Intégrale, longueur, aire," and Jordan's work "Cours d'Analyse." Many propositions in this chapter are straightforward to prove, and I have intentionally left them for the reader.

1.1 Jordan Measurability

A Elementary Measure

As the name of elementary measure suggests, its subject matter is quite elementary, specifically, the union of finitely many boxes. However, in the works of Jordan and Peano, this concept was not actually introduced. Modern textbooks introduce it to facilitate the explanation of Jordan measure theory, which will be covered in the next section.

Definition 1.1.1. An ***interval*** on $\mathbf{R}$ is a subset of the form

$$[a,b] = \{x \in \mathbf{R} : a \le x \le b\}, (a,b) = \{x \in \mathbf{R} : a < x < b\},$$
$$(a,b] = \{x \in \mathbf{R} : a < x \le b\}, [a,b) = \{x \in \mathbf{R} : a \le x < b\},$$

where $a \le b$ are real numbers. We define the ***length*** ℓ of an interval $I = [a,b], (a,b], [a,b), (a,b)$ to be $\ell(I) := b - a$. Generally, a ***box*** in $\mathbf{R}^n$ is a Cartesian product $B = I_1 \times I_2 \times \cdots \times I_n$ of n intervals $I_1, \ldots, I_n$. We define the ***volume*** vol_n of a box to be $vol(B) := \ell(I_1) \times \ell(I_2) \times \cdots \times \ell(I_n)$. An ***elementary set*** is any subset of $\mathbf{R}^n$ which is the union of a finite number of boxes. We denote with $\mathcal{E}_n$ the clss of all elementary sets in $\mathbf{R}^n$.

Remark 1.1.1.

1. The family of elementary sets is closed under finite intersection, relative complement, and symmetric difference. Moreover, also the translation.

2. Any elementary set can be expressed as the finite union of disjoint boxes.

Now, we want to give each elementary set a measure, and the idea is simply to extend the concept of "volume of boxes" to elementary sets. However, an elementary set may be divided in different ways. Readers can verify that the measure of a elementary set, as defined below, is independent of its division.

Definition 1.1.2. Let $E \subset \mathcal{E}_n$. If E is partitioned as the finite union $B_1 \sqcup B_2 \sqcup \cdots \sqcup B_n$ of disjoint boxes, then we refer to $m(E) := vol(B_1) + vol(B_2) + \cdots + vol(B_n)$ with the ***elementary measure*** of E.

Now, we have a definition for the measure m, it is natural to see the properties of such a mapping. In particular, we would like that, for boxes, m is consistent with the previously defined volume. At the same time, since we know that an object does not change its volume when translated, m should also satisfy ***translation invariance***. Moreover, From the definition, it is clear that $m(E)$ is a

nonnegative real number for every elementary set E, especially $m(\emptyset) = 0$. And one may notice that $m(E \sqcup F) = m(E) + m(F)$, by induction it also implies

$$m(E_1 \sqcup E_2 \sqcup \cdots \sqcup E_n) = \sum_{k=1}^{n} m(E_k),$$

We refer to the preceding property as ***finite additivity***.

Proposition 1.1.1. The nonnegativity and finite additivity are intuitive and elementary. Then the following properties are also important.

1. $m(\emptyset) = 0$,
2. ***Nonnegativity***: $m(E) \geq 0$ for all $E \in \mathcal{E}_n$,
3. ***Finite Addivity***: $m(E \sqcup F) = m(E) + m(F)$,
4. ***Monotonicity***: $m(E) \leq m(F)$ whenever $E \subset F$,
5. ***Finite subadditivity***: $m(E_1 \cup E_2 \cup \cdots \cup E_n) \leq \sum_{k=1}^{n} m(E_k)$ whenever $E_1, E_2, \ldots, E_n$ are elementary sets.
6. ***Translation invariance***: $m(E + x) = m(E)$ for all elementary sets E and $x \in \mathbf{R}^n$.

B Jordan Measure (Content)

Giuseppe Peano and Camille Jordan can be considered the first two mathematicians to systematically study measure theory. Jordan capacity theory was initially developed to generalize the Riemann integral. Mathematicians Peano and Jordan were highly dissatisfied with the performance of the Riemann integral in higher dimensions. Specifically, at that time, it was widely believed that for a planar figure, a rectangular grid could be used to partition it. When the partition was sufficiently fine, the total area of the rectangles intersecting the boundary of the region could become arbitrarily small. This was the key to extending the Riemann integral, as it allowed for the "roughness" of the boundary to be ignored.

However, in reality, many situations did not meet this requirement. For example, Peano constructed a curve (known as the Peano curve) within a unit square that could cover the entire unit square. This meant that a one-dimensional curve could "cover" a two-dimensional figure, or in other words, the boundary—even if one-dimensional—might not be coverable by arbitrarily small rectangles. We cannot ignore the roughness of the boundary! To address the issue of the boundary of regions, Jordan developed capacity theory.

In fact, what Jordan developed was the theory of content (contenu), as presented in his work "Cours d'analyse" (1893-1894). His approach can be easily *summarized as "inner approximation and outer covering." For now, let us set aside the term "content" and refer to it simply as measure... After this brief introduction to the work of these two mathematicians, let us return to the main discussion.*

So far, we have arrived at an axiomatic concept for measuring elementary sets. Unfortunately, however, the objects of elementary measure are limited to elementary sets, which do not include some important figures. For instance, in the plane, neither triangles nor circles are elementary sets. Hence, a natural idea arises: we need to create a new concept whose scope can encompass a wider range of sets.

Peano and Jordan contributed the following work: since we are already familiar with elementary sets, why not use these sets to "approximate" other non-elementary sets? For example, for a planar figure, we may attempt to "cover" and "fill" it with elementary sets. We formalize this idea as follows

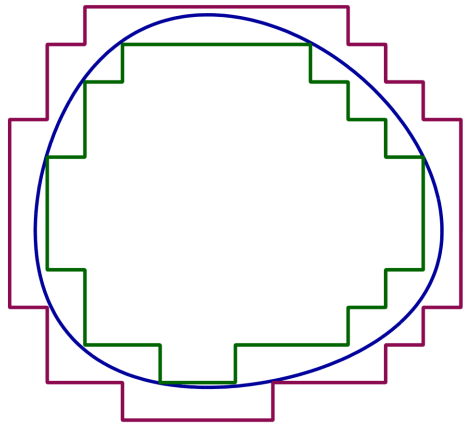
Definition 1.1.3. Let $E \subset \mathbf{R}^n$ be a bounded set.

- The ***Jordan outer measure*** $J_e(E)$ of E is defined as
$$J^*(E) := \inf\{m(A) : A \in \mathcal{E}_n, E \subset A\}$$
- The ***Jordan inner measure*** $J_i(E)$ of E is defined as
$$J_*(E) := \inf\{m(A) : A \in \mathcal{E}_n, E \supset A\}$$
- If $J^*(E) = J_*(E)$, then we say that E is ***Jordan measurable***, and call $J(E) := J^*(E) = J_*(E)$ the ***Jordan measure*** of E. We denote with $\mathcal{J}_n$ be the family of all Jordan measurable sets in $\mathbf{R}^n$.

Observe from the finite additivity and subadditivity of elementary measure that we can also write the Jordan inner measure and outer measure as

- $J_*(E) = \sup\left\{\sum_{k=1}^n m(B_k) : B_k \text{ is a box for each } k, E \supset \bigcup_{k=1}^n B_k\right\}$,
- $J^*(E) = \inf\left\{\sum_{k=1}^n m(B_k) : B_k \text{ is a box for each } k, E \subset \bigcup_{k=1}^n B_k\right\}$.

Indeed, the definitions of inner and outer measure we just provided are actually closer to those given by Jordan in his work ***Cours d'Analyse***: he primarily considered the case in the two-dimensional plane. For a set E, consider an inner domain S formed by squares lying entirely within E, while those squares lying entirely within E or intersecting its boundary together form a new outer domain $S + S'$. Jordan proved that as the side length of the squares used in the partition tends to 0, the (area) values of S and $S + S'$ have limits. He then defined a 'quarrable' set as a set S for which the limit of S' is 0. Here, 'quarrable' corresponds to what we call 'measurable', with the author specifically emphasizing measurability in the two-dimensional plane.



Under this definition, the family of Jordan-measurable sets becomes significantly larger than that of elementary sets. For example, it is straightforward to prove that in the plane, all triangles and disks are Jordan measurable.

Remark 1.1.2. The family of Jordan measurable sets is closed under finite intersection, relative complement, and symmetric difference. Moreover, also the translation.

Proposition 1.1.2. Jordan measure inherits all the properties of elementary measure: let $E, F \subset \mathbf{R}^n$ be Jordan measurable sets.

1. ***Nonnegativity***: $J(E) \geq 0$,
2. ***Finite additivity***: $J(E \sqcup F) = J(E) + J(F)$,
3. ***Monotonicity***: If $E \subset F$, then $J(E) \leq J(F)$,
4. ***Finite subadditivity***: $J(E_1 \cup E_2 \cup \ldots \cup E_n) \leq \sum_{k=1}^{n} J(E_k)$,
5. ***Translation invariance***: $J(E + x) = J(E)$ for any $x \in \mathbf{R}^n$.

Theorem 1.1.1. Let E be a bounded subset of $\mathbf{R}^n$, show that $J_n^*(E) = J_n^*(\overline{E})$, where $\overline{E}$ is the closure of E in the Euclidean metric sense. Similarily, $J_{n*}(E) = J_{n*}(E^\circ)$, where E° is the interior of E.

Proof. It is clear that $J_n^*(E) \leq J_n^*(\overline{E})$. To establish the converse direction, it suffices to show that if $\bigcup_{k=1}^n B_k$ covers E, then $\bigcup_{k=1}^n \overline{B_k}$ covers $\overline{E}$, this because $m(B_k) = m(\overline{B_k})$. Assume that $\bigcup_{k=1}^n \overline{B_k}$ does not cover $\overline{E}$, then there exists $x \in \overline{E}$ such that $x \notin \bigcup_{k=1}^n \overline{B_k}$. By the definition of "closure", $\overline{E} = E \cup E'$ where E' is the set of all limit points of E. Thus, for every $\epsilon > 0$, $(x-\epsilon, x+\epsilon) \cap E \neq \emptyset$, but $E \subset \bigcup_{k=1}^n \overline{B_k}$ which implies that $(x-\epsilon, x+\epsilon) \cap \bigcup_{k=1}^n \overline{B_k} \neq \emptyset$. Therefore x is a limit point of $\bigcup_{k=1}^n \overline{B_k}$.

However, since $\bigcup_{k=1}^n \overline{B_k}$ is the finite union of closed boxes, $\bigcup_{k=1}^n \overline{B_k}$ is closed and must contain all of its limit points, so $x \in \bigcup_{k=1}^n \overline{B_k}$, which is a contradiction.

Therefore, if $E \subset \bigcup_{k=1}^n B_k$, then $\overline{E} \subset \bigcup_{k=1}^n \overline{B_k}$, which implies that $J^*(E) \geq J^*(\overline{E})$. Thus, $J^*(E) = J^*(\overline{E})$. □

►**Corollary 1.** Let $\mathbf{Q} \cap [0,1] = \{r_1, r_2, \ldots\}$, then $J_1^*(\mathbf{Q} \cap [0,1]) = 1$. Since $\mathbf{Q} \cap [0,1]$ is dense in $[0,1]$, that is $\overline{\mathbf{Q} \cap [0,1]} = [0,1]$, $J_1^*(\mathbf{Q} \cap [0,1]) = J_1^*([0,1]) = 1$.

Here is a more intuitive equivalent definition of the concept of measurable sets and this proposition is the concept of measurability that Mr. Jordan most wanted to express:

Proposition 1.1.3. Let $E \subset \mathbf{R}^n$ be a bounded set. E is Jordan measurable ***iff*** the boundary ∂E has Jordan outer measure zero, i.e., $J_n^*(\partial E) = 0$.

1.2 Borel Measurability

Around the same time, specifically in 1898, just a few years after Jordan published his "Cours d'analyse", Émile Borel also published his own work "Leçons sur la théorie des fonctions" (1898). According to Borel himself, his approach to measure theory was entirely different. Jordan's focus was on constructing inner and outer content, which is the idea of defining measurability through "inner approximation and outer covering". Borel, on the other hand, started from the sets themselves, perhaps influenced by Cantor's set theory, and introduced the concept of countability into measure theory...

1.3 Lebesgue Measurability

Next, we turn to a pivotal figure in measure theory, Henri Lebesgue. As a central contributor to the field, his theory of measure was systematically presented in his 1902 doctoral thesis, "Intégrale, longueur, aire." Influenced by the work of both Borel and Jordan, Lebesgue extended the concept of outer content directly to the countable case. As for the refinement of inner content, he adopted an exceptionally subtle approach... Let us now return to the main text.

If we only study Jordan measurable sets, then Jordan content theory is already quite sufficient. However, unfortunately, there are still many common sets that are not Jordan measurable, such as some bounded open sets, compact sets, and others that interest us (at least in the topological sense). Moreover, it does not include sets obtained from countable intersections or countable unions of known Jordan measurable sets, which is quite critical. For example, in the context of

Riemann integrals, we have the following examples.

Example 1.3.1.

1. Countable union $\bigcup_{k=1}^{\infty} E_k$ or countable intersection $\bigcap_{k=1}^{\infty} E_k$ of Jordan measurable sets $\{E_k\} \subset \mathbf{R}$ need not be Jordan measurable.
2. Suppose that $\mathbf{Q} \cap [0, 1] = \{r_1, r_2, \dots\}$ and define a sequence of functions
$$f_n(x) = \begin{cases} 1, & x = r_1, r_2, \dots, r_n \\ 0, & other. \end{cases}$$
It is clear that $|f_n(x)| \leq 1$ for all n and x and f_n is Riemann integrable for each n and $f_n(x) \to f$ where f is the Dirichlet function that is not Riemann integrable
$$f(x) = \lim_{n\to\infty} f_n(x) = \begin{cases} 1, & x \in \mathbf{Q} \cap [0, 1] \\ 0, & x \in [0, 1] \setminus \mathbf{Q}. \end{cases}$$
Readers may find this confusing. We will learn in Section 1 of Chapter 4 why this example regarding Riemann integrals is related to Jordan measurability. For now, readers only need to know that Jordan measurability is profoundly connected to Riemann integrals. (That is, roughly speaking: a bounded function is Riemann integrable if and only if the corresponding two-dimensional region is Jordan measurable.)

Therefore, we must further refine our concepts to enable the study of more sets. Lebesgue took on this task and, in fact, presented this work in his doctoral dissertation titled Integral, Length, Area, a groundbreaking masterpiece that marked a new era in mathematics.

Definition 1.3.1. (In fact, Lebesgue studied measurability on the real line $\mathbf{R}$, but we can directly begin by defining it in n-dimensional Euclidean space.) Let $E \subset \mathbf{R}^n$ be a bounded set.

- The ***Lebesgue outer measure*** $m^*(E)$ of E is defined as
$$m^*(E) := \inf \left\{ \sum_{k=1}^{\infty} m(B_k) : B_k \text{ is a box for each } k, E \subset \bigcup_{k=1}^{\infty} B_k \right\}.$$
- The ***Lebesgue inner measure*** $m_*(E)$ of E is defined as
$$m_*(E) := m^*(A) - m^*(A \setminus E), \text{ for any box } A \supset E.$$
- If $m^*(E) = m_*(E)$, then we say that E is ***Lebesgue measurable***, and call $m(E) := m^*(E) = m_*(E)$ the ***Lebesgue measure*** of E.

It is worth noting that this definition requires the set to be *bounded*, because when a set has both inner and outer measures that are infinite, we cannot determine whether the set is measurable.

The reader might wonder: since Lebesgue outer measure can be directly generalized from Jordan outer content, why is Lebesgue inner measure defined using a different approach? In short: if inner measure were defined simply by replacing finite unions with countable unions, it would not offer any substantive advantage or enhancement over Jordan inner content. Specifically, while using countably many boxes to cover a set, rather than finitely many, and makes the covering much more “fine,” attempting to fill a set with countably many boxes runs into problems. Many complex sets, such as the set of irrational numbers or later important example, the Cantor set, contain no nonempty boxes in their interior that can be used for filling. Under such a definition, these sets would yield the same result as under Jordan inner content: a value of 0, even though the set itself may have positive outer measure. Therefore, it is entirely reasonable to define inner measure in terms of outer measure itself.

Clearly, based on the definition above, it is easy to deduce the relationship $\lambda_n^*(E) \leq J_n^*(E)$. Additionally, the Lebesgue outer measure is sometimes not only smaller than the Jordan outer measure but can also be significantly smaller. The following example illustrates this:

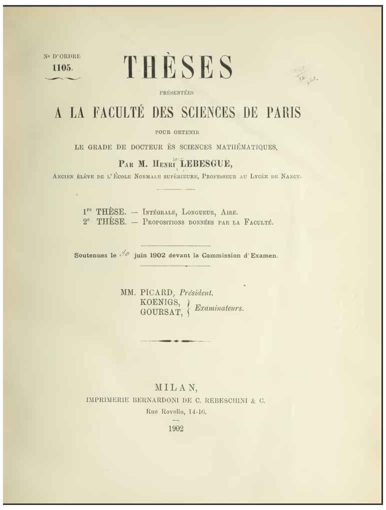
Example 1.3.2. Let $E = \{x_1, x_2, \ldots\} \subset \mathbf{R}^n$ be a countable set. From corollary 1 before, we know that the Jordan outer content of E can be quite large. On the other hand, all countable sets E have Lebesgue outer measure zero. Indeed, for every $\epsilon > 0$, $\{x_i\} \in B(x_i, \epsilon/2^i)$ for each i. Since $E \subset \bigcup_{i=1}^{\infty} B(x_i, \epsilon/2^i)$, $m^*(E) \leq \epsilon$ for every $\epsilon > 0$. Therefore, $\mu^*(E) = 0$. This example further demonstrates that for any countable set, its Lebesgue outer measure is 0.

Lebesgue provided an equivalent definition of a Lebesgue measurable set, which is intuitively obvious, and the proof can be left to the reader: The set $E \subset \mathbf{R}$ is Lebesgue measurable ***iff*** for any $\epsilon > 0$, there exist two sequences of open intervals $\{I_i\}$ and $\{J_j\}$ such that $E \subset \bigcup_{i=1}^{\infty} I_i, A \setminus E \subset \bigcup_{j=1}^{\infty} J_j$ and $m\left(\bigcup_{i=1}^{\infty} I_i \cap \bigcup_{j=1}^{\infty} J_j\right) < \epsilon$. He then proved that the union of any countable collection of Lebesgue measurable sets is also Lebesgue measurable. Thus, through Example 1.2.1, we learn that the family of Lebesgue measurable sets is larger than the family of Jordan measurable sets.

Let us pause our study of Lebesgue measure here, as the purpose of this chapter is simply to make it clear to the reader what we aim to achieve in this section: we seek to find a method of measurement that attempts to assign a measure to every subset of the real numbers.

Now, let us return to where we started. We have introduced a new concept called Lebesgue measure, but is it the ideal measure we originally sought? Can Lebesgue measure assign a measure to every subset that satisfies the conditions we hope for? (Perhaps readers may already know the answer.) The answer will be revealed later, as we currently lack the necessary tools to address these questions.

While the significant transformation in measure theory have been sparked by Lebesgue's seminal paper, it was Carathéodory who played a crucial role in its development by axiomatizing and standardizing the theory. ”Vorlesungen über reelle Funktionen” (1918), compiled by Carathéodory, presents measure theory in an axiomatic framework, fundamentally establishing it as a new branch of mathematics. The equivalent definition of Lebesgue measurability introduced in this work is particularly significant. Most subsequent research and literature on measure theory have adopted the definitions and theorems from this seminal text. In the next chapter (which is also the core section of this part), we will study measure theory from the perspective of modern measure theory.



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THÈSES

PRÉSENTÉES

A LA FACULTÉ DES SCIENCES DE PARIS

POUR OBTENIR

LE GRADE DE DOCTEUR ÈS SCIENCES MATHÉMATIQUES,

PAR M. HENRI LEBESGUE,

ANCIEN ÉLÈVE DE L'ÉCOLE NORMALE SUPÉRIEURE, PROFESSEUR AU LYCÉE DE NANCY.

1re THÈSE. — INTÉGRALE, LONGUEUR, AIRE.
2e THÈSE. — PROPOSITIONS DONNÉES PAR LA FACULTÉ.

Soutenues le 30 juin 1902 devant la Commission d'Examen.

MM. PICARD, *Président*.
KOENIGS, GOURSAT, *Examinateurs*.

MILAN,
IMPRIMERIE BERNARDONI DE C. REBESCHINI & C.
Rue Rovello, 14-16.
1902

Figure 1.1: Intégrale, longueur, aire.

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2

Outer Measure and Lebesgue Measure

2.1 Families of Sets

In modern mathematical research, no matter which branch it belongs to, work is often carried out within a specific space, this has become a fundamental recognition in the mathematical community. The main content of this subsection is to establish a basic structure — family of sets, in a space.

A Semi-ring, Ring and Sigma-ring

Let's begin by introducing families of sets, since the objects of measure theory are sets rather than individual points. We'll start with simple families of sets. Generally speaking, a semiring can be considered the "simplest" family with special closure properties (though a π-system, which we'll discuss later, is in some sense even simpler. However, a π-system does not necessarily include some important and essential sets, such as the empty set). So, we will begin with semirings.

Definition 2.1.1. Let $\mathscr{S}_r$ be a nonempty family of subsets of X, $\mathscr{S}_r$ is called a ***semi-ring*** if it satisfying

- $\emptyset \in \mathscr{S}_r$
- If $A, B \in \mathscr{S}_r$, then $A \cap B \in \mathscr{S}_r$.
- If $A, B \in \mathscr{S}_r$, then there exist pairwise disjoint sets $C_1, C_2, \ldots, C_n \in \mathscr{S}_r$ such that $A \setminus B = \bigcup_{k=1}^{n} C_k$.

Example 2.1.1. Any family of pairwise disjoint subsets of a set together with the empty set (in particular any partition of a set together with the empty set) is automatically a semiring. Another important example of a semi-ring is the family $\mathscr{S}$ of all half-open boxes in $\mathbf{R}^n$ defined by

$$\mathscr{S} = \{[a_1, b_1) \times \cdots \times [a_n, b_n) : a_i, b_i \in \mathbf{R}\},$$

where $[a_k, b_k) = \emptyset$ if $b_i \leq a_i$. The family $\mathscr{S}$ is a semi-ring. This semi-ring will reappear in Section 4 of this chapter during the construction of the Lebesgue measure.

Proof. We hereby prove only that this family of sets on the real line is a semi-ring. By the definition of $\mathscr{S}$, it is clear that $\emptyset \in \mathscr{S}$. We first show that $\mathscr{S}$ is closed under finite intersections, let $A = [a_1, b_1), B = [a_2, b_2)$, then

$$A \cap B = [a_1, b_1) \cap [a_2, b_2) = [\max(a_1, a_2), \min(b_1, b_2)).$$

Thus, $A \cap B \in \mathscr{S}$. To establish the third property, if $A \cap B = \emptyset$, then $A \setminus B = A$, as described. If $A \cap B \neq \emptyset$, then $A \setminus B = [a_1, a_2) \cup [b_2, b_1) \in \mathscr{S}$. Therefore, $\mathscr{S}$ is a semiring. □

Definition 2.1.2. Let $\mathscr{R}$ be a nonempty family of subsets of X, $\mathscr{S}$ is called a ***ring*** if it satisfying

- $\emptyset \in \mathscr{R}$
- If $A, B \in \mathscr{R}$, then $A \setminus B \in \mathscr{R}$.
- If $A, B \in \mathscr{R}$, then $A \cup B \in \mathscr{R}$.

Since a ring $\mathscr{R}$ being nonempty by definition, contains some set A, it follows that $\emptyset = A \setminus A \in \mathscr{R}$, so the first condition can be omitted and the simplest example of a ring is just $\{\emptyset\}$. Moreover,

$$A \cap B = A \setminus (A \setminus B) \text{ and } A \triangle B = (A \setminus B) \cup (B \setminus A),$$

we see that every ring is closed under pairwise intersections and symmetric differences. On the other hand, from the identities,

$$A \cup B = (A \triangle B) \triangle (A \cap B) \text{ and } A \setminus B = A \triangle (A \cap B),$$

if follows that a nonempty family $\mathscr{R}$ of subsets of a set X that is closed under symmetric differences and finite intersections is a ring.

Example 2.1.2. There is a simple example, we define the family as

$$\mathscr{R} = \{A \subset X : A \text{ is finite}\}.$$

This is because the finite union of finite sets remains a finite set, and the difference of two finite sets is also a finite set.

Indeed, a ring can be constructed from a semiring. Note that in the third property of a semiring, we do not require that the union of the disjoint sets $\bigcup_{k=1}^{n} C_k$ itself belongs to the semiring. In other words, the key difference between a ring and a semiring lies in the closure under finite disjoint unions. Therefore, we can construct the following family of sets to naturally satisfy closure under finite disjoint unions and then examine whether it forms a ring.

Theorem 2.1.1. Let $\mathscr{S}_r$ be a semi-ring on X, then the family $\mathscr{R}$ of all sets that can be written as a finite union of pairwise disjoint elements of $\mathscr{S}_r$ is the smallest ring containing $\mathscr{S}_r$. That is

$$\mathscr{R} = \mathscr{R}(\mathscr{S}_r) = \left\{ \bigcup_{k=1}^{n} A_k : A_k \in \mathscr{S}_r, A_k \cap A_j = \emptyset \text{ for } k \neq j \right\}.$$

We refer to $\mathscr{R}(\mathscr{S}_r)$ as ***the ring generated by*** $\mathscr{S}_r$.

Proof. Let $E, F \in \mathscr{R}$ and $E = \bigcup_{k=1}^{n} A_k, F = \bigcup_{j=1}^{m} B_j$, where $\{A_k\}, \{B_j\}$ are sequences of pairwise disjoint sets of $\mathscr{S}_r$. We first show that $\mathscr{R}$ is closed under relative complementation,

$$E \setminus F = \bigcup_{k=1}^{n}\left(A_k \setminus \bigcup_{j=1}^{m} B_j\right) = \bigcup_{k=1}^{n}\bigcap_{j=1}^{m}\left(A_k \setminus B_j\right).$$

Since $A_k, B_j \in \mathscr{S}_r$, $A_k \setminus B_j$ can be written as a finite union of pairwise disjoint sest of $\mathscr{S}_r$. Consider $\bigcap_{j=1}^{m}(A_k \setminus B_j)$, let

$$A_k \setminus B_j = \bigcup_{i=1}^{j_1} C_i^j, \text{ where } C_i^j \in \mathscr{S}.$$

Since $(A_k \setminus B_1) \cap (A_k \setminus B_2) = (A_k \setminus B_1) \setminus B_2$ and $C_i^j \setminus B_2$ also can be written as a finite union of elements of $\mathscr{S}_r$, $\bigcap_{j=1}^{m}(A_k \setminus B_j)$ can be written as a finite union of pairwise disjoint sets of $\mathscr{S}$. Thus, $E \setminus F$ is the finite union of pairwise disjoint sets of $\mathscr{R}$, i.e., $E \setminus F \in \mathscr{R}$. To establish the third property, we say that

$$\begin{aligned} A \cup B &= (A \setminus B) \cup (B \setminus A) \cup (A \cap B) \\ &= (A \setminus B) \cup (B \setminus A) \cup (A \setminus (A \setminus B)) \end{aligned}$$

By the preceding discuss, $A \cup B \in \mathscr{R}$, as desired. □

Definition 2.1.3. Let Σ_r be a nonempty family of susbets of X, Σ_r is called a ***σ-ring*** if it satisfying

- $\emptyset \in \Sigma_r$,
- If $A, B \in \Sigma_r$, then $A \setminus B \in \Sigma_r$,
- If $A_1, A_2, \cdots \in \Sigma_r$, then $\bigcup_{k=1}^{\infty} A_k \in \Sigma_r$.

Since our inspiration for constructing a ring from a semiring comes from the key property of rings being closed under finite unions, can we use a similar approach to construct a sigma-ring from a semiring? Specifically, recalling that the difference between a sigma-ring and a ring lies in the sigma-ring being closed under countable unions, it naturally leads us to consider constructing the following collection of sets:

$$\mathscr{F} = \left\{\bigcup_{k=1}^{\infty} A_k : A_k \in \mathscr{S}_r, A_k \cap A_j = \emptyset \text{ for } k \neq j\right\}.$$

Unfortunately, it can ve proven that this family of sets is not a σ-ring, it may not even be a ring, thus its structure is quite “bad”. Therefore, we take a step back and adopt the following method to generate a σ-ring, though this method will be rather complex in technical.

Theorem 2.1.2. Let $\{\Sigma_\gamma\}, \gamma \in \Gamma$ be a family of σ-rings on X, then $\bigcap_\gamma \Sigma_\gamma$ is also a σ-ring on X. In particular, let $\mathscr{F}$ be a nonempty family of subsets of

X. Then the intersection of all σ-rings that include $\mathscr{F}$ is the smallest σ-ring including $\mathscr{F}$, The σ-ring is denoted $\sigma_r(\mathscr{F})$, in other words,

$$\sigma_r(\mathscr{F}) = \bigcap \left\{ C \subset 2^X : \mathscr{F} \subset C \text{ and } C \text{ is a } \sigma\text{-ring} \right\}.$$

We refer to $\sigma_r(\mathscr{F})$ as ***the σ-ring generated by $\mathscr{F}$***.

Apart from this, there is another method. The reader should be aware of the following properties of functions: let $f : X \to Y$ be a function on X. We know that operation of taking inverse images of mappings is very well-behaved, as it preserves fundamental Boolean operations, such as the union, intersection, and difference of sets, among others.

1. $f^{-1}(\emptyset) = \emptyset, f^{-1}(Y) = X$
2. $f^{-1}\left(\bigcap_{\gamma\in\Gamma} A_\gamma\right) = \bigcap_{\gamma\in\Gamma} f^{-1}(A_\gamma)$
3. $f^{-1}\left(\bigcup_{\gamma\in\Gamma} A_\gamma\right) = \bigcup_{\gamma\in\Gamma} f^{-1}(A_\gamma)$
4. $f^{-1}(A \setminus B) = f^{-1}(A) \setminus f^{-1}(B)$.

The inverse image of a function is compatible with all set operations, which is a highly desirable property. We can use this to generate a σ-ring.

Theorem 2.1.3. Let $f : X \to Y$ be a function on X and Σ_r is a σ-ring on Y, denote

$$f^{-1}(\Sigma_r) = \left\{ f^{-1}(A) : A \in \Sigma_r \right\},$$

then $f^{-1}(\Sigma_r)$ is a σ-ring on X.

We close the subsection by presenting two technical properties of semi-rings that are of use in the later.

Proposition 2.1.1. Let $\mathscr{S}_r$ be a semi-ring on X then we have the following

1. If $A_1, A_2, \ldots, A_n, A \in \mathscr{S}$, then $A \setminus \bigcup_{k=1}^n A_k$ can be written as a finite union of a pairwise disjoint sets of $\mathscr{S}_r$.
2. If $\{A_n\}$ is a sequence in $\mathscr{S}_r$, then there exists a pairwise disjoint sequence $\{C_k\}$ in $\mathscr{S}_r$ satisfying $\bigcup_{n=1}^\infty A_n = \bigcup_{k=1}^\infty C_k$ and such that for each k there exists some n with $C_k \subset A_n$.

Proof. For (1), the case $n = 1$ is the definition of semi-ring. So assume the statement is true for n and let $A_1, A_2, \ldots, A_n, A_{n+1}, A \in \mathscr{S}_r$. By the induction hypothesis, there are pairwise disjoint sets $C_1, \ldots, C_m \in \mathscr{S}_r$ wuch that $A \setminus \bigcup_{k=1}^n A_k = \bigcup_{j=1}^m C_j$. And

$$A \setminus \bigcup_{k=1}^{n+1} A_k = \left(A \setminus \bigcup_{k=1}^{n} \right) \setminus A_{n+1} = \left(\bigcup_{j=1}^{m} C_j \right) \setminus A_{n+1} = \bigcup_{j=1}^{m} \left(C_j \setminus A_{n+1} \right).$$

For each j, there exist pairwise disjoint sets $D_1^j, D_2^j, \dots, D_{r_j}^j \in \mathscr{S}_r$ satisfying $C_j \setminus A_{n+1} = \bigcup_{i=1}^{r_j} D_i^j$, and then $A \setminus \bigcup_{k=1}^{n+1} = \bigcup_{j=1}^{m} \bigcup_{i=1}^{r_j} D_i^j$.

For (2), let $A = \bigcup_{n=1}^{\infty} A_n$ and $B_1 = A_1, B_{n+1} = A_{n+1} \setminus \bigcup_{i=1}^{n} A_i$. Then $B_i \cap B_j = \emptyset$ for $i \neq j$ and $A = \bigcup_{n=1}^{\infty} B_n$. By part (1), each B_n can be written as a union of a finite pairwise disjoint $\{C_k\}$ sets of $\mathscr{S}_r$ that satisfies the desired properties. $\square$

B Semi-algebra, Algebra, Sigma-algebra

After introducing semirings, rings, and sigma-rings, we will now present some more important families of sets. The reader may have noticed that semirings, rings, and sigma-rings all share the property of including the empty set. However, there is another important set that may not necessarily be part of their elements—the universal set 'X'. If we now add the universal set 'X' to the corresponding families of sets, we will obtain some new and more complex families of sets.

Definition 2.1.4. Let $\mathscr{S}_a$ be a nonempty family of subsets of X, $\mathscr{S}_a$ is called a ***semi-algebra*** if it is a semi-ring and $X \in \mathscr{S}_a$. Specifically,

- $X, \emptyset \in \mathscr{S}_a$
- If $A, B \in \mathscr{S}_a$, then $A \cap B \in \mathscr{S}_a$.
- If $A, B \in \mathscr{S}_a$, then there exist pairwise disjoint sets $C_1, C_2, \dots, C_n \in \mathscr{S}_a$ such that $A \setminus B = \bigcup_{k=1}^{n} C_k$.

Definition 2.1.5. Let $\mathscr{A}$ be a nonempty family of subsets of X, $\mathscr{A}$ is called a ***algebra*** if it is a ring and $X \in \mathscr{A}$. Specifically,

- $X, \emptyset \in \mathscr{A}$
- If A, then $A^c = X \setminus A \in \mathscr{A}$. (It is worth noting that closure under complement implies closure under difference.)
- If $A, B \in \mathscr{A}$, then $A \cup B \in \mathscr{A}$.

Since $\mathscr{A}$ is nonempty, there exists some $A \in \mathscr{A}$, so $X = A \cup A^c, \emptyset = X^c$, then the first condition can be omitted and the simplest example of an algebra is $\{\emptyset, X\}$. Since an algebra is itself a ring, it follows that an algebra is also closed under countable intersections and symmetric differences.

Similarly, the method of constructing a ring from a semiring can be applied here to construct an algebra generated by a semialgebra. We will not provide the proof here (it is left as an exercise for the reader):

$$\mathscr{A} = \mathscr{A}(\mathscr{S}_a) = \left\{ \bigcup_{k=1}^{n} A_k : A_k \in \mathscr{S}_a, A_k \cap A_j = \emptyset \text{ for } k \neq j \right\}.$$

Definition 2.1.6. Let Σ_a be a nonempty family of susbets of X, Σ_a is called a ***σ-algebra*** if it is a σ-ring and $X \in \Sigma_a$. Specifically,

- $X, \emptyset \in \Sigma_a$,
- If A, then $A^c = X \setminus A \in \Sigma_a$,
- If $A_1, A_2, \cdots \in \Sigma_a$, then $\bigcup_{k=1}^{\infty} A_k \in \Sigma_a$.

Using the same idea as in Theorem 2.1.2, we can generate a σ-algebra through the following method:

Theorem 2.1.4. Let $\left\{\Sigma_\gamma\right\}, \gamma \in \Gamma$ be a family of σ-algebras on X, then $\bigcap_\gamma \Sigma_\gamma$ is also a σ-algebra on X. In particular, let $\mathscr{F}$ be a nonempty family of subsets of X. Then the intersection of all σ-algebras that include $\mathscr{F}$ is the smallest σ-algebra including $\mathscr{F}$. The σ-algebra is denoted $\sigma(\mathscr{F})$, in other words,

$$\sigma(\mathscr{F}) = \bigcap \left\{C \subset 2^X : \mathscr{F} \subset C \text{ and } C \text{ is a } \sigma\text{-algebra}\right\}.$$

We refer to $\sigma(\mathscr{F})$ as ***the σ-algebra generated by $\mathscr{F}$***.

Similarly, we can also construct a new sigma-algebra using the inverse images of functions from an existing sigma-algebra.

Theorem 2.1.5. Let $f : X \to Y$ be a mapping on X and Σ is a σ-algebra on Y, then $f^{-1}(\Sigma)$ is a σ-algebra on X.

►**Corollary 1.** Let $f : X \to Y$ be a mapping on X and $\mathscr{F}$ is a nonempty family of subsets of X, then

$$f^{-1}\left(\sigma(\mathscr{F})\right) = \sigma\left(f^{-1}(\mathscr{F})\right).$$

Proof. By theorem 1.1.5, $f^{-1}\left(\sigma(\mathscr{F})\right)$ is the σ-algebra contains $f^{-1}(\mathscr{F})$, so $\sigma\left(f^{-1}(\mathscr{F})\right) \subset f^{-1}\left(\sigma(\mathscr{F})\right)$. To establish the converse, let

$$\Sigma = \left\{A \in \sigma(\mathscr{F}) : f^{-1}(A) \in \sigma\left(f^{-1}(\mathscr{F})\right)\right\}.$$

Note that Σ is the σ-algebra that contains $\mathscr{F}$, so $\sigma(\mathscr{F}) = \Sigma$. Thus, $f^{-1}\left(\sigma(\mathscr{F})\right) = f^{-1}(\Sigma) \subset \sigma\left(f^{-1}(\mathscr{F})\right)$, as desired. □

C λ-System, π-System and Monotone Class

A σ-algebra is usually most conveniently described in terms of a generating fam ily. In this section we study families of sets possessing certain monotonicity prop erties that are of interest mostly for technical reasons relating to the σ-algebras they generate.

Definition 2.1.7. Let X be a set and $\mathscr{D}$ be nonempty family of subsets of X. $\mathscr{D}$ is called a ***λ-system*** (or Dynkin System) if it satisfies the properties,

- $X \in \mathscr{D}$,
- If $A, B \in \mathscr{D}, A \subset B$, then $B \setminus A \in \mathscr{D}$,
- If $A_1, A_2, \cdots \in \mathscr{D}$ and $A_n \uparrow A$, then $A \in \mathscr{D}$.

Definition 2.1.8. A ***π-system*** is a nonempty family of subsets of a set that is closed under finite intersections.

As we can see, the structure of a π-system is quite simple, so much so that a semi-ring is also a π-system. Additionally, we can construct a λ-system in a similar manner. The proof of this is omitted below.

Theorem 2.1.6. Let $\{\mathscr{D}_\gamma\}, \gamma \in \Gamma$ be a family of λ-systems on X, then $\bigcap_\gamma \mathscr{D}_\gamma$ is also a λ-system on X. In particular, let $\mathscr{F}$ be a nonempty family of subsets of X. Then the intersection of all λ-systems that include $\mathscr{F}$ is the smallest λ-system including $\mathscr{F}$. The λ-system is denoted $\lambda(\mathscr{F})$, in other words,

$$\lambda(\mathscr{F}) = \bigcap \left\{C \subset 2^X : \mathscr{F} \subset C \text{ and } C \text{ is a } \lambda\text{-system}\right\}.$$

We refer to $\lambda(\mathscr{F})$ as ***the λ-system generated by $\mathscr{F}$***.

Next, we will present two important theorems that reveal the structural characteristics of sigma-algebras and their relationships with other families of sets. These theorems will be used repeatedly in subsequent studies.

π-λ System Theorem. A nonempty family of subsets of X is a σ-algebra ***iff*** it is both a λ-system and a π-system. In other words,

$$\sigma\text{-algebra} = \lambda\text{-system} + \pi\text{-system}.$$

In particular, if $\mathscr{P}$ is a π-system on X, then $\sigma(\mathscr{P}) = \lambda(\mathscr{P})$.

Proof. It is clear that σ-algebra is both a λ-system and a π-system. To establish the converse, let $\mathscr{D}$ be a λ-system that is also closed under finite intersections. Note that $\mathscr{D}$ is closed under complements and finite unions, so $\mathscr{D}$ is an algebra. To see that it is a σ-algebra, let $A = \bigcup_{n=1}^\infty A_n$ with $A_1, A_2, \cdots \in \mathscr{D}$ and let $B_n = \bigcup_{k=1}^n A_n \in \mathscr{D}$, then $B_n \uparrow A$, we obtain that $A \in \mathscr{D}$.

For the last statement, let $\mathscr{P}$ be a π-system on X. It is clear that $\mathscr{P} \subset \lambda(\mathscr{P}) \subset \sigma(\mathscr{P})$. Let

$$\mathscr{F}_1 = \{A \in \lambda(\mathscr{P}) : A \cap B \in \lambda(\mathscr{P}) \text{ for each } B \in \mathscr{P}\}.$$

Then $\mathscr{F}_1$ is the λ-system that including $\mathscr{P}$, hence $\mathscr{F}_1 = \lambda(P)$. Let

$$\mathscr{F}_2 = \{A \in \lambda(\mathscr{P}) : A \cap B \in \lambda(\mathscr{P}) \text{ for each } B \in \lambda(\mathscr{P})\}.$$

Then $\mathscr{F}_2$ is the λ-system that including $\mathscr{P}$, hence $\mathscr{F}_2 = \lambda(P)$. This implies that $\lambda(\mathscr{P})$ is a π-system, then $\lambda(\mathscr{P})$ is a σ-algebra by the first statement, so $\lambda(\mathscr{P}) \supset \sigma(\mathscr{P})$, as desired. □

► **Corollary 1. Dynkin's Lemma.** If $\mathscr{D}$ is a λ-system and $\mathscr{P} \subset \mathscr{D}$ is a π system, then $\sigma(\mathscr{P}) \subset \mathscr{D}$. That is, if $\mathscr{P}$ is a π-system, then $\sigma(\mathscr{P}) = \lambda(\mathscr{P})$ is the smallest λ-system that including $\mathscr{P}$.

The proof method of the above theorem is called the *good sets principle*. That is, to prove that a family $\mathscr{F}$ of sets that possesses a certain property, one first gathers all sets in $\mathscr{F}$ that have this property into a subfamily $\mathscr{G}$, and then proves that $\mathscr{G} = \mathscr{F}$.

Definition 2.1.9. Let X be a set and $\mathscr{M}$ be a nonempty family of subsets of X. $\mathscr{M}$ is called a ***monotone class*** if a sequence $\{A_n\} \subset \mathscr{M}$ satisfies $A_n \uparrow A$ or $A_n \downarrow A$, then $A \in \mathscr{M}$.

Theorem 2.1.8. Let $\{\mathscr{M}_\gamma\}, \gamma \in \Gamma$ be a family of monotone classes on X, then $\bigcap_\gamma \mathscr{M}_\gamma$ is also a monotone class on X. In particular, let $\mathscr{F}$ be a nonempty family of subsets of X. Then the intersection of all monotone classes that include $\mathscr{F}$ is the smallest monotone class including $\mathscr{F}$. The monotone class is denoted $M(\mathscr{F})$, in other words,

$$M(\mathscr{F}) = \bigcap \left\{C \subset 2^X : \mathscr{F} \subset C \text{ and } C \text{ is a monotone class}\right\}.$$

We refer to $M(\mathscr{F})$ as ***the monotone class generated by*** $\mathscr{F}$.

Readers may have noticed that if an algebra is closed under limits of monotone sequences, then it becomes a sigma-algebra. Thus, we have the following theorem.

Monotone Class Theorem. An algebra is a σ-algebra ***iff*** it is a monotone class. In other words,

$$\sigma\text{-algebra} = \text{algebra} + \text{monotone class}.$$

In particular, if $\mathscr{A}$ is an algebra on X, then $\sigma(\mathscr{A}) = M(\mathscr{A})$.

Proof. It is clear that a σ-algebra is both an algebra and a monotone class. To establish the converse, let $A_1, A_2, \dots \in \mathscr{A}$ and let $B_n = \bigcup_{k=1}^n A_n$, then $B_n \in \mathscr{A}$ and $B_1 \subset B_2 \subset \dots$ and $\bigcup_{n=1}^\infty B_n = \bigcup_{n=1}^\infty A_n$, i.e., $B_n \uparrow A = \bigcup_{n=1}^\infty A_n$. Since $\mathscr{A}$ is a monotone class, $\bigcup_{n=1}^\infty B_n \in \mathscr{A}$, so, $\bigcup_{n=1}^\infty A_n \in \mathscr{A}$.

For the last statement, let $\mathscr{A}$ be an algebra on X. It is clear that $\mathscr{A} \subset M(\mathscr{A}) \subset \sigma(\mathscr{A})$. Let

$$\mathscr{F}_1 = \{A \in M(\mathscr{A}) : A^c \in M(\mathscr{A}), A \cap B \in M(\mathscr{A}) \text{ for each } B \in \mathscr{A}\}.$$

Then $\mathscr{F}_1$ is a monotone class that including $\mathscr{A}$, hence $\mathscr{F}_1 = M(\mathscr{A})$. Let

$$\mathscr{F}_2 = \{A \in M(\mathscr{A}) : A \cap B \in M(\mathscr{A}) \text{ for each } B \in M(\mathscr{A})\}.$$

Then $\mathscr{F}_2$ is a monotone class that including $\mathscr{A}$ (since $\mathscr{F}_1 = M(\mathscr{A})$), hence $\mathscr{F}_2 = M(\mathscr{A})$. Thus, we have

$$\begin{aligned}
&\text{If } A \in M(\mathscr{A}), \text{ then } A^c \in M(\mathscr{A}) \text{ (because } M(\mathscr{A}) = \mathscr{F}_1);\\
&\text{If } A, B \in M(\mathscr{A}), \text{ then } A \in M(\mathscr{A}). \text{ (because } M(\mathscr{A}) = \mathscr{F}_2).
\end{aligned}$$

If follows that $M(\mathscr{A})$ is an algebra. Then, $\mathscr{A}$ is a σ-algebra by the first statement, so, $M(\mathscr{A}) \supset \sigma(\mathscr{A})$, as desired. □

This subsection focuses on the π-λ system theorem and the Monotone Class Lemma. We distill these two results into the following theorem, which we call the Monotone Class Theorem,

Class Theorem. Let $\mathscr{F}$ be a nonempty family of subsets of X,

1. If $\mathscr{F}$ is an algebra, then $M(\mathscr{F}) = \sigma(\mathscr{F})$,

2. If $\mathscr{F}$ is a π-system, then $\lambda(\mathscr{F}) = \sigma(\mathscr{F})$.

This theorem states that to vertify whether a σ-algebra possesses a certain property, it suffices to check one of the following conditions:

1. There exists an algebra that generates the σ-algebra, has the property in question, and is a monotone class, or

2. There exists a π-system that generates the σ-algebra, has the property in question, and is a λ-system.

This theorem plays a fundamental role in measure theory, and we will discuss its applications in this book.

I believe the reader has already discerned the relationships among these nine families of sets during the reading process. Let us conclude this section by presenting their inclusion relationships.

$$\begin{tikzcd}
 & \text{semi-ring} \arrow[r, Rightarrow] & \text{ring} \arrow[r, Rightarrow] & \sigma\text{-ring} & \\
\pi\text{-system} \arrow[ur, Rightarrow] \arrow[drr, Rightarrow] & & & \text{monotone class} \arrow[u, Rightarrow] \arrow[r, Rightarrow] & \lambda\text{-system} \arrow[d, Rightarrow] \\
 & & \text{semi-algebra} \arrow[r, Rightarrow] & \text{algebra} \arrow[r, Rightarrow] & \sigma\text{-algebra}
\end{tikzcd}$$

2.2 Charge and Measure

In this section, we will formally understand what a measure is, as well as the basic properties of measures. Lebesgue proposed three axioms for a measure: nonnegativity and the measure of the empty set being zero, additivity, and translation invariance. These three axioms align with our everyday intuition. However, the inclusion of the third axiom imposes many restrictions and makes operations inconvenient. Therefore, for now, we define a measure as a set function that satisfies the first two properties. To distinguish it from Lebesgue's three axioms, we refer to a measure that satisfies all three of Lebesgue's axioms (i.e., a measure that is translation-invariant) as a translation-invariant measure.

Definition 2.2.1. Let $\mathscr{F}$ be a nonempty family of subsets of X. And we say a set function $\mu : \mathscr{F} \longrightarrow [-\infty, \infty]$ is

1. ***monotone*** if $A, B \in \mathscr{S}, A \subset B$, then $\mu(A) \leq \mu(B)$.
2. ***additive*** if $\bigcup_{k=1}^{n} A_k \in \mathscr{S}$ for disjoint sets $A_1, A_2, \ldots, A_n$ in $\mathscr{S}$, then
$$\mu\left(\bigcup_{k=1}^{n} A_k\right) = \sum_{k=1}^{n} \mu(A_k)$$
3. ***σ-additive*** if $\bigcup_{k=1}^{\infty} A_k \in \mathscr{S}$ for countable sequence $\{A_n\}$ of pairwise disjoint sets in $\mathscr{S}$, then
$$\mu\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \mu\left(A_k\right).$$
4. ***subadditive*** if $\bigcup_{k=1}^{n} A_k \in \mathscr{S}$ for $\{A_1, \ldots, A_n\} \subset \mathscr{S}$, then
$$\mu\left(\bigcup_{k=1}^{n} A_k\right) \leqslant \sum_{k=1}^{n} \mu\left(A_k\right)$$
5. ***σ-subadditive*** if $\bigcup_{k=1}^{\infty} A_k \in \mathscr{S}$ for a sequence $\{A_n\} \subset \mathscr{S}$, then
$$\mu\left(\bigcup_{k=1}^{n} A_k\right) \leqslant \sum_{k=1}^{n} \mu\left(A_k\right)$$
6. ***finite*** if $\mu(A) < \infty$ for each $A \in \mathscr{F}$
7. ***σ-finite*** if for every $A \in \mathscr{F}$, there exist $A_1, A_2, \cdots \in \mathscr{F}$ such that $A \subset \bigcup_{n=1}^{\infty} A_n$ and $\mu(A_n) < \infty$ for each n.

We will focus on the following two set functions: charges (finite additive measures) and measures. The author may use the terms "charge" and "finite additive measure" interchangeably in the subsequent content to refer to the same concept within this note.

Definition 2.2.2. Let $\mu : \mathscr{F} \longrightarrow [-\infty, \infty]$ be a set function on a family $\mathscr{F}$ of sets of X, we say μ is a ***charge*** (finitely additive measure) if

- μ is nonegative.
- If $\emptyset \in \mathscr{F}$, then $\mu(\emptyset) = 0$,
- μ is additive.

Definition 2.2.3. Let $\mathscr{S}$ be a semiring of a set X. And let $\mu : \mathscr{S} \longrightarrow [-\infty, \infty]$ be a ***set function*** on a $\mathscr{S}$, it is a ***measure*** if

- μ is nonegative.
- If $\emptyset \in \mathscr{F}$, then $\mu(\emptyset) = 0$,
- μ is σ-additive.

Example 2.2.1.

1. Let $X = (0, 1]$ and $\mathscr{A}$ consists of $\emptyset$ and all finite unions of halt-open intervals $(a, b]$ contained in X. Let f be an arbitrary function on $[0, 1]$. Define $\nu_f((a, b]) = f(b) - f(a)$, and extend ν_f to be additive on $\mathscr{A}$. Then ν_f is an additive charge.

2. Let Ω be a set and Σ be a σ-algebra on Ω with a σ-finite measure μ. Suppose $\{\Omega_n\}$ is a partition of Ω and $0 < \mu(\Omega_n) < \infty$. Define
$$\nu(A) = \sum_{n=1}^{\infty} \frac{\mu(A \cap \Omega_n)}{2^n \mu(\Omega_n)}, A \in \Sigma.$$
Then ν is a measure on Σ and $\nu(\Omega) = 1$.

3. Let X be a set define a set function $\mu : 2^X \to [-\infty, \infty]$ by $\mu(A) = \#A, A \in 2^X$. Then μ is a measure, and we refer to μ as ***counting measure***.

4. Let $\mathscr{F} = \{A \subset \mathbf{N} : A \text{ or } A^c \text{ is finite}\}$ and define a set function by
$$\mu(A) = \begin{cases} 0, & A \text{ is finite,} \\ \infty, & A \text{ is infinite.} \end{cases}$$
Then μ is a charge but not a measure.

Note that, under the general definition, the concepts of "charge" and "measure" are mutually exclusive. That is, a charge is not necessarily a measure, and a measure is not necessarily a charge. Perhaps the first statement is quite intuitive, but the second one may seem puzzling. Let me provide an example.

Example 2.2.2. Let $\mathscr{F} = 2^{\mathbf{R}} \setminus \emptyset$, define a set function $\mu : \mathscr{F} \to [-\infty, \infty]$ by

$$\mu(A) = \begin{cases} 0, & A = \{1,2\}, \\ \#A, & A \neq \{1,2\}, \end{cases}$$

It is clear that μ is a measure on $\mathscr{F}$, but not a charge ($\mu(\{1\}) + \mu(\{2\}) = 2$ and $\mu(\{1,2\}) = 0$). If the given family of sets includes the empty set, then a measure must be a finitely additive measure: let $A_1, A_2, \ldots, A_n \in \mathscr{F}$, then

$$A_1 \cup A_2 \cup \ \ldots \ \cup A_n \ = A_1 \cup A_2 \cup \ \ldots \ \cup A_n \ \cup \emptyset \cup \emptyset \cup \ldots.$$

So, measure is itself a charge on every semiring, ring, algebra, σ-algebra and so on. However, for the set function defined in this example, it is not even a measure.

In our following study in this book, we will focus on the families of sets that include empty set, like semi-rings, rings, algebras, and σ-algebras. So, unless specified, we will always assume any "charge" or "measure" is defined on a semiring. Now, their fundamental properties are presented below.

Proposition 2.2.1. Every charge is monotone and subadditive. Every measure is monotone and σ-subadditive.

Proof. Let $\mathscr{S}$ be a semiring and let $\mu : \mathscr{S} \longrightarrow [0, \infty]$ be a charge. Let $A, B \in \mathscr{S}$ with $A \subset B$. Then there exist pairwise disjoint sets $C_1, C_2, \ldots, C_n \in \mathscr{S}$ such that $B \setminus A = \bigcup_{k=1}^{n} C_k$. Since the sets $A, C_1, C_2, \ldots, C_n$ are pairwise disjoint and $B = A \cup C_1 \cup C_2 \cup \ \ldots \ \cup C_n$,

$$\mu(A) \leq \mu(A) + \mu(C_1) + \cdots + \mu(C_n) = \mu(A \cup C_1 \cup \cdots \cup C_n) = \mu(B).$$

Let $A_1, A_2, \ldots, A_n \in \mathscr{S}$ satisfy $A = \bigcup_{k=1}^{n} A_k \in \mathscr{S}$. Take $B_1 = A_1, B_2 = A_2 \setminus A_1, \ldots, B_n = A_n \setminus \bigcup_{k=1}^{n-1} A_k$. Then $B_1, \ldots, B_n$ are pairwise disjoint and $A = \bigcup_{k=1}^{n} B_k$. From proposition 1.2.7, we can write $B_k = \bigcup_{j=1}^{n_k} C_j^k$ where $C_j^k \in \mathscr{S}$ and are pairwise disjoint, then

$$\mu(A) = \mu\left(\sum_{k=1}^{n} \sum_{j=1}^{n_k} C_j^k\right) = \sum_{k=1}^{n} \sum_{j=1}^{n_k} \mu(C_j^k).$$

Next, we claim that

$$\sum_{j=1}^{n_k} \mu(C_j^k) \leq \mu(A_k), \ \text{for each } k.$$

From proposition 1.2.7, we can write $A_k \setminus \bigcup_{j=1}^{n_k} C_j^k = \bigcup_{i=1}^{l} D_i$, where $D_i \in \mathscr{S}$ and are pairwise disjoint, and then $A_k = \bigcup_{i=1}^{l} D_i \cup \left(\bigcup_{j=1}^{n_k} C_j^k\right)$, which implies that

$$\sum_{j=1}^{n_k} \mu(C_j^k) \leq \mu(A_k),$$

thus, $\mu(A) \leq \sum_{k=1}^{n} \mu(A_k)$, as desired. □

Furthermore, a measure possesses several more important properties, which originate from its countable additivity, as shown below We will use these properties extensively throughout this book.

The Continuity of Measure. Let $\mu : \mathscr{S} \longrightarrow [0, \infty]$ be a measure on a semiring, and let $\{A_n\}$ be a sequence in $\mathscr{S}$, we have

1. If $A_n \uparrow A$ and $A \in \mathscr{S}$, then $\mu(A_n) \uparrow \mu(A)$.
2. If $A_n \downarrow A$ and $A \in \mathscr{S}$, $\mu(A_k) < \infty$ for some k, then $\mu(A_n) \downarrow \mu(A)$.

Proof. For (1), let $\{A_n\}$ be a sequence of sets in $\mathscr{S}$ with $A_n \subset A_{n+1}$ for each n and $A = \bigcup_{n=1}^{\infty} A_n \in \mathscr{S}$. If $\mu(A_n) = \infty$ for some n, then $\mu(A_n) \uparrow \mu(A)$ is trivial. So assume that $\mu(A_n) < \infty$ for each n. We write each set $A_k \setminus A_{k-1}$ as a finite pairwise disjoint union of sets in $\mathscr{S}$, that is $A_k = A_{k-1} \cup C_1^k \cup \cdots \cup C_{m_k}^k$ (assume $A_0 = \emptyset$). Then $\sum_{j=1}^{m_k} \mu(C_j^k) = \mu(A_k) - \mu(A_{k-1})$. Thus

$$\begin{aligned}\mu(A) &= \sum_{k=1}^{\infty} \sum_{j=1}^{m_k} \mu(C_j^k) = \sum_{k=1}^{\infty} [\mu(A_k) - \mu(A_{k-1})] \\ &= \lim_{n\to\infty} \sum_{k=1}^{n} [\mu(A_k) - \mu(A_{k-1})] = \lim_{n\to\infty} \mu(A_n).\end{aligned}$$

For (2), WLOG, assume that $\mu(A_1) < \infty$. Then $A_1 \setminus A = \bigcup_{k=1}^{\infty} (A_k \setminus A_{k+1})$. Once again we can write each set $A_k \setminus A_{k+1}$ as a finite pairwise disjoint union of sets in $\mathscr{S}$, that is $A_k = A_{k+1} \cup C_1^k \cup \cdots \cup C_{m_k}^k$. Thus,

$$\begin{aligned}\mu(A_1) - \mu(A) &= \sum_{k=1}^{\infty} \sum_{j=1}^{m_k} \mu(C_j^k) = \sum_{k=1}^{\infty} [\mu(A_k) - \mu(A_{k-1})] \\ &= \lim_{n\to\infty} \sum_{k=1}^{n} [\mu(A_k) - \mu(A_{k+1})] = \mu(A_1) - \lim_{n\to\infty} \mu(A_n).\end{aligned}$$

□

►**Corollary 1.** Let $\mu : \mathscr{S} \longrightarrow [0, \infty]$ be a measure on a semiring, and let $\{A_n\}$ be a sequence in $\mathscr{S}$, we have

1. $\mu(\liminf_{n\to\infty} A_n) \leq \liminf_{n\to\infty} \mu(A_n)$.
2. If $\mu(\bigcup_{n=1}^{\infty} A_n) < \infty$, then $\limsup_{n\to\infty} \mu(A_n) \leq \mu(\limsup_{n\to\infty} A_n)$.

3. If $\lim\limits_{n\to\infty} A_n$ exists and $\mu(\bigcup_{n=1}^{\infty} A_n) < \infty$, then $\lim\limits_{n\to\infty} \mu(A_n) = (\lim\limits_{n\to\infty} A_n)$.

The following theorem describes the relationship between finitely additive measures and measures. It is a very important theorem that will be encountered multiple times later: when we need to prove that a set function is a measure, but can only directly prove that it satisfies finite additivity, and cannot directly prove its σ-additivity, it suffices to demonstrate that it satisfies σ-subadditivity.

Theorem 2.2.2. Let $\mathscr{S}$ be a semiring on X and let μ be a charge on $\mathscr{S}$, then μ is a measure ***iff*** μ is a σ-subadditive charge.

Proof. Assume that μ is σ-subadditive and let $A_1, A_2, \cdots \in \mathscr{S}$ be a sequence of pairwise disjoint sets with $A = \bigcup_{n=1}^{\infty} A_n \in \mathscr{S}$. Then

$$\mu(A) \leq \sum_{n=1}^{\infty} \mu(A_n).$$

On the other hand, $A = \left(\bigcup_{k=1}^{n} A_k\right) \cup \left(A \setminus \left(\bigcup_{k=1}^{n} A_k\right)\right)$ for each n. By proposition 1.1.1, there exist pairwise disjoint sets $C_1^n, C_2^n, \ldots, C_{m_n}^n$ such that

$$A \setminus \left(\bigcup_{k=1}^{n} A_k\right) = \bigcup_{j=1}^{m_n} C_j^n \text{ for each } n.$$

Therefore, $\mu(A) = \sum_{k=1}^{n} \mu(A_k) + \sum_{j=1}^{m_n} \mu(C_j^n) \geq \sum_{k=1}^{n} \mu(A_k)$ for each n. Letting $n \to \infty$, we obtain $\mu(A) \geq \sum_{n=1}^{\infty} \mu(A_n)$, as desired. □

Theorem 2.2.3. Let $\mathscr{A}$ be an algebra on X and let μ be a finite charge on $\mathscr{A}$, then μ is a measure ***iff*** it satisfies $\mu(A_n) \downarrow 0$ whenever $A_n \downarrow \emptyset$.

Proof. Let $\{B_n\}$ be a sequence of pairwise disjoint sets in $\mathscr{A}$ with $\bigcup_{n=1}^{\infty} B_n \in \mathscr{A}$. Suppose $A_m = \bigcup_{n=m+1}^{\infty} B_n$, then $A_m \downarrow \emptyset$. Moreover,

$$\begin{aligned}
\mu\left(\bigcup_{n=1}^{\infty} B_n\right) &= \mu\left(\left(\bigcup_{n=1}^{m} B_n\right) \cup A_m\right) \\
&= \mu\left(\bigcup_{n=1}^{m} B_n\right) + \mu(A_m) \\
&= \sum_{n=1}^{m} \mu(B_n) + \mu(A_m).
\end{aligned}$$

Letting $m \to \infty$, we obtain $\mu\left(\bigcup_{n=1}^{m} B_n\right) = \sum_{n=1}^{\infty} \mu(B_n)$, as desired. □

The following theorem carries an implication of extension. It states that if two measures (defined on the same family $\mathscr{F}$ of sets) assign equal values to the sets within a subfamily under certain conditions, then they are equivalent measures (two measures are called equivalent if they assign the same value to every set in the family $\mathscr{F}$).

Theorem 2.2.4. Let Σ be a σ-algebra on X and let $\mathscr{P}$ be a π system on X such that $\Sigma = \sigma(\mathscr{P})$. If μ and ν are finite measures on Σ that satisfy $\mu(X) = \nu(X)$ and $\mu(P) = \nu(P)$ for each $P \in \mathscr{P}$, then $\mu = \nu$.

Proof. Let $\mathscr{D} = \{A \in \Sigma : \mu(A) = \nu(A)\}$. It is clear that $\mathscr{D}$ is a λ system. By Dynkin's Lemma, $\sigma(\mathscr{P}) \subset \mathscr{D}$ since $\mathscr{P} \subset \mathscr{D}$. Thus, $\mu(A) = \nu(A)$ holds for every $A \in \Sigma$, and the proof is complete. □

►**Corollary 1.** Let Σ be a σ-algebra on X and let $\mathscr{P}$ be a π system on X such that $\Sigma = \sigma(\mathscr{P})$. If μ and ν are measures on Σ that agree on $\mathscr{P}$ and if there exists an increasing sequence $\{P_n\} \subset \mathscr{P}$ have finite measure under μ and ν, and satisfy $\bigcup_n P_n = X$, then $\mu = \nu$.

Proof. For each n, define $\mu_n(A) = \mu(A \cap P_n)$ and $\nu_n(A) = \nu(A \cap P_n)$ for $A \in \Sigma$. It is clear that $\mu_n(P) = \nu_n(P)$ for every $P \in \mathscr{P}$. Theorem 2.2.4 implies that for each n we have $\mu_n = \nu_n$. Moreover,

$$\mu(A) = \lim_{n\to\infty} \mu_n(A) = \lim_{n\to\infty} \nu_n(A) = \nu(A)$$

holds for every $A \in \Sigma$, the measures μ and ν must be equal. □

2.3 Construction of Measures — Act I

Although semirings are intuitive and simple to construct, their structure is too limited for some critical applications, such as probability and integration (which we will study in Chapter 4). Specifically, semirings are not closed under countable unions, countable intersections, or complements. This limitation has significant consequences: in probability, if A represents an event, then A^c represents its complement—the event that A does not occur. However, A^c may not lie within the semiring and thus cannot be properly described. Therefore, we must extend our study to richer structures, such as algebras and σ-algebras.

Defining a measure on an algebra or σ-algebra is difficult or even impossible (although one could assign measure zero to all sets, such a definition is pointless). Fortunately, defining a measure or charge on a semiring is straightforward. A standard example is the semiring of half-open intervals on the real line, where ”length” serves as a natural measure. Therefore, the only thing we can do is to construct an algebra or a σ-algebra from the original semiring and refine the original charge or measure to preserve its properties on this new family.

A From a Semiring to a Ring

We will first discuss how to construct a ring from a given semiring. (The method for constructing an algebra from a semialgebra is identical, so we will focus on the case of rings.) Recall the method of constructing a ring from a semiring (Theorem 1.1.1)? This offers a key insight, and we can attempt a similar approach to see how to refine the original charge so that its properties are preserved on the new ring.

Theorem 2.3.1. Let $\mathscr{S}$ be a semi-ring on X, then the family $\mathscr{R}$ of all sets that can be written as a finite union of pairwise disjoint elements of $\mathscr{S}$ is the smallest ring containing $\mathscr{S}$. That is

$$\mathscr{R} = \mathscr{R}(\mathscr{S}) = \left\{ \bigcup_{k=1}^{n} A_k : A_k \in \mathscr{S}_r, A_k \cap A_j = \emptyset \text{ for } k \neq j \right\} \text{ is a ring.}$$

Moreover, if the semiring $\mathscr{S}$ is itself a semi-algebra, then $\mathscr{R}$ is an algebra.

Theorem 2.3.2. Let μ be a charge on a semiring. Then there exists a unique charge $\overline{\mu}$ on the ring $\mathscr{R} = \mathscr{R}(\mathscr{S})$ that extends μ. This extension is given explicitly by: for any $A \in \mathscr{R}$ with a pairwise disjoint representation $A = \bigcup_{k=1}^{n} A_k$, where $A_k \in \mathscr{S}$, we define

$$\tilde{\mu} = \sum_{k=1}^{n} \mu(A_k).$$

Proof. We begin by vertifying the correctness of the definition, i.e., it does not depend on the choic of the partion of A in $\mathscr{R}(\mathscr{S})$. Let $A = \bigsqcup_{k=1}^{n} A_k = \bigsqcup_{j=1}^{m} B_j$. Let $C_{ij} = A_i \cap B_j$. Then $A_i = \bigsqcup_{j=1}^{m} C_{ij}$ and $B_j = \bigsqcup_{i=1}^{n} C_{ij}$. Thus,

$$\sum_{i=1}^{n} \mu(A_i) = \sum_{i=1}^{n} \left(\sum_{j=1}^{m} \mu(C_{ij}) \right) = \sum_{j=1}^{m} \left(\sum_{i=1}^{n} \mu(C_{ij}) \right) = \sum_{j=1}^{n} \mu(B_j)$$

Then the definition is well defined.

For the finite additivity of $\tilde{\mu}$, let $A, B \in \mathscr{R}(\mathscr{S})$ be disjoint sets, and $A = \bigsqcup_{i=1}^{n} A_i$ and $B = \bigsqcup_{j=1}^{m} B_j$. Then $A \sqcup B$ is the finite union of pairwise disjoint elements of $\mathscr{S}$. Therefore,

$$\tilde{\mu}(A \sqcup B) = \sum_{j=1}^{n} \mu(A_i) + \sum_{j=1}^{m} \mu(B_j) = \tilde{\mu}(A) + \tilde{\mu}(B).$$

For the monotonicity, let $A, B \in \mathscr{R}(\mathscr{S})$ with $A \subset B$, then $B \setminus A \in \mathscr{R}(\mathscr{S})$ and $B = A \sqcup (B \setminus A)$. By the finite additivity,

$$\tilde{\mu}(B) = \tilde{\mu}(A) + \tilde{\mu}(B \setminus A) \geq \tilde{\mu}(A).$$

Finally, to prove the uniqueness, let ν is another charge on the ring $\mathscr{R}(\mathscr{S})$ that extends μ, i.e., $\nu(A) = \mu(A)$ for all $A \in \mathscr{S}$. Let $A \in \mathscr{R}(\mathscr{S})$ be any set and $A = \bigsqcup_{k=1}^{n} A_k$. Since ν is a charge and agrees with μ on S, we have

$$\nu(A) = \nu\left(\bigsqcup_{k=1}^{n} A_k\right) = \sum_{k=1}^{n} \nu(A_k) = \sum_{k=1}^{n} \mu(A_k) = \tilde{\mu}(A).$$

Therefore, $\nu(A) = \tilde{\mu}(A)$ for all $A \in \mathscr{R}(\mathscr{S})$. This shows that the extension is unique. □

B Method I of the Construction of Measure

As already mentioned, the most natural domain of definition for a countably additive measure is not an algebra, but a σ-algebra of sets. Hence, it is highly desirable to know how to extend a countably additive measure to the σ-algebra generated by the algebra.

Motivated by the successful extension from a semiring to a ring in Theorems 1.3.1 and 1.3.2—which was achieved by taking finite disjoint unions, we now explore whether taking countable disjoint unions can further extend the semialgebra $\mathscr{S}$ to a family that is closed under countable operations:

$$\mathscr{F} = \left\{\bigcup_{k=1}^{\infty} A_k : A_k \in \mathscr{S}_r, A_k \cap A_j = \emptyset \text{ for } k \neq j\right\}.$$

and extend the charge μ to $\mathscr{F}$ by the definition $\overline{\mu}(A) = \sum_{n=1}^{\infty} \mu(A_n)$ for $A = \sum_{n=1}^{A_n} \in \mathscr{F}$. We have already indicated, however, that $\mathscr{F}$ does not form a σ-algebra, or even an algebra. Specifically, it is not closed under relative complements.

Therefore, even if one could overcome the technical challenges of defining the measure on C correctly, this new domain would be too narrow and fragile to serve as a proper domain for a measure. And then we would be confronted with the problem of how to define the measure for the complements of sets in C and for other sets generated by more complex operations. This “patching” approach to extension would quickly become unworkable.

To circumvent these intricate difficulties entirely, we instead adopt a more powerful and systematic mathod – the ***Outer Measure*** approach. This method abandons the attempt to construct the domain explicitly. Instead, it first defines an “outer measure” on all subsets. It then carefully selects those subsets that behave well, i.e., those that satisfy the additivity condition – the ***measurable sets***. This selected family natrually forms the ideal domain for the measure: σ-algebra. The method first appeared in Lebesgue’s 1902 doctoral thesis, though it was not extensively explored there.

Definition 2.3.1. Let X be a set and $\mu^* : 2^X \longrightarrow [0, \infty]$ be a set function defined on the power set of X. Then the funtion μ^* is said to be an ***outer measure*** if it is *monotone, σ-subadditive, and satisfies* $\mu^*(\emptyset) = 0$.

In chapter 1, we introduced the Jordan outer content, it not be an outer measure. It obeys the empty set and monotonicity properties, but is only finitely subadditive rather than countably subadditive. In the later section, we will study Lebesgue outer measure, it is a model example pf an abstract outer measure.

At the beginning of this subsection, we informed the readers about the method of selecting "measurable sets" using outer measures to define "measure". Before learning Carathéodory's method, when trying to define a measure and measurable sets related to this outer measure, one naturally tends to imitate the approach used by Monsieur Lebesgue, which is based on the equality of inner and outer measures. However, this method struggles to support abstract outer measures, as illustrated by the following example.

Example 2.3.1. Let $X = \{1, 2, 3\}$ and let $\mu^*(\emptyset) = 0, \mu^*(X) = 2, \mu^*(A) = 1$ for every $A \subsetneq X$. It is clear that μ^* is an outer measure. Suppose now that we use the procedure that worked so well for the Lebesgue measure. We define the inner measure for this example as

$$\mu_*(A) = \mu^*(X) - \mu^*(X \setminus A) = 2 - \mu^*(X \setminus A).$$

If we then call A measurable if $\mu^*(A) = \mu_*(A)$, and let $\mu(A) = \mu^*(A)$. We find that all the subsets of X are measurable by this definition, but μ is clearly not additive on 2^X. *The classical inner-outer measure procedure completely fails to work in this simple example !* In fact, for $E = \{1, 2\} \supset A = \{1\}$, we see that

$$\mu^*(A) + \mu^*(E \setminus A) = 2 > 1 = \mu^*(E),$$
$$\mu^*(A) + \mu^*(X \setminus A) = 2 = 2 = \mu^*(X).$$

Thus, the definition of the inner measure in this example is not well defined. The reason lies precisely in the fact that the definition of outer measure in this example does not follow the definition of Lebesgue outer measure, which makes the definition of inner measure ill-posed.

This, however, exposes the significant constraints of defining measurability through the equality of inner and outer measures.

This consideration lead natrually to the following criterion of measurablility. It is created by Carathéodory. In chapter 1, we have already known the Lebesgue's notion of a measurable set, defined using inner and outer measures. Carathéodory's definition is more general and avoids the introduction of inner measures.

Definition 2.3.2. Let X be a set and $\mu^* : 2^X \longrightarrow [0, \infty]$ be an outer measure. Then a subset A of X is said to be ***μ^*-measurable*** if for every $S \subset X$,

$$\mu^*(S) = \mu^*(S \cap A) + \mu^*(S \cap A^c)$$

The Carathéodory condition is something about "all sets" decomposing properly. Carathéodory explains in his notes: a known characterization of Lebesgue measurability involved saying "all intervals" decompose properly. So he generalized to settings other than the real line by using "all sets".

Lemma 2.3.1. Let $\mu^* : 2^X \longrightarrow [0, \infty]$ be an outer measure. Then the family Σ_{μ^*} of sets that are μ^*-measurable is a σ-algebra.

Proof. First, since $\mu^*(\emptyset) = 0$, for any $S \subset X$ we have

$$\mu^*(S) = \mu^*(S) + 0 = \mu^*(S \cap X) + \mu^*(S \cap X^c),$$

which implies $X \in \Sigma_{\mu^*}$. To show the closure under complementation, let $A \in \Sigma_{\mu^*}$, then for any $S \subset X$,

$$\mu^*(S) = \mu^*(S \cap A^c) + \mu^*(S \cap (A^c)^c) = \mu^*(S \cap A^c) + \mu^*(S \cap A).$$

To show the closure under finite unions, let $A_1, A_2, \ldots, A_n \in \Sigma_{\mu^*}$. WLOG assume that they are pairwise disjoint. And for any subset $S \subset X$, we have,

$$\mu^*(S) = \mu^*(S \cap A_1) + \mu^*(S \cap A_1^c).$$
$$\mu^*(S \cap A_1^c) = \mu^*((S \cap A_1^c) \cap A_2) + \mu^*((S \cap A_1^c) \cap A_2^c).$$

Thus, $\mu^*(S) = \mu^*(S \cap A_1) + \mu^*(S \cap A_2) + \mu^*(S \cap (A_1 \cup A_2)^c)$. Continuing this process, we obtain

$$\mu^*(S) = \sum_{k=1}^{n} \mu^*(S \cap A_k) + \mu^*\left(S \cap \left(\bigcup_{k=1}^{n} A_k\right)^c\right).$$

Since outer measure is subadditive, $\sum_{k=1}^{n} \mu^*(S \cap A_k) \geq \mu^*\left(S \cap \left(\bigcup_{k=1}^{n} A_k\right)\right)$ which implies that $\bigcup_{k=1}^{n} A_k \in \Sigma_{\mu^*}$.

To show the closure under countable unions, let $\{A_n\} \subset \Sigma_{\mu^*}$. Without loss of generality, assume that they are pairwise disjoint. Let $A = \bigsqcup_{n=1}^{\infty} A_n$. For any $S \subset X$, we have

$$\mu^*(S) = \mu^*\left(S \cap \left(\bigcup_{k=1}^{n} A_k\right)\right) + \mu^*\left(S \cap \left(\bigcup_{k=1}^{n} A_k\right)^c\right) \text{ for all } n \in \mathbf{N}^+.$$

Since $\bigcup_{k=1}^{\infty} A_k \supset \bigcup_{k=1}^{n} A_k$, by the monotonicity of outer measure,

$$\begin{aligned}\mu^*(S) &= \mu^*\left(S \cap \left(\bigcup_{k=1}^{n} A_k\right)\right) + \mu^*\left(S \cap \left(\bigcup_{k=1}^{n} A_k\right)^c\right) \text{ for all } n \in \mathbf{N}^+. \\ &\geq \sum_{k=1}^{n} \mu^*(S \cap A_k) + \mu^*(S \cap A) \text{ for all } n \in \mathbf{N}^+.\end{aligned}$$

Taking the limit as $n \longrightarrow \infty$, $\mu^*(S) \geq \sum_{k=1}^{\infty} \mu^*(S \cap A_k) + \mu^*(S \cap A^c) \geq \mu^*(S \cap A) + \mu^*(S \cap A^c)$. where the last inequality follows from the countable subadditivity of μ^*. Therefore, $A \in \Sigma_{\mu^*}$. Completing the proof. $\square$

Lemma 2.3.2. The restriction of the outer measure μ^* to the σ-algebra Σ_{μ^*} is countably additive (so a measure).

Proof. According to the preceding Theorem, take $S = A = \bigsqcup_{k=1}^{\infty} A_k \in \Sigma_{\mu^*}$,

$$\mu^*(A) \geq \sum_{k=1}^{\infty} \mu^*(A_k) + \mu^*(\emptyset) = \sum_{k=1}^{\infty} \mu^*(A_k).$$

According to the proposition 2.2.4, outer measure is σ-subadditive, so

$$\mu^*(A) \leq \sum_{k=1}^{\infty} \mu^*(A_k).$$

Thus, $\mu^*|_{\Sigma_{\mu^*}}$ is countably additive. □

Method I construction of outer measure. Let X be a set and $\mathscr{F}$ be nonempty family of subsets of X, $\mu : \mathscr{F} \longrightarrow [0,\infty]$ be a set function on $\mathscr{F}$. Define a set function $\mu^*_{\mathscr{F}} : 2^X \longrightarrow [0,\infty]$ by $\mu^*_{\mathscr{F}}(\emptyset) = 0$ and for $A \subset X, A \neq \emptyset$,

$$\mu^*_{\mathscr{F}}(A) = \inf\left\{\sum_{k=1}^{\infty} \mu(A_k), A_k \in \mathscr{F}, A \subset \bigcup_{k=1}^{\infty} A_k\right\},$$

where an empty infimum is taken as ∞. Then $\mu^*_{\mathscr{F}}$ is an outer measure, which is called the ***outer measure with respect to*** $\mu : \mathscr{F} \to [0,\infty]$ or induced by μ.

Proof. It is clear that $\mu^*_{\mathscr{F}}(\emptyset) = 0$ and that $\mu^*_{\mathscr{F}}$ is monotone. To vertify the σ-additivity, let $\{A_n\}$ be a sequence of subsets of X. We show that

$$\mu^*_{\mathscr{F}}\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu^*_{\mathscr{F}}(A_n).$$

If $\mu^*_{\mathscr{F}}(A_k) = \infty$ for any n, there is nothing to prove, wo we suppose that each is finite. Let $\epsilon > 0$, there exists a sequence $\{F_{n,k}\}$ of sets of $\mathscr{F}$ such that $A_n \subset \bigcup_{k=1}^{\infty} F_{n,k}$ for each n, then $\bigcup_{n=1}^{\infty} A_n \subset \bigcup_n \bigcup_k F_{n,k}$ and

$$\sum_{k=1}^{\infty} \mu(F_{k,n}) \leq \mu^*_{\mathscr{F}}(A_n) + \frac{\epsilon}{2^n} \text{ for each } n.$$

Thus,

$$\mu^*_{\mathscr{F}}\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty}\sum_{k=1}^{\infty} \mu(F_{n,k}) \leq \sum_{n=1}^{\infty} \mu^*_{\mathscr{F}}(A_n) + \epsilon.$$

we conclude that

$$\mu^*_{\mathscr{F}}\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu^*_{\mathscr{F}}(A_n)$$

□

Method I is very useful and important, but it can have an important flaw when X is a metric space. In section 3.2. we will discuss this flaw and see a variant, called Method II, overcomes this problem.

From this point forward, to simplify notation, we will often denote the outer measure $\mu^*_{\mathscr{F}}$ simply by μ^*, when the underlying family $\mathscr{F}$ and set function μ are clear from the context.

Definition 2.3.3. Let $\mu : \mathscr{F} \to [0, \infty]$ be a set function defined on a nonempty family $\mathscr{F}$ and μ^* be the outer measure induced by μ. The measure $\tilde{\mu} = \mu^*|_{\Sigma_{\mu^*}}$ on the σ-algebra Σ_{μ^*} is called the ***Carathéodory measure*** induced by μ.

C From an Algebra to a Sigma-Algebra

Let us now return to the central thread of our discussion. Our goal is to use the outer measure approach, which allows us to obtain a σ-algebra from the original semiring and, ultimately, a measure from the original charge.

Theorem 2.3.4. Let $\mu : \mathscr{R} \longrightarrow [0, \infty]$ be a charge on a ring $\mathscr{R}$ of X. Then the *Carathéodory measure* $\tilde{\mu} : \Sigma_{\mu^*} \longrightarrow [0, \infty]$ induced by μ is an extension of μ.

Proof. Let $A \in \mathscr{R}, S \subset X$, and let $\epsilon > 0$ then there exists a sequence $\{A_k\} \subset \mathscr{R}$ such that

$$\sum_{k=1}^{\infty} \mu(A_k) \leq \mu^*(S) + \epsilon \text{ and } S \subset \bigcup_{k=1}^{\infty} A_k$$

Since $\mathscr{R}$ is a ring, $A_k \cap A \in \mathscr{R}$. Therefore,

$$\mu(A_k) = \mu(A_k \cap A) + \mu(A_k \cap A^c) \text{ for each } k.$$

Then we obtain,

$$\sum_{k=1}^{\infty} \mu(A_k) = \sum_{k=1}^{\infty} \mu(A_k \cap A) + \sum_{k=1}^{\infty} \mu(A_k \cap A^c).$$

Observe that $\{A_k \cap A\}$ and $\{A_k \cap A^c\}$ cover $S \cap A$ and $S \cap A^c$, respectively,

$$\sum_{k=1}^{\infty} \mu(A_k \cap A) \geq \mu^*(S \cap A), \quad \sum_{k=1}^{\infty} \mu(A_k \cap A^c) \geq \sum_{k=1}^{\infty} \mu^*(S \cap A^c).$$

Finally, we obtain,

$$\mu^*(S) + \epsilon \geq \mu^*(S \cap A) + \mu^*(S \cap A^c)$$

Thus, $\mu^*(S) = \mu^*(S \cap A) + \mu^*(S \cap A^c)$ since μ is σ-subadditive. So $\mu(A) = \mu^*(A)$ for each $A \in \mathscr{R}$. By Lemma 2.2.5.b., $\mu(A) = \mu^*(A) = \tilde{\mu}(A)$. □

The Carathéodory-Hahn Theorem.
If $\mu : \mathscr{S} \longrightarrow [0,\infty]$ is a charge on a semiring $\mathscr{S}$ of X. Then the Carathéodory measure $\tilde{\mu} : \Sigma_{\mu^*} \longrightarrow [0,\infty]$ induced by μ is an extension of μ. Moreover, if μ is σ-finite, then $\tilde{\mu}$ is the only measure on Σ_{μ^*} that extends μ.

Proof. By Lemma 2.2.2.b, $\tilde{\mu}$ is the unique charge on $\mathscr{R}\{\mathscr{S}\}$ that extends μ. Moreover, by Lemma 2.2.2.a, $\mathscr{R}(\mathscr{S})$ is a ring. Then we use the theorem 2.2.9., the Carathéodory measure induced by $\tilde{\mu} : \mathscr{R}(\mathscr{S}) \longrightarrow [0,\infty]$ is an extension of μ. Let $\tilde{\mu}^*$ be the outer measure respect to $\mathscr{R}(\mathscr{S})$ and $\tilde{\mu}$. We claim that $\tilde{\mu}^* = \mu^*$. Indeed, since $\mathscr{S} \subset \mathscr{R}(\mathscr{S})$, $\tilde{\mu}^*(E) \leq \mu^*(E)$ for every set $E \in X$. For the opposite direction, choose $\{B_1, B_2, \ldots, B_n\} \subset \mathscr{R}(\mathscr{S})$ such that $E \subset \bigcup_{k=1}^{n} B_k$, and $B_k = \bigcup_{l=1}^{n_k} A_{kl}$ where $A_{kl} \in \mathscr{S}$. Then,

$$\sum_{k=1}^{n} \tilde{\mu}(B_k) = \sum_{k}^{n} \sum_{l}^{n_k} \mu(A_{kl}) \geq \mu^*(E).$$

Thus, $\tilde{\mu}^*(E) \geq \mu^*(E)$. Therefore, $\Sigma_{\mu^*} = \Sigma_{\tilde{\mu}^*}$ and hence $\mathscr{S} \subset \Sigma_{\mu^*}$ and $\sigma(\mathscr{S}) \subset \Sigma_{\mu^*}$. Existence proof completed.

To prove the uniqueness, let $\nu : \Sigma_{\mu^*} \longrightarrow [0,\infty]$ be a measure that extends μ. It suffice to show that $\tilde{\mu}(A) = \nu(A)$ for each $A \in \Sigma_{\mu^*}$. First, we consider $\tilde{\mu}(A) < \infty$. Since ν is an outer measure on X, let $A \subset X$ and $\{A_n\} \subset \mathscr{S}$ satisfies $A \subset \bigcup_{k=1}^{\infty} A_k$, then

$$\nu(A) \leq \sum_{k=1}^{\infty} \nu(A_k) = \sum_{k=1}^{\infty} \mu(A_k),$$

the equality in the above formula holds because the outer measure ν induced by μ, when restricted to $\mathscr{S}$, is equal to μ. So $\nu(A) \leq \mu^*(A)$ for each subset A of X. So in order to show that $\nu = \mu^*$ on Σ_{μ^*}, it suffice to show that $\mu^*(A) \leq \nu(A)$ for each $A \in \Sigma_{\mu^*}$. Let $A \in \Sigma_{\mu^*}$ with $\mu^*(A) < \infty$. Let $\epsilon > 0$, then there exists a sequence $\{A_n\} \subset \mathscr{S}$ satisfying $A \subset \bigcup_{k=1}^{\infty} A_k$ and $\sum_{k=1}^{\infty} \mu(A_k) < \mu^*(A) + \epsilon$. From proposition 1.2.7., there exists a pairwise disjoint sequence $\{C_n\} \subset \mathscr{S}$ such that $B = \bigcup_{k=1}^{\infty} A_k = \bigsqcup_{k=1}^{\infty} C_k$. Since μ^* and ν are both measures on Σ_{μ^*}, we have

$$\mu^*(B) = \sum_{k=1}^{\infty} \mu^*(C_k) = \sum_{k=1}^{\infty} \mu(C_k) = \sum_{k=1}^{\infty} \nu(C_k) = \nu\,(B)\,.$$

Moreover, $\nu(B \setminus A) \leq \mu^*(B \setminus A) = \mu^*(B) - \mu^*(A) \leq \sum_{k=1}^{\infty} \mu(A_k) - \mu^*(A) < \epsilon$. The equality in this formula holds because $\tilde{\mu}(A) < \infty$. So $\mu^*(A) \leq \mu^*(B) = \nu(B) = \nu(A) + \nu(B \setminus A) < \nu(A) + \epsilon$, which shows that $\mu^*(A) \leq \nu(A)$. Thus, $\nu(A) = \mu^*(A)$ for each $A \in \Sigma_{\mu^*}$ with $\tilde{\mu}(A) < \infty$.

In case $\tilde{\mu}(A) = \infty$, let $\{X_n\}$ be the sequence of disjoint sets of X such that $X = \bigcup_{k=1}^{\infty} X_k$ and $\mu^*(X_k) < \infty$ for each k. Then $A = \bigcup_{k=1}^{\infty}(X_n \cap A)$ where $\mu^*(A \cap X_n) < \infty$. Thus, $\mu^*(A \cap X_n) \leq \nu(A \cap X_n)$ from the preceding case.

Moreover, $\nu(A) = \nu(\bigcup_{k=1}^{\infty}(A \cap X_n)) = \bigcup_{k=1}^{\infty} \nu(A \cap X_n)$. So,

$$\infty = \mu^*(A) = \sum_{k=1}^{\infty} \mu^*(A \cap X_n) \leq \nu(A).$$

□

Remark 2.3.1. Finally, there is an important reminder that may be useful later: $\tilde{\mu}$ is not only the unique extension of μ on Σ_{μ^*}, but also the unique extension on $\sigma(\mathscr{S})$. The reason is that the proof of uniqueness still holds if $\sigma(\mathscr{S})$ is substituted in place of Σ_{μ^*}. More generally, this claim reamins valid for any σ-algebra Σ satisfying $\mathscr{S} \subset \Sigma \subset \Sigma_{\mu^*}$.

Remark 2.3.2. If μ is not σ-finite, then the Carathéodory extension μ^* is the unique extension of μ to a measure on Σ_{μ^*}. Now consider: if μ is a general charge, is the extension of μ to a measure on Σ_{μ^*} unique ? The answer is no. Here is a simple example.

Consider $X = \{0, 1\}$, $\mathscr{S} = \{\emptyset, \{0\}\}$ is a semiring and define $\mu : \mathscr{S} \longrightarrow [0, \infty]$ by $\mu(\emptyset) = 0$ and $\mu(\{0\}) = 1$ (so a charge). Note that $\Sigma_{\mu^*} = \{\emptyset, \{0\}, \{1\}, X\}$. We have $\mu^*(\{1\}) = \inf \emptyset = \infty$. In particular, μ is not σ-finite. Now, define a set function $\nu : \Sigma_{\mu^*} \longrightarrow [0, \infty]$ by $\nu(\emptyset) = 0, \nu(\{0\}) = 1, \nu(\{1\}) = \alpha, \nu(X) = 1 + \alpha$ for any $\alpha \in [0, \infty]$, is a measure that agrees with μ on $\mathscr{S}$.

D The Completion of Measure Space

Definition 2.3.4. A ***measure space*** is a triplet (X, Σ, μ), where Σ is a σ-algebra of subsets of X and $\mu : \Sigma \longrightarrow [0, \infty]$ is measure. In this case, we say the sets of Σ are ***measurable***. A measure space (X, Σ, μ) is ***complete*** if $E \in \Sigma, \mu(E) = 0 \Rightarrow A \in \Sigma$ for all $A \subset E$. In this case, μ is called ***complete***.

Theorem 2.3.6. The measure space $(X, \Sigma_{\mu^*}, \mu^*|_{\Sigma_{\mu^*}})$ is complete.

Proof. Let E be the μ^*-measurable set such that $\mu^*(E) = 0$ and let $N \subset E$. By the monotonicity of μ^*, we have

$$\begin{aligned} \mu^*(N^c \cap A) &= \mu^*(E \cap N^c \cap A) + \mu^*(E^c \cap N^c \cap A) \\ &= \mu^*(E^c \cap A) \text{ for every } A \subset X. \end{aligned}$$

Thus,

$$\begin{aligned} \mu^*(A) &= \mu^*(E \cap A) + \mu^*(E^c \cap A) \\ &\geq \mu^*(N \cap A) + \mu^*(N^c \cap A) \text{ for every } A \subset X. \end{aligned}$$

It follows that $N \in \Sigma_{\mu^*}$, as desired. □

From the monotonicity of outer measure, we know that the outer measure of a null set is also 0. Does this imply that all measure spaces are complete? This is clearly not possible. A simple example is given below,

Example 2.3.2. Let $X = \{1, 2, 3\}$ and $\Sigma = \{\emptyset, 1, 2, 3, X\}$. Define a set function μ: $\mu(\emptyset) = \mu(\{1, 2\}) = 0$ and $\mu(\{3\}) = \mu(X) = 1$. It is clear that (X, Σ, μ) is a measure space. However, $\{1\} \notin \Sigma$ and hence, μ is not complete.

If a measure space is not complete, then for a subset of a measurable set with measure zero, intuitively, its "measure" should also be zero. However, in reality, such subsets cannot be assigned a measure, which is inconvenient. Fortunately, we can transform any measure space into a complete space naturally by extending μ to a measure $\overline{\mu}$ defined on the new σ-algebra generated by Σ and the family of subsets of sets of measure zero.

Theorem 2.3.7. Let (X, Σ, μ) be a measure sapce. Let

$$\mathcal{N} = \{N \in \Sigma : \mu(N) = 0\} .$$

Let $\overline{\Sigma} = \{E \cup Z : E \in \Sigma, Z \subset N \in \mathcal{N}\}$. For each $E \cup Z \in \overline{\Sigma}$, define $\overline{\mu}(E \cup Z) = \mu(E)$ on $\overline{\Sigma}$. Then

1. $\overline{\Sigma}$ is a σ-algebra that contains Σ and all subsets of μ-null sets.
2. $\overline{\mu}$ is well-defined on $\overline{\Sigma}$.
3. $\overline{\mu}$ is a measure on $\overline{\Sigma}$ that coincides wtih μ on Σ.
4. $\overline{\mu}$ is complete.

We refer to $(X, \overline{\Sigma}, \overline{\mu})$ as the ***completion*** of the measure space (X, Σ, μ). And one should check that the measure $\overline{\mu}$ is independent with the forms of a set in $\overline{\Sigma}$, that is, $E_1 \cup Z_1 = E_2 \cup Z_2$ implies $\overline{\mu}(E_1 \cup Z_1) = \overline{\mu}(E_2 \cup Z_2)$.

Proof. For (1). It is clear that $\overline{\Sigma}$ contains X and $\emptyset$. To show that $\overline{\Sigma}$ is closed under complementation, let $A = E \cup Z$ with $E \in \Sigma, Z \subset N$ and $\mu(N) = 0$. Then

$$\begin{aligned}(E \cup Z)^c = E^c \cap Z^c &= E^c \cap Z^c \cap (N \cup N^c) \\ &= (E^c \cap N^c) \cup (E^c \cap Z^c \cap N).\end{aligned}$$

Since $E^c \cap N^c \in \Sigma$ and $N \cap E^c \cap Z^c \subset N$, $(E \cup Z)^c \in \overline{\Sigma}$. Next, we show that $\overline{\Sigma}$ is closed under countable unions. Let $\{A_n = (E_n \cup Z_n)\}$ be a sequence of sets in $\overline{\Sigma}$ with $E_n \in \Sigma, Z_n \subset N_n \in \mathcal{N}$. Then

$$\bigcup_{n=1}^{\infty}(E_n \cup Z_n) = \left(\bigcup_{n=1}^{\infty} E_n\right) \cup \left(\bigcup_{n=1}^{\infty} Z_n\right).$$

Since $E_n \in \Sigma$, $\bigcup E_n \in \Sigma$. Moreover, $\bigcup Z_n \subset \bigcup N_n \in \mathcal{N}$ since $\mu(\bigcup N_n) \leq \sum \mu(N_n) = 0$. Thus, $\bigcup_{n=1}^{\infty}(E_n \cup Z_n) \in \overline{\Sigma}$. Therefore, $\overline{\Sigma}$ is a σ-algebra. It is straightforward to prove the minimality of $\overline{\Sigma}$.

For (2). Suppose that $A = E_1 \cup Z_n = E_2 \cup Z_2$ with $E_1, E_2 \in \Sigma, Z_1 \subset N_1 \in \mathcal{N}, Z_2 \subset N_2 \in \mathcal{N}$. We show that $\mu(E_1) = \mu(E_2)$. We observe that

$$E_1 \subset A = E_2 \cup Z_2 \subset E_2 \cup N_2.$$

Thus,

$$\mu(E_1) \leq \mu(E_2) + \mu(N_2) = \mu(E_2).$$

Similarly, $\mu(E_2) \leq \mu(E_1)$. Therefore, $\overline{\mu}$ is well defined.

For (3). We first show that $\overline{\mu}$ is a measure on $\overline{\Sigma}$. Let $\{A_n = E_n \cup Z_n\}$ be a sequence of sets in $\overline{\Sigma}$. Note that $\bigcup_{n=1}^{\infty} Z_n \subset \bigcup_{n=1}^{\infty} N_n \in \mathcal{N}$, thus

$$\begin{aligned}\overline{\mu}\left(\bigcup_{n=1}^{\infty} A_n\right) &= \overline{\mu}\left(\left(\bigcup_{n=1}^{\infty} E_n\right) \cup \bigcup_{n=1}^{\infty} Z_n\right)\\ &= \mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu\left(E_n\right) = \sum_{n=1}^{\infty} \overline{\mu}\left(A_n\right).\end{aligned}$$

Thus, $\overline{\mu}$ is a measure on $\overline{\Sigma}$. And it is clear that $\mu = \overline{\mu}$ on Σ.

For (4), let $A \in \overline{\Sigma}$ with $\overline{\mu}(A) = 0$. Write $A = E \cup Z, E \in \Sigma, Z \subset N \in \mathcal{N}$. Since $\overline{\mu}(A) = \overline{\mu}(E \cup Z) = 0$, $\mu(E) = 0$, so $A = E \cup Z \subset N_0 \in \mathcal{N}$. Therefore, any subset B of A satisfies $B \subset A \subset N_0 \in \mathcal{N} \subset \overline{\Sigma}$, and so $\overline{\mu}$ is complete. $\square$

Now there is another question: Is this completion the "smallest" completion? The following proposition addresses this issue.

Proposition 2.3.1. $(X, \overline{\Sigma}, \overline{\mu})$ is the smallest completion of (X, Σ, μ).

Proof. Suppose that $(X, \overline{\overline{\Sigma}}, \overline{\overline{\mu}})$ is another completion of (X, Σ, μ). It suffices to vertify that $\overline{\Sigma} \subset \overline{\overline{\Sigma}}$. Indeed, let $E \cup Z \in \overline{\Sigma}$, where $E \in \Sigma$ and $Z \subset N \in \mathcal{N}$. Since $\overline{\overline{\mu}}|_{\Sigma} = \mu$, $\overline{\overline{\mu}}(N) = \mu(N) = 0$. Moreover, $(X, \overline{\overline{\Sigma}}, \overline{\overline{\mu}})$ is complete, so $Z \in \overline{\overline{\Sigma}}$. Which implies that $\overline{\Sigma} \subset \overline{\overline{\Sigma}}$. $\square$

2.4 Lebesgue Measure

Section 2.3 provides the fundamental method for constructing measures. We can now utilize Carathéodory's method to reconstruct the Lebesgue measure. It is worth noting that the sets Lebesgue originally considered were bounded (the reasons for this were explained in Chapter 1), while we will not impose this restriction here. In fact, it can be proven that for bounded sets, their measurability under Lebesgue's method (based on inner and outer measures) is equivalent to their measurability under Carathéodory's method. Furthermore, after successfully constructing the Lebesgue measure, we will return to the introduction in Chapter 1 and demonstrate that the Lebesgue measure satisfies the three axioms of measure proposed by Lebesgue: non-negativity (the measure of the empty set is 0), additivity, and translation invariance. The first two axioms are inherent properties of the measure's definition, while the third requires a detailed proof.

A Lebesgue Measure

Although Mr. Lebesgue studied the Lebesgue measure on Euclidean space $\mathbf{R}$, similar ideas can be directly and unquestionably extended to $\mathbf{R}^n$.

Definition 2.4.1. Let I_i be an interval of real numbers. The ***length*** of I_i is defined by $\ell(I_i)$ to be ∞ if I is unbounded and otherwise to be the difference of its end-points. A subset I of $\mathbf{R}^n$ is the n-fold *Cartesian Product* of bounded intervals of real numbers is called a ***box***. If $I = I_1 \times I_2 \times \cdots \times I_n$, we define the ***volume*** of I by

$$vol(I) = \ell(I_1) \cdot \ell(I_2) \cdot \cdots \cdot \ell(I_n).$$

Proposition 2.4.1. The family $\mathscr{I}_n$ of all intervals in $\mathbf{R}^n$ is a semiring of subsets of $\mathbf{R}^n$. Moreover, the set function $vol : \mathscr{I}_n \longrightarrow [0, \infty)$ is a measure (so is a charge) on the semiring $\mathscr{I}_n$ of boxes in $\mathbf{R}^n$.

Proof. For the first statement of the proposition, refer to Example 2.1.1 in the first section of this chapter. Now, we proceed to prove the second statement. Let $B \in \mathscr{I}_1$ and $B = \bigcup_{k=1}^n \{B_k\} \in \mathscr{I}_1$ where B_k are disjoint intervals. And let a_k, b_k be the endpoints of B_k. Then $a_{k+1} = b_k$ for $k = 1, 2, \ldots, n-1$. Then $\sum_{k=1}^n vol(B_k) = \sum_{k=1}^n (b_k - a_k) = b_n - a_1 = vol(B)$. To see the countable subadditivity, let $B \subset \bigcup_{k=1}^\infty B_k$ where $B, B_k \in \mathscr{I}_1$ and let a, b be the endpoints of the interval B and a_k, b_k the endpoints of B_k. For every $\epsilon > 0$, find the intervals $B' \subset B$ and $B'_k \supset B_k$ such that B' is closed and B'_k are open intervals and

$$vol(B) - vol(B') + \sum_{k=1}^{\infty} \left(vol(B'_k) - vol(B_k)\right) < \epsilon. \tag{1}$$

For these new intervals, we have that $B' \subset \bigcup_{k=1}^\infty B'_k$. Now, we obtain a cover of a compact set. Let us choose a finite subcover, WLOG, assume that $B' \subset \bigcup_{k=1}^N B'_k$. Therefore we have that

$$vol(B') \leq \sum_{k=1}^{N} vol(B'_k) \leq \sum_{k=1}^{\infty} vol(B'_k).$$

By (1), we see that $vol(B) \leq \sum_{k=1}^\infty vol(B_k) + \epsilon$. By theorem 2.2.2, vol is a measure on the semiring $\mathscr{I}_1$. $\square$

Now, we can define the n-dimensional Lebesgue measure on $\mathbf{R}^n$ by Method I. The outer measure μ^* induced by the charge $vol : \mathscr{I}_n \longrightarrow [0, \infty)$ on the semiring $\mathscr{I}_n$ of intervals in $\mathbf{R}^n$ is called ***Lebesgue outer measure*** on $\mathbf{R}^n$,

$$\lambda^*(E) = \inf \left\{ \sum_{k=1}^{\infty} vol\,(B_k) : E \subset \bigcup_{k=1}^{\infty} B_k, B_k \text{ are boxes} \right\}.$$

The family $\Sigma_{\lambda_n^*}$ of λ_n^*-measurable sets is denoted by $\mathcal{L}^n$. And the restriction of λ_n^* to $\mathcal{L}^n$ is denoted by λ_n , called the ***Lebesgue measure***. The measure space $(\mathbf{R}^n, \mathcal{L}^n, \lambda_n)$ is called the Lebesgue measure space.

Every open subset O of $\mathbf{R}^n$ is Lebesgue measurable. We can directly prove this fact using topological sense. Therefore, we can conclude that all closed sets, half-open intervals (boxes), singleton sets, the set of rational numbers on the real line, and so on, are Lebesgue measurable sets. Indeed, let

$$\mathscr{F} = \{B \subset \mathbf{R}^n : B \text{ are open boxes with rational end-points and } B \subset O\}.$$

$\mathscr{F}$ is countable since the end-points are rational numbers. For every $x \in O$, there exists open box B_x with rational end-points such that $x \in B_x \subset O$ since O is open. Thus,

$$O = \bigcup_{B \in \mathscr{F}} B,$$

that is a countable union of Lebesgue measurable sets. Moreover, every closed set which is the complement of an open set is Lebesgue measurable.

Theorem 2.4.1. We now provide an equivalent definition of Lebesgue outer measure, which is also a very important property, known as the "outer regularity" of Lebesgue outer measure. Let λ_n^* be the Lebesgue outer measure. If $A \in 2^{\mathbf{R}^n}$, then

$$\lambda^*(A) = \inf \{\lambda^*(U) : U \supset A, U \text{ is open}\}.$$

Proof. If $\lambda^*(A) = \infty$, by the monotonicity of λ_n^*, we have $\lambda_n^*(U) = \infty$ for any open set $U \supset A$ and so the infimum is also ∞. Now, we assume that $\lambda_n^*(A) < \infty$. Let $\epsilon > 0$, then there exists a sequence $\{B_k\}$ of boxes that cover A and $\sum_{k=1}^{\infty} vol\,(B_k) < \lambda_n^*(A) + \epsilon/2$. Choose a sequence of open boxes $\{B'_k\}$ such that

$$B_k \subset B'_k \text{ and } vol(B'_k) - vol(B_k) < \frac{\epsilon}{2^{k+1}} \text{ for all } k.$$

Then we have, $A \subset U = \bigcup_{k=1}^{\infty} B'_k$ and $\sum_{k=1}^{\infty} vol\,(B'{}_k) < \lambda_n^*(A) + \epsilon$. And hence,

$$\lambda_n^*(A) \le \lambda_n^*(U) \le \sum vol(B'_k) < \lambda_n^*(A) + \epsilon.$$

It follows that $\lambda_n^*(A) = \inf \left\{\lambda_n^*(U) : A \subset U, U \text{ is open} \right\}$. □

►**Corollary 1.** Let λ^* be the Lebesgue outer measure. A is Lebesgue measurable ***iff*** for every $\epsilon > 0$, there exist an open set U and a closed set F such that $F \subset A \subset U$ and $\lambda^*(A \setminus F) < \epsilon$ and $\lambda^*(U \setminus A) < \epsilon$.

Proof. First, we assume that A is Lebesgue measurable and $\lambda^*(A) = \lambda(A) < \infty$, by theorem 2.4.3, there exists an open set U such that $\lambda(U) < \lambda(A) + \epsilon$, that is $\lambda(U \setminus A) < \epsilon$. If $\lambda^*(A) = \infty$, consider $A_n = A \cap (B(0,n) \setminus B(0,n-1))$ for every n, the first case shows that there exists an open set $U_n \supset A_n$ such that $\lambda(U_n \setminus A_n) < \epsilon/2^n$. Let $U = \bigcup_{n=1}^{\infty} U_n$, then U is open and $U \supset A$ and

$$\lambda(U \setminus A) \le \bigcup_{n=1}^{\infty} \lambda(U_n \setminus A_n) < \epsilon.$$

To see the converse, let $k > 0$, and there exists an open set U_k such that $U_k \supset A$ and $\lambda^*(U_k \setminus E) < 1/k$. Let $G = \bigcap_{k=1}^{\infty} U_k$. It is clear that G is measurable and $A \subset G$ and $G \setminus A \subset U_k \setminus A$ for each k, hence

$$\lambda^*(G \setminus A) \le \lambda^*(U_k \setminus A) < \frac{1}{k}.$$

It follows that $\lambda^*(G\setminus A) = 0$, and hecne $G\setminus A$ is measurable, so is $A = G\setminus(G\setminus A)$. Where G is a G_δ set.

For the "closed" case, there exists an open set U such that $U \supset \mathbf{R}^n \setminus A$ and $\lambda(U \setminus A^c) < \epsilon$. Define $F = U^c$, then F is a closed subset of A and

$$A \setminus F = A \setminus (\mathbf{R}^n \setminus U) = A \cap U = U \setminus (\mathbf{R}^n \setminus A).$$

Therefore, $\lambda(A \setminus F) = \lambda(U \setminus A^c) < \epsilon$. □

Remark 2.4.1. Similarly, Lebesgue outer measure also possesses *inner regularity*, though its formulation differs slightly from that of outer regularity: if $A \in \mathcal{L}^n$, then

$$\lambda^*(A) = \sup\{\lambda(K) : K \subset A : K \text{ is compact}\}.$$

Proof. By the monotonicity of outer measure, $\lambda^*(A) \ge \sup\{\lambda^*(K) : K \subset A\}$. To see the converse, first we assume that A is bounded, so $\lambda^*(A) < \infty$. For $\epsilon > 0$, there exista a closed set $K \subset A$ such that $\lambda^*(A\setminus K) < \epsilon$ by theorem before. Since A is bounded and $A \supset K$, we have K is compact. And $\lambda(K) = \lambda(A) - \lambda(A\setminus K) > \lambda(A) - \epsilon$. If follows that $\lambda(A) = \lambda^*(A) = \sup\{\lambda^*(K) : K \subset A, K \text{ is compact}\}$. For the case where A is unbounded, see the method we used in theorem 2.4.1. □

Note that there are two distinctions here compared to outer regularity: first, the target set is required to be measurable; second, the approximation here is from the inside using compact sets (in Euclidean space $\mathbf{R}^n$, compact sets are equivalent to closed and bounded sets) rather than just closed sets. This definition of inner regularity is not coincidental: the real reasons will be explained in Part III. For now, it can be noted that bounded closed sets, compared to general closed sets, ensure finite measure, which makes them more convenient when handling countable addition or countable union operations.

Furthermore, referring to the form of inner measure defined by Mr. Lebesgue in Chapter 1, we define the Lebesgue inner measure of a bounded set A as follows (When discussing Lebesgue inner measure, we only consider bounded sets. This is done to align with Mr. Lebesgue's definition of measurability, for reasons that have been mentioned in Chapter 1.) :

$$\lambda_*(A) = \lambda^*(E) - \lambda^*(E \setminus A) \text{ for every elementary set } E \supset A.$$

We first prove that it is well-defined, that is, assuming A and A' are two elementary sets containing E, then $\lambda^*(A) - \lambda^*(A \setminus E) = \lambda^*(A') - \lambda^*(A' \setminus E)$. Moreover, we must also prove that a set E is measurable ***iff*** $\lambda^*(E) = \lambda_*(E)$.

Theorem 2.4.2. Let $A \subset \mathbf{R}^n$ be a bounded set, the Lebesgue inner measure is well defined and A satisfies the Carathéodory condition ***iff*** $\lambda^*(A) = \lambda_*(A)$.

Proof. First, let E, F be elementary sets such that $A \subset F \subset E$ and $\lambda(E) < \infty$. For $\epsilon > 0$, let $\{B_n\}$ be a sequence of boxes such that $E \setminus A \subset \bigcup_{n=1}^{\infty} B_n$ and $\sum_{n=1}^{\infty} vol(B_n) < \lambda^*(E \setminus A) + \epsilon$. Note that

$$F \setminus A \subset (E \setminus A) \setminus (E \setminus F) \subset \bigcup_{n=1}^{\infty} (B_n \setminus (E \setminus F)),$$

$$E \setminus F \subset \bigcup_{n=1}^{\infty} (B_n \cap (E \setminus F)).$$

It follows that

$$\begin{aligned} \lambda(E \setminus F) + \lambda^*(F \setminus A) &\leq \sum_{n=1}^{\infty} \lambda^*(B_n \cap (E \setminus F)^c) + \lambda^*(B_n \cap (E \setminus F)) \\ &= \sum_{n=1}^{\infty} vol(B_n) \leq \lambda^*(E \setminus A) + \epsilon. \end{aligned}$$

Letting $\epsilon \to 0$, we have $\lambda(E) - \lambda^*(E \setminus A) \leq \lambda(F) - \lambda^*(F \setminus A)$. On the other hand, $E \setminus A \subset (E \setminus F) \cup (F \setminus A)$, so

$$\begin{aligned} \lambda(E) - \lambda^*(E \setminus A) &\geq \lambda(E) - \lambda(E) + \lambda(F) - \lambda^*(F \setminus A) \\ &= \lambda(F) - \lambda^*(F \setminus A). \end{aligned}$$

Now, assume that E, F are elementary sets such that $A \subset F$ and $A \subset E$ and $\lambda(E) < \infty, \lambda(F) < \infty$, we have $A \subset E \cap F$. Thus,

$$\begin{aligned} \lambda(E) - \lambda^*(E \setminus F) &= \lambda(E \cap F) - \lambda^*((E \cap F) \setminus A) \\ &= \lambda(F) - \lambda^*(F \setminus A). \end{aligned}$$

Therefore λ_* is well defined.

Assume that $\lambda^*(A) = \lambda^*(A \cap E) + \lambda^*(A \setminus E)$ for every $A \subset \mathbf{R}^n$. Now, let A be the elementary set with $A \supset E$, then $\lambda^*(A) = \lambda^*(E) + \lambda^*(A \setminus E)$, that is $\lambda^*(E) = \lambda_*(E)$. On the other hand, assume $\lambda^*(E) = \lambda_*(E)$, for $\epsilon > 0$, there exists an open set U such that $A \subset U$ and $\lambda^*(U) \leq \lambda^*(A) + \epsilon$ by theorem proceeding. Note that

$$\lambda^*(U) - \epsilon \leq \lambda^*(A) = \lambda_*(A) = \lambda^*(U) - \lambda^*(U \setminus A).$$

So, $\lambda^*(U \setminus A) < \epsilon$, which implies that A is measurable in Carathéodory sense. $\square$

Remark 2.4.2. Lebesgue inner measure also has an equivalent definition. We generally do not specifically refer to this property of inner measure as the "inner regularity" of Lebesgue inner measure, because it is essentially the difference between two outer measure values. Moreover, its proof relies on the outer regularity of Lebesgue outer measure, so adding another term would be redundant. Let λ_* be the Lebesgue inner measure. If A is bounded, then

$$\lambda_*(A) = \sup \{\lambda(K) : K \subset A, K \text{ is compact}\}.$$

Proof. First we assume that A is bounded and choose a bounded elementary closed set E such that $E \supset A$. From theorem 2.4.3,

$$\begin{aligned}
\lambda_*(A) &= \lambda^*(E) - \lambda^*(E \setminus A) \\
&= \lambda^*(E) - \inf \{\lambda^*(U) : U \supset E \setminus A, U \text{ is open}\} \\
&= \sup \{\lambda^*(E) - \lambda^*(U) : U \supset E \setminus A\} \\
&= \sup \{\lambda^*(E) - \lambda^*(E \setminus K) : E \setminus K \supset E \setminus A, E \setminus K \text{ is open}\} \\
&= \sup \{\lambda^*(K) : K \subset A, K \text{ is compact}\}.
\end{aligned}$$

K is closed since $K = E \setminus (E \setminus K) = E \cap U^c$, where E, U^c are closed. K is bounded since $E \supset K$ is bounded. Thus, K is compact. $\square$

However, if we rely solely on the method of defining measurable sets based on the equality of inner and outer measures as introduced by Mr. Lebesgue, then the measurable sets defined in this way are only applicable to the Lebesgue outer measure and Lebesgue inner measure, or to similar cases (readers may note that defining measurable sets through the equality of inner and outer measures essentially depends on the outer regularity of the Lebesgue outer measure, yet not all outer measures possess such a property). As illustrated by an example introduced at the beginning of this section, this approach is not suitable for modern abstract measure theory. On the other hand, due to the inherent limitations of the inner measure itself, it can only be applied to the study of bounded sets.

►**Corollary 1.** A subset $E \subset \mathbf{R}^n$ is Lebesgue measurable ***iff*** $E = G \setminus N$, where G is a G_δ set (the intersection of countable open subsets of $\mathbf{R}^n$) and N is a set of Lebesgue measure zero i.e., $\lambda_n(N) = 0$.

Proof. Let E be Lebesgue measurable. By the preceding theorem, for each k, there is an open set G_k such that $\lambda_n(G_k \setminus E) < 1/k$. Define $G = \bigcap_{k=1}^{\infty} G_k$. Since $G \setminus E \subset G_k \setminus E$ for every k, $\lambda_n(G \setminus E) = 0$. Thus, $E = G \setminus (G \setminus E)$ where $\lambda_n(G \setminus E) = 0$. Conversly, since $\lambda_n(N) = 0$, N is Lebesgue measurable. Thus, $E = G \setminus N$ is Lebesgue measurable. □

►**Corollary 2.** A subset $E \subset \mathbf{R}^n$ is Lebesgue measurable ***iff*** $E = F \cup N$, where F is a $F_\sigma$set (the union of countable closed subsets of $\mathbf{R}^n$) and N is a set of Lebesgue measure zero i.e., $\lambda_n(N) = 0$.

Proof. Let E be Lebesgue measurable. Then $\mathbf{R}^n \setminus E = G \setminus N$, where G is a G_δ set and $\lambda_n(N) = 0$. Thus, $E = (\mathbf{R}^n \setminus G) \cup N$. Conversly, since $\lambda_n(N) = 0$, N is Lebesgue measurable. Thus, $E = F \cup N$ is Lebesgue measurable. □

Let $\mathscr{S}$ be a family of subsets of X. The smallest σ-algebra that contains the open subsets of $\mathbf{R}^n$ is called the ***Borel σ-algebra***, denoted by $\mathcal{B}(\mathbf{R}^n)$. The element of the Borel σ-algebra is called a ***Borel set***. We can see that all the G_δ sets and F_σ sets are Borel sets. Since all the open sets are Lebesgue measurable, $\mathcal{B}(\mathbf{R}^n) \subset \mathcal{L}^n$ is true.

►**Corollary 3.** The family $\mathcal{L}^n$ of Lebesgue measurable subsets of $\mathbf{R}^n$ is the smallest σ-algebra that contains the Borel σ-algebra and all sets of Lebesgue outer measure zero, that is $\mathcal{L}^n = \overline{\mathcal{B}(\mathbf{R}^n)}$.

While the collection of Lebesgue measurable sets is very extensive — including all open sets, closed sets, countable sets, and their various combinations. A natural question arises: does it include every subset of $\mathbf{R}^n$? Vitali's Theorem in section 2.5. provides a negative answer. It proves that for any set with positive outer measure, there exists a subset within it that is not Lebesgue measurable.

Before we discuss the non-measurable sets, we check that the three baisc properties (nonnegativity, translation invariance and countable additivity) on $\mathcal{L}^n$. We have known that Lebesgue measure is constructed by Method I, automatically, nonnegativity and countable additivity are ture. It remains to check the translation invariance:

Lemma 2.4.1. Let $A, B \in \mathbf{R}^n$, then for every $x \in \mathbf{R}^n$,

1. $A \cap (B + x) = (A - x) \cap B + x$;
2. $A^c + x = (A + x)^c$.

Theorem 2.4.3. If $E \subset \mathbf{R}^n$, then $\lambda^*(E) = \lambda^*(E + x)$ for all $x \in \mathbf{R}^n$. Moreover, if $E \in \mathcal{L}^n$, then $E + x \in \mathcal{L}^n$.

Proof. For the first statement, let $E \subset \bigcup_{k=1}^{\infty} B_k$ where B_k are disjoint boxes. Then $E+x \subset \bigcup_{k=1}^{\infty}(B_k+x)$ for every $x \in \mathbf{R}^n$ and $vol(B_k+x) = vol(B_k)$ by the definition of the volume. Hence $\lambda^*(E) = \lambda^*(E+x)$. Now, assume that $E \in \mathcal{L}^n$, then $\lambda^*(A) = \lambda^*(A-x) \geq \lambda^*(E \cap (A-x)) + \lambda^*(E^c \cap (A-x))$. Moreover,

$$\lambda^*(E \cap (A-x)) = \lambda^*((E+x) \cap A - x) = \lambda^*((E+x) \cap A).$$
$$\lambda^*(E^c \cap (A-x)) = \lambda^*((E^c+x) \cap A - 1) = \lambda^*((E+x)^c \cap A).$$

And hence, $\lambda^*(A) \geq \lambda^*((E+x) \cap A) + \lambda^*((E+x)^c \cap A)$, which implies that $E+x \in \mathcal{L}^n$ and so $\lambda(E) = \lambda(E+x)$. □

If we consider translation from the perspective of mapping, we actually gain more insights. For example, if we consider more general mappings, does a measurable set remain measurable after a mapping? How does the measure behave under such changes ?

B Non-measurable Sets

The first non-measurable set was proposed by Vitali in 1906, though it also relied on the Axiom of Choice.

Vitali Set.

We can now return to the question posed in Chapter 1: whether the measure problem has a solution ? That is, whether it is possible to define a set function m on $2^{\mathbf{R}}$ such that, under the condition $m([0,1]) = 1$, it satisfies the three axioms of measure and assigns a measure value to every subset. The answer is surprisingly disappointing: we cannot find such a set function that meets all these conditions. The proof is remarkably straightforward, and we summarize it in the following theorem.

Theorem 2.4.5. There does no exist a set function μ with all the following properties.

1. $\mu([0,1]) = 1$.

2. μ is a set function from $2^{\mathbf{R}}$ to $[0, \infty]$.

3. μ is σ-additive.

4. μ is translation invariant.

2.5 The Cantor Set

Every countable set has outer measure 0. A reasonable question arises about whether the converse holds. In other words, is every set with outer measure 0 countable? The Cantor set, which is introduced in this section, provides the answer to this question.

The Cantor set commonly referred to was discovered by Henry John Stephen Smith in 1874 and later mentioned by the German mathematician Georg Cantor in 1883. The most well-known example is the following Cantor ternary set.

Consider the closed, bounded interval $C_0 = [0, 1]$. The first step in the constructing the Canotr set is to divide C_0 into three intervals of equal length 1/3 and remove the interior of the middle interval, that is, remove the interval $G_1 = (1/3, 2/3)$ and obtain the closed set C_1, which is the union of two disjoint closed intervals, i.e.,

$$C_1 = [0, 1/3] \cup [2/3, 1].$$

G_1 is shown in red.

In the second step, we apply the same procedure as in the first step to each closed interval in C_1. This removes the union of two open intervals, that is, $G_2 = (1/9, 2/9) \cup (7/9, 8/9)$, and obtains the closed set C_2, which is the union of 2^2 disjoint closed intervals. i.e.,

$$C_2 = [0, 1/9] \cup [2/9, 3/9] \cup [6/9, 7/9] \cup [8/9, 1].$$

$G_1 \cup G_2$ is shown in red.

In the third step, we repeat the process on each closed intervals in C_2. This removes the union of four open intervals, that is, G_3, and obtains a closed set C_3, which is the disjoint union of 2^3 closed intervals.

$G_1 \cup G_2 \cup G_3$ is shown in red.

Continue this removal operation countably many times to obtain a countable collection of sets $\{C_k\}$ and $\{G_k\}$. Finally, we define the ***Cantor set*** **C** by

$$\mathbf{C} = \bigcap_{k=1}^{\infty} C_k = [0, 1] \setminus \bigcup_{k=1}^{\infty} G_k.$$

Ternary representations provide a useful way to think about the Cantor set. Every number $x \in [0,1]$ can be represented as a ternary (base-3) expansion: $x = 0.x_1x_2x_3\ldots_3$ where each digit x_i is either 0, 1 or 2. This notation means

$$x = \frac{x_1}{3} + \frac{x_2}{3^2} + \frac{x_3}{3^3} + \cdots = \sum_{k=1}^{\infty} x_k, \text{ each } x_k \text{ is 0,1 or 2.}$$

For example, $0.1_3 = \frac{1}{3} = 0.333\ldots_{10}$, $0.01_3 = \frac{0}{3} + \frac{1}{3^2} = \frac{1}{9}$, $0.12_3 = \frac{1}{3} + \frac{2}{3^2} = \frac{5}{9}$. This expansion is unique except when

$$x \in S_{fin} = \left\{ \frac{a}{3^n} = \sum_{k=1}^{n} \frac{x_k}{3^k} \in (0,1) : a, n \in \mathbf{Z}^+, 3 \nmid k \right\}$$

More precisely, S_{fin} consists of numbers that have a finite ternary expansion ending with a non-zero digit. For example

$$\frac{1}{3} = 0.1_3 = 0.0\overline{2}_3, \quad \frac{2}{3} = 0.2_3 = 0.1\overline{2}_3.$$

The ambiguity arises because a number in S_{fin} has two representations: one finite form and one infinite form ending with repeating 2's. To facilitate subsequent discussions of the Cantor set, we must eliminate this ambiguity. Our approach is as follows:

- If $x_n = 2$, we will use this expansion to represent x,
- If $x_n = 1$, we will use the following representation for x:

$$x = \frac{x_1}{3} + \frac{x_2}{3^2} + \cdots + \frac{x_{n-1}}{3^{n-1}} + \frac{0}{3^n} + \sum_{k=n+1}^{\infty} \frac{2}{3^k}.$$

Thus, with this approach, each number $x \in [0,1]$ has a unique ternary expansion. The digits in the ternary expansion have a direct geometric interpretation for locating $x \in [0,1]$. Specifically,

- If $x_1 = 0$, then $x \in \left[0, \frac{1}{3}\right]$
- If $x_1 = 1$, then $x \in \left[\frac{1}{3}, \frac{2}{3}\right]$
- If $x_1 = 2$, then $x \in \left[\frac{2}{3}, 1\right]$

$$\xrightarrow{x \in \left[0, \frac{1}{3}\right]}$$

- If $x_2 = 0$, then $x \in \left[0, \frac{1}{9}\right]$
- If $x_2 = 1$, then $x \in \left[\frac{1}{9}, \frac{2}{9}\right]$
- If $x_2 = 2$, then $x \in \left[\frac{2}{9}, \frac{3}{9}\right]$

$$\Big\downarrow \; x \in \left[0, \frac{1}{9}\right]$$

- If $x_3 = 0$, then $x \in \left[0, \frac{1}{27}\right]$
- If $x_3 = 1$, then $x \in \left[\frac{1}{27}, \frac{2}{27}\right]$
- If $x_3 = 2$, then $x \in \left[\frac{2}{27}, \frac{3}{27}\right]$

$$\cdots\cdots\cdots \quad \xleftarrow[x \in \left[0, \frac{1}{27}\right]]{}$$

From the analysis above, we can easily derive the following interesting theorem, which also reveals the unifying characteristic of the elements in the Cantor set.

Theorem 2.5.1. The Cantor set C is a subset of $[0,1]$ that every $x \in C$ can be representaed as a ternary expansion

$$x = \sum_{k=1}^{\infty} \frac{a_k}{3^k},\ a_k \in \{0,2\} \text{ for each } k.$$

The following presents some fundamental properties of the Cantor set. Some of these properties may appear contradictory—for example, the Cantor set is totally disconnected yet has no isolated points. This makes the Cantor set a particularly peculiar and significant example for study.

Proposition 2.5.1.

1. The Cantor set is a closed and bounded (compact) subset of $\mathbf{R}$.
2. The Cantor set is totally disconnected.
3. The Cantor set has no isolated points.
4. The Cantor set has Lebesgue measure 0.

Proof. For (2). Since $x < y$, there exists N such that $|x-y| > \frac{1}{3^N}$. Moreover, $x, y \in C = \bigcap_{k=1}^{\infty} C_k$ which implies that $x, y \in C_N$. Now, C_N consists of 2^N disjoint closed intervals of length $\dfrac{1}{3^N}$. The inequality $|x-y| > \dfrac{1}{3^N}$ implies that x, y must lie in two different closed intervals C_N, say I_x, I_y with $I_x \cap I_y = \emptyset$. Between I_x and I_y, there was an open middle-third interval $G_{N,k} = (a,b)$. Now we define

$$V = C \cap [0,a], W = C \cap [b,1]$$

One may check that V, W form a disconnection of C.

For (3). Let x be an arbitrary point in C, and let its ternary expansion be $0, a_1a_2a_3\ldots$ with $a_k \in \{0,2\}$. Let $\epsilon > 0$, we will find another point $y \in C$ such that $|y-x| < \epsilon$. We can choose N large enough so that $\dfrac{1}{3^N} < \epsilon$. Now, construct a new point y by altering the Nth digit of x:

- If $a_N = 0$, define $y = 0.a_1a_2\ldots a_{N-1}2a_{N+1}\cdots \in C$,
- If $a_N = 2$, define $y = 0.a_1a_2\ldots a_{N-1}0a_{N+1}\cdots \in C$.

Then $|y-x| = \dfrac{2}{3^N} < \epsilon$. This proves that no point in C is isolated. □

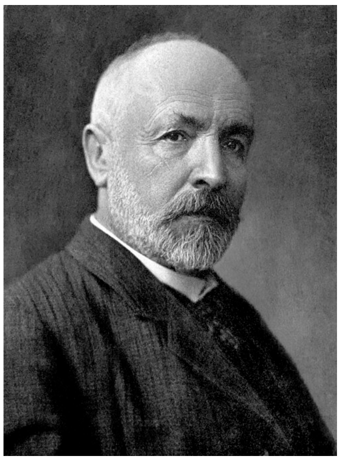

Figure 2.1: Georg Cantor
See page for author, Public domain, via Wikimedia Commons

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3

Metric Outer Measure and Hausdorff Measure

Abstract measure theory on measure space provides a magnificent foundation. The only ingredients are a set X, a σ-algebra of X, and a measure defined on the σ-algebra. However, it is static and devoid of geometry. It lacks any concept of "closeness", "continuity", or "approximation". This makes it ill-suited for analysis, which is fundamentally about limits and approximation. So, we consider to put a topology on X.

3.1 Borel Outer Measure

Given a topological space X, if we wish to study measure theory on this space, we first need to define an outer measure. However, this outer measure cannot be chosen arbitrarily, it must satisfy a fundamental requirement: all open sets should be measurable under this outer measure. Otherwise, the study would lose its topological relevance. Based on this requirement, we define an important outer measure as follows.

Definition 3.1.1. Let X be a topological space. The ***Borel σ-algebra*** of X is the smallest σ-algebra of X containing the open subsets of X, denoted as $\mathscr{B}(X)$. A ***Borel subset*** of X is an element of the Borel σ-algebra. Now we give a set function $\mu : \mathscr{F} \to [0, \infty]$, define the outer measure μ^* with respect to $\mathscr{F}$ and μ. If $\mathscr{B}(X) \subset \Sigma_{\mu^*}$, then we refer to μ^* as a ***Borel outer measure***.

The above definition provides the definition of the Borel outer measure. Next, we present the definition of the Borel measure. Readers might recall the previous definitions of Lebesgue outer measure and Lebesgue measure, whereby restricting the Borel outer measure to the corresponding generated sigma-algebra yields what is called the Borel measure. While this approach is indeed reasonable, the standard historical definition is as follows:

Definition 3.1.2. Let X be a topological space. The family of Borel subsets of X is the smallest σ-algebra that contains all the open sets in X, denoted as $\mathscr{B}(X)$. And the measure defined on $(X, \mathscr{B}(X))$ is called a ***Borel measure***.

The reason for such a definition, in my view, is that studying topological structures does not require an overly complete space, and introducing too many irregular measurable sets may actually be detrimental to the research. However, to distinguish between the two, we refer to the measure obtained via the Carathéodory method as the "induced Borel measure."

It is convenient to have other expressions for the Borel sets. The family of Borel sets can be seen to be the smallest σ-algebra that contains all the closed sets in X. But for some applications we shall need the following characterization.

Theorem 3.1.1. The family of Borel sets $\mathscr{B}(X)$ of a metric space X is the smallest class $\mathscr{F}$ of subsets of X with the properties,

- If $E_1, E_2, \cdots \in \mathscr{F}$, then so does $\bigcup_{k=1}^{\infty} E_k$.
- If $E_1, E_2, \cdots \in \mathscr{F}$, then so does $\bigcap_{k=1}^{\infty} E_k$.
- $\mathscr{F}$ contains all the closed sets in X.

Note that Lebesgue outer measure is a Borel outer measure, and it possesses excellent properties, namely, outer regularity and inner regularity. These properties allow Lebesgue measurable sets to be approximated by topologically relevant sets, such as open sets, closed sets, and compact sets. This advantage stems from the favorable topological characteristics of Euclidean space $\mathbf{R}^n$ as a topological space. For general topological spaces, however, do Borel outer measures also enjoy such excellent topologically related properties? Unfortunately, this does not hold in general (See).

This example also demonstrates that even for well-behaved topological spaces such as locally compact Hausdorff spaces (including Euclidean space $\mathbf{R}^n$, which is one such space), these two excellent properties are not guaranteed. Given this, we further restrict our scope to metric spaces, a class of spaces more familiar to us, and note that $\mathbf{R}^n$ itself is also a metric space and then reexamine these two properties.

Theorem 3.1.2. Let X be a metric space, μ^* be a Borel outer measure on X with $\mu^*(X) < \infty$. Let B be a Borel set in X. Theo for every $\epsilon > 0$, there exist an open set U and a closed set F such that $F \subset B \subset F$ and $\mu^*(U \setminus B) < \epsilon$ and $\mu^*(U \setminus F) < \epsilon$.

Proof. In fact, we only need to consider Borel measure. Let μ be a Borel measure. Similarly, many of the subsequent conclusions can be restricted to the $\mathcal{B}(X)$. Let $\mathscr{F} = \{E \subset X : \forall \epsilon > 0, \exists \text{ closed set } F \subset E, \mu(E \setminus F) < \epsilon\}$.We claim that all Borel set $B \subset X$ is a member of $\mathscr{F}$. It is clear that $\mathscr{F}$ contains all closed sets. We show that $\mathscr{F}$ contains the closed sets and it is closed under countable unions and closed under countable intersections. Suppose that $E_1, E_2, \cdots \in \mathscr{F}$. There exist closed sets $F_k \subset E_k$ with $\mu(E_k \setminus F_k) < \epsilon/2^k$. Then we have,

$$\mu\left(\bigcap_{k=1}^{\infty} E_k \setminus \bigcap_{k=1}^{\infty} F_k\right) \leq \mu\left(\bigcup_{k=1}^{\infty} (E_k \setminus F_k)\right) < \epsilon.$$

Thus $\bigcap_{k=1}^{\infty} F_k$ is a closed subset of $\bigcap_{k=1}^{\infty} E_k$.

Since the countable union of closed sets is not necessarily closed, we require an extra step. Note that

$$\begin{aligned}\lim_{n\to\infty} \mu\left(\bigcup_{k=1}^{\infty} E_k \setminus \bigcup_{k=1}^{n} F_k\right) &= \mu\left(\bigcup_{k=1}^{\infty} E_k \setminus \bigcup_{k=1}^{\infty} F_k\right)\\ &\leq \mu\left(\bigcup_{k=1}^{\infty} (E_k \setminus F_k)\right) < \epsilon.\end{aligned}$$

The reason the first equality holds in the above equation is that $\mu(X) < \infty$. Thus, for sufficiently large n, we must have

$$\mu\left(\bigcup_{k=1}^{\infty} E_k \setminus \bigcup_{k=1}^{n} F_k\right) < \epsilon.$$

and in this case, $\bigcup_{k=1}^{n} F_k$ is a closed subset of $\bigcup_{k=1}^{\infty} E_k$. Thus, we conclude that $\mathscr{F}$ contains all the Borel sets of X by theorem 3.1.1.

For the approximation from outside, for a Borel set $B \in \mathcal{B}(X)$, consider B^c, then $B^c \in \mathcal{B}(X) \subset \mathscr{F}$, then there exists a closed set $F \subset B^c$ such that $\mu(B^c \setminus F) < \epsilon$. Then F^c is open and $F^c \supset B$,

$$\mu(F^c \setminus B) = \mu(B \setminus F^c) < \epsilon.$$

Then $G = F^c$ is open and G contains B, and $\mu(G \setminus B) < \epsilon$. □

It is worth noting that the above conclusions can be generalized further, as demonstrated in the following two theorems. For the inner approximation property with closed sets, we do not necessarily require the entire space X to have finite measure; it is sufficient to require the target measurable set to have finite measure. For the outer approximation with open sets, we do not even need the target measurable set to have finite measure; we only require that it can be covered by countably many open sets of finite measure. The latter is a very powerful conclusion (although restricted to metric spaces), and we will use this result in many subsequent discussions.

Theorem 3.1.3. Let X be a metric space, μ^* be a Borel outer measure on X. And let B_0 be a Borel set with $\mu^*(B_0) < \infty$. Then for every $\epsilon > 0$, there exists a closed set F such that $F \subset B_0$ and $\mu^*(B_0 \setminus F) < \epsilon$.

Proof. Define a new measure μ_0 by $\mu_0(E) = \mu(B_0 \cap E)$ for all $E \subset X$. Then μ_0 is a finite Borel measure on X. By the prove we have just proved, all Borel sets can be approximated from inside by closed sets. In particular, there is a closed set $F \subset B_0$ such that

$$\mu_0(B_0 \setminus F) < \epsilon.$$

Since $\mu_0(B_0 \setminus F) = \mu(B_0 \setminus F)$, we have done. □

Theorem 3.1.4. Let X be a metric space, μ^* be a Borel outer measure on X. And let B_0 be a Borel set that is contained in the union of countable many open sets each of finite outer measure. Then for every $\epsilon > 0$, there exists an open set U such that $B_0 \subset U$ and $\mu^*(U \setminus B_0) < \epsilon$.

Proof. Suppose that $B_0 \subset \bigcup_{i=1}^{\infty} O_i$ where O_i are open sets of finite measure. Since $\mu(O_i \setminus B_0) \le \mu(O_i) < \infty$, there exists a closed set $F_i \subset O_i \setminus B_0$ such that

$$\mu\left((O_i \setminus F_i) \setminus B_0\right) = \mu\left((O_i \setminus B_0) \setminus F_i\right) < \frac{\epsilon}{2^i}.$$

Here $B_0 \cap O_i$ is a subset of the open set $O_i \setminus F_i$. Now, define

$$G = \bigcup_{i=1}^{\infty} (O_i \setminus F_i).$$

Then G is open, G contains B_0 and $\mu(G \setminus B_0) < \epsilon$. □

The theorem mentioned above may differ somewhat from the outer and inner regularity properties of Lebesgue outer measure. Moreover, the scope of the study is limited to measurable sets, excluding non-measurable sets, though studying non-measurable sets is of little practical significance.

In the previous chapter, after introducing the approximation theorem for the Lebesgue measure, we further used more general sets — G_δ sets and F_σ sets — for approximation. The Borel measure also possesses this property.

Theorem 3.1.5. Let X be a metric space, μ be a Borel measure on X with $\mu(X) < \infty$. For every $\epsilon > 0$ and $B \in \mathcal{B}(X)$, there exist F_σ set F and G_δ set G such that

$$\mu(B) = \mu(F) = \mu(G).$$

Since even the Borel outer measure in a metric space struggles to achieve the desired properties, we are compelled to introduce a new definition specifically for Borel outer measures that satisfy those two special properties. Let us revisit the inner regularity and outer regularity of the Lebesgue outer measure once again, from which we can distill the following definition.

Remark 3.1.1. Let μ^* be a Borel outer measure on the topological space X $(\Sigma_{\mu^*} \supset \mathcal{B}(X))$. And let $A \subset X$. Then we say that A is a

- ***outer regular set***, if $\mu^*(A) = \inf \{\mu^*(U), U \supset A, U \text{ open }\}$.
- ***inner regular set***, if $\mu^*(A) = \sup \{\mu^*(K), K \subset A, K \text{ compact }\}$.

Let $\mathscr{S} \subset 2^X$, we say that μ^* is ***outer (inner) regular on*** $\mathscr{S}$ if all the sets in $\mathscr{S}$ is outer (inner) regular.

These two definitions are entirely derived from the properties of the Lebesgue outer measure. It is clear that the Lebesgue outer measure is outer regular on its family of measurable sets and inner regular on all open sets (in fact, on its Borel sigma-algebra $\mathcal{B}(\mathbf{R}^n)$). We extract these two properties and define a Borel outer measure on a topological space that satisfies all two as a ***regular Borel outer measure***. That is we say that μ^* is a regular Borel outer measure if

- For every subset $A \in 2^X$, A is outer regular.
- For every subset $A \in \mathscr{T}$, A is inner regular.

3.2 Construction of Measures — Act II

A Metric Outer Measure

Borel outer measures are a major object of study in the theory of measures on topological spaces. However, just as we encountered challenges in the measure construction theory discussed in Chapter 2, it is also relatively difficult to directly construct a (non-trivial) Borel outer measure on a topological space. Method I provides an excellent approach for constructing outer measures. We can then examine whether an outer measure constructed in this way meets our requirements, specifically, whether it is a Borel outer measure. The following example illustrates this:

Example 3.2.1. Let $X = \mathbf{R}^2$ and let $\mathscr{T}$ be the family of open squares in X, and choose a set function $\tau : \mathscr{T} \to [0, \infty]$ to be the diameter of $T \in \mathscr{T}$. We apply the Carathéodory Construction to obtain an outer measure μ^* and then a space $(\mathbf{R}^2, \Sigma_{\mu^*}, \mu)$.

Let $T_0 \in \mathscr{T}$ have side length 3, and let $T_1, T_2, T_3, T_3 \in \mathscr{T}$, each with side length 1 and as shown as the follow figure. Then $\tau(T_0) = 3\sqrt{2}$, $\tau(T_i) = \sqrt{2}$ for $i = 1, 2, 3, 4$. Then

$$\mu^*\left(\bigcup_{i=1}^{4} T_i\right) \leq \mu^*(T_0) = \tau(T_0) = 3\sqrt{2} < 4\sqrt{2} = \sum_{i=1}^{4} \tau(T_i) = \sum_{i=1}^{4} \mu^*(T_i).$$

It is easy to say that $T_i, i = 1, 2, 3, 4$ are not μ^*-measurable. Therefore, any square of $\mathbf{R}^2$ can not be measurable.

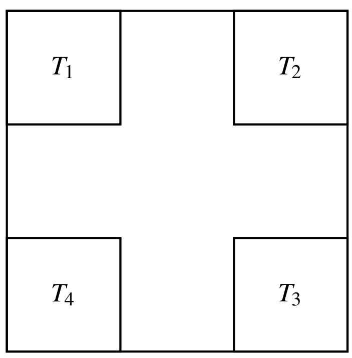

Figure 3.1: Square in Square

This example reveals a serious problem: The outer measure constructed by Method I is not necessarily a Borel outer measure. In fact, even some simple open sets (such as the square in this case) may not be measurable. Therefore, we must return to the Lebesgue outer measure and reexamine its special properties, which enable it to be a Borel outer measure. The following is an important property regarding the Lebesgue outer measure.

Proposition 3.2.1. Recall that in a metric space (X, d) we use $d(A, B) = \inf\{d(x, y) : x \in A, y \in B\}$ as the distance between A and B. If $d(A, B) > 0$, then we say that A and B are ***separated*** sets. For example, the sets T_i appearing in example 3.2.1. are pairwise separated. For the Lebesgue outer measure λ^*, suppose $A, B \subset R^n$ and $d(A, B) > 0$, then $\lambda^*(A \cup B) = \lambda^*(A) \cup \lambda^*(B)$.

Proof. Omitted. □

In fact, this highly reasonable and intuitive property in metric spaces also fails for the outer measure in Example 3.2.1. This further demonstrates that Method I is insufficient to support our modern topological research. We give a name to outer measures that satisfy such a property:

Definition 3.2.1. Let μ^* be an outer measure on a metric space X. If $\mu^*(A \cup B) = \mu^*(A)+\mu^*(B)$whenever A, B are separated subsets of X. Then μ^* is called a ***metric outer measure***.

Next, we can prove that in a metric space, a metric outer measure is indeed a Borel outer measure. Before proceeding, we first prove the following lemma. Since the concept of a metric outer measure is inherently defined on metric spaces, in subsequent discussions about metric outer measures, it will be assumed by default that the topological space in question is a metric space.

Lemma 3.2.1. Let μ^* be a metric outer measure on X, G be an open subset of X and let $F = X \setminus G$. Let $A \subset G$. Define $A_n = \left\{x \in A : \operatorname{dist}(x, F) \geq \frac{1}{n}\right\}$ Then

$$\mu^*(A) = \lim_{n\to\infty} \mu^*(A_n)$$

Proof. Since $A_n \subset A$ for all n, $\mu^*(A) \geq \lim_{n\to\infty} \mu^*(A_n)$. To esdablish the reverse inequality, let $B_n = A_{n+1} \setminus A_n = \{x : 1/n + 1 \leq d(x, F) < 1/n\}$. Since G is open, $d(x, F) > 0$ for all $x \in A$, so there exists n such that $x \in A_n$. It follows that $A = \bigcup_{n=1}^{\infty} A_n$. Thus,

$$A = A_{2n} \cup \bigcup_{k=2n}^{\infty} B_k = A_{2n} \cup \bigcup_{k=n}^{\infty} B_{2k} \cup \bigcup_{k=n}^{\infty} B_{2k+1}.$$

which implies that

$$\mu^*(A) \leq \mu^*(A_{2n}) + \sum_{k=n}^{\infty} \mu^*(B_{2k}) + \sum_{k=n}^{\infty} \mu^*(B_{2k+1}).$$

If $\sum_{k=1}^{\infty} \mu^*(B_{2k})$ and $\sum_{k=1}^{\infty} \mu^*(B_{2k+1})$ are convergent, then we have $\mu^*(A) \leq \lim_n \mu^*(A_{2n}) = \lim_n \mu^*(A_n)$. If one of the series diverges, say $\sum_{k=1}^{\infty} \mu^*(B_{2k}) = \infty$. Since $d(B_{2k}, B_{2k+1}) \geq \frac{1}{2k+1} - \frac{1}{2k+2} > 0$ for each k and μ^* is a metric outer measure,

$$\mu^*\left(\bigcup_{k=1}^{n-1} B_{2k}\right) = \sum_{k=1}^{n-1} \mu^*(B_{2k}).$$

Moreover, $A_{2n} \supset \bigcup_{k=1}^{n-1} B_{2k}$, so $\mu^*(A_{2n}) \geq \sum_{k=1}^{n-1} \mu^*(B_{2k})$. whcih implies that $\lim_{n\to\infty} \mu^*(A_{2n}) = \infty$, so

$$\lim_{n\to\infty} \mu^*(A_n) \geq \mu^*(A).$$

□

Theorem 3.2.1. Let μ^* be an outer measure on a metric space X. Then every Borel set in X is measurable ***iff*** μ^* is a metric outer measure. In other words, the outer measure on a metric space X is Borel iff metric.

Proof. First, assume that μ^* is a metric outer measure. Since the the family of Borel sets is a σ-algebra generated by the closed sets, it suffices to show that every closed set is measurable. Let F be a nonempty closed set and let $G = X \setminus F$. Then G is open. Let E be a subset of X. We show that $\mu^*(E) = \mu^*(E \cap F) + \mu^*(E \setminus F)$. Let $A = E \setminus F$ and define

$$A_n = \left\{x \in A : \text{dist}(x, F) \geq \frac{1}{n}\right\}$$

Since μ^* is a metric outer measure and A_n and F are separated for each n,

$$\begin{aligned} \mu^*(E) = \mu^*\left((E \cap F) \cup F\right) &\geq \mu^*\left((E \cap F) \cup A_n\right) \\ &= \mu^*(E \cap F) + \mu^*(A_n), \end{aligned}$$

By lemma 3.2.1, $\lim_{n\to\infty} \mu^*(A_n) = \mu^*(E \setminus F)$, thus,

$$\mu^*(E) \geq \mu^*(E \cap F) + \mu^*(E \setminus F).$$

Thus, F is measurable. To see the converse, assume that all Borel sets are measurable. Let A_1, A_2 be separated and $dist(A_1, A_2) = \delta > 0$. For each $x \in A_1$, let

$$G(x) = \{z : \rho(x, z) < \delta/2\}, \; G = \bigcup_{x \in A_1} G(x).$$

It is clear that G is open and $A_1 \subset G, A_2 \cap G = \emptyset$. Since G is measurable, we have

$$\begin{aligned} \mu^*(A_1 \cup A_2) &= \mu^*\left((A_1 \cup A_2) \cap G\right) + \mu^*\left((A_1 \cup A_2) \cap G^c\right). \\ &= \mu^*(A_1) + \mu^*(A_2), \end{aligned}$$

□

B Method II of the Construction of Measure

Although we now know that a metric outer measure is a Borel outer measure, it remains challenging to directly construct a metric outer measure. As we have seen, the Method I construction applied in a metric space can fail to produce a metric outer measure. We now seek to modify Method I in such a manner so as to guarantee that the resulting outer measure is metric.

Return to example 3.2.1. Consider the problem of measuring the "size" of New Zealand. The following image is a map of the main island of New Zealand under the Mercator projection, with a distance of 1400 km between the latitude line at the southernmost point (Stewart Island) and the latitude line at the northernmost point (Cape Reinga). Our tool is the outer measure μ^* constructed by the set function τ throuth the Method I. Then by the definition of μ^*. According to the definition of this outer measure, we only need to select a huge square that covers the main island of New Zealand while also covering a much larger area of ocean. At this point, the information expressed by the outer measure is that the "size" of these seawater areas and the main islands of New Zealand is 1400 km, which is clearly not what we want. Next, we restrict this outer measure so that when we use it to measure size, we cannot use excessively large squares but must cover the area with squares no larger than 250 km in diameter. At this point, according to the definition of the outer measure, we use 15 squares of 250 km in diameter to cover the main islands of New Zealand, and the outer measure, being the infimum, equals 3750 km. The information it expresses now mostly reflects the "size" of the main islands. As we impose further restrictions on the outer measure, the "size" measured will convey more precise information.

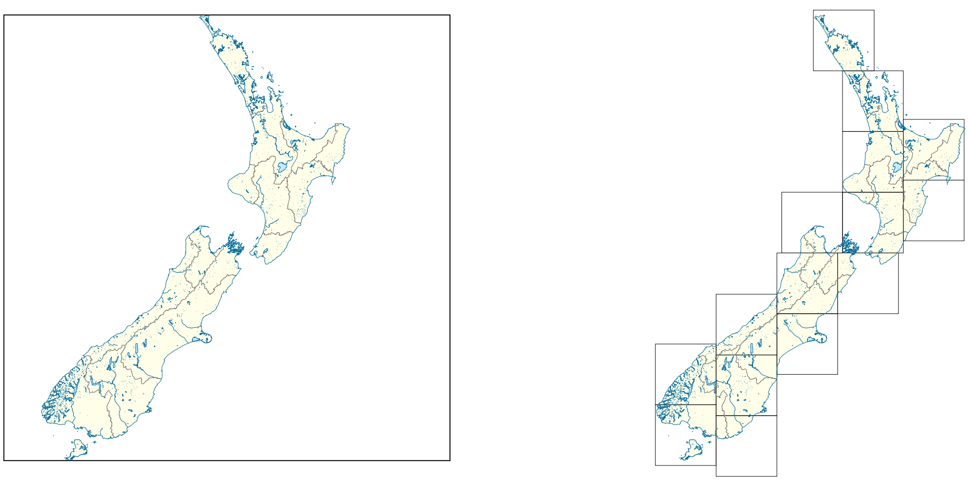

Figure 3.2: New Zealand

The example of measuring the New Zealand, we can summarize it into the following approach, which forms the main content of this section. This corresponds to ***Method I*** for constructing measures in Chapter 2, while the method discussed here will be referred to as ***Method II***.

Definition 3.2.2. Let X be a metric soace and $\mathscr{T}$ be a family of subsets of X. And let $\mu : \mathscr{F} \to [0, \infty]$ be a set function on $\mathscr{F}$. For each $\delta > 0$, let

$$\mathscr{F}_\delta = \{F \in \mathscr{F} : \text{diameter}(F) \le \delta\}.$$

We define an outer measure $\mu_\delta^* : 2^X \to [0, \infty]$ (Method I) with $\mu_\delta^*(\emptyset) = 0$ and for $A \subset X, A \neq \emptyset$,

$$\mu_\delta^*(A) = \inf\left\{\sum_{k=1}^{\infty} \mu(F_k) : A \subset \bigcup_{k=1}^{\infty} F_i \text{ and } F_k \in \mathscr{F}_\delta\right\}.$$

Let $\delta_1 < \delta_2$, then $\mathscr{F}_{\delta_1} \subset \mathscr{F}_{\delta_2}$. So, $\mu_{\delta_1}^*(A) \ge \mu_{\delta_2}^*(E)$ for each $A \subset X$. Thus the limit of $\mu_\delta^*(A)$ approaches a finite or infinite limit. We define

$$\mu_0^*(A) = \lim_{\delta \to 0} \mu_\delta^*(A), A \subset X.$$

It is called the ***Method II outer measure with respect to*** μ.

Theorem 3.2.2. Let X be a metric space. Let μ_0^* be the Method II outer measure with respect to $\mu : \mathscr{F} \to [0, \infty]$. Then μ_0^* is a metric outer measure.

Proof. One can show that μ_0^* is an outer measure; this is left to the reader. Here we will show that it is also a metric outer measure. Let A and B be separated and $dist(A, B) = \rho > 0$. If $\mu_0^*(A \cup B) = \infty$, there is nothing to prove, so we assume that $\mu_0^*(A \cup B) < \infty$. For $\epsilon > 0$, choose $0 < \delta < \rho$. By the definition of outer measure, there exists a sequence $\{C_n\} \subset \mathscr{F}_\delta$ such that

$$\sum_{k=1}^{\infty} \mu(C_k) \le \mu_\delta^*(A \cup B) + \epsilon.$$

Since $\rho > \delta$, the covering $\{C_n\}$ splits into two families,

$$\mathscr{C}_1 = \{C \in \{C_n\} : C_n \cap A \neq \emptyset\},$$
$$\mathscr{C}_2 = \{C \in \{C_n\} : C_n \cap B \neq \emptyset\}.$$

Then

$$\mu_\delta^*(A) \le \sum_{C \in \mathscr{C}_1} \mu(C),$$
$$\mu_\delta^*(B) \le \sum_{C \in \mathscr{C}_2} \mu(C).$$

Therefore, $\mu_\delta^*(A) + \mu_\delta^*(B) \le \sum_{C \in \{C_n\}} \mu(C) = \sum_{k=1}^{\infty} \mu_\delta^*(A \cup B) + \epsilon$. We could conclude that $\mu_0^*(A) + \mu_0^*(B) \le \mu_0^*(A \cup B)$. Thus, μ_0^* is a metric outer measure. □

Theorem 3.2.3. Let μ_0^* be the Method II outer measure with respect to μ and $\mathscr{F}$ and let μ^* be the Method I outer measure respect to μ and $\mathscr{F}$. Then $\mu^* = \mu_0^*$ ***iff*** for each $\epsilon > 0, F \in \mathscr{F}, n \in \mathbf{N}$ there exists a sequence $\{F_k\} \subset \mathscr{F}_n$ such that $F \subset \bigcup_{k=1}^{\infty} F_k$ and $\sum_{k=1}^{\infty} \mu(F_k) \le \mu(F) + \epsilon$.

3.3 Hausdorff Measure

The Hausdorff outer measure is a core concept in modern geometric measure theory, and it is precisely a metric outer measure. Here, we only present it as an example, focusing on some key conclusions and concepts rather than delving into a detailed study. Interested readers are encouraged to consult textbooks on geometric measure theory for further exploration.

A Hausdorff Measure

Let us return to our example 3.2.1.. Method II gives us a way to construct a "better" outer measure – metric outer measure which solved problem (i) – the additivity over separated sets, and also allows all Borel sets to be measurable (ii). However, some issues still remain: let T_0 be a square of unit diameter, and let $m, n \in \mathbf{N}$. We use $(m-1)^2$ squares to fill T_0, each of diameter $1/m$. Then

$$\mu^*_{1/m}(T_0) \geq \sum_{i=1}^{(m-1)^2} \tau(T_i) = \frac{(m-1)^2}{m}.$$

Consequently, each measure has $\mu^*_{1/m}(T_0) = \infty$ as $m \to \infty$ and

$$\mu^*_0(T_0) = \lim_{m\to\infty} \mu^*_{1/m}(T_0) = \infty.$$

Which means that the metric outer measure of any nonempty square is infinite. This is meaningless for our study. Perhaps from the very beginning, the reader has doubted whether it is appropriate to assign a measure to a two-dimensional concept – a square with using a one-dimensional concept – its diameter. Do not doubt it: it is indeed inappropriate. Similarly, if we use a three-dimensional concept – the cube of the diameter of a square – to assign a measure to this two-dimensional concept, we would also obtain similarly meaningless results: suppose that we take $\tau(T) = (diameter(T))^3$. We use $(n+1)^2$ squares fo cover T_0, each of diameter $1/n$. Then we find for all $m \leq n$ that

$$\mu^*_{1/m}(T_0) \leq \sum_{i=1}^{(n+1)^2} \tau(T_i) = \frac{(n+1)^2}{n^3}.$$

Consequently, each measure has $\mu^*_{1/m}(T_0) = 0$ as $n \to \infty$ and

$$\mu^*_0(T_0) = \lim_{m\to\infty} \mu^*_{1/m}(T_0) = 0.$$

The same is true of any open square. Consider now a further choice of the set function $\tau = (diameter(T))^2$, which is intermediate between the two preceding examples. A similar process shows that

$$\mu^*_{1/m}(T_0) \leq \sum_{i=1}^{(n+1)^2} \tau(T_i) = \frac{(n+1)^2}{n^2}.$$

so $\mu_0^*(T_0) \leq 1 = \tau(T_0) = 2\mu_2(T_0)$, where μ_2 denotes the Lebesgue measure. On the other hand, take $\{T_k\} \subset \mathscr{T}_n$ such that $T_0 \subset \bigcup_{k=0}^{\infty} T_k$, then

$$\begin{aligned}\sum_{k=1}^{\infty} \tau(T_k) = \sum_{k=1}^{\infty} (diameter(T_k))^2 \geq \sum_{k=1}^{\infty} \mu_2(T_k) \\ \geq \mu_2\left(\bigcup_{k=1}^{\infty} T_k\right) \geq \mu_2(T_0).\end{aligned}$$

so, $\mu_n^*(T_0) \geq \mu_2(T_0)$ for each n. Thus, $\mu_0^*(T_0) \geq \mu_2(T_0)$. Combine the preceding inequality,

$$\mu_2(T_0) \leq \mu_0^*(T_0) = \mu_0(T_0) \leq 2\mu_2(T_0).$$

We are now ready to define Hausdorff measure. Suppose that (X, ρ) is a metric space. Let $\mathscr{F}$ be the family of all open subsets of X. Define a set function $\mu : \mathscr{F} \to [0, \infty]$ by

$$\mu(F) = (diameter(T))^s, \ s > 0.$$

Then the outer measure μ_0^* obtained from μ and $\mathscr{F}$ by Method II is called the ***Hausdorff s-dimensional outer measure***. And the measure $\mu_0 = \mu_0^*|_{\Sigma_{\mu_s^*}}$ called the ***Hausdorff s-dimensional measure***. Specifically, for each $\delta > 0$, we define the outer measure

$$\mathscr{H}_\delta^s(A) = \mu_\delta^*(A) = \inf\left\{\sum_{k=1}^{\infty} (diam(F_k))^s : A \subset \bigcup_{k=1}^{\infty} F_k, diam(F_k) \leq \delta\right\}.$$

The outer measure $\mathscr{H}^s(A) = \lim_{\delta \to 0} \mathscr{H}_\delta^s(A) = \mu_0^*(A)$ is called the Hausdorff s-dimensional outer measure of A.

Proposition 3.3.1. By proposition 3.2.6.. The Hausdorff outer measure is a metric outer measure. Which implies it is a Borel outer measure by theorem 3.2.1. (every Borel set is measurable)

Proposition 3.3.2. Let $f : A \subset \mathbf{R}^n \to \mathbf{R}^n$ be Hölder or Lipshitz with exponent $\alpha \in (0, 1]$ and Hölder constant M. Then we have

$$\mathscr{H}^{\frac{s}{\alpha}}(f(A)) \leq M^{\frac{1}{\alpha}} \mathscr{H}^s(A).$$

Proof. Let $A \subset \bigcup_{k=1}^{\infty} F_k$ where $diam(F_k) \leq \delta$, then we have $|f(A \cap F_k)| \leq M|F \cap F_k|^\alpha \leq M|F_k|^\alpha$, it follows that $f(A) \subset \bigcup_{k=1}^{\infty} f(A \cap F_k)$, $diam(f(A \cap F_k)) \leq \epsilon = M\delta^\alpha$. Thus $\sum_{k=1}^{\infty} |f(A \cap F_k)|^{s/\alpha} \leq M^{s/\alpha} \sum_{k=1}^{\infty} |F_k|^s$, so that $\mathscr{H}_\epsilon^{s/\alpha}(f(A)) \leq M^{s/\alpha} \mathscr{H}_\delta^s(A)$. Letting $\delta \to 0$, so that $\epsilon \to 0$, as desired. $\square$

In particular, $\alpha = 1$, that is, $|f(x) - f(y)| \leq M|x - y|$, when f is called a Lipshitz mapping. And then $\mathscr{H}^s(f(A)) \leq M^s \mathscr{H}^s(A)$. Moreover, if f is isometry, i.e., $|f(x) - f(y)| = |x - y|$, then $\mathscr{H}^s(f(A)) = \mathscr{H}^s(A)$.

B Hausdorff Dimension

Dimension is a core concept in fractal geometry. Unlike the concept of dimension in topology, the dimension here reflects the degree of local complexity. A curve that is topologically one-dimensional may have a non-integer dimension in fractal geometry. The Hausdorff dimension can be said to be the oldest dimension concept (compared to other dimension concepts in analytic geometry). It is proposed based on measure theory, which makes it widely applicable in modern analysis. Perhaps the notion of ”local complexity” is still somewhat abstract. However, since this note does not focus on analytic geometry, several examples are provided to offer a general understanding: suppose there is a set F and a point $x \in F$, we enclose this point with a small ball $B(x, r)$.

- For a smooth curve, when r is sufficiently small, the intersection of the ball with the curve approximates a line segment, as r increases, its measure (length) grows proportionally to r^1.
- For a smooth surfacem, when r is sufficiently small, the intersection approximates a disk, as r increases, its measure (area) grows proportionally to r^2.
- For the Koch curve, however, the measure of the intersection grows proportionally to $r^{1.262}$.

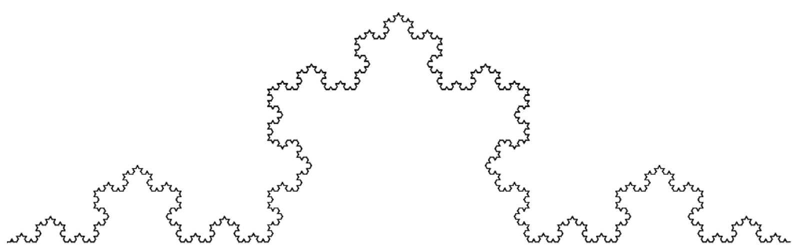

Figure 3.3: Koch Curve
José Luis Llopis Fabra, CC BY-SA 4.0, via Wikimedia Commons

Proposition 3.3.3. If $\mathscr{H}^s(A) < \infty$ and $t > s$, then $\mathscr{H}^t(A) = 0$. If $\mathscr{H}^s(A) > 0$ and $t < s$, then $\mathscr{H}^t(A) = \infty$.

Proof. For the first statement, fix $\delta > 0$, there exists a sequence $\{B_n\}$ covering A with $diamB_k \leq \delta$ and $\sum_{k=1}^{\infty}(diamB_k)^s \leq \mathscr{H}^s(A) + 1$. If $t > s$, then

$$\sum_{k=1}^{\infty}(diamB_k)^t \leq \delta^{t-s}\sum_{k=1}^{\infty}(diamB_k)^s \leq \delta^{s-t}(\mathscr{H}^s(A) + 1),$$

which implies that $\mathscr{H}^t_\delta(A) \leq \delta^{t-s}(\mathscr{H}^s + 1)$. Taking $\delta \to 0$, which gives the desired result. For the second statement, if $\mathscr{H}^t(A) < \infty$, then $\mathscr{H}^s(A) = 0$ since $s > t$, this is a contradiction. □

Definition 3.3.1. For any $A \subset X$ we have the following formulation by proposition 3.3.3,

$$\begin{aligned}\dim_{\mathscr{H}}(A) &= \sup\{s : \mathscr{H}^s(A) > 0\} = \sup\{s : \mathscr{H}^s(A) = \infty\} \\ &= \inf\left\{t : \mathscr{H}^t(A) < \infty\right\} = \inf\left\{t : \mathscr{H}^t(A) = 0\right\}.\end{aligned}$$

And the common value $dim_{\mathscr{H}}(A)$ is called the ***Hausdorff dimension*** of A.

Proposition 3.3.4. Hausdorff dimension satisfies the following properties

1. (monotonicity) If $E \subset F$, then $dim_{\mathscr{H}}(E) \leq dim_{\mathscr{H}}(F)$.
2. (countable stability) If $\{E_n\}$ is a sequence of sets, then $dim_{\mathscr{H}}(\bigcup_k F_k) = \sup\{dim_{\mathscr{H}}(F_k) : k \geq 1\}$.
3. (countable sets) If E is countable, then $dim_{\mathscr{H}}(E) = 0$.

Proposition 3.3.5. Let $f : A \subset \mathbf{R}^n \to \mathbf{R}^n$ be Hölder with exponent $\alpha \in (0, 1]$. Then we have $dim_{\mathscr{H}}(f(A)) \leq (1/\alpha)dim_{\mathscr{H}}(A)$. In particular, if f is Lipshitz, then $dim_{\mathscr{H}}(f(A)) \leq dim_{\mathscr{H}}(A)$.

Proof. If $s > \dim_{\mathscr{H}}(A)$, then $\mathscr{H}^{s/\alpha}(f(A)) \leq M^{s/\alpha}\mathscr{H}^s(A) = 0$ by proposition 3.3.3. It follows that $dim_{\mathscr{H}}(f(A)) \leq s/\alpha$ for all $s > dim_{\mathscr{H}}(A)$. □

In general, the dimension of a set alone tells us little about its topological properties. However, any set of dimension less than 1 is necessarily so sparse as to be totally disconnected; that is, no two of its points lie in the same con nected component

Theorem 3.3.1. A set $A \subset \mathbf{R}^n$ with $dim_{\mathscr{H}}(A) < 1$ is totally disconnected.

Proof. Let x, y be distinct points of A. Define a mapping $f : \mathbf{R}^n \to [0, \infty)$ by $f(z) = |z - x|$. Then $|f(z) - f(\omega)| = ||z - x| - |\omega - x|| \leq |(z - \omega)|$, then $dim_{\mathscr{H}}(f(A)) \leq dim_{\mathscr{H}}(A) < 1$. Thus $\mathscr{H}^1(f(A)) = 0$. Consider the interval $(0, f(y))$, if $(0, f(y)) \subset f(A)$, then $\mathscr{H}^1(f(A)) > 0$, which implies there exists $r \in (0, f(y))$ with $r \notin f(A)$. Since $r \notin f(A)$, there are no points $z \in \mathbf{R}^n$ for whcih $|z - x| = r$. Thus,

$$A = \{z \in A : |z - x| < r\} \cup \{z \in A : |z - x| > r\}.$$

Thus, A is contained in two disjoint open sets. $x \in \{z \in A : |z - x| < r\}$ since $|x - x| = 0$. And $y \in \{z \in A : |z - x| > r\}$ since $f(y) = |y - x| > r$, so that x, y lie in diferent connected components of A. □

The Hausdorff dimension describes the degree of local complexity. A locally complex set well-known in this note is the Cantor set. We conclude this section by calculating the Hausdorff dimension of the Cantor set.

Theorem 3.3.2. The Hausdorff dimension of C is $\gamma = \log 2/\log 3$.

Proof. The Cantor set $C = \bigcap_{k=1}^{\infty} C_k$ where $C_k = \bigcup_{j=1}^{2^k}(C_{k,j})$ with $diam(C_{k,j}) = 1/3^k$. Since $C \subset C_k$ for each k,

$$\mathscr{H}^{\gamma}_{1/3^k} \leq \sum_{j=1}^{2^k}(diam C_{k,j})^{\gamma} = 2^k \frac{1}{3^{k\cdot\gamma}} = \left(\frac{2}{3^{\gamma}}\right)^k = 1.$$

On the other hand, the Cantor-Lebesgue function f is the uniform limit of $\{f_n\}$ and $|f_{n+1}(x) - f_n(x)| \leq \frac{1}{2^n}$. Therefore, $|f_m(x) - f_n(x)| \leq \sum_{j=0}^{m-n} 1/2^{n+j} < \sum_{j=0}^{\infty} = 1/2^{n-1}$. Taking $m \to \infty$, we obtain $|f(x) - f_n(x)| \leq 1/2^{n-1}$. Moreover, we have known $|f_n(x) - f_n(y)| \leq (3/2)^n |x - y|$. Thus,

$$\begin{aligned}|f(x) - f(y)| &\leq |f(x) - f_n(x)| + |f_n(x) - f_n(y)| + |f_n(y) - f(y)| \\ &\leq \left(\frac{3}{2}\right)^n |x - y| + \frac{4}{2^n} \\ &= \frac{1}{2^n}(3^n |x - y| + 4), \text{ for each } n.\end{aligned}$$

For $x, y \in [0, 1]$, choose n such that $\frac{1}{3^n} \leq |x - y| \leq \frac{1}{3^{n-1}}$. Then $|f(x) - f(y)| \leq 7/2^n = 7/3^{n\gamma} \leq 7|x - y|^{\gamma}$. Since $f(C) = [0, 1]$ by proposition 3.3.2, $1 = \mathscr{H}^1(f(C)) \leq 7^{\frac{1}{\gamma}}\mathscr{H}^{\gamma}(C)$. Thus, $\mathscr{H}^{\gamma}(C) \in (0, 1]$ and thne $\dim_{\mathscr{H}}(C) = \log 2/\log 3$. ☐

C Kakeya Conjecture

Now that you've come to study the Hausdorff dimension, why not learn about the Kakeya conjecture? It so happens that this conjecture is quite popular recently, and it is itself very interesting.

Definition 3.3.2. A subset K of a metric space is called ***totally bounded*** if for any $\epsilon > 0$, it can be covered by a finite number of balls of diameter ϵ. For Euclidean spaec, this is the same as being a bounded sset. For a totally bounded set K, let $N(K, \epsilon)$ denote the minimal number of sets of diameter at most ϵ needed to cover K. We define the ***upper Minkowski dimension*** as

$$\overline{\dim}_{\mathscr{M}}(K) = \limsup_{\epsilon \to 0} \frac{\log N(K, \epsilon)}{\log 1/\epsilon},$$

and the ***lower Minkowski dimension***

$$\underline{\dim}_{\mathscr{M}}(K) = \liminf_{\epsilon \to 0} \frac{\log N(K, \epsilon)}{\log 1/\epsilon}.$$

If $\overline{\dim}_{\mathscr{M}}(K) = \underline{\dim}_{\mathscr{M}}(K)$, the common value is simply called the ***Minkowski dimension*** of K and denoted by $\dim_{\mathscr{M}}(K)$.

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II Integration and Signed Measure

4

Integration of Measurable Functions

We should point out that in Chapter 2, our discussion of measures is quite abstract — not limited to metric spaces on $\mathbf{R}^n$. Still, the learning process won't be difficult and should be easy for beginners to follow. However, since these notes are designed for undergraduate students, we need to set some limits when studying integration theory. Modern integration has become very abstract, but the main problem is: if we start directly with the abstract theory, it's hard to build basic understanding of integrals. Right now, we still need to think of integration as ”finding area”. So we'll limit our study of integration to metric spaces in the real number system (the easiest topological space to understand). We will still handle function domains in an abstract way, but in a simple manner that's easy to accept.

The main thread of this chapter is to study integration theory, just as we did in mathematical analysis. We will first introduce a new type of function — measurable functions — which correspond to the continuous functions in mathematical analysis (in fact, it can be proven that continuous functions are a type of measurable function). Then we will proceed to the formal study of integration (the main approach still follows H. Lebesgue's research ideas and the introduction of integration theory in modern analysis textbooks). Let us begin by understanding H. Lebesgue's research on integration.

Moreover, starting from this chapter, we need to emphasize once again that unless otherwise specified, the real number space $\mathbf{R}^n$ mentioned in following text will be equipped with its standard Euclidean metric (i.e., it will be considered as a Euclidean space).

4.1 Integral Problem

We first recall the definition and some elementary facts concerning Riemann integration. *In the fourth section of Riemann's inaugural dissertation "Über die Darstellbarkeit einer Function durch eine trigonometrische Reihe" (1854), Riemann defined the Riemann integral. The following is a restatement of its content in modern terminology:*

Suppose $[a, b]$ is a closed interval in $\mathbf{R}$. By a partition $\mathcal{P}$ of $[a, b]$ we mean a ***finite*** set of points $x_1, x_2, \ldots, x_n$ such that $a = x_1 < x_2 < \ldots < x_n = b$. Let

$$|\mathcal{P}| = \max\{x_{i+1} - x_i : 1 \leq i \leq n\}.$$

For each i, let x_i^* be an arbitrary point of the interval $[x_i, x_{i+1}]$. A bounded function $f : [a, b] \to \mathbf{R}$ is ***Riemann integrable*** if

$$\lim_{|\mathcal{P}|\to 0} \sum_{i=1}^{n} f(x_i^*)(x_{i+1} - x_i) \text{ exists.}$$

In which case, the value is the Riemann integral of f over $[a, b]$, denoted by

$$\int_a^b f(x)dx.$$

Darboux, in 1875, gave a equivalent definition of Riemann integrable as follows: Given a partition $\mathcal{P}$ of $[a, b]$, and let

$$U(\mathcal{P}) = \sum_{i=1}^{n} \left[\sup_{x\in[x_i,x_{i+1}]} f(x) \right] (x_{i+1} - x_i),$$

$$L(\mathcal{P}) = \sum_{i=1}^{n} \left[\inf_{x\in[x_i,x_{i+1}]} f(x) \right] (x_{i+1} - x_i).$$

Then the bounded function f is Riemann integrable if and only if $\lim_{|\mathcal{P}|\to 0} U(\mathcal{P}) = \lim_{|\mathcal{P}|\to 0} L(\mathcal{P})$. In fact, many modern textbooks prefer using the latter approach to explain Riemann integrability due to its intuitive clarity.

For instance, let us first consider only positive bounded function $f : [a, b] \to \mathbf{R}$. By employing rectangles parallel to the x-axis and y-axis to cover or fill the planar set E associated with f, as the partition tends to zero, if the total area S of the rectangles used for covering (referred to as the upper sum) equals the total area s of the rectangles used for filling (referred to as the lower sum), then the value $S = s$ is defined as the definite integral of the bounded function f over the interval $[a, b]$,

The reader may have already noticed that this description aligns precisely with the definition of Jordan measurability. Indeed, Riemann integrability and Jordan measurability are closely related. We present the following proposition:

Proposition 4.1.1. Let $f : [a.b] \to \mathbf{R}$ be a bounded function. And let $E_+ = \{(x, t) : x \in [a, b], 0 \leq t \leq f(x)\}$ and $E_- = \{(x, t) : x \in [a, b], f(x \leq t \leq 0)\}$.

Then f is Riemann integrable iff E_+ and E_- are both Jordan measurable, in which case,

$$\int_a^b f(x)dx = c(E_+) - c(E_-).$$

The reader has likely already grasped what comes next: he inspiration for the Lebesgue integral precisely stems from the proposition mentioned above. That is, if E_+ and E_- are Lebesgue measurable (so is E), then the integral of f over $[a,b]$ is defined as $\mu_2(E_+) - \mu_2(E_-)$.

Lemma 4.1.1. The intersection of a planar Borel measurable set with a line parallel to one of the coordinate axes is a linear Borel measurable set.

Proof. In the topological sense, this is easily proven and left to the reader. There is a another method to prove, see subsection 2.2.D. “the monotone class theorem” □

Let’s first see what we can obtain when E is Lebesgue measurable. To make the discussion simpler, suppose $f : [a,b] \to \mathbf{R}$ is a positive bounded function. By the regularity of Lebesgue measure, there exists a G_δ set E_1 such that $E \subset E_1$ and $\mu_2(E_1) = \mu_2(E)$ and there exists a F_σ set E_2 such that $E \supset E_2$ and $\mu_2(E_2) = \mu_2(E)$. Let

$$e = E \cap \{(x,t) : x \in [a,b], t > m > 0\},$$
$$e_i = E_i \cap \{(x,t) : x \in [a,b], t > m > 0\}, i = 1,2.$$

It is clear that e_1, e_2 are Borel measurable and e is Lebesgue measurable and $\mu_2(e) = \mu_2(e_1) = \mu_2(e_2)$. Now, let $L = \{(x,t) : x \in [a,b], t = m + h, h > 0\}$

$$s = e \cap L.$$
$$s_i = e_i \cap L, i = 1,2.$$

Then s_1, s_2 are linear Borel measurable by Lemma 4.1.1. We claim that s is Lebesgue measurable (left to the reader). Thus, $\{x : f(x) > m\}$ is Lebesgue measurable for every $m > 0$. Similarly if f is a negative bounded function, $\{x : f(x) < m < 0\}$ is Lebesgue measurable. Thus, for any bounded function $f; [a,b] \to \mathbf{R}$, $\{x : \alpha f(x) < \beta\}$ is Lebesgue measurable for every pair $\alpha < \beta$.

What about the converse ? That is, if for every pair $\alpha < \beta$, the set $\{x : \alpha < f(x) < \beta\}$ is Lebesgue measurable, then the planar set E associated with f is a Lebesgue measurable set ?

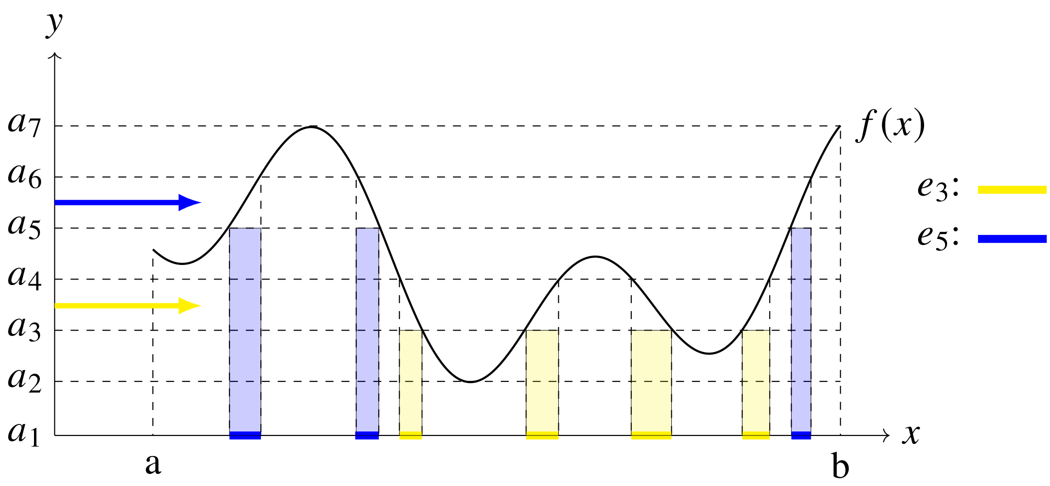
Suppose $f : [a,b] \to \mathbf{R}$ is a positive bounded function and let m, M denote the greatest lower bound and the least upper bound f on $[a,b]$, let $\mathcal{P}$ denote the ***partition*** such that $m = a_1 < a_2 < \ldots < a_n = M$. And consider the set e_i of all x such that

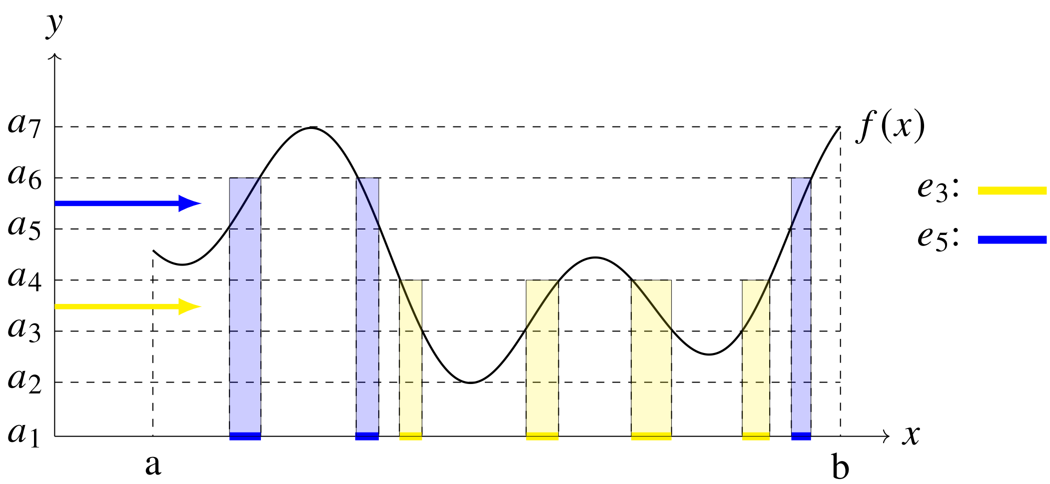
$$a_i < f(x) < a_{i+1}.$$

Then $e_1, e_2, \ldots, e_{n-1}$ partition $[a, b]$. And the planar set E lies between two generalized rectangles with measures

$$L(\mathcal{P}) = \sum_{i=1}^{n-1} a_i \mu_1(e_i) \text{ and } U(\mathcal{P}) = \sum_{i=1}^{n-1} a_{i+1} \mu_1(e_i),$$

and the difference is $\sum_{i=i}^{n-1} (a_{i+1} - a_i)\mu_1(e_1) \leq |\mathcal{P}| (b - a)$.





Thus,

$$\lim_{|P| \to 0} \sum_{i=1}^{n-1} (a_{i+1} - a_i)\mu_1(e_i) = 0,$$

which implies E is Lebesgue measurable. For any bounded function f, we only need a discussion the E_- similar to E_+, then we conclude the following proposition:

Proposition 4.1.2. For a bounded function $f : [a, b] \to \mathbf{R}$, the set $\{x : \alpha < f(x) < \beta\}$ is Lebesgue measurable for any pair $\alpha < \beta$ ***iff*** the planar set E associated with f is Lebesgue measurable. In this case, we call f is a ***measurable function***, and

$$\int_a^b f(x)dx = \mu_2(E_+) - \mu_2(E_-).$$

4.2 Measurable Functions

From the previous section, we have already learned the definition of a Measurable function: if for any pair of numbers $\alpha < \beta$, the set $\{x : \alpha < f(x) < \beta\}$ is (Lebesgue) measurable, then f is called a measurable function. Of course, this is the definition of a measurable function on an interval in $\mathbf{R}$. To broaden the scope of our study, we adopt the following definition. Suppose X and Y are topological spaces. A mapping $f : X \to Y$ is continuous ***iff*** $f^{-1}(U)$ is open whenever $U \subset Y$ is open. Then we define a new notation — measurable mapping analogously.

Definition 4.2.1. Let $\mathcal{F}_X, \mathcal{F}_Y$ be nonempty families of subsets of X and Y, respectively. A mapping $f : X \to Y$ is called ***($\mathcal{F}_X$,$\mathcal{F}_Y$)-measurable*** if

$$f^{-1}(A) \in \mathcal{F}_X \text{ for each } A \in \mathcal{F}_Y.$$

In fact, this definition is fully applicable to the previous notion of measurable functions, specifically: we first partition the range into finite intervals, and these intervals belong to the family $\mathcal{B}(\mathbf{R})$. If for any pair $\alpha < \beta$, the set $\{x : \alpha < f(x) < \beta\} \in \mathcal{L}^1$, then f is called a measurable function. Although we currently only require this condition for all intervals, which is not yet sufficiently to satisfy our definition 4.2.1, we will later prove that for such a measurable function, this relation holds for all sets in $\mathcal{B}(\mathbf{R})$.

A The Baisc Properties of Measurable Functions

Throughout this chapter, we will consider an abstract measure space (X, Σ, μ), where μ is a measure defined on the σ-algebra Σ.

Virtually all the material in this first section depends only on the σ-algebra and not on the measure μ. This is a reflection of the fact that the elementary properties of measurable functions are "set-theoretic" and are not related to μ.

We will consider functions $f : X \to \overline{\mathbf{R}}$, where $\overline{\mathbf{R}} = \mathbf{R} \cup \{-\infty\} \cup \{+\infty\}$ is the set of ***extend real numbers***. The arithmetic operations on $\overline{\mathbf{R}}$ are subject to the following conventions: for $x \in \mathbf{R}$, we define

$$x + (\pm\infty) = (\pm\infty) + x = \pm\infty, \tag{4.1}$$

$$(\pm\infty) + (\pm\infty) = \pm\infty, (\pm\infty) - (\mp\infty) = \pm\infty, \tag{4.2}$$

$$(\pm\infty) + (\mp\infty) \text{ and } (\pm\infty) - (\pm\infty) \text{ are undefined}, \tag{4.3}$$

$$x \cdot (\pm\infty) = (\pm\infty) \cdot x = \begin{cases} \pm\infty, & x > 0 \\ 0, & x = 0 \\ \mp\infty, & x < 0 \end{cases}, \tag{4.4}$$

$$(\pm\infty) \cdot (\pm\infty) = \pm\infty \text{ and } (\pm\infty) \cdot (\mp\infty) = -\infty. \tag{4.5}$$

We equip $\mathbf{R}$ and $\overline{\mathbf{R}}$ with their Borel σ-algebra $\mathcal{B}(\mathbf{R})$ and $\mathcal{B}(\overline{\mathbf{R}})$. And define

$$\mathcal{B}(\overline{\mathbf{R}}) = \left\{E \subset \overline{\mathbf{R}} : E \cap \mathbf{R} \in \mathcal{B}(\mathbf{R})\right\}.$$

Definition 4.2.2. A ***measurable space*** is a pair (X, Σ), where X is a set and Σ is a σ-algebra of subsets of X.

To emphasize that the measurablility of a mapping $f : X \to Y$ depends on the σ-algebras Σ_X, Σ_Y, we sometimes equip the spaces with their σ-algebras and denote the map as $f : (X, \Sigma_X) \to (Y, \Sigma_Y)$. However, for convenience of notation, we will retain the original notation, i.e., $f : X \to Y$, when f is known to be (Σ_X, Σ_Y)-measurable. In a special case, if $f : (X, \Sigma) \to (Y, \mathcal{B}(Y))$ is $(\Sigma, \mathcal{B}(Y))$-measurable, we refer to f as Σ-measurable. In particular, if $\Sigma_X \supset \mathcal{B}(X)$ and $\Sigma_Y = \mathcal{B}(Y)$, then f is called a ***Borel measurable mapping***. If $X = \mathbf{R}^n$ and $\Sigma_X = \mathcal{L}^n, \Sigma_Y = \mathcal{B}(Y)$, then f is called a ***Lebesgue measurable mapping***.

One may wonder why the notation in 2.4.1 is defined as it is. There is a reason that is to ensure that continuous mappings be measurable. Specifically,

Proposition 4.2.1. Every continuous function defined on a topological space to a topological space is a Borel measurable mapping.

Proof. Let X, Y be topological spaces and $f : X \to Y$ be a continuous function. Then $f^{-1}(A) \in \mathcal{B}(X)$ for each A is open in Y. Now, we define

$$\Sigma = \left\{A \subset Y : f^{-1}(A) \in \mathcal{B}(X)\right\}.$$

It is clear that Σ is a σ-algebra that contains all the open subsets of Y. However, $\mathcal{B}(Y)$ is the smallest σ-algebra contains all the open subsets of Y, which implies $\mathcal{B}(Y) \subset \Sigma$. Thus, $f^{-1}(A) \in \mathcal{B}(X)$ for every element $A \in \mathcal{B}(Y)$. Therefore, f is Borel measurable. □

As the study of continuous functions provided a foundation for Riemann integration, this notation will be essential for understanding Lebesgue integration.

The definition of an Σ-measurable function requires the inverse image of every Borel subset of $\mathbf{R}$ to be in Σ. The following statements shows that to verify that a function is Σ-measurable, we can check the inverse image of a much smaller collection of subsets of $\mathbf{R}$.

Theorem 4.2.1. Let $f : X \to Y$ be a mapping between measurable spaces, and let $\Sigma_Y = \sigma(\mathcal{C})$. Then f is (Σ_X, Σ_Y)-measurable ***iff*** $f^{-1}(C) \in \Sigma_X$ for each $C \in \mathcal{C}$.

Proof. First, assume that f is measurable, then $f^{-1}(A) \in \Sigma_X$ for each $A \in \Sigma_Y = \sigma(\mathcal{C})$. In particular, $\mathcal{C} \subset \Sigma_Y$, thus $f^{-1}(C) \in \Sigma_X$ for each $C \in \mathcal{C}$. For the converse, it is clear that Σ_X is the σ-algebra that contains $f^{-1}(C)$, so

$$\sigma(f^{-1}(C)) \subset \Sigma_X.$$

Thus, we obtain $f^{-1}(\sigma(C)) \subset \Sigma_X$, i.e., $f^{-1}(\Sigma_Y) \subset \Sigma_X$ by the corollary 1 of theorem 1.1.5. Completing the proof. □

Our study in this part will focus on measurable mappings of the form $f : (X, \Sigma) \to (\mathbf{R}, \mathcal{B}(\mathbf{R}))$ or $(\overline{\mathbf{R}}, \mathcal{B}(\overline{\mathbf{R}}))$. To differentiate them from general measurable mappings and for the sake of expository convenience, we specifically refer to this type of mapping as a ***measurable function***.

►**Corollary 1.** For the real-valued function $f : X \to \mathbf{R}$ on a measurable space, let $\mathcal{I}$ be any one of the following families:

$$\begin{aligned} &\mathcal{I}_1 = \{(a,b) : a < b\}, \quad \mathcal{I}_2 = \{[a,b] : a < b\}, \quad \mathcal{I}_3 = \{[a,b) : a < b\}, \\ &\mathcal{I}_4 = \{(a,b] : a < b\}, \quad \mathcal{I}_5 = \{(a,\infty) : a \in \mathbb{R}\}, \quad \mathcal{I}_6 = \{(-\infty,b) : b \in \mathbb{R}\} \\ &\mathcal{I}_7 = \{[a,\infty) : a \in \mathbb{R}\}, \quad \mathcal{I}_8 = \{(-\infty,b] : b \in \mathbb{R}\}. \end{aligned}$$

Then f is Σ-measurable ***iff*** $f^{-1}(I) \in \Sigma$ for each $I \in \mathcal{I}$.

Proof. It is clear that $\sigma(\mathcal{I}) = \mathcal{B}(\mathbf{R})$. □

►**Corollary 2.** For the extended real-valued function $f : X \to \overline{\mathbf{R}}$ on a measurable space, let $\mathcal{I}$ be any one of the following families:

$$\begin{aligned} &\overline{\mathcal{I}}_5 = \{(a,\infty] : a \in \mathbb{R}\}, \quad \overline{\mathcal{I}}_6 = \{[-\infty,b) : b \in \mathbb{R}\}, \\ &\overline{\mathcal{I}}_7 = \{[a,\infty] : a \in \mathbb{R}\}, \quad \overline{\mathcal{I}}_8 = \{[-\infty,b] : b \in \mathbb{R}\}. \end{aligned}$$

Then f is Σ-measurable ***iff*** $f^{-1}(I) \in \Sigma$ for each $I \in \mathcal{I}$.

Therefore, some textbooks define measurable functions as follows: a function $f : X \to \mathbf{R}$ is called Σ-measurable if $f^{-1}((-\infty,a)) = \{x \in X : f(x) < a\} \in \Sigma$ or $f^{-1}((a,\infty)) \in \Sigma$. To simplify notation, we simply write $[f > a]$ to denote $\{x \in X : f(x) > a\}$.

We now proceed to show that measurablility is preserved under elementary arithmetic operations on measurable functions. For this, the following will be useful.

Proposition 4.2.2. Let $f, g : X \to \overline{\mathbf{R}}$ be Σ-measurable functions. Then the following sets are measurable:

$$[f > g], \quad [f \geq g], \quad [f = g]$$

Proof. If $f(x) > g(x)$, then there is a rational number r such that $f(x) > r > g(x)$. Therefore, it follows that

$$[f > g] = \bigcup_{r \in \mathbf{Q}} [f > r] \cap [g < r] \in \Sigma.$$

For $[f \geq g]$, it is the complement of $[f > g]$ with f and g interchanged, so it is measurable. And $[f = g]$ is the intersection of two measurable sets $([f \geq g], [f \leq g])$, and so it too is measurable. □

Proposition 4.2.3. Let $f, g : X \to \mathbf{R}$ be Σ-measurable functions and let $\alpha \in \mathbf{R}$. The following functions are measurable:

$$\alpha f, \quad f+g, \quad f^2, \quad fg, \quad f/g, \quad |f|, \quad f^+, \quad f^-$$

Proof. The proof of the first is trivial. To vertify the second statement, we observe that $f(x)+g(x) > a$ ***iff*** there is a rational number q satisfying $f(x) > q > a - g(x)$. Thus$[f + g > a] = \bigcup_{q\in\mathbf{Q}} [f > q] \cap [g > a - q] \in \Sigma$. Thus, $f + g$ is measurable. To vertify the thrid and forth statement, we observe that $[f^2 > a] = [f > \sqrt{a}] \cup [f < -\sqrt{a}] \in \Sigma$ and $[f/g > a] = ([g > 0] \cap [f > ag]) \cup ([g < 0] \cap [f < ag])$. Thus, $f^2, f/g$ are measurable. For the remaining statements, we observe that

$$fg = \frac{(f+g)^2 - f^2 - g^2}{2}, f^+ = \frac{|f| + f}{2}, f^- = \frac{|f| - f}{2}.$$

Thus, they are measurable. □

Proposition 4.2.4. Let $\{f_n : X \to \mathbf{R}\}$ be a sequence of measurable functions then the following functions are measurable:

$$\inf f_n, \quad \sup f_n, \quad \liminf_{n\to\infty} f_n, \quad \limsup_{n\to\infty} f_n$$

Proof. Observe that $[\inf f_n > a] = \bigcup_{n=1}^{\infty} [f_n > a]$ and $[\sup f_n > a] = \bigcap_{n=1}^{\infty} [f_n > a]$. The result for the liminf and limsup also follows from the two observations,

$$\liminf_{n\to\infty} f_n = \sup_{k\to\infty} \left(\inf_{n\geq k} f_n \right), \limsup_{n\to\infty} f_n = \inf_{k\to\infty} \left(\sup_{n\geq k} f_n \right)$$

□

Proposition 4.2.5. If $\{f_n : X \to \mathbf{R}\}$ is a sequence of Σ-measurable functions that converges pointwise on X to the function $f : X \to \mathbf{R}$, then f is Σ-measurable.

Proof. Note that $[f > a] = \bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} \left[f_n > a + \frac{1}{k} \right]$. □

An analogous result applies to extended real valued function $f : X \to \overline{\mathbf{R}}$ if they are well-defined. For example, $f + g$ is measurable if $f(x), g(x)$ are not simultaneously equal to ∞ and $-\infty$, and fg is measurable if that $f(x), g(x)$ are not simultaneously equal to 0 and $\pm\infty$. This difficulty is overcome if we define

$$(f+g)(x) = \begin{cases} f(x) + g(x), & x \in X \setminus B, \\ \alpha, & x \in B, \end{cases}$$

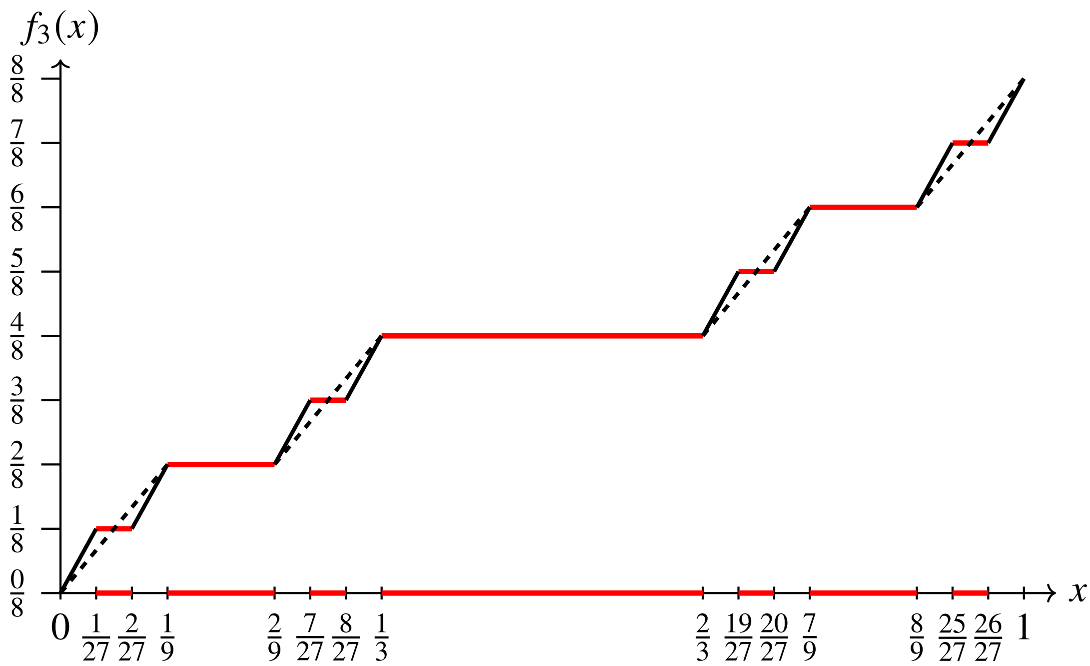
where $\alpha \in \overline{\mathbf{R}}$ is chosen arbitrarily and $B = (f^{-1}\{\infty\} \cap g^{-1}\{-\infty\}) \cup (f^{-1}\{-\infty\} \cap g^{-1}\{\infty\})$.

Theorem 4.2.2. Let $f : (X, \Sigma_X) \to (Y, \Sigma_Y)$ and $g : (Y, \Sigma_Y) \to (Z, \Sigma_Z)$ be measurable. Then the composition $g \circ f : (X, \Sigma_X) \to (Z, \Sigma_Z)$ is also measurable.

Because of the proposition, one might be tempted to conclude that the composition of Lebesgue measurable functions is again Lebesgue measurable. Suppose $f, g : \mathbf{R} \to \mathbf{R}$ are Lebesgue measurable functions. Thus, if $g \circ f$ were to be Lebesgue measurable, it would be necessary that $f^{-1}(g^{-1}(A)) \in \mathcal{L}$ for each $A \in \mathcal{B}(\mathbf{R})$. The definitions imply that it would be necessary for $g^{-1}(A) \in \mathcal{B}(\mathbf{R})$ for each $A \in \mathcal{B}(\mathbf{R})$. The following example shows that this is not generally true.

Example 4.2.1. Recall that the Cantor set C can be expressed as $C = \bigcap_{j=1}^{\infty} C_j$ where C_j is the union of the 2^j closed intervals that remain after the jth step of the construction. Each of these intervals has length $1/3^j$. And the set $G_1 \cup \cdots \cup G_j$ consists of the $2^j - 1$ open intervals that are deleted at the jth step. Let these intervals be denoted by $G_{j,k}$, $k = 1, 2, \ldots, 2^{j-1}$ such that $G_1 = G_{1,1}$, $G_2 = G_{2,1} \cup G_{2,2}, \ldots, G_j = G_{j,1} \cup G_{j,2} \cup \cdots \cup G_{j,2^{j-1}}$. Now define a continuous function f_j on $[0, 1]$ by

$$f_j(0) = 0, f_j(1) = 1, f_j(x) = \frac{k}{2^i} \text{ for } x \in G_{i,k} (i \leq j),$$
$$f_j(x) \text{ linearly on each interval of } C_j.$$



Cantor function up to step 3

The function f_j is continuous and nondecreasing, and it satisfies

$$\left|f_j(x) - f_{j+1}(x)\right| < \frac{1}{2^j} \text{ for } x \in [0,1]$$

Since $\left|f_j - f_{j+m}\right| < \sum_{i=i}^{j+m-1} \frac{1}{2^i} < \frac{1}{2^{j-1}}$, it follows that $\{f_j\}$ is converges uniformly to a continuous function f called ***Cantor-Lebesgue function***, or more picturesquely, the ***Devil's staircase***. The Cantor function is not Lipschitz, but it does satisfy a weaker condition

Definition 4.2.3. A mapping $f : \mathbf{R}^n \to \mathbf{R}^m$ is ***Hölder*** with exponent $\alpha > 0$ if there exists a constant $C > 0$ such that

$$|f(x) - f(y)| \leq C\,|x-y|^\alpha \text{ for all } x, y \in \mathbf{R}^n$$

And the smallest such C called the ***Hölder constant for*** f. Thus, Lipschitz mapping is a Hölder mapping with exponent $\alpha = 1$.

Proposition 4.2.6. The Cantor-Lebesgue function is Hölder on $[0,1]$ precisely for exponents $\alpha \in (0, \log_3 2]$. It follows that Cantor-Lebesgue function is not Lipschitz.

Proposition 4.2.7. Let f be the Cantor-Lebesgue function and define a function h on $[0,1]$ by $h(x) = f(x) + x$. Then h is a strictly increasing continuous function that maps $[0,1]$ to $[0,2]$,

- maps the Cantor set to a Lebesgue measurable set of positive measure and
- maps a measurable set, a subset of the Cantor set, to a nonmeasurable set.

Proof. The first statement is straightforward and h maps the Cantor set to a set P of measure 1. We show that last statement, let N be a nonmeasurable subset of P. Then, there exists a nonmeasurable subset of P. Let $A = h^{-1}(N)$, we have $A \subset C$, and therefore A is Lebesgue measurable since $(\mathbf{R}, \mathcal{L}^1, \mu)$ is complete. Thus, we see that h maps a measurable set to a nonmeasurable set. □

Note that h^{-1} is Lebesgue measurable, since it is continuous. Let $F := h^{-1}$. Observe that A is not a Borel set (otherwise, $F^{-1}(A) = h(A) = N$ is a Borel set.) Now, χ_A is a Lebesgue measurable function, since A is a Lebesgue measurable set. Let $g := \chi_A$, then $g^{-1}(1) = A$, and thus, g is an example of a Lebesgue measurable function that does not preserve Borel sets. Moreover,

$$g \circ F = \chi_A \circ h^{-1} = \chi_N,$$

which shows that this composition of Lebesgue measurable functions is not Lebesgue measurable. Although the example shows that Lebesgue measurable functions are not generally closed under composition, a positive result can be obtained if the outer function in the composition is assumed to be Borel measurable.

Theorem 4.2.3. Let $f : X \to \mathbf{R}$ be a Σ-measurable function and $g : \mathbf{R} \to \mathbf{R}$ is a Borel measurable function. Then $g \circ f$ is Σ-measurable. In particular, if $X = \mathbf{R}^n$ and f is Lebesgue measurable, then $g \circ f$ is Lebesgue measurable.

B Convergence

Several forms of convergence of a sequence of functions are important in the theory of integration. We begin this subsection with two forms *pointwise convergence* and *uniformly convergence* that readers should be known.

Definition 4.2.4. Let X be a set, $\{f_n : X \to \mathbf{R}\}$ be a sequence of functions and $f : X \to \mathbf{R}$.

1. The sequence $\{f_n\}$ ***converges pointwise*** on X to f if
$$\lim_{n\to\infty} f_n(x) = f(x) \text{ for all } x \in X.$$
We denote it as $f_n \to f$ for all $x \in X$.

2. The sequence $\{f_n : X \to \mathbf{R}\}$ ***converges uniformly*** on X to f if for every $\epsilon > 0$, there exists $N \in \mathbf{N}$ such that $|f_n(x) - f(x)| < \epsilon$ for all $n > N$ and $x \in X$. We denote it as $f_n \xrightarrow{u} f$ for all $x \in X$.

We discuss two other forms in this subsection: *almost everywhere convergence* and *convergence in measure*.

Let (X, Σ, μ) be a mesaure space and $E \subset X$, and let $\mathscr{P}$ be some proposition related to points in E. If there exists a null set $E_0 \subset X$ ($\mu(E_0) = 0$), such that proposition $\mathscr{P}$ holds for all $x \in E \setminus E_0$, then proposition $\mathscr{P}$ is said to hold ***μ-almost everywhere*** on E. For example, let (X, Σ, μ) be a measure space and let $f, g : X \to \overline{\mathbf{R}}$, then we say that $f = g$ μ-almost everywhere if there is a null set $N \in \Sigma$ such that $f(x) = g(x)$ for all $x \in X \setminus N$, denoted as $f \overset{\text{a.e.}}{=\!=} g$. If the measure μ is clear from the context, the reference to μ will be omitted.

Definition 4.2.5. Let (X, Σ, μ) be a measure space and let $\left\{f, f_n : X \to \overline{\mathbf{R}}\right\}$ be a sequence of finite a.e. measurable functions. If $\lim_{n\to\infty} f_n \overset{\text{a.e.}}{=\!=} f$, we say that $\{f_n\}$ ***converges almost everywhere*** to f, denoted as $f_n \xrightarrow{\text{a.e.}} f$. The sequence of functions is said to ***converge almost uniformly*** to f if for every $\epsilon > 0$, there exists a mesaurable set E such that $\mu(X \setminus E) < \epsilon$ and $f_n \xrightarrow{u} f$ on E, denoted as $f_n \xrightarrow{a.u.} f$.

Proposition 4.2.8. Let (X, Σ, μ) be a complete measure space.

1. Let $f, g : X \to \overline{\mathbf{R}}$. If f is measurable and $f = g$ a.e., then g is measurable.
2. Let $\left\{f_n : X \to \overline{\mathbf{R}}\right\}$ be a sequence of finite a.e. measurable functions and $f_n \xrightarrow{\text{a.e.}} f$, then f is measurable.

Proof. For (1), let $E = [f = g]$. Then E^c is a null set. For $a \in \mathbf{R}$, we have

$$[g > a] = ([g > a] \cap E) \cup ([g > a] \cap E^c).$$

Moreover,

$$[g > a] \cap E = [f > a] \cap E = [f > a] \setminus ([f > a] \cap E^c).$$

Since Σ is complete and E^c is a null set, $[g > a] \cap E^c, [f > a] \cap E^c$ are measurable. This shows that $[g > a]$ is measurable for every $a \in \mathbf{R}$. Therefore, g is a measurable functions.

For (2), there exists a null set N such that $f_n \to f$ for all $x \in X \setminus N$. Let $E = X \setminus N, g = f\chi_E, g_n = f_n\chi_E$ for each n, then $f = g$ for all $x \in E$. Since each f_n is a measurable functions and E is a mesaurable set, each g_n is measurable. Moreover, $g_n \to g$ for all $x \in X$, hence g is measurable. Since $f = g$ for $x \in E = X \setminus N$, f is measurable by (1). □

The following theorem reveals the essence of these two modes of convergence and can serve as their equivalent definitions. From this, it is also easy to observe that: $f_n \xrightarrow{a.u.} f$ implies that $f_n \xrightarrow{a.e.} f$. This aligns with the results of our previous study.

Theorem 4.2.4. Let (X, Σ, μ) be a measure space and let $\{f, f_n : X \to \overline{\mathbf{R}}\}$ be a sequence of finite a.e. mesaurable functions. Then,

1. $f_n \xrightarrow{a.e.} f$ ***iff*** for every $\epsilon > 0$, $\mu\left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} [|f_k - f| \geq \epsilon]\right) = 0$.
2. $f_n \xrightarrow{a.u.} f$ ***iff*** for every $\epsilon > 0$, $\lim_{n\to\infty} \mu\left(\bigcup_{k=n}^{\infty} [|f_k - f| \geq \epsilon]\right) = 0$.

Proof. For (1), it is clear that

$$[f_n \to f] = \bigcap_{i=1} \bigcup_{n=1} \bigcap_{k=n} \left[[f_k - f] < \frac{1}{i} \right].$$

It follows that $f_n \xrightarrow{a.e.} f$ ***iff***

$$\mu\left(\bigcup_{i=1} \bigcap_{n=1} \bigcup_{k=n} \left[[f_k - f] \geq \frac{1}{i} \right]\right) = 0,$$

which means for every $i \geq 1$, $\mu\left(\bigcap_{n=1} \bigcup_{k=n} \left[[f_k - f] \geq \frac{1}{i} \right]\right) = 0$.

For (2), first we assume that $f_n \xrightarrow{a.u.} f$, then for every $\delta > 0$, there exists a measurable set E such that $\mu(X \setminus E) < \delta$ and $f \xrightarrow{u} f$ on E. Therefore, for every $\epsilon > 0$, there exists N, such that

$$|f_k(x) - f(x)| < \epsilon \text{ for all } k \geq N \text{ and } x \in E.$$

It follows that $\bigcup_{k=N}^{\infty} [|f_k - f| \geq \epsilon] \subset X \setminus E$, so

$$\limsup_{n\to\infty} \mu\left(\bigcup_{k=n} [|f_k - f| \geq \epsilon]\right) \leq \mu(X \setminus E) < \delta.$$

To establish the converse, for every $\delta > 0$ and $i \geq 1$, there exists N_i such that

$$\mu\left(\bigcup_{k=N_i}^{\infty} \left[|f_k - f| \geq \frac{1}{i}\right]\right) \leq \frac{\delta}{2^i}.$$

Let

$$F = \bigcup_{i=1}^{\infty} \bigcup_{k=N_i}^{\infty} \left[|f_k - f| \geq \frac{1}{i}\right].$$

Then $\mu(F) < \delta$ and

$$X \setminus F = \bigcap_{i=1}^{\infty} \bigcap_{k=N_i}^{\infty} \left[|f_k - f| < \frac{1}{i}\right],$$

which means $f_n \xrightarrow{u} f$ on $X \setminus F$, so, $f \xrightarrow{a.u.} f$ on X, as desired. □

In previous studies, we have learned that the uniform convergence of f_n f implies pointwise convergence of f_n to f, but the converse does not necessarily hold. Does this conclusion also apply to the two new modes of convergence? The following theorem addresses this question.

Egorov's Theorem for Finite Case. Let (X, Σ, μ) be a measure space with $\mu(X) < \infty$ and let $\{f, f_n : X \to \overline{\mathbf{R}}\}$ be a sequence of finite a.e. measurable functions such that $f_n \xrightarrow{a.e.} f$. Then $f_n \xrightarrow{a.u.} f$. In other words, $f \xrightarrow{a.e.} f$ ***iff*** $f \xrightarrow{a.u.} f$ whenever $\mu(X) < \infty$.

Proof. This result follows from an application of Theorem 4.2.4 and the continuity of measures. □

Remark 4.2.1. The theorem illustrates that in a finite measure space, almost everywhere convergence is equivalent to almost uniform convergence. It is worth noting that the condition of a finite measure space cannot be removed, as demonstrated by the following example: consider $f_n(x) = \chi_{(0,n)}(x), n = 1, 2, \ldots, x \in (0, \infty)$. It is clear that $f_n \xrightarrow{a.e.} f \equiv 1$ but $f_n \nrightarrow f$ a.u..

For measurable functions, considering only pointwise convergence and almost everywhere convergence is likewise insuffcient. Let us provide an example to illustrate the reason specifically,

Example 4.2.2. For any nonnegative integers m, we can write as $m = 2^n + k$, where n is an nonnegative integer and $0 \le k < 2^n$. Given a sequence of functions on $[0, 1]$:

$$f_m = \begin{cases} f_1(x) = \chi_{[0,1]}, & m = 1 \\ f_m(x) = \chi_{\left[\frac{k}{2^n}, \frac{k+1}{2^n}\right]}, & m = 2, 3, \dots \end{cases}$$

We see that

$$\begin{aligned} &f_2 = \chi_{[0,\frac{1}{2}]}, \ f_3 = \chi_{[\frac{1}{2},1]}, \\ &f_4 = \chi_{[0,\frac{1}{4}]}, \ f_5 = \chi_{[\frac{1}{4},\frac{1}{2}]}, \ f_6 = \chi_{[\frac{1}{2},\frac{3}{4}]}, \ f_7 = \chi_{[\frac{3}{4},1]}, \\ &f_8 = \chi_{[0,\frac{1}{8}]}, \ \cdots \end{aligned}$$

Every point $x_0 \in [0, 1]$ belongs to infinitely many of the sets $\left[\frac{k}{2^n}, \frac{k+1}{2^n}\right]$, and so

$$\limsup_m f_m(x) = 1, \text{ and } \liminf_m f_m(x) = 0.$$

Thus, $\{f_m\}$ disconverges to any point in $[0, 1]$ and is also dose not converge almost everywhere. Therefore, considering only pointwise convergence provides us with no information. Although there are infinitely many functions in this sequence that take the values 1 and 0 for any given x, the reader should note that as we traverse all numbers from 0 to 1, for ant given function, the frequency of taking the value 0 is very high. Moreover, this frequency gradually approaches 1 as m increases. This indicates that f_m tends to cluster closer to 0 and rarely approaches 1. Therefore, the reader should easily observe the fact:

$$\mu_1([|f_m - 0| \ge \epsilon]) = \frac{1}{2^n} \to 0 \text{ as } m \to \infty \text{ where } m = 2^n + k.$$

In this case, we say that f_m converges in measure to $f = 0$. Now, let us provide the precise definition,

Definition 4.2.6. Let (X, Σ, μ) be a measure space and let $\{f, f_n : X \to \overline{\mathbf{R}}\}$ be a sequence of finite a.e. measurable functions. If for every $\epsilon > 0$, we have

$$\lim_{n\to\infty} \mu([|f_n - f| \ge \epsilon]) = 0, \tag{$\bigstar$}$$

we say that $\{f_n\}$ ***converges in measure*** to f, denoted as $f_n \xrightarrow{\mu} f$.

Remark 4.2.2. ($\bigstar$) can be replaced by $\lim_{n\to\infty} \mu([|f_n - f| > \epsilon]) = 0$. It is clear that $lim_{n\to\infty}\mu([|f_n - f| \ge \epsilon]) = 0$ implies $\lim_n \mu([|f_n - f| > \epsilon]) = 0$ since $[|f_n - f| \ge \epsilon] \supset [|f_n - f| > \epsilon]$. On the other hand, note that

$$[|f_n - f| \ge \epsilon] = \bigcap_{m=1}^{\infty} [|f_n - f| > \epsilon - 1/m].$$

It follows that $[|f_n - f| \geq \epsilon] \subset [|f_n - f| > \epsilon - 1/m]$ for each $m \in \mathbf{N}$ (with $\epsilon - 1/m > 0$). Therefore, $\lim_n \mu([|f_n - f| > \epsilon] - 1/m) = 0$ implies that $\lim_n \mu([|f_n - f| \geq \epsilon]) = 0$, as desired.

According to the definition of convergence in measure and the equivalent definition of almost uniform convergence, it is easy to see that a sequence of functions converging almost uniformly must also converge in measure. That is, we have the following theorem:

Theorem 4.2.6. Let (X, Σ, μ) be a measure space and let $\{f, f_n : X \to \overline{\mathbf{R}}\}$ be a sequence of finite a.e. measurable functions. If $f_n \xrightarrow{a.u.} f$, then $f_n \xrightarrow{\mu} f$.

The reader should realize that almost everywhere convergence and convergence in measure share a common characteristic: the exclusion of a "bad set". However, there is a fundamental difference between the two: almost everywhere convergence emphasize the convergence if function values at points, excluding a fixed "bad set" of points. The latter does not emphasize convergence at specific points, but only requires that the measure of the "bad set" tends to 0, regardless of the specific location of the "bad set". In fact, as m changes, the "bad set" also changes. Let us now see the specific relationship between the two.

Theorem 4.2.7. Let (X, Σ, μ) be a *finite* measure space ($\mu(X) < \infty$) and let $\{f, f_n : X \to \overline{\mathbf{R}}\}$ be a sequence of finite a.e. mesaurable functions. If $f_n \xrightarrow{a.e.} f$, then $f_n \xrightarrow{\mu} f$.

The converse is not necessarily true, as is evident from the equivalent definition of "a.e.". However, the two are always related. We will attempt to investigate this connection using subsequences, starting with the following proposition.

Lemma 4.2.1. Let (X, Σ, μ) be a measure space, and let $\{f, f_n : X \to \overline{\mathbf{R}}\}$ be a sequence of finite a.e. mesaurable functions. Then, $f_n \xrightarrow{\mu} f$ ***iff*** for any subsequence $\{f_{n_k}\}$, there exists a subsubsequnce $\{f_{n_{k_j}}\}$ such that $f_{n_{k_j}} \xrightarrow{a.u.} f$.

Proof. We first assume that $f \xrightarrow{\mu} f$, and $\{f_{n_k}\}$ is a subsequence of $\{f_n\}$, and then $f_{n_k} \xrightarrow{\mu} f$ as $k \to \infty$, so, for each $j = 1, 2, \ldots$, there exists N_j such that

$$\mu\left(\left[\left|f_{n_k} - f\right| \geq \frac{1}{j}\right]\right) \leq \frac{\epsilon}{2^j} \text{ whenever } n_k \geq N_j.$$

Choose a number n_{k_j} satisfies $n_{k_j} \leq N_j$ for each $j = 1, 2, \ldots$. Then we obtain a sequence $\{f_{n_{k_j}}\}$ such that

$$\mu\left(\left[\left|f_{n_{k_j}} - f\right| \geq \frac{1}{j}\right]\right) \leq \frac{\epsilon}{2^j} \text{ for each } j.$$

Therefore,

$$\mu\left(\bigcup_{j=r}^{\infty}\left[\left|f_{n_{k_j}} - f\right| \geq \frac{1}{j}\right]\right) \leq \sum_{j=r}^{\infty}\frac{\epsilon}{2^j} = \frac{\epsilon}{2^{r-1}}.$$

By theorem 1.2.4, $f_{n_{k_j}} \xrightarrow{a.u.} f$.

To establish the converse, we assume f_n is not converge to f in measure. Then there exists $\epsilon > 0$, such that

$$\limsup_{n\to\infty} \mu\left([|f_n - f| \geq \epsilon]\right) \geq \delta > 0.$$

Then we can choose a subsequence $\{f_{n_k}\}$ such that $\mu\left(\left[\left|f_{n_k} - f\right|\right] \geq \epsilon\right) \geq \delta$ for each k. By theorem 1.2.4, there is no subsequence of $\{f_{n_{k_j}}\}$ such that $f_{n_{k_j}} \xrightarrow{a.u.} f$, which is a contradiction. □

►**Corollary 1.** Let (X, Σ, μ) be a *finite* measure space and let $\{f, f_n : X \to \overline{\mathbf{R}}\}$ be a sequence of finite a.e. mesaurable functions. Then $f_n \xrightarrow{\mu} f$ ***iff*** for every subsequence f_{n_k} there exists a subsubsequnce $f_{n_{k_j}}$ such that $f_{n_{k_j}} \xrightarrow{a.e.} f$

The necessity part in the Corollary 1 does not require "μ to be a finite measure." This necessity part is known as Riesz's theorem, which can be stated as follows:

Riesz's Theorem. Let (X, Σ, μ) be a measure space and let $\{f, f_n : X \to \overline{\mathbf{R}}\}$ be a sequence of finite a.e. mesaurable functions. If $f_n \xrightarrow{\mu} f$, then there exists a subsequence $\{f_{n_k}\}$ such that $f_{n_k} \xrightarrow{a.e.} f$.

The following figure depicts the relationships between different modes of convergence in a finite measure space. In a general measure space, two of these implications fail. Can you identify which two?

$$\begin{tikzcd}
& f_n \to f \text{ unif.} \arrow[d] & \\
& f_n \to f \text{ a.u.} \arrow[dl, Leftrightarrow] \arrow[dr] & \\
f_n \to f \text{ a.e.} \arrow[rr] & & f_n \to f\ \mu
\end{tikzcd}$$

Figure 4.1: In a finite measure space

C Approximation

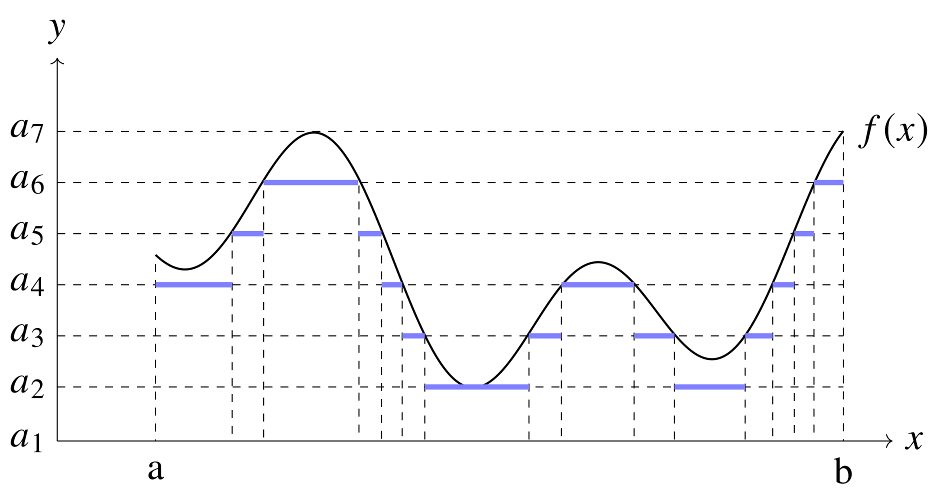
Definition 4.2.7. Let (X, Σ) be a measurable space. A function $\varphi : X \to \mathbf{R}$ is said to be ***Σ-simple*** (or simple) if φ is Σ-measurable and takes only a finite number of values, say $c_1, c_2, \ldots, c_n$ with $A_i = \varphi^{-1}(\{c_i\}) \in \Sigma$. (It is clear that A_k is measurable for each k and $X = \bigcup_{k=1}^{n} A_k$ and $A_i \cap A_j = \emptyset$ whenever $i \neq j$).

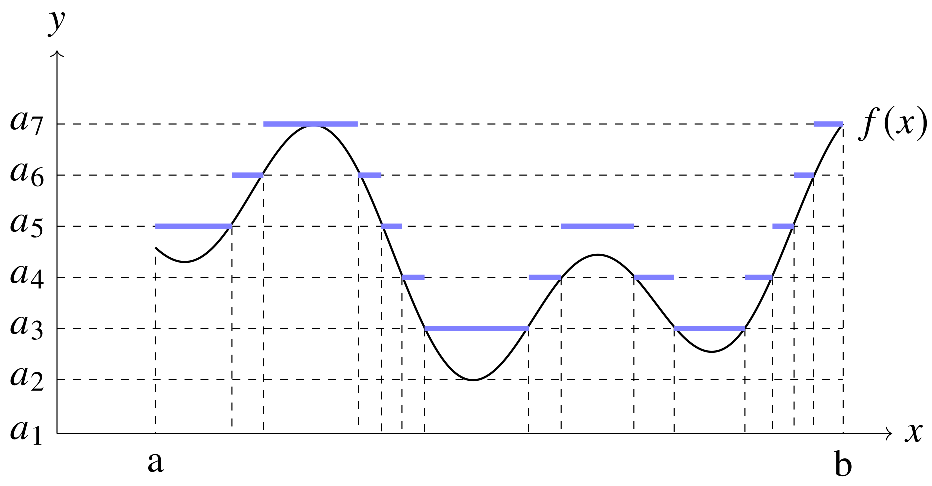
It is worth noting that simple functions are real-valued functions, so discussing their linear operations is always well-defined. As the name suggests, a simple function, being a measurable function, has a very straightforward structure. Specifically, it can be represented as follows: Suppose E is a subset of a set X. We say the ***characteristic function*** of E is the function

$$\chi_E(x) = \begin{cases} 1 & \text{if } x \in E \\ 0 & \text{if } x \notin E \end{cases}$$

It is easily to check that χ_E is measurable ***iff*** $E \in \Sigma$. Then we may write φ in its ***standard representation*** $\varphi = \sum_{i=1}^{n} c_i \chi_{A_i}$. The reader may note that the representation of a simple function is not unique. For instance, consider $\phi = \chi_{A_1} + 2\chi_{A_2}$. The same function can also be written as $\phi = \chi_{A_{1,1}} + \chi_{A_{1,2}} + 2\chi_{A_2}$, provided that $A_1 = A_{1,1} \cup A_{1,2}$ and $A_{1,1} \cap A_{1,2} = \emptyset$.

Do you remember the expressions for the upper sum and lower sum we provided in section 4.1 ? If we plot them in the graph, we obtain some rectangles. If we then keep only the top edges of these rectangles, we obtain the following two figures.





The reader may have noticed that the blue lines form two simple functions ! Perhaps these figures are rather special (since it represents a continuous function), but I believe the readers can grasp the deeper meaning behind it. As we take finer and finer partitions of the range, these simple functions appear to "approximate" the target function more closely. Perfect — this is precisely the key insight from Lebesgue's 1902 thesis. We now summarize this idea in the following theorems.

Theorem 4.2.9. Let (X, Σ, μ) be a measure space and $f : X \to [0, \infty]$ be a Σ-measurable function. Then there exists a sequence $\{\varphi_n : X \to [0, \infty)\}$ of nonnegative Σ-simple functions such that $\varphi_n(x) \uparrow f(x)$ for all $x \in X$.

Proof. Let f is $(\Sigma, \sigma[a, b))$-measurable. Let $I_n = [n, \infty]$ and partition the interval $[0, n)$ into $n2^n$ half-open intervals of length $1/2^n$,

$$I_{n,k} = \left[\frac{k-1}{2^n}, \frac{k}{2^n}\right), k = 1, 2, \ldots, n2^n.$$

Then $[0, \infty]$ is the union $\left(\bigcup_{k=1}^{n2^n} I_{n,k} \cup I_n\right)$ of disjoint sets. For each $k = 1, 2, \ldots,$ $n2^n$ let

$$A_n = [f \geq n] \text{ and } A_{n,k} = \left[\frac{k-1}{2^n} \leq f < \frac{k}{2^n}\right].$$

Since f is measurable, $A_n, A_{n,k}$ are Σ-measurable. And X is the union $\bigcup_{k=1}^{n2^n}$ $A_{n,k} \cup A_n$. of disjoint sets. Define the function

$$\varphi_n = \sum_{k=1}^{n2^n} \frac{k-1}{2^n} \chi_{A_{n,k}} + n\chi_{A_n}.$$

It is clear that φ_n is a simple function from X to $\mathbf{R}$.

If $x \in X$ and $f(x) = \infty$, then $x \in A_n$ for all n. Hence, $\varphi_n(x) = n$ for all n. It follows that $\lim_{n\to\infty} \varphi_n(x) = \infty = f(x)$. If $f(x) < \infty$ and $f(x) \geq n$, then $x \in A_n$ and $\varphi_n(x) = n \leq f(x)$. If $f(x) < n$, then $x \in A_{n,k}$ for a unique $k \in \{1, 2, \ldots, n2^n\}$, one may check that $\varphi_n(x) \uparrow f(x)$. $\square$

In fact, we can also use a decreasing sequence of simple functions to achieve convergence. However, in this case, the target function needs to be bounded (perhaps this is why Lebesgue required f to be bounded in his early study of integration). Readers can think the reason for this.

Theorem 4.2.10. Let (X, Σ, μ) be a measure space and $f : X \to \mathbf{R}$ be a Σ-measurable function. Then there exists a sequence $\{\varphi_n : X \to \mathbf{R}\}$ of Σ-simple functions such that $\varphi_n(x) \to f(x)$ for all $x \in X$.

Proof. Note that $f = f^+ - f^-$ where $f^+, f^- : X \to [0, \infty)$ are Σ-measurable functions. Then there exists two sequences $\{\psi_n\}$ and $\{\eta_n\}$ such that

$$\psi_n(x) \uparrow f^+(x) \text{ and } \eta_n(x) \uparrow f^-(x).$$

Moreover, $\varphi_n = \psi_n - \eta_n$ are Σ-simple functions. Thus, $\varphi_n(x) \to f(x)$. □

Theorem 4.2.11. Let X be a set and $f : X \to \mathbf{R}$ be a Σ-measurable function. Then there exists a sequence $\{\varphi_n : X \to \mathbf{R}\}$ of Σ-simple functions such that $|\varphi_n(x)| \uparrow |f(x)|$ for all $x \in X$.

Proof. We claim that the sequence $\{|\varphi_n|\}$ satisfies $|\varphi_n(x)| = \psi_n(x) + \eta_n(x)$ for all x and n.

If $f(x) \geq 0$, then $f^-(x) = 0$. Since η_n is a sequence of nonnegative simple functions converging increasingly to $f^-(x) = 0$, it must be that $\eta_n(x) = 0$ for all n. Similarly, if $f(x) \leq 0$, then $f^+(x) = 0$ and hence $\psi_n(x) = 0$ for all n.

Therefore, in either case, we have:

$$|\varphi_n(x)| = |\psi_n(x) - \eta_n(x)| = \psi_n(x) + \eta_n(x).$$

Since both $\{\psi_n(x)\}$ and $\{\eta_n(x)\}$ are nonnegative and increasing sequences, their sum $\psi_n(x) + \eta_n(x)$ is also increasing and converges to $f^+(x) + f^-(x) = |f(x)|$. Hence, $|\varphi_n(x)| \uparrow |f(x)|$. □

While simple functions, as measurable functions, have straightforward structures, measurable sets are after all a relatively complex class of sets. A class of functions we are more familiar with is continuous functions. Furthermore, the reader might perceive that there is also a relationship between measurable functions and continuous functions, namely, that a measurable function is almost a continuous function. Of course, this relationship is influenced by the topological space; this perception is built upon the favorable properties of metric spaces, as some topological spaces do not support this intuition. Regarding the relationship between continuous and measurable functions, we will only discuss Borel measures on metric spaces, i.e., the measure space $(X, \mathcal{B}(X), \mu)$. This is reasonable because continuity is closely related to topological structure, and Borel measures are also an important tool in the measure theory of topological spaces. The intention behind this will also be clarified in the proof process.

Luzin's Theorem. Let (X, Σ, μ) be a measure space with the Borel measure $\mu(X) < \infty$, and let f be a finite a.e. and measurable function on X. Then for every $\epsilon > 0$, there exists a closed set F_ϵ with $\mu(X \setminus F_\epsilon) < \epsilon$ and such that $f|_{F_\epsilon}$ is continuous.

Proof. WLOG, assume that f is real-valued, because $\mu([f = \pm\infty]) = 0$. First, we consider the special case where f is a simple function $f = \sum_{k=1}^{n} a_k \chi_{A_k}$ with $A_k = [f = a_k]$ for each $k = 1, 2, \ldots, n$ and A_k are disjoint sets. Let $\epsilon > 0$, by the theorem 3.1.2, there exists a closed set $F_k \subset A_k$ for each k such that

$$\mu(A_k \setminus F_k) < \epsilon/n.$$

Let $F_\epsilon = \bigcup_{k=1}^{n} F_k$, then F_ϵ is closed in X, $\mu(X \setminus F_\epsilon) \leq \sum_{k=1}^{n} \mu(A_k \setminus F_k) < \epsilon$. The function $f|_{F_k}$ is continuous for each k. Then $f|_F$ is continuous since F_k are disjoint.

Now consider an arbitrary measurable function f. By theorem 4.2.5, there exists a sequence of simple functions $\{\varphi_n : X \to \mathbf{R}\}$ such that $\varphi_n \to f$. For each φ_n, there is a closed set F_ϵ^n such that

$$\mu\left(X \setminus F_\epsilon^n\right) < \epsilon/2^{n+1}$$

and $\varphi_n|_{F_\epsilon^n}$ is continuous. By Egorov's, there exists a measurable set E such that

$$\mu(X \setminus E) < \epsilon/2, \text{ and } \varphi_n \xrightarrow{u} f \text{ on } E.$$

Let $F_\epsilon = E \cap \bigcap_{n=1}^{\infty} F_\epsilon^n$, then $\mu(X \setminus F_\epsilon) \leq \mu(X \setminus E) + \sum_{n=1}^{\infty} \mu(X \setminus F_\epsilon^n) < \epsilon$. Since φ_n is continuous on F_ϵ and $\varphi_n \xrightarrow{u} f$ on $F_\epsilon \subset E$, $f|_{F_\epsilon}$ is continuous. □

In some textbooks, Luzin's theorem is introduced after the Riesz representation theorem, leading to a different version of Luzin's theorem. Furthermore, the two versions have different requirements for the measure space and the measure, yet the underlying idea they convey is consistent: measurable functions on certain topological measure spaces can be approximated by continuous functions.

D The Monotone Class Theorem

Recall that in section 2.1, we discussed the Monotone Class Theorem for families of sets. A similar principle applies to functions. Specifically, in some cases, it suffices to verify that a certain property holds for a special class of measurable functions on a measurable space, one can then conclude that all measurable functions on this measurable space possess that property.

Monotone Class Theorem. $\mathcal{F}$ be a π system on X, and let $\mathcal{H}$ be linear space of all real-valued functions on X with the following properties

1. $1 \in \mathcal{H}$.
2. If $\{f_n \geq 0\} \subset \mathcal{H}$ with $f_n \uparrow f$ and f is real-valued, then $f \in \mathcal{H}$.
3. For each $A \in \mathcal{F}$, $\chi_A \in \mathcal{H}$.

Then $\mathcal{H}$ contains all the $\sigma(\mathcal{F})$-measurable rea-valued functions on X.

Proof. Let $\mathcal{D} = \{A \subset X : \chi_A \in \mathcal{H}\}$, then

- $X \in \mathcal{D}$ since $1 \in \mathcal{H}$,
- Let $A, B \in \mathcal{D}$ with $A \supset B$, then $\chi_{A \setminus B} = \chi_A - \chi_B \in \mathcal{H}$, so $A \setminus B \in \mathcal{D}$,
- Let $\{A_n\} \subset \mathcal{D}$ with $A_n \uparrow A$, then $\chi_{A_n} \in \mathcal{H}$ and $\chi_{A_n} \uparrow \chi_A$ and real -valued

It follows that $\mathcal{D}$ is a λ-system. Moreover condition 3 implies that $\mathcal{F} \subset \mathcal{D}$, $\sigma(\mathcal{F}) \subset \mathcal{D}$ by Dynkin's Lemma. Which means that $\chi_A \in \mathcal{H}$ for each $A \in \sigma(\mathcal{F})$. Let f be a $\sigma(\mathcal{F})$-measurable function on X. By theorem 2.2.6, there exists a sequence $\{\phi_n : X \to [0, \infty)\}$ of simple functions where every simple function is formed with countable characteristic functions such that $\phi_n \uparrow f$, thus, $\{\phi_n\} \subset \mathcal{H}$ and then $f \in \mathcal{H}$ by condition 2. □

Example 4.2.3. Let $f(x, y)$ be a $\mathcal{B}(\mathbf{R}^2)$-measurable function, then for each $x_0 \in \mathcal{R}$, we define

$$f_{x_0}(y) := f(x_0, y)$$

Then f_{x_0} is a $\mathcal{B}(\mathbf{R})$-measurable function. (See Lemma 4.1.1)

Proof. Hint: For a fixed point x_0, define $T_{x_0} : \mathbf{R} \to \mathbf{R}^2$ by $T_{x_0}(y) = (x_0, y)$. It is clear that T_{x_0} is continuous, so, T_{x_0} is a $\mathcal{B}(\mathbf{R}^2)$-measurable. Define a π system $\mathcal{F} = \{A \times B : A, B \in \mathcal{B}(\mathbf{R})\}$ and a linear space

$$\mathcal{H} = \left\{f : \mathbf{R}^2 \to \mathbf{R} : f \text{ is } \mathcal{B}(\mathbf{R}^2)\text{-measurable and } f \circ T_{x_0} \text{ is } \mathcal{B}(\mathbf{R})\text{-measurable}\right\}$$

Then $\sigma(\mathcal{F}) = \mathcal{B}(\mathbf{R}^2)$ and we show that $\mathcal{H}$ contains all the $\sigma(\mathcal{F}) = \mathcal{B}(\mathbf{R}^2)$-measurable functions. □

4.3 The Construction of Integration

From section 4.1, we have generally understood the motivation and basic approach of Lebesgue integration. After learning about measurable function, the next step is to elaborate on and extend the theory outlined in section 4.1. Looking back at section 4.1, now, we can easily see that Lebesgue's Method essentially uses simple functions to approximate the original function, where the upper and lower sums correspond precisely to the integrals of these simple functions. Therefore, following Lebesgue, we will first study the integral of nonnegative simple functions.

A Integrals of Nonnegative Measurable Functions

Let (X, Σ, μ) be a measure space. Unless otherwise specified, all measurable functions in this section are extended real-valued functions. We will denote by $\mathscr{S}$ the family of all μ-simple functions on X and by $\mathscr{S}^+$ be the family of nonnegative functions in $\mathscr{S}$. Generally speaking, if we construct the integral on the measure space $(\mathbf{R}, \mathcal{L}^1, \lambda)$, we refer to it as the Lebesgue integral.

Definition 4.3.1. Let (X, Σ, μ) be a measure space, and let $\varphi \in \mathscr{S}^+$. If $\varphi = \sum_{k=1}^{n} a_k \chi_{E_k}$, then we define the ***integral*** of φ with respect to μ by

$$\int_X \varphi d\mu = \sum_{k=1}^{n} a_k \mu(E_k).$$

If for some k, $a_k = 0$ and $\mu(E_k) = \infty$, we define $a_k\mu(E_k) = 0$. If $A \in \Sigma$, then $\phi\chi_A = \sum a_k \chi_{A\cap E_k}$ is also simple and we define

$$\int_A \phi d\mu := \int_X \phi\chi_A d\mu.$$

Proposition 4.3.1. Let $\phi, \psi \in \mathscr{S}^+$ and $c \geq 0$. Then

1. (homogenenity) $\int_X c\phi = c\int_X \phi$.
2. (additivity) $\int_X (\phi + \psi) = \int_X \phi + \int_X \psi$.
3. (monotonicity) If $\phi \leq \psi$, then $\int_X \phi \leq \int_X \psi$.

Proof. (1) is trivial. For (2), let $\phi = \sum_{k=1}^{n} a_k \chi_{E_k}, \psi = \sum_{j=1}^{m} b_j \chi_{F_j}$. Then $\phi + \psi$ is a nonnegative simple function. Let $C_{kj} = E_k \cap F_j$, it is clear that

$$\bigcup_{k,j} C_{kj} = X \text{ and } E_k = \bigcup_{j=1}^{m} (E_k \cap F_j), F_j = \bigcup_{k=1}^{n} (E_k \cap F_j).$$

Then $\phi + \psi$ is constant $a_k + b_j$ on each C_{kj}, thus,

$$\begin{aligned}
\int_X (\phi + \psi) d\mu &= \sum_{k,j} (a_k + b_j)\mu(C_{kj}) \\
&= \sum_{k,j} a_k \mu(C_{kj}) + \sum_{k,j} b_j \mu(C_{kj}) \\
&= \int_X \phi d\mu + \int_X \psi d\mu.
\end{aligned}$$

For (3), on the set C_{kj}, $\phi = a_k \leq b_j = \psi$, thus,

$$\int_X \phi d\mu = \sum_{k,j} a_k \mu(C_{k,j}) \leq \sum_{k,j} b_j \mu(C_{kj}) = \int_X \psi d\mu.$$

□

We refer to the above three properties (linearity and monotonicity) as the fundamental properties of integration, and these three properties continue to hold when integration is extended to general integrable functions.

In the study of mathematical analysis, our definite integrals satisfy the property: $\int_a^b f(x)dx + \int_b^c f(x)dx = \int_a^c f(x)dx$. This property also holds in the integral theory discussed here. In fact, this indicates an important fact, which we summarize as the following theorem:

Theorem 4.3.1. Let (X, Σ, μ) be a measure space and $\phi \in \mathscr{S}^+$. Define a set function ν by $\nu(A) = \int_A \phi$ for each $A \in \Sigma$, then ν is a measure on (X, Σ).

Proof. It is clear that ν is negative and $\nu(\emptyset) = 0$. To establish the σ-countability, let $\{A_n\}$ be a disjoint sequence in Σ and $A = \bigcup_{k=1}^{\infty} A_k$, then

$$\int_A \phi d\mu = \sum_j a_j \mu(A \cap E_j) = \sum_{k,j} a_j \mu(A_k \cap E_j) = \sum_k \int_{A_k} \phi d\mu.$$

Hence, ν is a measure on (X, Σ). □

For the needs of the following discussion (see Definition 4.3.2), we have the following lemma. This lemma serves no other purpose than to demonstrate that the definition in 4.3.2 is well-defined. However, the proof process itself does offer some insight.

Lemma 4.3.1. Let (X, Σ, μ) be a measure space and $\{\phi_n\}, \{\psi_n\} \subset \mathscr{S}^+$ be two sequences, and let ϕ, ψ belongs to $\mathscr{S}^+$. Then,

1. If $\phi_n \uparrow$ and $\lim_{n\to\infty} \phi_n \geq \phi$, then $\lim_{n\to\infty} \int \phi_n \geq \int \phi$,
2. If $\phi_n \uparrow \phi, \psi_n \uparrow \phi$, then $\lim \int \phi_n = \lim \int \psi_n$.

Proof. For (1), WLOG, assume that $\phi > 0$, and let

$$M = \max \phi(x), \quad m = \min \phi(x).$$

Since ϕ is a simple function, $0 < m \leq M < \infty$. For every $\epsilon \in (0, m)$, define

$$A_n = [\phi_n > \phi - \epsilon], n \geq 1,$$

It is clear that $A_n \uparrow X$. If $\mu(X) = \infty$, then

$$\int_X \phi_n d\mu \geq \int_X \phi_n \chi_{A_n} d\mu \geq \int_X (\phi - \epsilon)\chi_{A_n} d\mu \geq (m - \epsilon)\mu(A_n).$$

Note that $\mu(A_n) \uparrow \mu(X) = \infty$, so, $\lim_{n\to\infty} \int \phi_n = \infty$, as desired. Now, we assume $\mu(X) < \infty$, and let $B_n = X \setminus A_n$, then $B_n \downarrow \emptyset$, therefore

$$\begin{aligned}\int_X \phi_n d\mu \geq \int_X (\phi - \epsilon)\chi_{A_n} d\mu &= \int_X \phi\chi_{A_n} d\mu - \epsilon\mu(A_n) \\ &= \int_X \phi d\mu - \int_X \phi\chi_{B_n} d\mu - \epsilon\mu(A_n) \\ &\geq \int_X \phi d\mu - M\mu(B_n) - \epsilon\mu(X).\end{aligned}$$

Since $B_n \downarrow \emptyset$, $\mu(B_n) \downarrow 0$, therefore

$$\lim_{n\to\infty} \int_X \phi_n d\mu \geq \int_X \phi - \epsilon\mu(X).$$

Since $\mu(X) < \infty$, $\lim_{n\to\infty} \int \phi_n \geq \int \phi$. For (2), note that $\lim_{n\to\infty} \phi_n \geq \psi_m$ and $\lim_{n\to\infty} \psi_n \geq \phi_m$ for every $m \geq 1$. By part (1),

$$\lim_{n\to\infty} \int_X \phi_n d\mu \geq \int_X \psi_m d\mu, \quad \lim_{n\to\infty} \int_X \psi_n d\mu \geq \int_X \phi_m d\mu.$$

Letting $m \to \infty$, we obtain the statement. □

Now, let f be an arbitrarily nonnegative measurable function. Referring to Lebesgue's Method of integrating nonnegative measurable functions, the essence lies in approximating the original function using simple functions, and then representing the integral of the original nonnegative measurable function by the integrals of these approximating simple functions. Let us formalize this procedure symbolically.

Just as in theorem 4.2.10, we can define the integral of a nonnegative measurable function using the intgerals of an increasing sequence of simple functions, naturally as follows:

Definition 4.3.2. Let (X, Σ, μ) be a measure space, and let f be a nonnegative measurable function on X. By theorem 4.2.10, there exists a sequence of increasing simple functions $\{\phi_n\}$ such that $\phi_n \uparrow f$ for all $x \in X$. Then we define the ***integral*** of f with respect to μ by

$$\int_X f d\mu := \lim_{n\to\infty} \int_X \phi_n d\mu.$$

The definition of integral of f with respect to μ is well-defined (See lemma 4.3.1). We say that f is ***integrable with respect to*** μ if $\int_X |f| \, d\mu < \infty$. For $A \in \Sigma$, we define

$$\int_A f d\mu := \int_X f\chi_A d\mu.$$

Proposition 4.3.2. Let (X, Σ, μ) be a measure space, and let f, g be nonnegative measurable functions and $c \geq 0$, then

1. (homogenenity) $\int_X cf = c\int_X f$.
2. (additivity) $\int_X (f+g) = \int_X f + \int_X g$.
3. (monotonicity) $\int_X f \le \int_X g$ if $f \le g$.

Proof. For (1) and (2), let $\{\phi_n\}$ and $\{\psi_m\}$ be sequences of simple functions such that $\phi_n \uparrow f$ and $\psi_m \uparrow g$. Then $c\phi_k + b\psi_k$ is simple for each k and $a\phi_k + b\psi_k \uparrow af + bg$ for $a, b \ge 0$. For (3), note that

$$\lim_{k\to\infty} \int_X \min\{\phi_k, \psi_k\}\, d\mu = \int_X f d\mu,$$
$$\lim_{k\to\infty} \int_X \max\{\phi_k, \psi_k\}\, d\mu = \int_X g d\mu.$$

where $\min\{\phi_k, \psi_k\}$ and $\max\{\phi_k, \psi_k\}$ are simple functions with $\min\{\phi_k, \psi_k\} \uparrow f, \max\{\phi_k, \psi_k\} \uparrow g$. Therefore, $\int_X f \le \int_X g$. (Or see lemma 4.3.1.) □

In fact, most modern textbooks on real analysis or measure theory do not define nonnegative measurable functions as in definition 4.3.2. Instead, they adopt the following form:

Remark 4.3.1. Let (X, Σ, μ) be a measure space, and let f be a nonnegative measurable function on X. The ***integral*** of f with respect to μ, denoted by $\in_X f d\mu$, is

$$\int_X f d\mu = \sup\left\{\int_X \phi d\mu : 0 \le \phi \le f, \phi \text{ simple}\right\}.$$

However, I do not believe such a definition offers strong pedagogical insight or seamless continuity with prior content. Even though definition 4.3.2 involves numerous technical intricacies and its implied convergence properties of integrals are not known in advance, I consider it more imortant to understand the motivation behind the concepts and to follow the path of earlier explorations (indeed, when Lebesgue studied the integration of nonnegative measurable functions (bounded), he approximated the target function using convergent sequences of simple functions). We retain this definition as a proposition and prove that it is equivalent to definition 4.3.2

Proof. Since $\phi \le f$ for every simple function $0 \le \phi \le f$, $\int \phi d \le \int f$ by the proposition 4.3.2. And hence $\sup\left\{\int \phi : 0 \le \phi \le f, \phi \text{ simple}\right\} \le \int f$. On the other hand, there exists a sequence $\{\phi_n\}$ of nonnegative simple functions such that $\phi_n \uparrow f$, then $\int_f = \lim_{n\to\infty} \int \phi_n \le \sup\left\{\int \phi : 0 \le \phi \le f, \phi \text{ simple}\right\}$. □

Our definition employs simple functions as approximators; consequently, it naturally extends to approximation using general nonnegative measurable functions. The following theorem is referred to as the Monotone Convergence Theorem for Nonnegative Measurable Functions. This result can be further generalized to measurable functions with arbitrary sign.

Monotone Convergence Theorem for Nonnegative Case. Let (X, Σ, μ) be a measure space, and let $\{f_n\}$ be a sequence of nonnegative measurable functions, we have

1. If $f_n \uparrow f$, then $\lim_n \int f_n = \int f$.
2. If $f_n \downarrow f$ and f_k are integrable for some k, then $\lim_n \int f_n = \int f$.

Proof. For (1), $\int_X f_n \leq \int_X f$ for all n, by the monotonicity of integral, so, $\lim_{n\to\infty} f_n \leq \int_X f$. To establish the reverse inequality, fix $\alpha \in (0, 1)$. By theorem 4.2.10 and definition 4.3.2, there exists a sequence $\{\phi_n\}$ of simple functions such that $\phi_n \uparrow f$ and

$$\int_X f d\mu = \lim_{n\to\infty} \int_X \phi_m d\mu.$$

Fix m_0, define a set

$$E_n = \left[f_n \geq \alpha\phi_{m_0}\right] = \left\{x \in X : f_n(x) \geq \alpha\phi_{m_0}(x)\right\}.$$

Then $\{E_n\}$ is an increasing sequence of measurable sets with $E_n \uparrow X$, and

$$\int_X f_n d\mu \geq \int_{E_n} f_n d\mu \geq \int_{E_n} \alpha\phi_{m_0} d\mu = \alpha \int_{E_n} \phi_{m_0} d\mu.$$

By proposition 4.3.2. and the continuity of measure,

$$\lim_{n\to\infty} \int_X f_n d\mu \geq \alpha \lim_{n\to\infty} \int_{E_n} \phi_{m_0} d\mu = \alpha \int_X \phi_{m_0} d\mu.$$

Since this is true for all $\alpha < 1$,

$$\lim_{n\to\infty} \int_X f_n d\mu \geq \int_X \phi_{m_0} d\mu.$$

And let $m_0 \to \infty$, we obtain

$$\lim_{n\to\infty} \int_X f_n d\mu \geq \lim_{m_0\to\infty} \int_X \phi_{m_0} d\mu = \int_X f d\mu.$$

For (2), WLOG, aasume that f_1 is integrable. Since $0 \leq f \leq f_1$, f is integrable, let $g_n = f_1 - f_n, n = 1, 2, \ldots$. Then $\{g_n\}$ is an increasing sequence and $g_n \uparrow (f_1 - f)$ and

$$\lim_{n\to\infty} \int_X g_n d\mu = \int_X (f_1 - f)\, d\mu = \int_X f_1 d\mu - \int_X f d\mu.$$

Note that $f_1 = (f_1 - f_n) + f_n$, we have

$$\int_X f_1 d\mu = \int_X (f_1 - f_n)\, d\mu + \int_X f_n d\mu.$$

Thus,

$$\int_X f_1 d\mu - \int_X f d\mu = \lim_{n\to\infty} \int_X (f_1 - f_n) d\mu = \lim_{n\to\infty} \left(\int_X f_1 d\mu - \int_X f_n d\mu \right).$$

Since f_1 is integrable, $\int_X f d\mu = \lim_{n\to\infty} \int_X f_n d\mu$, as required. □

►**Corollary 1.** Let (X, Σ, μ) be a measure space and let $\{f_n\}$ be a sequence of nonnegative measurable functions on X. we have

1. $\int_X \left(\sum_{n=1}^{\infty} f_n\right) = \sum_{n=1}^{\infty} \int_X f_n$.
2. Define ν by $\nu(A) = \int_A f$ for each $A \in \Sigma$, then ν is a measure.

Proof. For (1), let $f = \sum_{n=1}^{\infty} f_n$. For each $k \in \mathbf{N}$, let

$$S_k = f_1 + f_2 + \cdots + f_k.$$

It is clear that $\{S_k\}$ is an increasing sequence of nonnegative measurable functions and $S_k \uparrow f$. By the monotone convergence theorem, we have

$$\int_X f d\mu = \lim_{k\to\infty} \int_X S_k d\mu. \tag{1}$$

Now, for each $k \in \mathbf{N}$,

$$\int_X S_k d\mu = \int_X f_1 d\mu + \int_X f_2 d\mu + \cdots + \int_X f_k d\mu = \sum_{n=1}^{k} \int_X f_n d\mu.$$

Thus, by (1),

$$\int_X f d\mu = \lim_{k\to\infty} \int_X S_k d\mu = \lim_{k\to\infty} \sum_{n=1}^{k} \int_X f_n d\mu = \sum_{n=1}^{\infty} \int_X f_n d\mu.$$

as required. For (2), let $\{A_k\}$ be a sequence of disjoint measurable sets in Σ and $A = \sum_{k=1}^{\infty} A_k$, then

$$\begin{aligned}
\sum_{k=1}^{\infty} \nu(A_k) = \sum_{k=1}^{\infty} \int_{A_k} f_k d\mu = \int_{A_k} \left(\sum_{k=1}^{\infty} f_k \right) d\mu = \int_X \left(\sum_{k=1}^{\infty} f \chi_{A_k} \right) d\mu \\
= \int_X \left(f \sum_{k=1}^{\infty} \chi_{A_k} \right) d\mu = \int_X f \chi_A d\mu \\
= \int_A f d\mu = \nu \left(\bigcup_{k=1}^{\infty} A_k \right).
\end{aligned}$$

which establishes (2). □

When it comes to problems of almost everywhere, we can prove that the integrals of two nonnegative measurable functions that are equal almost everywhere are completely identical. The following proposition provides more facts.

Proposition 4.3.3. Let (X, Σ, μ) be a measure space and let f, g be nonnegative measurable functions. Then

1. If $f = g$ a.e., then $\int f = \int g$.
2. If N is a null set, then $\int_N f = \int (f\chi_N) = 0$,
3. If $f \leq g$ a.e., then $\int f \leq \int g$
4. $\int f = 0$ ***iff*** $f = 0$ a.e..

Proof. (1). Let $N = [f \neq g]$, then $N \in \Sigma$ and $\mu(N) = 0$. Suppose that $h_n = n\chi_N$, then $\int h_n d\mu$=0 and suppose that

$$h(x) = \begin{cases} \infty, & x \in N \\ 0, & x \notin N. \end{cases}$$

It is clear that $h_n \uparrow h$ and hence $\int h = \lim_{n\to\infty} \int h_n = 0$. Note that $f \leq g + h$, and then

$$\int_X f d\mu \leq \int_X (g + h) d\mu = \int_X g d\mu + \int_X h d\mu = \int_X g d\mu.$$

Similarily, $\int g \leq \int f$, therefore, $\int f = \int g$.
For (2). If $x \in N^c$, then $f(x)\chi_N(x) = 0$, which means $N^c \subset [f\chi_N = 0]$ and hence $[f\chi_N \neq 0] \subset N$, so $[f\chi_N \neq 0]$ (measurable) is a null set, that is $f\chi_N = 0$ a.e.. By part (1), $\int_N f = 0$.
For (3), let $N = [f > g]$, then $\mu\,(N) = 0$, so

$$\int_X f d\mu = \int_X f\chi_N d\mu + \int_X f\chi_{N^c} d\mu = \int_X f\chi_{N^c} d\mu,$$
$$\int_X g d\mu = \int_X g\chi_N d\mu + \int_X g\chi_{N^c} d\mu = \int_X g\chi_{N^c} d\mu.$$

By proposition 4.3.2, $\int f \leq \int g$.
For (4), we first let $f = 0$ a.e., it is clear that $\int f = 0$ by part (1), to establish the converse, let $E_n = [f \geq 1/n]$, then $[f > 0] = \bigcup_{n=1}^{\infty} E_n$. Assume that $\mu([f > 0]) > 0$,

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \mu\,([f > 0]) > 0.$$

Thus, there exist some k such that $\mu(E_k) > 0$. However, then, $f \geq 1/n\chi_{E_k}$, so $\int f \geq 1/n\mu(E_k) > 0$. $\square$

►**Corollary 1.** Let (X, Σ, μ) be a measure space, and let $\{f_n\}$ be a sequence of nonnegative measurable functions, we have

1. If $f_n \uparrow f$ a.e., then $\lim_n \int f_n = \int f$.
2. If $f_n \downarrow f$ a.e. and f_k are integrable for some k, then $\lim_n \int f_n = \int f$.

B Integrals of General Measurable Functions

Now, we have understood the definition of the intergal for nonnegative measurable functions. The next natural step is to extend it to general measurable functions. Just as with the relationship between theorem 4.2.9 and theorem 4.2.10, since we know that both f^+ and f^- are nonnegative measurable functions when f is measurable, we can define the integral of a general measurable function as follows:

Definition 4.3.3. Let (X, Σ, μ) be a measure space and f be a measurable function, We say that the integral of f with respect to μ exists if at least one of $\int_X f^+$ and $\int_X f^-$ is finite, then we define the integral of f by

$$\int_X f d\mu = \int_X f^+ d\mu - \int_X f^- d\mu.$$

We say that f is ***integrable with respect to*** μ if $\int_X |f|\, d\mu < \infty$, in this case, $\int_X f^+$ and $\int_X f^-$ are both finite since $|f| = f^+ + f^-$. For $A \in \Sigma$, we define

$$\int_A f d\mu := \int_X f\chi_A d\mu = \int_X (f\chi_A)^+ d\mu - \int_X (f\chi_A)^- d\mu.$$

One may check that if the integral of f exists (or $f \in \mathscr{L}^1(\mu)$), then the integral of $f\chi_A$ exists (or $f\chi_A \in \mathscr{L}^1(\mu)$) (See Exercise 3). We define $\mathscr{L}^1(X, \Sigma, \mu)$ (or sometimes simply $\mathscr{L}^1(\mu)$) to be the set of all integrable functions on X. It is easy to check that $\mathscr{L}^1(\mu)$ is a vector space:

Proposition 4.3.4. Let (X, Σ, μ) be a measure space and $f, g \in \mathscr{L}^1(X, \Sigma, \mu)$ and $c \in \mathbf{R}$. Then

1. $cf \in \mathscr{L}^1(\mu)$ and $\int_X cf = c \int_X f$.
2. $f + g \in \mathscr{L}^1(\mu)$ and $\int_X (f + g) = \int_X f + \int_X g$.
3. $\int_X f \le \int_X g$ if $f \le g$.

Proof. For (1), if $c \ge 0$, then $(cf)^+ = cf^+$ and $(cf)^- = cf^-$, thus, $(cf)^+, (cf)^-$, and hence cf, are integrable and

$$\begin{aligned}\int_X cf d\mu &= \int_X (cf)^+ d\mu - \int_X (cf)^- d\mu \\ &= c\int_X f^+ d\mu - c\int_X f^- d\mu = c\int_X f d\mu.\end{aligned}$$

If $c < 0$, then $(cf)^+ = -cf^-$ and $(cf)^- = -cf^+$, then we also can use the preceding argument to show that cf is integrable and $\int cf = c \int f$.

For (2), note that $(f+g)^+ \leq f^+ + g^+$ and $(f+g)^- \leq f^- + g^-$, thus,

$$\int_X (f+g)^+ d\mu \leq \int_X f^+ d\mu + \int_X g^+ d\mu < \infty;$$
$$\int_X (f+g)^- d\mu \leq \int_X f^- d\mu + \int_X g^- d\mu < \infty,$$

and so $f+g$ is integrable. Since $(f+g) = (f+g)^+ - (f+g)^- = f^+ + g^+ - (f^- + g^-)$, it follows that

$$\int_X (f+g)^+ d\mu + \int_X (f^- + g^-) d\mu = \int_X (f+g)^- d\mu + \int_X (f^+ + g^+) d\mu.$$

Since $f+g, f^+, g^+, f^-, g^-$ are integrable,

$$\int_X (f+g)^+ d\mu - \int_X (f+g)^- d\mu = \int_X (f^+ + g^+) d\mu - \int_X (f^- + g^-) d\mu.$$

Therefore $\int (f+g) = \int f + \int g$.

For (3), if $f \leq g$ holds on X, then $g - f$ is a nonnegative integrable function, so, $\int (g-f) \geq 0$, and so $\int g - \int f = \int (g-f) \geq 0$, as desired. □

Remark 4.3.2. The three properties above hold not only for integrable functions. In fact, any measurable function whose integral exists also satisfies these properties.

Remark 4.3.3. Comparing corollary 1 of the preceding theorem 4.3.2 and theorem 4.3.1, we can consider that if the integration of a nonnegative measurable function with respect to a measure is treated as an operator (more commonly viewed as a mapping) on measurable sets, then it is itself a measure. However, this conclusion does not necessarily hold for general integrable functions (or their integrations are well defined), because they do not satisfy the nonnegativity property of a measure. However, we observe that, apart from failing to satisfy nonnegativity, the other two properties are satisfied (i.e., it assigns a value of 0 to the emptyset and satisfies σ-additivity). This will lead to a new concept, which we will study in the next chapter.

Proposition 4.3.5. Let (X, Σ, μ) be a measure space and let f, g be measurable functions. Then

1. If $f = g$ a.e. and one of either f or g exists (and is integrable, correspondingly), then the integral of the other also exists (and is integrable, correspondingly) and $\int f = \int g$.
2. If N is a null set, then $\int_N f = \int f \chi_N = 0$.
3. If the integral of g exists and $f \leq g$ a.e., then $\int f \leq \int g$.
4. If the integral of f exists, then $\int |f| = 0$ iff $f = 0$ a.e..

The following three theorems are the core of this section: the Monotone Convergence Theorem, Fatou's Lemma, and the Lebesgue Dominated Convergence Theorem. Together, they elucidate the convergence properties of the integral and will serve as the foundational stones for subsequent theoretical development.

Monotone Convergence Theorem. Let (X, Σ, μ) be a measure space and let $\{f_n\}$ be a sequence of measurable functions for which the integrals exist.

1. If $f_n \uparrow f$ a.e and $\int f_k > -\infty$ for some $k \in \mathbf{N}$, then the integral of f exists and $\lim_n \int f_n = \int f$ ($\int f_n \uparrow \int f$).
2. If $f_n \downarrow f$ a.e. and $\int f_k < \infty$ for some $k \in \mathbf{N}$, then the integral of f exists and $\lim_n \int f_n = \int f$ ($\int f_n \downarrow \int f$).

Proof. Here we only prove the first statement, as the second one can be shown similarly. Note that $f_n^+ \uparrow f^+$ a.e. and $f_n^- \downarrow f^-$ a.e. By the monotone convergence theorem for nonnegative measurable function, we obtain that $\int f_n^+ \uparrow \int f^+$ and $\int f_n^- \downarrow \int f^-$. Since $\int f_k > -\infty$, $\int f_k^- < \infty$ and hence the integral of f exists. So

$$\int_X f_n d\mu = \left(\int_X f_n^+ d\mu - \int_X f_n^- d\mu\right) \uparrow \left(\int_X f^+ d\mu - \int_X f^- d\mu\right) = \int_X f d\mu.$$

□

Now, we introduce Fatou's Lemma. The Fatou's Lemma did not emerge from dedicated theoretical research; rather, it was derived out of necessity by Fatou in his doctoral dissertation *Séries trigonométriques et séries de Taylor (1906)* while studying Poisson integrals and boundary value problems. Moreover, the original version primarily focused on bounded integrable functions. Nonetheless, this result remains one of the central topics in modern real analysis.

Fatou's lemma. Let (X, Σ, μ) be a measure space and let $\{g, h, f_n\}$ be measurable functinos for which the integrals exist.

1. If $\int g > -\infty$ and $f_n \geq g$ for each n, then the integral of $\liminf_n f_n$ exists and $\int \liminf_n f_n \leq \liminf_n \int f_n$.
2. If $\int h < \infty$ and $f_n \leq h$ for each n, then the integral of $\limsup_n f_n$ exists and $\int \limsup_n f_n d\mu \geq \limsup_n \int f_n$.

Proof. It suffices to prove the first statement, the second one follows by considering $\{-f_n\}$. Note that $\inf_{k \geq n} f_k \uparrow \liminf_n f_n$ a.e. and $\int \inf_{k \geq n} f_k \geq \int g > -\infty$. By the monotone convergence theorem, the integral of $\liminf_n f_n$ exists and

$$\int_X \liminf_{n \to \infty} f_n d\mu = \lim_{n \to \infty} \int_X \inf_{k \geq n} f_k d\mu \leq \liminf_{n \to \infty} \int_X f_n d\mu.$$

□

Lebesgue Dominated Convergence Theorem. Let (X, Σ, μ) be a measure space, and let $\{f_n\}$ be a sequence of measurable functions such that $f_n \xrightarrow{a.e.} f$. If there exist functions $g, h \in \mathscr{L}^1(X, \Sigma, \mu)$ such that $g \leq f_n \leq h$ a.e. for all n and $x \in X$, then $f \in \mathscr{L}^1(X, \Sigma, \mu)$ and

$$\int_X f d\mu = \lim_{n\to\infty} \int_X f_n d\mu.$$

Proof. Since $f_n \xrightarrow{a.e.} f$, f is measurable. It is clear that $g \leq f \leq h$ a.e., then $f^+ \leq h^+, f^- \leq g^-$, and hence $\int |f| \leq \int h^+ + \int g^- < \infty$ so f is integrable. Note that $\limsup_n f_n = f = \liminf_n f_n$ a.e.. By Fatou's lemma,

$$\begin{aligned}\limsup_{n\to\infty} \int_X f_n d\mu &\leq \int_X \limsup_{n\to\infty} f_n d\mu = \int_X f d\mu \\ &= \int_X \liminf_{n\to\infty} f_n d\mu \leq \liminf_{n\to\infty} \int_X f_n d\mu.\end{aligned}$$

□

►**Corollary 2.** Let (X, Σ, μ) be a measure space, and let $\{f_n\}$ be a sequence of measurable functions such that $f_n \xrightarrow{a.e.} f$. If there exists a function $g \in \mathscr{L}^1(X, \Sigma, \mu)$ such that $|f_n| < g$ for all n and $x \in X$, then

$$\lim_{n\to\infty} \int_X |f_n - f| \, d\mu = 0$$

Proof. It is clear that f is integrable. Note that $0 \leq |f_n - f| \leq 2g$ a.e., and $|f_n - f| \xrightarrow{a.e.} 0$, by the theorem before, $\int_X |f_n - f| \, d\mu = 0$. Moreover, since $\left|\int f_n - \int f\right| \leq \int |f_n - f|$, which implies $\lim_n \int f_n = \int f$. □

►**Corollary 3.** Let (X, Σ, μ) be a measure space, and let $\{f_n\}$ be a sequence of measurable functions such that $f_n \xrightarrow{\mu} f$. If there exists a function $g \in \mathscr{L}^1(X, \Sigma, \mu)$ such that $|f_n| < g$ for all n and $x \in X$, then

$$\lim_{n\to\infty} \int_X |f_n - f| \, d\mu = 0$$

Proof. Since $f_n \xrightarrow{\mu} f$, there exists a subsubsequnce $\{f_{n_{k_j}}\}$ of every subsequence $\{f_{n_k}\}$ such that $f_{n_{k_j}} \xrightarrow{a.e.} f$. Then $\lim_{j\to\infty} \int_X \left|f_{n_{k_j}} - f\right| d\mu = 0$. By the arbitrarity of $\{f_{n_k}\}$, $\lim_{n\to\infty} \int_X |f_n - f| \, d\mu = 0$. □

Reviewing the content of this section, the process of learning integration theory can roughly be divided into three major steps: the first step is to study the integration of nonnegative simple functions, the second step is to study the integration of nonnegative measurable functions, and the third step is to study the integration of general measurable functions. In fact, for greater detail, one can even start by studying characteristic functions before moving on to nonnegative simple functions. Many problems related to integration can be approached using this step-by-step procedure, which forms a typical method in integration theory. Specifically, the steps are as follows:

1. Verify the correctness of the target conclusion for characteristic functions.
2. Verify the correctness of the nonnegative simple functinos.
3. Verify the correctness of the nonnegative measurable functinos.
4. Verify the correctness of the measurable functinos.

The transition from the first step to the second relies on the linear relationships between characteristic functions and simple functions, the progression from the second to the third step depends on the Monotone Convergence Theorem, and the shift from the third to the fourth step is based on the decomposition of $f = f^+ - f^-$. The typical methods of integration will continue to be applied frequently in future studies. We present the following two theorems as examples of the typical methods in integration theory.

Theorem 4.3.6. Let (X, Σ) be a measurable space and μ_1, μ_2 be measures on (X, Σ). If $f \in \mathscr{L}^1(\mu_1) \cap \mathscr{L}^1(\mu_2)$, then $f \in \mathscr{L}^1(\mu_1 + \mu_2)$ and

$$\int_X f d(\mu_1 + \mu_2) = \int_X f d\mu_1 + \int_X f d\mu_2.$$

Proof. First we assume that $f = \chi_A$ for some $A \in \Sigma$, then

$$\begin{aligned}\int_X \chi_A d(\mu_1 + \mu_2) &= (\mu_1 + \mu_2)(A) = \mu_1(A) + \mu_2(A) \\ &= \int_X \chi_A d\mu_1 + \int_X \chi_A d\mu_2\end{aligned}$$

Next, we assume that $f = \sum_{i=1}^n a_i \chi_{A_i}$ where $a_i \geq 0$. Then

$$\begin{aligned}\int_X \sum_{i=1}^n a_i \chi_{A_i} d(\mu_1 + \mu_2) &= \sum_{i=1}^n a_i \int_X \chi_{A_i} d(\mu_1 + \mu_2) \\ &= \sum_{i=1}^n a_i \int_X \chi_{A_i} d\mu_1 + \sum_{i=1}^n a_i \int_X \chi_{A_i} d\mu_2 \\ &= \int_X \left(\sum_{i=1}^n a_i \chi_{A_i} \right) d\mu_1 + \int_X \left(\sum_{i=1}^n a_i \chi_{A_i} \right) d\mu_2.\end{aligned}$$

Now, we assume that f is a nonnegative measurable function and choose a sequence $\{f_n\}$ of nonnegative simple functions such that $f_n \uparrow f$, and hence

$$\begin{aligned}\int_X f d(\mu_1 + \mu_2) &= \lim_{n \to \infty} \int_X f_n d(\mu_1 + \mu_2) = \lim_{n \to \infty} \left(\int_X f_n d\mu_1 + \int_X f_n d\mu_2 \right) \\ &= \lim_{n \to \infty} \int_X f_n d\mu_1 + \lim_{n \to \infty} \int_X f_n d\mu_2 = \int_X f d\mu_1 + \int_X f d\mu_2.\end{aligned}$$

Finally, assume that f is a general measurable function, than

$$\begin{aligned}\int_X f d(\mu_1+\mu_2) &= \int_X f^+ d(\mu_1+\mu_2) - \int_X f^- d(\mu_1+\mu_2)\\ &= \int_X f^+ d\mu_1 + \int_X f^- d\mu_2 - \int_X f^- d\mu_1 - \int_X f^- d\mu_2\\ &= \int_X f d\mu_1 + \int_X f d\mu_2.\end{aligned}$$

□

Theorem 4.3.7. Let (X, Σ, μ) be a measure space and let f be a nonnegative measurable function on X. Then the function $\nu(A) = \int_A f d\mu$ is a measure on (X, Σ). Let g be a measurable function on X. Then the integral of g with respect to ν exists ***iff*** the integral of gf with respect to μ exists and

$$\int_A g d\nu = \int_A g f d\mu \text{ for every } A \in \Sigma.$$

Proof. First we assume that $g \in \mathscr{S}^+$ and $g = \sum_{i=1}^n a_i \chi_{A_i}$. Then

$$\begin{aligned}\int_A g d\mu &= \int_A \left(\sum_{i=1}^n a_i \chi_{A_i}\right) d\nu = \sum_{i=1}^n a_i \int_A \chi_{A_i} d\nu = \sum_{i=1}^n a_i \nu(A \cap A_i)\\ &= \sum_{i=1}^n a_i \int_{A\cap A_i} f d\mu = \sum_{i=1}^n a_i \int_A f \chi_{A_i} d\mu = \int_A g f d\mu.\end{aligned}$$

Next, we aasume g is a nonnegative measurable function, choose a sequence $\{g_n\} \subset \mathscr{S}^+$ such that $g_n \uparrow g$, it is clear that $g_n f \uparrow gf$, and hence

$$\int_A g d\nu = \lim_{n\to\infty} \int_A g_n d\nu = \lim_{n\to\infty} \int_A g_n f d\mu = \int_A g f d\mu.$$

Finally, assume that g is a measurable function, then

$$\begin{aligned}\int_A (gf)^+ d\mu = \int_A g^+ f d\mu = \int_A g^+ d\nu,\\ \int_A (gf)^- d\mu = \int_A g^- f d\mu = \int_A g^- d\nu.\end{aligned}$$

Therefore, if the integral of g with respect to ν exists, i.e., $\int g^+ d\nu < \infty$ or $\int g^- d\nu < \infty$, then $\int (gf)^+ d\mu < \infty$ or $\int (gf)^- d\mu < \infty$, that is the integral of gf with respect to μ exists and

$$\int_A g d\nu = \int_A g^+ d\nu - \int_A g^- d\nu = \int_A (gf)^+ d\mu - \int_A (gf)^- d\mu = \int_A g f d\mu.$$

□

C Convergence in the Mean

In Section 4.2, we discussed several modes of convergence of a sequence of measurable func tions, and we indicated implications that exist among them. We now use our knowledge of the integral to obtain some further convergence theorems. We begin by defining a new notion of convergence for a sequence of integrable functions

Definition 4.3.4. Let (X, Σ, μ) be a measure space and let $\{f_n : X \to \overline{\mathbf{R}}\}$ be a sequence of integrable functinos. If there exists $f \in \mathscr{L}^1(X, \Sigma, \mu)$ such that

$$\lim_{n\to\infty} \int_X |f_n - f| \, d\mu = 0,$$

we say that f_n ***converge to*** f ***in the mean*** and write $f_n \xrightarrow{mean} f$.

Theorem 4.3.8. Let (X, Σ, μ) be a measure space and let $\{f_n\}$ be a sequence of measurable functions. If $f_n \xrightarrow{mean} f$, then $f \xrightarrow{\mu} f$.

Proof. Let $A_\epsilon = [|f_n - f| \geq \epsilon]$, then $\int_X |f_n - f| \, d\mu \geq \int_{A_\epsilon} |f_n - f| \, d\mu \geq \epsilon\mu(A_\epsilon)$. It follows that $f_n \xrightarrow{mean} f$ implies that $f_n \xrightarrow{\mu} f$. □

Theorem 4.3.9. Let (X, Σ, μ) be a measure space and let $\{f_n\}$ be a sequence of measurable functions. If $f_n \xrightarrow{\mu} f$ and there exists $g \in \mathscr{L}^1(X, \Sigma, \mu)$ such that $|f_n| \leq g$ a.e. for each n, then $f_n \xrightarrow{mean} f$.

Proof. See the corollary 2 of LDCT. □

Egorov's Theorem for Dominated Case. Let (X, Σ, μ) be a measure space and let $\{f, f_n : X \to \overline{\mathbf{R}}\}$ be a sequence of finite a.e. measurable Functions such that $f_n \xrightarrow{a.e.} f$. If there exists $g \in \mathscr{L}^1(X, \Sigma, \mu)$ such that $|f_n| \leq g$ a.e. for each n, then $f_n \xrightarrow{a.u.} f$.

Proof. $f_n \xrightarrow{a.e.} f$ implies that $\mu(\bigcap_{n=1}^\infty \bigcup_{k=n}^\infty [|f_k - f| \geq 1/m]) = 0$ for each $m \in \mathbf{N}$. Our hypothesis imply that $|f| \leq g$ a.e.. Thus,

$$|f_k - f| \leq 2g \text{ a.e. for every } k \in \mathbf{R}.$$

Note that $\bigcup_{n=k}^\infty [|f_k - f| \geq 1/m] \subset S \cup T$ where $S = [2g \geq 1/m]$ and $T = \bigcup_{k=n}^\infty [|f_k - f| \geq 2g]$. Since $|f_k - f| \leq 2g$ a.e., $\mu(T) = 0$. Moreover, $g \in \mathscr{L}^1(X, \Sigma, \mu)$ implies that $\mu(S) < \infty$. Thus, it follows that $\mu(\bigcup_{k=n}^\infty [|f_k - f| \geq 1/m]) < \infty$. By the continuity of measure, $\lim_{k\to\infty} \mu(\bigcup_{k=n}^\infty [|f_n - f| \geq 1/m]) = 0$. That is $f_n \xrightarrow{a.u.} f$. □

Finally, we summarize the relationships among the four convergence modes in the dominated case with the following diagram. Readers may refer back to the relationships among the three convergence modes in the finite case to appreciate their connections. (Some unnecessary connecting lines have been omitted.)

$$
\begin{tikzcd}
 & f_n \xrightarrow{\text{unif}} f \arrow[dl] \arrow[dr] & \\
f_n \xrightarrow{\text{mean}} f \arrow[d, Leftrightarrow] & & f_n \xrightarrow{\text{a.u.}} f \arrow[ll] \arrow[d, Leftrightarrow] \\
f_n \xrightarrow{\mu} f & & f_n \xrightarrow{\text{a.e.}} f \arrow[ll]
\end{tikzcd}
$$

Figure 4.2: there exists $g \in \mathscr{L}^1(\mu)$ such that $|f_n| \le g$ for all n.

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5

Signed Measure and BV Functions

5.1 Signed Measure

In section 4.3, we have known that for a nonnegative measurable function f on X, the function $\nu(A) = \int_A \in f d\mu$ is a measure on the measurable space (X, Σ). However, this conclusion is not true for general integrable function $g \in \mathscr{L}^1(\mu)$. But we can check that the function $\nu(A) = \int_A g d\mu$ satisfies the σ-additivity and $\nu(\emptyset) = 0$. This lead us to define a new notation similarly to measure which except the nonnegativity.

Definition 5.1.1. Let (X, Σ) be a measurable space. A ***signed measure*** on (X, Σ) is a set function $\mu : \Sigma \to \overline{\mathbf{R}}$ such that

- μ takes on at most one of the value $-\infty$ or ∞
- $\mu(\emptyset) = 0$
- μ is σ-additive.

We say μ is ***finite*** if $-\infty < \mu(E) < \infty$ for every $E \in \Sigma$. The first condition is very important, if we allow the values to be ∞ and $-\infty$, then situations like this may arise: let μ be a signed measure on (X, Σ) and $A, B \in \Sigma$ with $\mu(A) = \infty, \mu(B) = -\infty$. Since $A \cup B = (A \setminus B) \cup B = (B \setminus A) \cup A$, we have

$$\mu(A \cup B) = \mu(A \setminus B) + \mu(B),$$
$$\mu(A \cup B) = \mu(B \setminus A) + \mu(A).$$

$\infty - \infty$ and $-\infty + \infty$ are undefined, so $\mu(A \setminus B) < \infty$ and $\mu(B \setminus A) > -\infty$. It follows that $\mu(A \cup B) = -\infty$ and $\mu(A \cup B) = \infty$, this is a contradiction.

Continuity of Signed Measure. Let μ be a signed measure on a measurable space (X, Σ), and let $\{A_n\}$ be a sequence of sets in Σ, then we have

1. If $A_n \uparrow A$, then $\mu(A_n) \to \mu(A)$
2. If $A_n \downarrow A$ and $\mu(A_k)$ is finite for some k, then $\mu(A_n) \to \mu(A)$.

Proof. See the Continuity of Measure in Chapter2. Notice, however, some key differences. The reader might know why ? □

Theorem 5.1.2. Let (X, Σ) be a measurable space and μ_1, μ_2 be measures on (X, Σ), if at least one of them is finite, then $\mu = \mu_1 - \mu_2$ is a signed measure.

Since the difference of two measures can give rise to a signed measure, what if we start with a signed measure, can it be decomposed into the difference of two positive measures ? The answer is yes, with just a bit of technical work. This will be the main focus of this subsection.

Definition 5.1.2. Let (X, Σ) be a measurable space and μ be a signed measure on (X, Σ). And let $P, N, M \in \Sigma$. These sets we say that

- P is positive if $\mu(E) \geq 0$ for all $E \in \Sigma$ with $E \subset P$,
- N is negative if $\mu(E) \leq 0$ for all $E \in \Sigma$ with $E \subset N$,
- M is null if $\mu(E) = 0$ for all $E \in \Sigma$ with $E \subset N$.

Proposition 5.1.1. Let μ be a signed measure on (X, Σ).

1. Let P be a positive set and if $A \in \Sigma$ is such that $A \subset P$, then A is positive.
2. Let $\{P_n\}$ be a sequence of positive sets, then $P = \bigcup_{k=1}^{\infty} P_n$ is also positive.
3. Let P be positive and N be negative, then $P \cap N$ is null.

Lemma 5.1.1. Let μ be a signed measure on the measurable space (X, Σ), and let $A \in \Sigma$ satisfies $-\infty < \mu(A) < 0$. Then there is a negative set $B \subset A$ with

$$\mu(B) \leq \mu(A).$$

Proof. Let $\delta_1 = \sup\{\mu(E) : E \in \Sigma \text{ and } E \subset A\}$, then $\delta_1 \geq \mu(\emptyset) = 0$. Choose a measurable subset A_1 of A that satisfies

$$\mu(A_1) \geq \min\left(\frac{1}{2}\delta_1, 1\right) \geq 0.$$

Define the sequences $\{\delta_n\}$ and $\{A_n\}$ by

$$\delta_n = \sup\left\{\mu(E) : E \in \Sigma \text{ and } E \subset \left(A \setminus \bigcup_{k=1}^{n-1} A_k\right)\right\},$$

choose a measurable subset A_n of $A \setminus \bigcup_{k=1}^{n-1} A_k$ that satisfies

$$\mu(A_n) \geq \min\left(\frac{1}{2}\delta_n, 1\right) \geq 0.$$

Now, define $A_\infty = \bigcup_{n=1}^{\infty} A_n$ and $B = A \setminus A_\infty$. Since A_n are pairwise disjoint and $\mu(A_n) \geq 0$, it follows that $\mu(A_\infty) \geq 0$, therefore

$$\mu(A) = \mu(A_\infty) + \mu(B) \geq \mu(B).$$

Since $\mu(A)$ is finite, $\mu(A_\infty)$ is finite, and hence $\mu(A_\infty) = \sum_{n=1}^{\infty} \mu(A_n)$ implies $\lim_{n\to\infty} \mu(A_n) = 0$. Thus, $\lim_{n\to\infty} \delta_n = 0$. Moreover, every measurable subsets E of B satisfies $\mu(E) < \delta_n$ for each n and so satisfies $\mu(E) \leq 0$, B is a negative set. $\square$

Hahn Decomposition Theorem. Let μ be a signed measure on a measurable space (X, Σ), then there exists a positive set P and the negative set $N = X \setminus P$.

Proof. Assume without loss of generality that μ does not take the value $-\infty$ (otherwise, consider $-\mu$). Define

$$L = \inf\{\mu(A) : A \text{ is a negative set}\}.$$

Choose a sequence $\{A_n\}$ of negative sets such that

$$L = \lim_{n\to\infty} \mu(A_n).$$

Let $N = \bigcup_{n=1}^{\infty} A_n$. It is easy to check that N is also a negative set. Therefore,

$$L \le \mu(N) \le \mu(A_n) \quad \text{for each } n,$$

which implies $L = \mu(N)$. Since μ does not take the value $-\infty$, we have $-\infty < \mu(N) \le 0$.

Now let $P = N^c$. Suppose, for contradiction, that there exists a measurable set $A \subset P$ such that $\mu(A) < 0$. Then by the previous lemma, there exists a negative set $B \subset A$ such that $\mu(B) < 0$. Hence,

$$\mu(N \cup B) = \mu(N) + \mu(B) < \mu(N) = L \quad (\text{since } \mu(N) \text{ is finite}).$$

However, $N \cup B$ is a negative set, which contradicts the definition of L as the infimum. Therefore, P must be a positive set. □

We refer to $\{P, N\}$ as a ***Hahn decomposition*** of X with respect to μ. The decomposition $X = P \cup N$ is usually not unique, but it leads to a canonical representation of μ as the difference of two measures. On the other hand, if $\{P_1, N_1\}, \{P_2, N_2\}$ are Hahn decomposition of μ, then we have $P_1 \setminus P_2 \subset P_1$ and $P_1 \setminus P_2 \subset N_2$, so that $P_2 \setminus P_2$ is both positive and negative, hence null. Likewise for $P_2 \setminus P_1$. Thus, the Hahn decomposition of μ is essentially unique.

Jordan Decomposition Theorem. Let μ be a signed measure on a measurable space (X, Σ), then there exist two positive measures μ^+ and μ^- such that $\mu = \mu^+ - \mu^-$, at least one of which is finite. Specifically, for $A \in \Sigma$,

$$\mu^+(A) = \sup\{\mu(B) : B \in \Sigma \text{ and } B \subset A\};$$
$$\mu^-(A) = \sup\{-\mu(B) : B \in \Sigma \text{ and } B \subset A\}.$$

Proof. Choose a Hahn decomposition $\{P, N\}$ for μ, and define

$$\mu^+(A) = \mu(A \cap P);$$
$$\mu^-(A) = -\mu(A \cap N).$$

It is clear that μ^+, μ^- are positive measures such that $\mu = \mu^+ - \mu^-$. Since $+\infty, -\infty$ cannot both occur among the values of μ, at least one of the values $\mu(P)$ and $\mu(N)$, and hence at least one of the measures μ^+, μ^-, must be finite. Each measurable subset B of A satisfies

$$\mu(B) = \mu^+(B) - \mu^-(B) \le \mu^+(B) \le \mu^+(A).$$

On the other hand, $\mu^+(A) = \mu(A \cap P)$ where $A \cap P \subset A$, therefore

$$\mu^+(A) = \sup \{\mu(B) : B \in \Sigma \text{ and } B \subset A\}.$$

Likewise the measure μ^- satisfies

$$\mu^-(A) = \sup \{-\mu(B) : B \in \Sigma \text{ and } B \subset A\}.$$

Thus μ^+, μ^- do not depend on the particular Hahn decomposition used in their construction. □

The measure μ^+ and μ^- are called the ***positive*** and ***negative variations*** of μ and we refer to $\{\mu^+, \mu^-\}$ as the ***Jordan decomposition*** of μ. And we define the ***variation*** of μ to be the measure $|\mu| = \mu^+ + \mu^-$. The ***total variation*** of μ is defined by $|\mu|(X)$. Moreover, we claim that

$$|\mu|(A) = \sup \left\{ \sum_{k=1}^{n} |\mu(A_k)| : A_k \in \Sigma, \mathcal{P} = \{A_k\} \text{ is a finite partition of } A \right\}$$

Indeed,

$$|\mu|(A) = \mu^+(A) + \mu^-(A) = |\mu(A \cap P)| + |\mu(A \cap N)| \leq \sup\{\dots\},$$

on the other hand,

$$\begin{aligned}
\sum_{k=1}^{n} |\mu(A_k)| &= \sum_{k=1}^{n} \left|\mu^+(A_k) - \mu^-(A_k)\right| \\
&\leq \sum_{k=1}^{n} \left(\mu^+(A_k) + \mu^-(A_k)\right) \\
&= \sum_{k=1}^{n} |\mu|(A_n) = |\mu|(A).
\end{aligned}$$

In fact, this is the original and most intuitive definition of variation. Indeed, we will encounter the concept of variation again later, though in the context of functions. At that point, we will discuss the meaning and properties of variation in detail.

Definition 5.1.3. Let μ be a signed measure on (X, Σ) and let f be a measurable function on (X, Σ). If the integrals of f wih respect to μ^+, μ^- exist and $\int_X f d\mu^+ - \int_X f d\mu^-$ is well defined, then we could define the integral of f with respect to the signed measure μ by

$$\int_X f d\mu := \int_X f d\mu^+ - \int_X f d\mu^-.$$

We say that f is ***integrable with respect to*** μ if $-\infty < \int_X f d\mu < \infty$, in this case, $\int_X f d\mu^+$ and $\int_X f d\mu^-$ are both finite, that is $f \in \mathscr{L}^1(\mu^+) \cap \mathscr{L}^1(\mu^-)$, then we refer to the space consists of such functions as $\mathscr{L}^1(\mu)$. For $A \in \Sigma$, we define

$$\nu(A) = \int_A f d\mu := \int_X f \chi_A d\mu.$$

One may check that if the integral of f exists (or $f \in \mathscr{L}^1(\mu)$), then the integral of $f\chi_A$ exists (or $f\chi_A \in \mathscr{L}^1(\mu)$)

Theorem 5.1.5. Let μ be a signed measure on (X, Σ) and let f be a measurable function on (X, Σ). Suppose the integral of f with respect to μ exists. Then the set function $\nu(A) = \int_A f d\mu$ is a signed measure on (X, Σ). Moreover, $\nu^+(A) = \int_A f^+ d\mu$, $\nu^-(A) = \int_A f^- d\mu$ and $|\nu|(A) = \int_A |f| d\mu$.

Proof. The first statement is straightforward. Now, we prove the last statement, let $A \in \Sigma$, then

$$\begin{aligned}\nu^+(A) &= \sup\{\nu(B) : B \in \Sigma \text{ and } B \subset A\} = \sup\left\{\int_B f d\mu : B \in \Sigma, B \subset A\right\} \\ &\le \sup\left\{\int_B f^+ d\mu : B \in \Sigma, B \subset A\right\} = \int_A f^+ d\mu.\end{aligned}$$

On the other hand,

$$\int_A f^+ d\mu = \int_{A\cap[f\ge 0]} f d\mu = \nu\left(A \cap [f \ge 0]\right),$$

where $A \cap [f \ge 0] \subset A$, hence $\int_A f^+ d\mu \le \nu^+(A)$. Therefore $\nu^+(A) = \int_A f^+ d\mu$. Similarily, $\nu^-(A) = \int_A f^{-1} d\mu$. So, $|\nu|(A) = \int_A |f| d\mu$. □

5.2 *Absolute Continuity and Singularity*

A *Absolute Continuity and Radon-Nikodym Theorem*

In the previous subsection, we concluded that, given any integrable function in a measure space, we can obtain a signed measure on the corresponding measurable space through integration. A natural follow-up question is: if we start with an arbitrary signed measure and a measurable space, can we reverse this process to obtain an integrable function ? The Radon-Nikodym theorem wil provide an answer to this. Before that, we need to introduce a new notation.

Definition 5.2.1. Let μ, ν be signed measures on (X, Σ), then ν is called ***absolutely continuous*** with respect to μ and write $\nu \ll \mu$ if $|\nu|(A) = 0$ for every $A \in \Sigma$ with $|\mu|(A) = 0$.

Remark 5.2.1. Let μ, ν be two signed measures on (X, Σ), we claim that $\nu \ll \mu$ ***iff*** $\nu(A) = 0$ for every $A \in \Sigma$ with $|\mu|(A) = 0$.

Proof. First, let $\nu \ll \mu$ and $\mu(A) = 0$, then $\nu(A) = 0$. To establish the converse, we assume that $\{P, N\}$ is a Hahn decomposition for ν. Let $|\mu|(A) = 0$, then $\mu(A \cap P) = \mu(A \cap N) = 0$. It follows that $\nu^+(A) = \nu(A \cap P) = 0, \nu^-(A) = \nu(A \cap N) = 0$. Therefore, $|\nu|(A) = 0$, as desired. □

Let (X, Σ, μ) be a measure space and f is a nonnegative function in $\mathscr{L}^1$. We have seen that the formula $\nu(A) = \int_A f d\mu$ defines a finite positive measure ν on Σ. If $\mu(A) = 0$, then $\nu(A) = 0$ by proposition 4.3.3. Thus ν is absolutely continuous with respect to μ.

The next proposition explains the terminology "absolutely continuous". It is worth noting that the " if " part of this proposition does not require the condition that ν is finite.

Proposition 5.2.1. Let μ be a signed measure on (X, Σ), and let ν be a finite signed measure on (X, Σ). Then $\nu \ll \mu$ ***iff*** for every $\epsilon > 0$, there exists $\delta > 0$ such that $|\nu|(A) < \epsilon$ whenever $|\mu|(A) < \delta$.

Proof. WLOG, we assume that μ, ν are positive measures. First, let A be measurable and $\mu(A) = 0$, then $\mu(A) < \delta$ holds for all δ, and so $\nu(A) < \epsilon$ for each ϵ, hence $\nu(A) = 0$. Thus, ν is absolutely continuous with respect to μ.

Next, we assume that there exists $\epsilon > 0$ for which there is no suitable δ. Then for each k with $\mu(A_k) \leq \frac{1}{2^k}$ and $\nu(A_k) \geq \epsilon$. Let $E_n = \bigcup_{k=n}^{\infty} A_k$ and $E = \bigcap_{n=1}^{\infty} E_n$. Then $\mu(E_n) \leq \sum_{k=n}^{\infty} \mu(A_k) < 1/2^{n-1}$, so $\mu(E) = 0$. However $\nu(E_n) \geq \nu(A_n) \geq \epsilon$ holds for each n, since ν is finite, $\nu(E) = \lim_{n\to\infty} \nu(E_n) \geq \epsilon$, which is a contradiction. □

Here are some properties of absolute continuity. Their proofs are straightforward and follow directly from the definition; hence, they are left as exercise for the reader.

Proposition 5.2.2. Let μ, ν_1, ν_2 be signed measures on (X, Σ),

1. If μ, ν_1 are both positive measures, then $\nu_1 \ll \nu_1 + \mu$.
2. If $\nu_1 \ll \nu_2, \nu_2 \ll \mu$, then $\nu_1 \ll \mu$.
3. If $\mu_1 \ll \mu, \nu_2 \ll \mu$ and $a\nu_1 + b\nu_2$ is itself a signed measure for some $a, b \in \mathbf{R}$, then $(a\nu_1 + b\nu_2) \ll \mu$.

Proof. For (1), for $A \in \Sigma$, if $(\mu + \nu_1)(A) = 0$, then $\mu(A) + \nu_1(A) = 0$, and hence, $\nu_1(A) = 0$, that is $\nu_1 \ll \mu + \nu_1$. For (2), for $A \in \Sigma$, if $|\mu|(A) = 0$, then $\nu_2(A) = 0$, and hence $\nu_1(A) = 0$, that is $\nu_1 \ll \mu$. For (3), for $A \in \Sigma$, if $|\mu|(A) = 0$, then $\nu_1(A) = \nu_2(A) = 0$, and hence $(a\nu_1 + b\nu_2)(A) = 0$, that is $a\nu_1 + b\nu_2 \ll \mu$. □

Radon-Nikodym Theorem. Let (X, Σ) be a measurable space and let μ be a σ-finite positive measure and ν be a σ-finite signed measure on (X, Σ). If ν is absolutely continuous with respect to μ, then there exists a Σ-measurable function $g < \infty$ a.e. such that

$$\nu(A) = \int_A g d\mu \text{ for each } A \in \Sigma.$$

The function g is unique up to μ-almost everywhere equality.

Proof. ***Case 1***: assume that μ, ν are finite positive measures. Let $\mathscr{F}$ be the set

$$\mathscr{F} = \left\{ f : X \to [0, \infty] \text{ be measurable} : \int_A f d\mu \leq \nu(A) \right\}.$$

First, we check that if $f_1, f_2 \in \mathscr{F}$, then $h = \max\{f_1, f_2\} \in \mathscr{F}$: to see this, let A be an arbitrary set in Σ and $A_1 = [f_1 > f_2]$ and $A_2 = [f_2 \geq f_1]$, then

$$\int_A h d\mu = \int_{A_1} f_1 d\mu + \int_{A_2} f_2 d\mu \leq \nu(A_1) + \nu(A_2) = \nu(A).$$

Now, let $a = \sup\left\{\int f d\mu : f \in \mathscr{F}\right\}$, note that $a < \nu(X) < \infty$, we choose a sequence $\{f_n\} \subset \mathscr{F}$ such that

$$\lim_{n\to\infty} \int f_n d\mu = a = \sup\left\{ \int f d\mu : f \in \mathscr{F} \right\}.$$

Let $g_n = \max\{f_1, f_2, \ldots, f_n\}$ and $g = \sup_n f_n$. Then $g_n \in \mathscr{F}$ and $g_n \uparrow g$. By the monotone convergence theorem,

$$\int_A g d\mu = \lim_{n\to\infty} \int_A g_n d\mu \leq \nu(A) < \infty.$$

It follows that $g \in \mathscr{F}$ and $\int g d\mu = a$. We turn to the proof that $\nu(A) = \int_A g d\mu$ holds for every $A \in \Sigma$. Since $g \in \mathscr{F}$, the formula $\nu_0(A) = \nu(A) - \int_A g d\mu$ defines a positive measure on Σ. Assume $\nu_0(A) = 0$. Then, since μ is finite, there exists $\epsilon > 0$ such that

$$\nu_0(X) > \epsilon\mu(X).$$

Let $\{P, N\}$ be a Hahn decomposition for the signed measure $\nu_0 - \epsilon\mu$. Note that for every $A \in \Sigma$, we have $\nu_0(A \cap P) \geq \epsilon\mu(A \cap P)$, so, we have

$$\begin{aligned}
\nu(A) = \int_A g d\mu + \nu_0(A) &\geq \int_A g d\mu + \nu_0(A \cap P) \\
&\geq \int_A g d\mu + \epsilon\mu(A \cap P) \\
&= \int_A (g + \epsilon\chi_P) d\mu.
\end{aligned}$$

So, $g + \epsilon\chi_P \in \mathscr{F}$. However, $\mu(P) > 0$ (otherwise, $\nu_0(P) = 0$ ($\nu \ll \mu$) and so $\nu_0(X) - \epsilon\mu(X) = (\nu_0 - \epsilon\mu)\ (N) \leq 0$, contradicting $\nu_0(X) > \epsilon\mu(X)$) and

$\int g d\mu \leq \nu(X) < \infty$, so $\int (g + \epsilon\chi_P) d\mu > \int g d\mu$, which is a contradiction. So, $\nu_0(A) = 0$ for every $A \in \Sigma$, i.e., $\int_A g d\mu = \nu(A)$ for every $A \in \Sigma$ where $g \in \mathscr{L}^1$ and $g < \infty$ a.e.. Suppose that $\tilde{g} : X \to [0, \infty]$ such that

$$\int_A \tilde{g} d\mu = \int_A g d\mu = \nu(A).$$

Let $E_1 = [g > \tilde{g}]$ and $E_2 = [g < \tilde{g}]$, then

$$\int_{E_1} g - \tilde{g} d\mu = \int_{E_1} g d\mu - \int_{E_1} \tilde{g} d\mu = \nu(E_1) - \nu(E_1) = 0;$$
$$\int_{E_2} g - \tilde{g} d\mu = \int_{E_2} g d\mu - \int_{E_2} \tilde{g} d\mu = \nu(E_2) - \nu(E_2) = 0.$$

Thus, $g \xlongequal{a.e.} \tilde{g}$ by proposition 4.3.3.

Case 2: we assume that μ, ν are both σ-finite positive. Then X is the union of a sequence $\{B_n\}$ of disjoint measurable sets, each of which has finite measure under μ and under ν. For each n, the first case of this proof provides a measurable function $g_n : B_n \to [0, \infty)$ such that $\nu(A) = \int_A g_n d\mu$ holds for each measurable subset A of B_n. So we define $g = \sum_{n=1}^{\infty} g_n \chi_{B_n}$ that agrees on each B_n with g_n and $g < \infty$ a.e., and

$$\begin{aligned} \nu(A) &= \sum_{n=1}^{\infty} \nu(A \cap B_n) = \sum_{n=1}^{\infty} \int_{A \cap B_n} g_n d\mu \\ &= \sum_{n=1}^{\infty} \int_{A \cap B_n} g d\mu = \int_A g d\mu. \end{aligned}$$

The uniqueness of g is determined as in the first case,

Case 3: we assume ν is a σ-finite signed measure, we apply the preceding argument to ν^+ and ν^- and substract the results,

$$\nu^+(A) = \int_A g_1 d\mu, \quad \nu^-(A) = \int_A g_2 d\mu, \text{ for every } A \in \Sigma,$$

where $g_1, g_2 < \infty$ μ-a.e. and nonnegative, then $g = g_1 - g_2 < \infty$ μ-a.e.. Since $\nu = \nu^+ - \nu^-$, at least one of them is finite, thus

$$\nu(A) = \nu^+(A) - \nu^-(A) = \int_A g d\mu \text{ is well defined.}$$

□

Remark 5.2.2. It is worth noting that the σ finiteness of μ in the theorem's assumptions cannot be omitted. An example can be provided to illustrate this fact: define the set function $\mu : \mathcal{B}(\mathbf{R}) \to \mathbf{R}$ by $\mu(A) = \#A$, then it is clear that μ is a measure but not σ-finite. One may check that $\lambda \ll \mu$. Assume there exists a measurable function f such that

$$\lambda(A) = \int_A f d\mu \text{ for each } A \in \mathcal{B}(\mathbf{R}).$$

Then for the single point set $\{a\}$, $0 = f(a)\mu(\{a\}) = f(a)$. Hence, $f \equiv 0$, and so, $\lambda \equiv 0$, which is a contradiction.

Remark 5.2.3. In fact, the above theorem can be further generalized by removing the σ-finiteness requirement on ν, that is, ν can be any signed measure. In this case, the associated function g need not be μ a.e. finite.

Proof. We only need to prove the conclusion for the case where μ is a finite measure and ν is a measure. Other cases can be handled using methods similar to those employed in the proof of theorem before.
Now, let $\mathscr{G} = \{G \in \Sigma : \nu \text{ is } \sigma\text{-finite on } G\}$ and $\gamma = \sup\{\mu(G) : G \in \mathscr{G}\}$. Then there exist $G_n \in \mathscr{G}$ such that $\lim_n \mu(G_n) = \gamma$. Let $G = \bigcup_{n=1}^{\infty} G_n$, then $G \in \mathscr{G}$ and $\mu(G) = \gamma < \infty$.
Define $\Sigma_1 := G \cap \Sigma = \{G \cap A : A \in \Sigma\}$. Then Σ_1 is the σ algebra on G. Hence $\nu_1 = \nu|_{\Sigma_1}$ is σ-finite on Σ_1 and $\nu_1 \ll \mu$. By theorem before, there exists a Σ_1-measurable function g_1 such that

$$\nu_1(A) = \int_A g_1 d\mu \text{ for every } A \in \Sigma_1.$$

Define $\Sigma_2 := G^c \cap \Sigma = \{G^c \cap A : A \in \Sigma\}$. Then Σ_2 is the σ-algebra on G^c. Then $\nu_2 = \nu|_{\Sigma_2} \ll \mu$ and we claim that $\nu_2(A) = \infty$ for all $A \in \Sigma_2$ with $\mu(A) > 0$. Otherwise, if $\nu_2(A) < \infty$, then $G \cup A \in \mathscr{G}$, hence

$$\mu(G \cup A) = \mu(G) + \mu(A) > \gamma,$$

contradicting the definition of γ. Now, define the function

$$g(x) = \begin{cases} g_1(x), & x \in G; \\ \infty, & x \in G^c. \end{cases}$$

It is clear that g is measurable and the integration $\int g d\mu$ is well defined. For every $A \in \Sigma$, $\nu(A) = \nu_1(A \cap G) + \nu_2(A \cap G^c)$. Moreover

$$\int_A g d\mu = \int_{A\cap G} g d\mu + \int_{A\cap G^c} g d\mu = \int_{A\cap G} g_1 d\mu + \int_{A\cap G^c} \infty d\mu$$

If $\mu(A \cap G^c) > 0$, then $\nu_2(A \cap G^c) = \infty$, hence $\nu(A) = \nu_1(A \cap C) + \infty = \infty$ and $\int_{A\cap G^c} \infty d\mu = \infty$. Hence $\int_A g d\mu = \int_{A\cap G} g_1 d\mu + \infty\infty = \nu(A)$. If $\mu(A \cap G^c) = 0$, then $\nu(A \cap G^c) = 0$ by $\nu \ll \mu$, that is $\nu_2(A \cap G^c) = 0$. So $\nu(A) = \nu_1(A \cap G) + 0 = \int_{A\cap G} g_1 d\mu$ and $\int_A g d\mu = \int_{A\cap G} g_1 d\mu + \int_{A\cap G^c} \infty d\mu = \int_{A\cap G} g_1 d\mu = \mu(A)$. □

In the Radon-Nikodym theorem, the function g is called the ***Radon-Nikodym derivative*** of ν with respect to μ and will be denoted by $d\nu/d\mu$, i.e., $g = d\nu/d\mu$ and

$$\nu(A) = \int_A \frac{d\nu}{d\mu} d\mu, \text{ for every } A \in \Sigma.$$

Recall theorem 5.1.5, let (X, Σ, μ) be a measure space and and the integral of f with respect to μ exists. Then we defined a signed measure $\nu(A) = \int_A f d\mu$ for $A \in \Sigma$. By the Radon-Nikodym theorem,

$$\frac{d\nu^+}{d\mu} = \left(\frac{d\nu}{d\mu}\right)^+ = f^+ \ \mu\text{-a.e. and } \frac{d\nu^-}{d\mu} = \left(\frac{d\nu}{d\mu}\right)^- = f^- \ \mu\text{-a.e.}$$

The reader may find that a new differential theory has quietly appeared. Recall in Mathematical Analysis, suppose that F is differentiable and $F' = f$ is continuous on $[a, b]$, we get the classic Fundamental theorem of Calculus:

$$F(b) - F(a) = \int_a^b f(t)dt,$$

we refer to f as the derivative of F with respect to x. But here, we are talking about "the derivative of a measure", not the "derivative of a function". However, the reader should remember that a measure is itself a kind of function. So, it's natural to think: if a function F can be represented as a signed measure ν and satisfying some conditons (like in Mathematical Analysis, F must be differentiable with a continuous derivative f) — and we will later prove that some functions F can indeed be represented this way — then, using this theorem, we can find a measurable function f that is finite μ-a.e. This function f would satisfy the integral equation and it makes perfect sense to call it the derivative. This gives us a more complete version of the Fundamental Theorem of Calculus.

But for now, let's put this idea aside. We still have more to learn about signed measures.

Proposition 5.2.3. Let μ be a σ-finite measure on (X, Σ) and ν_1, ν_2 be σ-finite signed measure on (X, Σ). If $\nu_1 \ll \mu, \nu_1 \ll \mu$ and the linear combination $a\nu_1 + b\nu_2$ (with $a, b \in \mathbf{R}$) is itself a signed measure, then the Radon-Nikodym derivatives satisfies $\dfrac{d(a\nu_1 + b\nu_2)}{d\mu} = a\dfrac{d\nu_1}{d\mu} + b\dfrac{d\nu_2}{d\mu}$

Proof. By Radon-Nikodym theorem, for $A \in \Sigma$, we have

$$\begin{aligned}\int_A \frac{d(a\nu_1 + b\nu_2)}{d\mu} d\mu &= a\nu_1(A) + b\nu_2(A) \\ &= a\int_A \frac{d\nu_1}{d\mu} d\mu + a_2 \int_A \frac{d\nu_2}{d\mu} d\mu \\ &= \int_A \left(a\frac{d\nu_1}{d\mu} + b\frac{d\nu_1}{d\mu}\right) d\mu.\end{aligned}$$

Therefore, $\dfrac{d(a\nu_1 + b\nu_2)}{d\mu} = a\dfrac{d\nu_1}{d\mu} + b\dfrac{d\nu_2}{d\mu}$ μ-a.e.. □

Theorem 5.2.2. Let μ be a σ-finite positive measure and ν be a σ-finite signed measure on (X, Σ) such that $\nu \ll \mu$. Let g be a measurable function on X. Then the integral of g with respect to μ exists ***iff*** the integral of $gd\nu/d\mu$ with respect to μ exists and

$$\int_A g d\nu = \int_A g \frac{d\nu}{d\mu} d\mu \text{ for every } A \in \Sigma.$$

Proof. It is clear that

$$\left(g\frac{d\nu}{d\mu}\right)^+ = g^+\frac{d\nu^+}{d\mu} + g^-\frac{d\nu^-}{d\mu},$$
$$\left(g\frac{d\nu}{d\mu}\right)^- = g^-\frac{d\nu^+}{d\mu} + g^+\frac{d\nu^-}{d\mu}.$$

First, we assume that the integral of g with respect to ν exists, so is to ν^+, ν^-. By theorem 4.3.3, the integrals of $gd\nu^+/d\mu, gd\nu^-/d\mu$ with respect to μ exist and

$$\int_A g d\nu^+ = \int_A \left(g\frac{d\nu^+}{d\mu}\right) d\mu,$$
$$\int_A g d\nu^- = \int_A \left(g\frac{d\nu^-}{d\mu}\right) d\mu.$$

Therefore,

$$\int_A g d\nu^+ = \int_A \left(g^+\frac{d\nu^+}{d\mu}\right) d\mu - \int_A \left(g^-\frac{d\nu^+}{d\mu}\right) d\mu,$$
$$\int_A g d\nu^- = \int_A \left(g^+\frac{d\nu^-}{d\mu}\right) d\mu - \int_A \left(g^-\frac{d\nu^-}{d\mu}\right) d\mu.$$

Since $\int gd\nu^+ - \int gd\nu^-$ is well defined, $\int (gd\nu/d\mu)^+ d\mu - \int (gd\nu/d\mu)^-$ is well defined, that is the integral of $gd\nu/d\mu$ with respect to μ exists and

$$\int_A g d\nu = \int_A g d\nu^+ - \int_A g d\nu^- = \int_A \left(g\frac{d\nu}{d\mu}\right) d\mu.$$

□

Proposition 5.2.4. Let μ, ν be σ-finite measures on (X, Σ) and λ be a σ-finite signed measure on (X, Σ). If $\lambda \ll \nu, \nu \ll \mu$, then

$$\frac{d\lambda}{d\mu} = \frac{d\lambda}{d\nu} \cdot \frac{d\nu}{d\mu} \ \mu\text{-a.e..}$$

Proof. It is clear that $\lambda \ll \mu$. By theorem 5.1.8,

$$\int_A \left(\frac{d\lambda}{d\nu} \cdot \frac{d\nu}{d\mu}\right) d\mu = \int_A \frac{d\lambda}{d\nu} d\nu = \lambda(A) = \int_A \frac{d\lambda}{d\mu} d\mu.$$

□

From the above theorems, we observe that the derivative of a sigend measure maintains the same basic operational properties as derivatives in mathematical analysis, including the chain rule.

B Singularity and Lebesgue Decomposition Theorem

Although the Radon-Nikodym theorem is highly powerful and important, its application requires a fundamental condition: the absolute continuity between signed measures. Often, the signed measure ν we examine may not necessarily satisfy this absolute continuity conditons. However, the Lebesgue Decomposition Theorem presented in this section shows that for the sigend measures ν and the positive measure μ, we can decompose ν into two parts such that one of them is absolutely continuous with respect to μ.

Definition 5.2.2. Let ν, μ be signed measures on (X, Σ), then μ and ν are ***mutually singular*** and we write $\nu \perp \mu$ if there exists a measurable set A such that

$$|\nu|(A) = 0, \quad |\mu|(A^c) = 0.$$

Proposition 5.2.5. Let μ, ν_1, ν_2 be signed measures on (X, Σ).

1. If $\nu_1 \ll \mu, \nu_1 \perp \mu$, then $\nu_1 = 0$.
2. If $\nu_1 \ll \mu, \nu_2 \perp \mu$, then $\nu_1 \perp \nu_2$.
3. If $\nu_1 \perp \mu, \nu_2 \perp \mu$ and $\nu_1 + \nu_2$ is itself a signed measure, then $(\nu_1 + \nu_2) \perp \mu$.

Proof. (1) and (2) are trival, left as an exercise to the reader. For (3), there exists $N_1 \in \Sigma$ such that $|\mu|(N_1) = 0$ and $\nu_1(N_1^c) = 0$, thus, $\nu_1|_{N_1^c} \ll \mu|_{N_1^c}$, and hence $\nu_1(A \cap N_1^c) = 0$ for all $A \in \Sigma$ by (1). Similarily, there exists $N_2 \in \Sigma$ such that $|\mu|(N_2) = 0$ and $\nu_2(A \cap N_2^c) = 0$ for all $A \in \Sigma$. Let $N = N_1 \cup N_2$, then $|\mu|(N) = 0$ and

$$\begin{aligned}(\nu_1 + \nu_2)(A \cap N^c) &= \nu_1(A \cap N^c) + \nu_2(A \cap N^c) \\ &= \nu_1\left(\left(A \cap N_2^c\right) \cap N_1^c\right) + \nu_2\left(\left(A \cap N_1^c\right) \cap N_2^c\right) \\ &= 0.\end{aligned}$$

Thus, $(\nu_1 + \nu_2) \perp \mu$. □

Lebesgue Decomposition Theorem. Let μ be a σ-finite positive measure and ν be a σ-finite signed measure on (X, Σ). Then there exist two signed measure ν_{ac}, ν_s such that

$$\nu = \nu_{ac} + \nu_s, \quad \nu_{ac} \ll \mu, \quad \nu_s \perp \mu.$$

Moreover, these signed measures are unique. And we refer to ν_{ac} as the continuous part of ν, ν_s as the singular part of ν.

*Proof. **Case 1***: we assume that ν is σ-finite positive measure, then $\nu + \mu$ so is. It is clear that $\mu \ll \mu + \nu$, so by the Radon-Nikodym theorem, there exists nonnegative measurable function $g \in \mathscr{L}^1$ such that

$$\begin{aligned}\mu(A) &= \int_A gd(\mu+\nu), \text{ for every } A \in \Sigma. \\ &= \int_A gd\mu + \int_A gd\nu.\end{aligned}$$

Let $B = [g > 0]$, and define $\nu_{ac} = \nu(A \cap B), \nu_s(A) = \mu(A \cap B^c)$. It is clear that ν_{ac}, ν_s are σ-finite positive measures and $\nu(A) = \nu_{ac}(A) + \nu_s(A)$. Assume $\mu(A) = 0$, then $\int_A gd\nu = 0$, thus $g \xlongequal{\nu-a.e.} 0$ on A. If follows that $\nu(A \cap B) = 0$ and hence $\nu_{ac}(A) = 0$, that is $\nu_{ac} \ll \mu$. Moreover, $\nu_s(B) = \mu(B^c) = 0$, that is $\nu_s \perp \mu$. To show the uniqueness, we first assume ν is a finite positive measure, if $\nu = \nu_{ac} + \nu_s = \nu'_{ac} + \mu'_s$ where $\nu_{ac} \ll \mu, \nu'_{ac} \ll \mu, \nu_s \perp \mu, \nu'_s \perp \mu$. Then $\gamma = \nu_{ac} - \nu'_{ac} = \nu'_s - \nu_s$ is such that $\gamma \ll \mu, \gamma \perp \mu$ and hence $\gamma = 0$, i.e., $\nu_{as} = \nu'_{as}, \nu_s = \nu'_s$.

Case 2: assume that ν is a σ-finite signed measure, then ν^+, ν^- are σ-finite, applying case (1) to ν^+, ν^-, respectively. For the uniqueness, assume that $X = \bigcup_{n=1}^{\infty} B_n$ where $\{B_n\}$ is pairwise disjoint and $|\nu|(B_n) < \infty$, and then applying case (1) to every B_n. □

5.3 Signed Measure and Function — Act I

By the Radon-Nikodym theorem, the derivative of a measure $d\nu/d\mu$ can be expressed as an integrable function f. However, the study of integration in Mathematical Analysis is often accompanied by the emergence of the antiderivative. Unfortunately, here ν is a sigend measure, not a function. So, can this measure itself be regarded as function F such that its derivative yeilds the integrable function f ? If not, that conditions must be satisfied for this to be possible ? To address this question, we must first explore the relationship between signed measures and functions before turning to the study of this problem. For convenience, we will begin by discussing positive measures on $\mathbf{R}$.

A Monotone Increasing Functions

For positive measures, they satisfy monotonicity. Therefore, it is very reasonable to first consider monotone increasing functions. Let us review some basic properties of monotone functions.

Lemma 5.3.1. If $f : \mathbf{R} \to \mathbf{R}$ is monotone, then it has at most countably many discontinuities, and they are all jump discontinuities.

Proof. Assume wlog that f is increasing. Then for every $x \in \mathbf{R}$, x is a discontinuity point of f iff $f(x+0) - f(x-0) > 0$. For a monotonic function, if x_1, x_2 are two distinct discontinuity points, it is clear that $(f(x_1 - 0), f(x_1 + 0)) \cap$

$(f(x_2 - 0), f(x_2 + 0)) = \emptyset$. Moreover, the family of pairwise disjoint open intervals of positive length on the real line is (at most) countable. Hence the set of discontinuity poitns of f is countable. □

The above lemmas are not directly relevant to this section and will be revisited later when discussing the differentiation of monotone functions. For now, we return to the main topic and explore the connection between monotone functions and signed measures.

Theorem 5.3.1. Let μ be a positive finite Borel measure on $(\mathbf{R}, \mathcal{B}(\mathbf{R}))$ and let $F_\mu : \mathbf{R} \to \mathbf{R}$ be defined by $F_\mu(x) = \mu((-\infty, x])$. Then F_μ is bounded, increasing and right continuous with $\lim_{x\to-\infty} F_\mu(x) = 0$.

Proof. It is clear that F_μ is monotone increasing and bounded. For $x \in \mathbf{R}$, define a sequence $\{x_n = x + 1/n\}$, then $(-\infty, x] = \bigcap_{n=1}^\infty (-\infty, x_n]$. By the continuity of measures, $F_\mu(x) = \lim_{n\to\infty} F_\mu(x_n)$. The right continuity of F_μ follows. A similar argument shows that $\lim_{x\to-\infty} F_\mu(x) = 0$. □

Theorem 5.3.2. For each bounded, increasing and right continuous function $F : \mathbf{R} \to \mathbf{R}$ with $\lim_{x\to-\infty} F(x) = 0$, there exists a unique finite Borel measure μ on $(\mathbf{R}, \mathcal{B}(\mathbf{R}))$ such that $F(x) = \mu((-\infty, x])$ for every $x \in \mathbf{R}$.

Proof. We use the Carathéodory extension theorem to construct this measure. First, define a finitely additive measure on a semiring, then prove that it is σ-subadditive on the semiring, thereby showing that this finitely additive set function is actually a measure. Next, we only need to show that it is a σ-finite measure on the semiring, so that the extension theorem guarantees it can be extended to a measure on $\mathcal{B}(\mathbb{R})$. Finally, proving that it is a finite measure completes the argument.

#1. A natural choice for the semiring is the family of all half-open intervals (which has been proven in Chapter 1 to indeed form a semiring) that is $\mathscr{S} = \{(a, b] : a, b \in \mathbf{R}, a \le b\}$. We define the set function $\mu_F : \mathscr{S} \to [0, \infty]$ as follows: $\mu_F((a, b]) = F(b) - F(a)$.

#2. We claim that μ_F is a charge on $\mathscr{S}$ and μ_F is σ-subadditive. Let $(a, b] = \bigcup_{i=1}^n I_i$ where $I_i \in \mathscr{S}$ and $I_i \cap I_j = \emptyset$. It is clear that μ_F is a charge. Now, suppose that $(a, b] \subset \bigcup_{n=1}^\infty I_n$ where $I_n = (a_n, b_n]$. Take $\epsilon > 0$ for which $\epsilon < b - a$. Since F is right continuous, that is $F(a + \delta) \to F(a^+) = F(a)$ as $\delta \downarrow 0$, it follows that there exists $\delta_0 > 0$ such that

$$\mu_F((a + \delta_0, b]) \ge \mu_F((a, b]) - \epsilon.$$

For every $n \ge 1$, take δ_n such that

$$\mu_F((a_n, b_n + \delta_n]) \le \mu_F(I_n) + \frac{\epsilon}{2^n}.$$

Let $J = [a+\epsilon, b]$ and $J_n = (a_n, b_n+\delta_n), n = 1, 2, \ldots$. Then $J \subset I \subset \bigcup_{n=1}^{\infty} I_n \subset \bigcup_{n=1}^{\infty} J_n$. By the Henie-Borel theorem (finite convering theorem), there exists $N \geq 1$ such that $J \subset \bigcup_{n=1}^{N} J_n$, that is $(a+\delta_0, b] \subset \bigcup_{n=1}^{N}(a_n, b_n+\delta_n]$. Since μ_F is a charge, $\mu_F((a+\delta_0, b]) \leq \sum_{n=1}^{N} \mu_F((a_n, b_n+\delta_n])$. And hence

$$\begin{aligned}\mu_F((a,b]) &\leq \mu_F((a+\delta_0,b]) + \epsilon \leq \sum_{n=1}^{N} \mu_F((a_n, b_n+\delta_n]) + \epsilon \\ &\leq \sum_{n=1}^{N}\left(\mu_F(I_n) + \frac{\epsilon}{2^m}\right) + \epsilon \leq \sum_{n=1}^{N} \mu_F(I_n) + 2\epsilon \leq \sum_{n=1}^{\infty} \mu_F(I_n) + 2\epsilon.\end{aligned}$$

#3. We claim that μ_F is σ-finite on $\mathscr{S}$. One should prove this. By the extension theorem, there is a uniqueness extension μ_F on $\mathscr{B}(\mathbf{R})$. While $\mu_F(\mathbf{R}) = \lim_{x\to\infty} F(x) - \lim_{x\to-\infty} F(x) = \lim_{x\to\infty} F(x) < \infty$, the measure is finite. □

The above two theorems demonstrate: a finite Borel measure on Euclidean space $\mathbf{R}$ can induce a bounded, monotone increasing, and right-continuous function. Conversely, a bounded, monotone increasing, and right-continuous function can also induce a finite Borel measure. We call F_μ is the ***distribution function induced by the finite Borel measure*** μ, and the measure μ_F is the ***measure induced by the Distribution function*** F. establishing a connection between measures and monotone functions, although here it is limited to finite Borel measures. After studying signed measures, it will also be easy to show that a finite Borel signed measure can be expressed as the difference of two bounded, monotone increasing, and right-continuous functions. This will be elaborated in the next section.

B BV Functions

From the previous subsection, the relationship between signed measures and two monotonically increasing functions was introduced. First, to analyze measures using analytical language, we cannot rely solely on the original definition of signed measures. Do you remember that we have a conclusion regarding the variation of signed measures ?

$$|\mu|(A) = \sup\left\{\sum_{i=1}^{n} |\mu(A_i)| : A = \bigsqcup_{i=1}^{n} A_i\right\}$$

From the previous subsection, we learned that finite Borel measures can be represented by an increasing function F. Therefore, we can similarly examine how increasing functions would be represented in terms of the variation of a signed measure:

$$F(x) = \mu((-\infty, x]) = \sup\left\{\sum_{i=1}^{n} |\mu(A_i)| : (-\infty, x] = \bigsqcup_{i=1}^{n} A_i\right\}$$
$$= \sup\left\{\sum_{i=1}^{n} |F(t_i) - F(t_{i-1})| : A_i = (t_{i-1}, t_i]\right\}$$

Of course, this holds only for specific monotone increasing functions F. For general functions, such an equality may not hold. However, we can use this idea to define a new concept: the variation of a function.

Definition 5.3.1. Let F be a real-valued function which domain includes the interval $[a, b]$. And let $\mathcal{P} = \{t_0, t_1, \ldots, t_n\}$ be a partition of $[a, b]$ such that $a = t_0 < t_1 < \cdots < t_n = b$. Set $V_F([a, b], \mathcal{P}) = \sum_{i=1}^{n} |F(x_i) - F(x_{i-1})|$ and define the ***variation of*** F ***over*** $[a, b]$ is

$$V_F[a, b] = \sup\left\{V_F([a, b], \mathcal{P}) : \mathcal{P} \text{ is a partition of } [a, b]\right\}.$$

The function F is ***of bounded variation*** on $[a, b]$ if $V_F[a, b] < \infty$, and we denote the space of all such F by $BV[a, b]$. And we could define the function $V_F(x) = V_F[a, x]$, it is clear that $V_F(x)$ is an increasing function.

From our introduction, it is easy to see that for a monotone increasing function F on $[a, b]$ with $F(a) = 0$, its variation function $V_F = F$. Thus, it can also be concluded that bounded monotone functions are necessarily functions of bounded variation.

The practical significance of the variation of a function on the interval $[a, b]$ is the sum of the vertical distance fluctuations along the curve of the function from the point $(a, F(a))$ to $(b, F(b))$. When the function oscillates intensely, the variation can become extremely large. We provide serveral examples.

Example 5.3.1. A similar class of functions that control the intensity of variations includes Lipschitz functions. Let F be a Lipschitz function on $[a, b]$, then $F \in BV[a, b]$ and $V_F[a, b] \leq c(b - a)$ where c is the Lipschitz constant.

Example 5.3.2. Define the function $F(x) = x\cos(\pi/2x), x \in (0, 1]$ and $F(0) = 0$. From the graph of the function, it can be seen that it oscillates extremely intensely near the point 0, clearly making it impossible to control its variation. Specifically, F is continuous on $[0, 1]$, and therefore bounded, but does not have finite variation. If we take the partition $P = \{0, 1/2n, 1/2n - 1$

, $1/2n-2, \ldots, 1/3, 1/2, 1\}$ of $[0,1]$, then

$$
\begin{aligned}
F(t_0) &= 0,\\
F(t_1) &= \frac{1}{2n}\cos\left(\frac{\pi}{2\,(1/2n)}\right) = \pm\frac{1}{2n},\\
F(t_2) &= \frac{1}{2n-1}\cos\left(\frac{\pi}{2\,(1/(2n-1))}\right) = 0,\\
F(t_3) &= \frac{1}{2n-2}\cos\left(\frac{\pi}{2\,(1/(2n-2))}\right) = \pm\frac{1}{2n-2}, \ldots
\end{aligned}
$$

So that

$$
\begin{aligned}
|F(t_1) - F(t_0)| &= \frac{1}{2n}, |F(t_2) - F(t_1)| = \frac{1}{2n},\\
|F(t_3) - F(t_2)| &= \frac{1}{2n-2}, |F(t_4) - F(t_3)| = \frac{1}{2n-2}, \ldots
\end{aligned}
$$

Thus, $V_F[0,1] \geq 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$, which diverges as n goes to ∞. Therefore, F is not of finite variation on $[0,1]$.

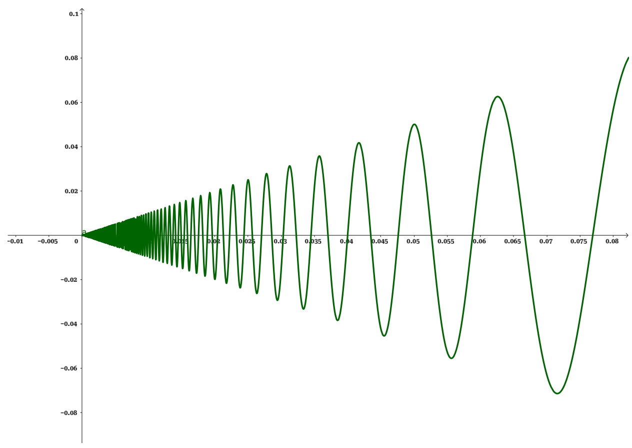

Figure 5.1: $F(x) = 2\cos(\pi/2x)$

Before formally discussing the main topic of this section, it is necessary to introduce the space of functions of bounded variation and the fundamental properties of such functions. The following proposition makes it easy to show that the space of functions of bounded variation forms a vector space:

Proposition 5.3.1. Let $F, G \in BV[a,b]$ and c, d be constant. Then

1. $F \in BV[c,d]$ where $[c,d] \subset [a,b]$,
2. $cF + dG \in BV[a,b]$,
3. $|F| \in BV[a,b]$,
4. $FG \in BV[a,b]$.

Proposition 5.3.2. Let $F : [a,b] \to \mathbf{R}$ be a function and let $c \in (a,b)$. If $F \in BV[a,c]$ and $F \in BV[c,b]$, then $F \in BV[a,b]$ and

$$V_F[a,b] = V_F[a,c] + V_F[c,b].$$

Proof. It is clear that $V_F[a,c] + V_F[c,b] \leq V_F[a,b]$. To see the converce, take the partition $\mathcal{P}_n$ of $[a,b]$, then $\mathcal{P}_n \cup \{c\}$ is also a partition of $[a,b]$ and it is clear that

$$\sum_{\mathcal{P}_n} |F(t_i) - F(t_{i-1})| \leq \sum_{\mathcal{P}_n \cup \{c\}} |F(s_i) - F(s_{i-1})|.$$

Now, let $\mathcal{P}^{(1)} = \{s_i \in \mathcal{P}_n \cup \{c\} : s_i \leq c\}$ and $\mathcal{P}^{(2)} = \{s_i \in \mathcal{P} \cup \{c\} : s_i \geq c\}$. Then $\mathcal{P}^{(1)}, \mathcal{P}^{(2)}$ are the partitions of $[a,c]$ and $[c,b]$ respectively and

$$\begin{aligned}\sum_{\mathcal{P}_n \cup \{c\}} |F(s_i) - F(s_{i-1})| &\leq \sum_{\mathcal{P}^{(1)}} |F(s_i) - F(s_{i-1})| + \sum_{(\mathcal{P}^{(2)})} |F(s_i) - F(s_{i-1})| \\ &\leq V_F[a,c] + V_F[c,b], \text{as desired.}\end{aligned}$$

□

Proposition 5.3.3. Let $F : [a,b] \to \mathbf{R}$ be a function. Then $V_F[a,b] = 0$ ***iff*** F is constant on $[a,b]$.

Proof. Assume F is constant, then F is monotone function and it is clear that $V_F[a,b] = F(b) - F(a) = 0$. To establish the converse, we proceed by contradiction, assume that F is not constant on $[a,b]$. There exist t_1, t_2 such that $F(t_1) \neq F(t_2)$. If we take these two points as a partition of $[a,b]$, we obtain

$$V_F[a,b] \geq |F(t_1) - F(a)| + |F(t_2) - F(t_1)| + |F(b) - F(t_2)| > 0.$$

We see that the sum is greater that zero, thus $V_F[a,b] \neq 0$, which is a contradiction. □

Proposition 5.3.4. Let $F \in BV[a,b]$, then the function $V_F(x) = V_F[a,x]$ is an increasing function.

Proof. We begin by introducing t_1 and t_2 such that $x_1 < x_2$. We want to show that $V_F(x_1) \leq V_F(x_2)$. By proposition 5.2.2,

$$V_F[a,x_2] = V_F[a,x_1] + V_F[x_1,x_2]$$

So, $V_F[a,x_2] - V_F[a,x_1] = V_F[x_1,x_2]$, i.e., $V_F(x_2) - V_F(x_2) = V_F[x_1,x_2]$. Since $V_F[x_1,x_2] \geq 0$, we see that $V_F(x_2) \geq V_F(x_1)$, as desired. □

Apart from defining the variation on closed bounded intervals, we can also define it on intervals of the form $(-\infty, b]$ and $(-\infty, \infty)$ using a similar approach. Similarily, the corresponding variation function can be defined as $V_F(x) = V_F(-\infty, x]$, and it is straightforward to see that this function is increasing.

Lemma 5.3.2. Let $F \in BV(\mathbf{R})$, then $V_F + F$ and $V_F - F$ are increasing.

Proof. We show that $T_F(y) + F(y) \geq T_F(x) + F(x)$ whenever $y > x$. For every $\epsilon > 0$, there is a partition $\mathcal{P} = \{t_0, t_1, \ldots, t_n\}$ such that $x_0 < \cdots < x_n = x$ and

$$V_F((-\infty, x], \mathcal{P}) = \sum_{i=1}^{n} |F(t_i) - F(t_{i-1})| \geq V_F(-\infty, x] - \epsilon = V_F(x) - \epsilon.$$

On the other hand, $\mathcal{P} \cup \{y\}$ is a partition for which $V_F((-\infty, y], \mathcal{P} \cup \{y\}) = \sum_{i=1}^{n} |F(t_i) - F(t_{i-1})| + |F(y) - F(x)| \leq V_F(y)$. Therefore

$$\begin{aligned} V_F(y) + F(y) \geq & \sum_{i=1}^{n} |F(t_i) - F(t_{i-1})| + |F(y) - F(x)| \\ & + (F(y) - F(x)) + F(x) \geq V_F(x) - \epsilon + F(x). \end{aligned}$$

And hence $V_F(y) + F(y) \geq V_F(x) + F(x)$. For the second case, the proof follows exactly the same process. □

The following theorem is an exceptionally elegant result, which also aligns perfectly with our expectations: a function of bounded variation can always be expressed as the difference of two monotone increasing functions. Conversely, the difference of two monotone functions is a function of bounded variation. Moreover, this result implies that the set of discontinuity points of a function of bounded variation is countable.

Jordan Decomposition Theorem. $F \in BV(\mathbf{R})$ ***iff*** $F = F_1 - F_2$ where F_1, F_2 are bounded increasing functions.

Proof. Note that $F = 1/2(V_F + F) - 1/2(V_F - F)$, and lemma 5.3.2 implies that $1/2(V_F + F), 1/2(V_F - F)$ are increasing. To see the boundedness, note that $V_F(y) + F(y) \leq V_F(x) + F(x)$ whenever $y > x$, it follows that

$$\begin{aligned} |F(y)| &\leq |F(y) - F(x_0)| + |F(x_0)| \\ &\leq V_F(y) - V_F(x_0) + |F(x_0)| \\ &\leq V_F(\infty) - V_F(-\infty) + |F(x_0)| < \infty \end{aligned}$$

So that F and hence $V_F \pm F$ are bounded, as desired. □

The representation $F = \frac{1}{2}(F + V_F) - \frac{1}{2}(V_F - F)$ of a real-valued functions $F \in BV$ is called the ***Jordan decomposition*** of F and $\frac{1}{2}(V_F + F), \frac{1}{2}(V_F - F)$ are called the ***positive*** and ***negative variations*** of F.

In fact, the Jordan decomposition theorem for functions of bounded variation is closer to the original version of the Jordan decomposition theorem. Indeed, this is also the theorem that Jordan first mentioned in *Sur la série de Fourier (1881)* and *Cours d'analyse*.

Lemma 5.3.3. Let $F \in BV(\mathbf{R})$, then $V_F(-\infty) = 0$. Moreover, if F is also right continuous, then so is V_F.

Proof. For every $\epsilon > 0$ and $x \in \mathbf{R}$, there is a partition $\mathcal{P}$ for which $t_0 < \cdots < t_n = x$ such that $\sum_{i=1}^{n} |F(t_i) - F(t_{i-1})| \geq V_F(x) - \epsilon$. And hence $V_F(x) - V_F(t_0) \geq V_F(x) - \epsilon$ by the definition of variation. So, $V_F(y) \leq \epsilon$ for $y \leq t_0$. Thus, $V_F(-\infty) = 0$. Now, suppose that F is right continuous, we show that $V_F(x+) - V_F(x) = 0$ for every $x \in \mathbf{R}$. Let $V_F(x+) - V_F(x) = \alpha$. For every $\epsilon > 0$, there exists $\delta > 0$ such that $|F(x+h) - F(x)| < \epsilon$ and $V_F(x+h) - V_F(x+) < \epsilon$ whenever $0 < h < \delta$. Choose the partition $\mathcal{P}_0 = \{t_0 = x, t_1, \ldots, t_n = x+h\}$ such that

$$\sum_{i=1}^{n} |F(t_i) - F(t_{i-1})| \leq \frac{3}{4}\left(V_F(x+h) - V_F(x)\right) \geq \frac{3}{4}\alpha.$$

And hence $\sum_{i=2}^{n} |F(t_i) - F(t_{i-1})| \geq \frac{3}{4}\alpha - |F(t_1) - F(t_0)| \geq 3/4\alpha - \epsilon$. Similarily, for the interval $[t_0 = x, t_1]$ there exists a partition $\mathcal{P}_1 = \{s_0 = x < \cdots < s_m = t_1\}$ such that $\sum_{i=1}^{m} |F(s_i) - F(t_{s-1})| \geq 3/4\alpha$, and hence

$$\begin{aligned} \alpha + \epsilon &> V_F(x+h) - V_F(x) \\ &\geq \sum_{i=1}^{m} |F(s_i) - F(t_{s-1})| + \sum_{i=2}^{n} |F(t_i) - F(t_{i-1})| \geq \frac{3}{2}\alpha - \epsilon. \end{aligned}$$

Thus, $\alpha < 4\epsilon$, as desired. □

Now returning to the main topic: based on the fact that a signed measure can be decomposed into the difference of two measures, and building upon the conclusions and insights from the previous subsection, we can derive the following two results.

Theorem 5.3.4. Let μ be a finite signed measure on $(\mathbf{R}, \mathcal{B}(\mathbf{R}))$ and let $F_\mu : \mathbf{R} \to \mathbf{R}$ be defined by $F_\mu(x) = \mu\left((-\infty, x]\right)$. Then F_μ is of finite variation, right continuous with $\lim_{x\to-\infty} F_\mu(x) = 0$.

Proof. Notice that

$$\sum_{i=1}^{n} \left|F_\mu(t_i) - F_\mu(t_{i-1})\right| = \sum_{i=1}^{n} \left|\mu\left((t_{i-1}, t_i]\right)\right| \leq |\mu|(\mathbf{R}).$$

It follows that $V_F(-\infty, \infty) \leq |\mu|(\mathbf{R})$ and hence that F_μ is of finite variation. Consider the Jordan decomposition μ^+, μ^- of μ, it is clear that F_μ is vanishes at $-\infty$ and is right continuous by theorem 5.2.1. □

Let $F \in BV(-\infty, \infty)$, and let F_1, F_2 be the Jordan decomposition of F. lemma 5.2.1 implies that if F is right continuous, then F_1 and F_2 are right continuous, and that if F vanishes at $-\infty$, then F_1, F_2 vanish at $-\infty$.

Theorem 5.3.5. For each right continuous function $F : \mathbf{R} \to \mathbf{R}$ of finite variation with $\lim_{x\to-\infty} F(x) = 0$, there exists a unique finite signed measure μ_F on $(\mathbf{R}, \mathcal{B}(\mathbf{R}))$ such that $F(x) = \mu_F((-\infty, x])$ for every $x \in \mathbf{R}$. Moreover, $|\mu_F| = \mu_{T_F}$.

Proof. By the jordan decomposition theorem before and the theorem 5.2.2, there exist two measures μ_1, μ_2 such that $F_1(x) = \mu_1(-\infty, x], F_2(x) = \mu_2(-\infty, x]$, So, $F(x) = F_1(x) - F_2(x) = \mu_1(-\infty, x] - \mu_2(-\infty, x] = \mu(-\infty, x]$ where $\mu = \mu_1 - \mu_2$ which is a signed measure. For the uniqueness, let μ, ν are finite signed Measures such that $F(x) = \mu(-\infty, x] = \nu(-\infty, x]$, where $\mu = \mu_1 - \mu_2, \nu = \nu_1 - \nu_2$ generated by F_1, F_2 and $\tilde{F}_1, \tilde{F}_2$. So, $F_1 + \tilde{F}_2 = \tilde{F}_1 + F_2$, it follows from theorem 5.2.2, $\mu_1 + \nu_2 = \mu_2 + \nu_1$ and hence $\mu = \nu$. To see the last statement, it suffices to show that $G(x) = |\mu_F|((-\infty, x]) = V_F(x)$. Note that $V_F(x) \le G(x)$ for every $x \in \mathbf{R}$. Now, we show that $|\mu_F((a, b])| \le \mu_{T_F}((a, b])$, indeed,

$$\begin{aligned}
|\mu_F((a,b])| &= |\mu_F((-\infty,a]) - \mu_F((-\infty,b])| = |F(b) - F(a)| \\
&\le V_F(-\infty,b] - V_F(-\infty,a] = V_F(b) - V_F(a) \\
&= \mu_{T_F}(-\infty,b] - \mu_{T_F}(-\infty,a] = \mu_{T_F}(a,b].
\end{aligned}$$

Now, consider $|\mu_F|(B)$ for $B \in \mathcal{B}(\mathbf{R})$, we have known that

$$\begin{aligned}
|\mu_F|(B) &= \sup\left\{\sum_{i=1}^n |\mu_F(B_i)| : B = \bigsqcup_{i=1}^n B_i, B_i \text{ interval}\right\} \\
&\le \sup\left\{\sum_{i=1}^n \mu_{T_F}(B_i)\right\} = \sup\left\{\mu_{T_F}(B)\right\} = \mu_{T_F}(B).
\end{aligned}$$

In particular, $G(x) = |\mu_F|(-\infty, x] \le \mu_{T_F}(-\infty, x] = V_F(x) - V_F(-\infty) = V_F(x)$. By the uniqueness of μ_{V_F}, $|\mu_F| = \mu_{T_F}$. $\square$

Remark 5.3.1. For a right continuous function $F \in BV[a, b]$, we can discuss similar problem. We can first standardize it as follows:

$$\tilde{F}(x) = \begin{cases} 0, & x \le a \\ F(x), & a < x \le b \\ F(b), & x > b. \end{cases}$$

Then $\tilde{F}$ is right continuous and of bounded variation on $\mathbf{R}$. By the theorem before, there exists a unique finite Borel sigend measure $\mu_{\tilde{F}}$ such that $\tilde{F}(x) = \mu_{\tilde{F}}((-\infty, x])$. And we define the finite sigend measure $\mu_F = \mu_{\tilde{F}} + a$. For example, consider $F(X) = x - a$ on $[a, b]$, we obtain that finite sigend measure $\mu_{\tilde{F}}$ and define $\mu_F = \mu_{\tilde{F}} + a$ on the space $[a, b]$, We claim that $\mu_F = \lambda\lfloor_{[a,b]}$ on $[a, b]$ (use theorem 2.2.4).

C *Absolutely Continuous Functions*

After discussing the relationship between functions of bounded variation and finite Borel signed measures, we can use measure theory to study functions of bounded variation. For sigend measures, we have focused on their properties of being absolutely continuous with respect to another sigend measure. We can now discuss the absolute continuity of one function of bounded variation with respect to another.

For any function of bounded variation $F \in BV[a,b]$, we can naturally identify another function of bounded variation, such as the uniform distribution function on the interval $G(x) = x - a \in BV[a,b]$. Undoubtedly, the measure induced by $G(x)$ is the Lebesgue measure restricted on $[a,b]$. We explore what properties the function of bounded variation possesses when it is known that a finite Borel sigend measure $\mu_F \ll \lambda$.

By the proposition 5.2.1, for every $\epsilon > 0$, there exists $\delta > 0$ such that $|\mu_F|(A) < \epsilon$ whenever $|\lambda|(A) = \lambda(A) < \delta$. On the other hand, we have known that $|\mu_F|(A) = \sup\left\{\sum_{i=1}^n \left|\mu_F(A_i) : A = \bigsqcup_{i=1}^n A_i\right|\right\}$. In particular, $A = [a,b] = \bigsqcup_{i=1}^n A_i$, where $A_i = (s_i, t_i]$ are intervals, then we have

$$\sum_{i=1}^n |\mu_F(A_i)| = \sum_{i=1}^n |F(s_i) - F(t_i)| < \epsilon$$

whenever $\lambda(A) = \sum_{i=1}^n \lambda(A_i) = \sum_{i=1}^n |t_i - s_i| < \delta$. That is, when the induced signed measure μ_F is absolutely continuous with respect to the Lebesgue measure λ, the bounded variation function has the following property: let $\epsilon > 0$, for every finite family $\{(s_i,t_i)\}_i^n$ of pairwise disjoint intervals lie in $[a,b]$ with $\sum_{i=1}^n |t_i - s_i| < \delta$ we have $\sum_{i=1}^n |F(s_i) - F(t_i)| < \epsilon$. Based on this, we summarize the following definition.

Definition 5.3.2. Let F be a real-valued function which domain includes the interval $[a,b]$. F is ***absolutely continuous*** on $[a,b]$ if for every $\epsilon > 0$, there exists $\delta > 0$ such that for every finite family $\{(s_i,t_i)\}_i^n$ of pairwise disjoint intervals lie in $[a,b]$ with $\sum_{i=1}^n |t_i - s_i| < \delta$ we have

$$\sum_{i=1}^n |F(s_i) - F(t_i)| < \epsilon.$$

We denote the space of all such F by $AC[a,b]$.

The definition of absolute continuity given here does not assume $F \in NBV$ in advance. However, it can be proved that an absolutely continuous function F must be of bounded variation.

We now consider some specific examples of an absolutely continuous function. It is clear if F is absolutely continuous, then F is uniformly continuous (when $n = 1$ in the definition).

Example 5.3.3. Let F be a Lipschitz function on $[a, b]$, then $F \in AC[a, b]$. However, continuous function is not necessarily absolutely continuous. The Cantor function $\phi : [0, 1] \to [0, 1]$ is continuous increasing but it is not absolutely continuous.

Proof. At the first step in the construction of the Cantor set, we obtain $[s_1, t_1] = [0, 1/3]$ and $[s_2, t_2] = [2/3, 1]$, then

$$\sum_{i=1}^{2}(t_i - s_i) = \frac{2}{3} \text{ and } \sum_{i=1}^{2} (\phi(t_i) - \phi(s_i)) = 1.$$

At the second step, we obtain four intervals of length $1/3^2$: $[s_1, t_1] = [0, 1/9]$, $[s_2, t_2] = [2/9, 3/9]$, $[s_3, t_3] = [6/9, 7/9]$, $[s_4, t_4] = [8/9, 1]$. Then,

$$\sum_{i=1}^{2^2}(t_i - s_i) = \left(\frac{2}{3}\right)^2 \text{ and } \sum_{i=1}^{2^2} (\phi(t_i) - \phi(s_i)) = 1.$$

In general at $n - th$ step, we obtain 2^n intervals of length $1/3^n$ and

$$\sum_{i=1}^{2^n}(t_i - s_i) = \left(\frac{2}{3}\right)^n \text{ and } \sum_{i=1}^{2^n} (\phi(t_i) - \phi(s_i)) = 1.$$

It follows that for $\epsilon = 1/2$, the condition for absolute continuity does not hold since we have a family of finitely many intervals with total measure $(2/3)^n$ which can be made as small as we which while variation of ϕ is 1. □

Proposition 5.3.5. The space $AC[a, b]$ is a vector space. Specifically, let F, G be absolutely continuous on $[a, b]$ and c, d be constant. Then

1. $cF + dG$ is absolutely continuous on $[a, b]$,
2. $|F|$ is absolutely continuous on $[a, b]$,
3. FG is absolutely condition on $[a, b]$.

Theorem 5.3.6. If $F \in AC[a, b]$, then $F \in BV[a, b]$.

Proof. Assume $F \in AC[a, b]$, let $\epsilon = 1$ and $\delta > 0$ be the corresponding positive number such that F satisfies the absolute continuity property, that is

$$\sum_{i=1}^{n} |t_i - s_i| < \delta \text{ and } \sum_{i=1}^{n} |F(t_i) - F(s_i)| < 1. \tag{1}$$

Let $N \in \mathbf{N}$ be such that $N > (b - a)/\delta$ and define the points $x_i = a + i\frac{b-a}{N}, i = 0, 1, \ldots, N$. Then we obtain a partition $P = \{x_0, x_1, \ldots, x_N\}$ of $[a, b]$

and $|x_{i+1} - x_i| < \delta$. Thus, $V_F[x_{i-1}, x_i] \leq 1$ by (1). There are at most N of these intervals and by proposition 5.4.2, we have

$$V_F[a,b] \leq \sum_{i=1}^{N} V_F[x_{i-1}, x_i] \leq N.$$

Therefore $F \in BV[a,b]$ and $AC[a,b] \subset BV[a,b]$. □

Lemma 5.3.4. If $F \in AC[a,b]$, then $V_F(x) = V_F[a,x] \in AC[a,b]$. In particular, an absolutely continuous function can be written as the difference of two increasing absolutely continuous functions.

Proof. Let $\epsilon > 0$ and let $\delta > 0$ such that F satisfies the absolute continuity property for the pair $\epsilon/2$. Let $\{(s_i, t_i)\}_{i=1}^n$ be a family of disjoint intervals of $[a,b]$ such that $\sum_{i=1}^n (t_i - s_i) < \delta$. For $i \in \{1, 2, \ldots, n\}$, let $P_i = \left\{(u_j, v_j)\right\}_{j=1}^m$ be a partition of (s_i, t_i). Then

$$\sum_{i=1}^{m} \left|F(u_j) - F(v_j)\right| \leq \frac{\epsilon}{2}.$$

Taking the supremum over each partition P_i of (s_i, t_i), we get

$$\sum_{i=1}^{n} |V_F(t_i) - V_F(s_i)| = \sum_{i=1}^{n} V_F[s_i, t_i] \leq \frac{\epsilon}{2} < \epsilon.$$

That is $V_F \in AC[a,b]$. Since $F \in AC[a,b] \subset BV[a,b]$, $F = F_1 - F_2$ where $F_1 = (V_F + F)/2$ and $F_2 = (V_F - F)/2$. Since $V_F, F \in AC[a,b]$, $F_1, F_2 \in AC[a,b]$ by proposition 5.3.5. □

Theorem 5.3.7. Let μ be a finite signed measure on $(\mathbf{R}, \mathcal{B}(\mathbf{R}))$, and let $F_\mu : \mathbf{R} \to \mathbf{R}$ be defined by $F_\mu(x) = \mu((-\infty, x])$, then $F_\mu \in AC(\mathbf{R})$ ***iff*** μ is absolutely continuous with respect to the Lebesgue measure λ (that is $\mu \ll \lambda$).

Proof. Assume $F_\mu : \mathbf{R} \to \mathbf{R} \in AC$ and $F_\mu(x) = \mu((-\infty, x])$. Then V_F is absolutely continuous by lemma before, and so $F_1 = (V_F + F)/2$ and $F_2 = (V_F - F)/2$ are absolutely continuous. Let μ_1, μ_2 be the finite Borel measures on $(\mathbf{R}, \mathcal{B}(\mathbf{R}))$ that correspond to F_1, F_2. Since $F_\mu = F_1 - F_2$, if follows that $\mu = \mu_1 - \mu_2$. Thus, we need only to show that $\mu_1 \ll \lambda$ and $\mu_2 \ll \lambda$. Let $\epsilon > 0$, and let $\delta > 0$ such that

$$\sum_{i=1} |F(t_i) - F(s_i)| < \epsilon \text{ whenever } \sum_{i} (t_i - s_i) < \delta,$$

where $\{(s_i, t_i)\}$ is a finite sequence of disjoint intervals. Suppose that $A \in \mathcal{B}(\mathbf{R})$ such that $\lambda(A) < \delta$. Then there exists an open set $G \supset A$ such that $\lambda(G) < \delta$. Since G can be written as a union of disjoint open intervals $\{(s_i, t_i)\}$, it follows that

$$\mu_1\left(\bigcup_{i=1}^{n}(s_i,t_i)\right)=\sum_{i=1}^{n}\left(F_1(t_i)-F_1(s_i)\right)<\epsilon \text{ for each } n.$$

Hence $\mu_1(G) = \mu_1\left(\bigcup_{i=1}^{\infty}(s_i,t_i)\right) \le \epsilon$. And so $\mu_1(A) \le \epsilon$. Thus, $\mu_1 \ll \lambda$. Similarly, $\mu_2 \ll \lambda$.
Now, assume that $\mu \ll \lambda$, let $\epsilon > 0$ be such that $|\mu|(A) < \epsilon$ whenever $\lambda(A) < \delta$ for some $\delta > 0$. If $\{(s_i,t_i)\}$ is a finite sequence of disjoint intervals such that $\sum_{i=1}^{n}(t_i - s_i) < \delta$, then $\lambda(\bigcup_{i=1}^{n}[s_i,t_i]) < \delta$ and

$$\begin{aligned}\sum_{i=1}^{n}|F(t_i)-F(s_i)| &= \sum_{i=1}^{n}|\mu([s_i,t_i])| = \sum_{i=1}^{n}|\mu(s_i,t_i]| \\ &\le |\mu|\left(\bigcup_{i=1}^{n}(s_i,t_i]\right) < \epsilon. \text{ as desired}\end{aligned}$$

□

The reader may feel somewhat disappointed, as the spaces and measures discussed in earlier sections of this chapter are relatively abstract, while here we only consider finite Borel signed measures on Euclidean space $\mathbb{R}$. It should be clarified that, while readers might be eager to extend these results to $\mathbb{R}^n$, we currently lack a well-defined total order structure to define $F(\mathbf{x}) = \mu((-\infty,\mathbf{x}])$. Precisely because of this limitation, our exploration so far is confined to $\mathbb{R}$. Nevertheless, the insights gained in this section are highly significant: they allow us to transform measure-theoretic problems into problems in functions. We will revisit the relationship between measures and functions in later discussions, at which point further generalizations can be suitably addressed.

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6

Linear Functional

6.1 Banach Spaces

A Normed Spaces and Sequence Spaces

Definition 6.1.1. A ***norm*** on a vector space X over $\mathbf{K}$ is a function $\|\cdot\| : X \to [0, \infty)$ such that

- (positive definite) $\|x\| = 0$ ***iff*** $x = 0$;
- (scalar homogeneity) $\|cx\| = |c|\,\|x\|$ for all $c \in \mathbf{K}$ and $x \in X$;
- (triangle inequality) $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in X$.

A ***normed vector space*** (or normed space) is a pair $(X, \|\cdot\|)$, where X is a vector space and $\|\cdot\|$ is a norm on X. When the norm $\|\cdot\|$ is understood we simply write X instead of $(X, \|\cdot\|)$. If we with to emphasis the role of X. we write $\|\cdot\|_X$ instead of $\|\cdot\|$.

Remark 6.1.1. Here, x refers to an element vector in a vector space. It can be a sequence of points, a function, a matrix, or any object that satisfies the eight axioms of a vector space. It is believed that the reader can readily name several common vector spaces: $\mathbf{R}^n, \mathbf{C}^n$ $(\mathbf{K}^n)$, polynomial space $P[x]$, continuous function space $C[a, b]$, matrix space, and $\mathscr{L}^1(X, \Sigma, \mu)$.

Remark 6.1.2. Recall that a metric space is a non-empty set X with a function (metric), $d : X \times X \to \mathbf{K}$ that satisfies the following conditions

- $d(x, y) \geq 0$ for all $x, y \in X$ and $d(x, y) = 0$ ***iff*** $x = y$.
- $d(x, y) = d(y, x)$ for all $x, y \in X$.
- $d(x, y) \leq d(x, z) + d(z, y)$ for all $x, y, z \in X$.

For a normed space $(X, \|\cdot\|)$, if we define $d(x, y) = \|x - y\|$ for $x, y \in X$, then X is a metric space under the metric d. We say that the topological space induced by the metric $d(x, y) = \|x - y\|$ is a topological space induced by the norm $\|\cdot\|$.

The above remark indicates that any normed space is a metric space. Therefore, we can study all the properties of metric spaces in normed spaces, including the direct use of related terminology, such as open sets, closed sets, compact sets, bounded sets, limits, convergence, continuity, and so on.

Recall that metrics have a concept of equivalence: we say that two metrics d_1, d_2 are equivalent on X if a set is open in the space (X, d_1) iff it is also open in (X, d_2). It can be shown that this is equivalent to the existence of positive numbers m, M such that

$$md_1(x, y) \leq d_2(x, y) \leq Md_1(x, y) \text{ for all } x, y \in X.$$

Indeed, we can define equivalence between norms using a similar approach. The following theorem and its corollary illustrate this. Additionally, the definition of a ball aligns with that in metric spaces and requires no further elaboration.

Theorem 6.1.1. Let X be a vector space and $\|\cdot\|_{(1)}$, $\|\cdot\|_{(2)}$ are two norms on X. The topology $\mathscr{T}_2$ induced by the norm $\|\cdot\|_{(2)}$ is weaker than the topology $\mathscr{T}_1$ induced by $\|\cdot\|_{(1)}$ (that is $\mathscr{T}_2 \subset \mathscr{T}_1$) iff there exists a number $M > 0$ such that

$$\|x\|_{(2)} \leq M \|x\|_{(1)} \text{ for all } x \in X.$$

Proof. First we assume that $\mathscr{T}_2 \subset \mathscr{T}_1$. The open ball $B_2(0,1)$ for the $\|\cdot\|_{(2)}$ is then open in $\mathscr{T}_1$. This means that $B_1(0,r) \subset B_2(0,1)$ for some $r > 0$ where $B_1(0,r)$ is the ball for $\|\cdot\|_{(1)}$. Since $(r/2)(x/\|x\|_{(1)}) \in B_1(0,r) \subset B_2(0,1)$, we obtain that

$$\left\| \frac{r}{2} \frac{x}{\|x\|_{(1)}} \right\|_{(2)} < 1.$$

Thus, $\|x\|_{(2)} < 2/r \cdot \|x\|_{(1)}$, as desired ($M = 2/r$). To see the converse, from $\|x-y\|_{(1)} \leq M \|x-y\|_{(2)}$, we obtain that $B_1(x, r/M) \subset B_2(x,r)$. So, that the interior point of a set A under the norm $\|\cdot\|_{(2)}$ will be an interior point of A under the norm $\|\cdot\|_{(1)}$. Thus, every open set under $\|\cdot\|_{(2)}$ will be an open set under $\|\cdot\|_{(1)}$, that is $\mathscr{T}_2 \subset \mathscr{T}_1$. □

►**Corollary 1.** Two norms $\|\cdot\|_{(1)}$ and $\|\cdot\|_{(2)}$ on the same vector space X result the same topology iff there exists numbers $0 < m \leq M$ such that

$$m \|x\|_{(1)} \leq \|x\|_{(2)} \leq M \|x\|_{(1)} \text{ for all } x \in X.$$

We say that $\|\cdot\|_{(1)}$ and $\|\cdot\|_{(2)}$ are ***equivalent norms***.

We present several important examples of normed spaces, which also introduce some of the most commonly encountered norms. The first norm is the p-norm. For a real number $1 \leq p < \infty$ and let $X = \mathbf{K}^n$ be the vector spaces. For all n-tuples $x = (x_1, x_2, \ldots, x_n) \in \mathbf{K}^n$ define the function $\|\cdot\|_p$:

$$\|x\|_p := \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}.$$

Now we prove that this function is indeed a norm defined on the vector space $\mathbf{K}^n$. In fact, the first two properties are very easy to prove, but the triangle inequality $\|x+y\|_p \leq \|x\|_p + \|y\|_p$ is more difficult. In fact, this inequality is called Minkowski's inequality.

Definition 6.1.2. Let $p, q \in \mathbf{R}$. If $p, q > 1$ and $1/p + 1/q = 1$, we say that p and q are ***Hölder conjugates***. The Hölder conjugate of $p = 1$ is defined to be $q = \infty$ and the conjugate of $p = \infty$ is $q = 1$.

W.H. Young's Inequality. Suppose $p, q > 1$ are Hölder conjugates. Then for any nonnegative real number a, b, we have

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Proof. Fix a and consider the function $f(b) = ab - b^q/q$ with derivative $f'(b) = a - b^{q-1}$. It is clear that f attains its maximum only at $b = a^{1/(q-1)} = a^{p/q}$. So for nonnegative b we have

$$ab - \frac{b^q}{q} = f(b) \leq f\left(a^{p/q}\right) = a^{p/q+1} - \frac{a^p}{q} = a^p - \frac{a^p}{q} = \frac{a^p}{p}.$$

□

Hölder's Inequality. Suppose $p, q > 1$ are Hölder conjugates. Then for any $x, y \in \mathbf{K}^n$, we have the inequality

$$\sum_{i=1}^{n} |x_i y_i| \leq \left(\sum_{i=}^{n} |x_i|^p\right)^{1/p} \left(\sum_{i=1}^{n} |y_i|^q\right)^{1/q} = \|x\|_p \|y\|_p .$$

The special case $p = q = 2$ is know as the Cauchy's inequality. Furthermore, for any $x = (x_1, x_2, \dots)$ and $y = (y_1, y_2, \dots)$,

$$\sum_{i=1}^{\infty} |x_i y_i| \leq |x_i y_i| \leq \left(\sum_{i=}^{\infty} |x_i|^p\right)^{1/p} \left(\sum_{i=1}^{\infty} |y_i|^q\right)^{1/q}$$

Proof. If $x = (0, 0, \dots, 0)$ or $y = (0, 0, \dots, 0)$, the result is clear. Otherwise, $\|x\|_p \neq 0$ and $\|y\|_q \neq 0$. By Young's inequality, for each $i = 1, 2, \dots, n$ with $a = x_i / \|x\|_p$ and $b = y_i / \|y\|_q$, we have

$$\frac{|x_i y_i|}{\|x\|_p \|y\|_q} \leq \frac{|x_i|^p}{p \cdot (\|x\|_p)^p} + \frac{|y_i|^q}{q \cdot (\|y\|_q)^q}.$$

Therefore, we have the inequality

$$\frac{\sum_{i=1}^n |x_i y_i|}{\|x\|_p \|y\|_q} \leq \frac{\sum_{i=1}^n |x_i|^p}{p \cdot (\|x\|_p)^p} + \frac{\sum_{i=1}^n |y_i|^q}{q \cdot (\|y\|_q)^q} = 1, \text{ as desired.}$$

For the last statement, let $n \to \infty$, first on the right side to obtain an upper bound, then on the left side to obtain tha desired result. □

Minkowski's Inequality. Suppose $p \geq 1$. Then for any $x, y \in \mathbf{K}^n$, we have

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p .$$

Furthermore, for any $x = (x_1, x_2, \dots)$ and $y = (y_1, y_2, \dots)$,

$$\left(\sum_{i=1}^{\infty} |x_i + y_i|^p\right)^{1/p} \leq \left(\sum_{i=1}^{\infty} |x_i|^p\right)^{1/p} + \left(\sum_{i=1}^{n} |y_i|^p\right)^{1/p} .$$

However, the inequality does not hold for $0 < p < 1$, one should show this.

Proof.

$$\begin{aligned}\sum_{i=1}^{n}|x_i+y_i|^p &= \sum_{i=1}^{n}|x_i+y_i|\,|x_i+y_i|^{p-1} \le \sum_{i=1}^{n}(|x_i|+|y_i|)\,|x_i+y_i|^{p-1}\\ &= \sum_{i=1}^{n}|x_i|\,|x_i+y_i|^{p-1} + \sum_{i=1}^{n}|y_i|\,|x_i+y_i|^{p-1}\\ &\le \left(\sum_{i=1}^{n}|x_i|^p\right)^{1/p}\left(\sum_{i=1}^{n}|x_i+y_i|^{(p-1)q}\right)^{1/q}\\ &+ \left(\sum_{i=1}^{n}|y_i|^p\right)^{1/p}\left(\sum_{i=1}^{n}|x_i+y_i|^{(p-1)q}\right)^{1/q} \quad \text{by Hölder's Inequality,}\end{aligned}$$

where $1/p + 1/q = 1$. Since $(p-1)q = p$, we have

$$\sum_{i=1}^{n}|x_i+y_i|^p \le (\|x\|_p + \|y\|_p)\left(\sum_{i=1}^{n}|x_i+y_i|^p\right)^{1/q}, \text{ as desired.}$$

For the last statement, let $n \to \infty$, first on the right side to obtain an upper bound, and then on the left side to obtain the desired result. □

Jensen's Inequality. If $0 < p < q$ and $x \in \mathbf{K}^n$, then

$$\|x\|_q \le \|x\|_p .$$

Proof. The case $x = 0$ is clear. Otherwise, we will show that $\|x\|_p / \|x\|_q \ge 1$ for $0 < p < q$. Let $A = \|x\|_q > 0$. Then for each i, we have $|x_i|/A \le 1$. Since $0 < p < q$, we have $(|x_i|/A)^p \ge (|x_i|/A)^q$. It follows that

$$\begin{aligned}\frac{\|x\|_p}{\|x\|_q} = \frac{\|x\|_p}{A} &= \left(\sum_{i=1}^{n}\left(\frac{|x_i|}{A}\right)^p\right)^{1/p}\\ &\ge \left(\sum_{i=1}^{n}\left(\frac{|x_i|}{A}\right)^q\right)^{1/p} = \left(\frac{\sum_{i=1}^{n}|x_i|^q}{A^q}\right)^{1/p} = 1 \text{ as desired.}\end{aligned}$$

□

Thus, we have proven that $(\mathbf{K}^n, \|\cdot\|_p)$ is indeed a normed space. In the proof of the above inequalities, we have demonstrated that they apply equally to sequences $x = (x_1, x_2, \dots)$, we denote the set of all these sequences by $\mathbf{K}^{\mathbf{N}} = \{x = (x_i) : x_i \in \mathbf{K}\}$ one should prove that this is a vector space. And we can fully extend the p-norm:

$$\|x\|_p := \left(\sum_{i=1}^{\infty}|x_i|^p\right)^{1/p}.$$

Unfortunately, $(\mathbf{K}^{\mathbf{N}}, \|\cdot\|_p)$ is not a normed space, observe that it does not meet the requirement for a "norm", specifically finiteness (for instance, taking the sequence $(1, 1, \dots)$, we have $\|(1, 1, \dots)\|_p = \infty$!). Therefore, if we insist on defining this norm on a certain space, we need to impose restrictions on the vector space. The most straightforward approach is the following cases:

$$\ell^p = \left\{ x = (x_1, x_2, \dots) \in \mathbf{K}^{\mathbf{N}} : \sum_{i=1}^{\infty} |x_i|^p < \infty \right\}.$$

One should prove that ℓ^p is a subspace of $\mathbf{K}^{\mathbf{N}}$, that is ℓ^p is a vector space and then $(\ell^p, \|\cdot\|_p)$ is a normed space. Although the norms in these two spaces are denoted by the same symbol, the vector spaces are different and the expressions of the two norms are similar. Therefore, using the same symbol is reasonable. And $(\ell^p, \|\cdot\|_p)$ is called the ***sequence spaces***.

Lemma 6.1.1. For every $x \in \mathbf{K}^n$, we have

$$\lim_{p\to\infty} \|x\|_p = \sup\{|x_i| : i = 1, 2, \dots, n\} := \|x\|_\infty .$$

Moreover, for every $x \in \mathbf{K}^{\mathbf{N}}$ such that $\|x\|_p < \infty$ for some $1 \le p < \infty$, then the equality also holds, that is $\lim_{q\to\infty} \|x\|_q = \sup\{|x_i| : i = 1, 2, \dots\} := \|x\|_\infty$.

Proof. Since $1 \le i \le n$, the supremum $\sup_i |x_i|$ is really a maximum, which means that there exists a j such that $|x_j| = \sup_i |x_i|$. Let $1 < p < \infty$. We have $|x_j|^p \le \sum_{i=1}^n |x_i|^p \le n\,|x_j|^p$ and thus,

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p} \ge \left(|x_j|^p\right)^{1/p} = |x_j| = \sup_i |x_i| .$$

For the inequality in the opposite direction, we see that

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p} \le \left(n\,|x_j|^p\right)^{1/p} = n^{1/p} \sup_i |x_i| , \text{ as desired.}$$

For the last statement, if $\|x\|_p < \infty$ for some $1 \le p < \infty$, by Jensen's inequality we have $\|x\|_q < \infty$ for all $q > p$. By the theory of convergent series, $x_i \to 0$, so just as in the proof of the first case, there exists a j such that $|x_j| = \sup_i |x_i|$ and $\|x\|_q \ge |x_j| = \sup_i |x_i| = \|x\|_\infty$. So the decreasing $\|x\|_q$ is bounded below and $\lim_{q\to\infty} \|x\|_q \ge \|x\|_\infty$. For the inequality in the opposite direction, given $\epsilon > 0$, choose $N > 0$ such that

$$\left(\sum_{i=N+1}^{\infty} |x_i|^p\right)^{1/p} < \epsilon.$$

Let $u_i = x_i$ for $1 \le i \le N$ and $u_i = 0$ for $i > N$, and let $v_i = x_i - u_i$. Minkowski's and Jensen's inequalities on $x_k = u_k + v_k$ show that for $q > p$,

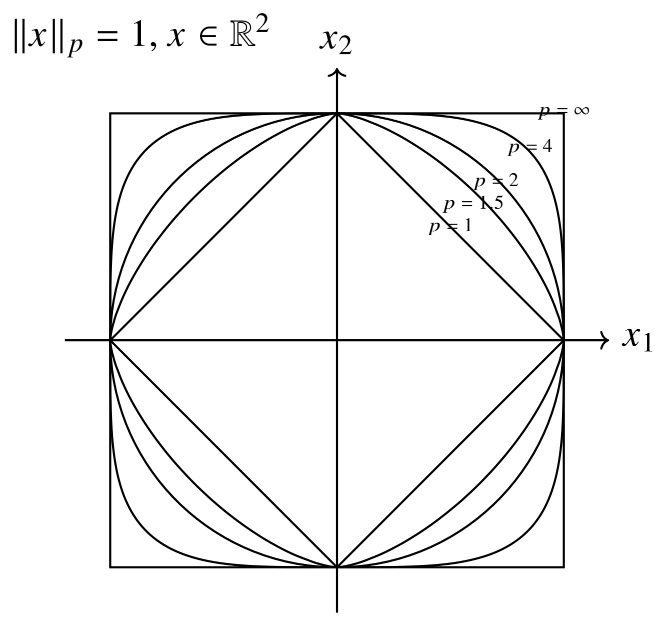
$$\|x\|_q = \left(\sum_{i=1}^{\infty} |u_i + v_i|^q\right)^{1/q} \leq \left(\sum_{i=1}^{N} |x_i|^q\right)^{1/q} + \left(\sum_{N+1}^{\infty} |x_i|^q\right)^{1/q}$$
$$\leq N^{1/q} \sup_i |x_i| + \epsilon, \text{ as desired.}$$

□

Based on the lemma above, we can similarly define a norm $\|\cdot\|_\infty$ on the vector space $\mathbf{K}^n$ by $\|x\|_\infty = \sup\{|x_i| : i = 1, 2, \ldots, n\}$ and a norm $\|\cdot\|_\infty$ on the vector space (one should prove this) $\ell^\infty = \left\{x = (x_i) \in \mathbf{K}^{\mathbf{N}} : \sup\{x_i\} < \infty\right\}$. And we refer to the norm $\|\cdot\|_\infty$ as the supremum norm. We can use Jensen's inequality to show that ℓ^p becomes larger with p, that is

$$\ell^1 \subset \ell^q \subset \ell^p \subset \ell^\infty \text{ for } 1 < q < p < \infty.$$

A classic illustrative figure is often used to demonstrate the differences between p-norms, with particular interest focused on the $\|\cdot\|_1$, $\|\cdot\|_2$ and $\|\cdot\|_\infty$.



Let us conclude this subsection by presenting several important examples, which we will frequently encounter in the future.

Example 6.1.1. $C[a, b]$ is a vector space of continuous functions from $[a, b]$ to $\mathbf{R}$. Now we define the function $\|\cdot\|_1$ on $C[a, b]$ by

$$\|f\|_1 = \int_a^b |f(t)|\, dt.$$

It is clear that $\|\cdot\|_1$ is a norm on $C[a, b]$. But this conclusion does not hold for integrable functions. The notation for the norm here is the same as that for the previous p-norm, but the expressions differ. We refer to the norm here as the $\mathscr{L}^1$-norm, yet this does not lead to ambiguity, as it applies specifically to functions, while the p-norm refers to sequences. Similarily, we can also define another norm on $C[a, b]$:

$$\|f\|_\infty = \sup\{f(x) : x \in [a, b]\}.$$

Example 6.1.2. As we known that $\mathscr{L}^1(X, \Sigma, \mu)$ is the vector space of measurable functions $f : X \to \mathbf{R}$ such that $\int_X |f| < \infty$. Define the function $\|\cdot\|_1$ on $\mathscr{L}^1(X, \Sigma, \mu)$ by

$$\|f\|_1 = \int_X |f|\, d\mu.$$

Then $\|\cdot\|_1$ satisfies the homogeneity condition and the triangle inequality on $\mathscr{L}^1(X, \Sigma, \mu)$. However, $\|\cdot\|_1$ is not a norm on the vector space because the positive definite condition is not staisfied. Specifically, if E is a μ-null set, then $\|\chi_E\|_1 = 0$ but $\chi_E \neq 0$.

The examples above illustrate that the Lebesgue spaces we have studied come very close to being normed spaces. However, we can also define the following spaces to generalize the concept of important function spaces like Lebesgue spaces — specifically, spaces equipped with seminorms.

Definition 6.1.3. A ***seminorm*** on a vector space X over $\mathbf{K}$ is a function $\|\cdot\| : X \to [0, \infty)$ such that

- $\|cx\|^* = |c|\, \|x\|^*$ for all $x \in X$ and $c \in \mathbf{K}$;
- $\|x + y\|^* \leq \|x\|^* + \|y\|^*$ for all $x, y \in X$.

A seminorm on a vector space differs from a norm only in the first condition. In fact, there is a related concept called a semimetric space, and a seminorm naturally induces a special type of semimetric space. (The definition of a semimetric space will not be provided here; interested readers may refer to specialized textbooks or literature for further details.) In this lecture, we focus on one key example of a seminorm: the seminorm on Lebesgue spaces. We will discuss the more general Lebesgue spaces in detail later on.

The following theorem illustrates the close relationship between seminorms and norms. Readers are expected to recall relevant knowledge from abstract algebra, which will not be elaborated here in detail. Readers are encouraged to attempt to apply this theorem to refine the Lebesgue space example mentioned above.

Theorem 6.1.6. Let X be a seminorm space and define the relation $\sim$, where $x \sim y$ holds iff $\|x \sim y\|^* = 0$. Then $\sim$ is an equivalence relation on X and the equivalence classes are $\langle x \rangle = \left\{y \in X : \|x - y\|^* = 0\right\}$. The equivalence classes form a normed space if we define $\|\langle x \rangle\| = \|x\|^*$.

B *Banach Spaces*

In addition to discussing open sets in metric spaces induced by norms, we can also explore concepts such as sequences and limits. For metric spaces, we are particularly interested in the relationship between Cauchy sequences and convergent sequences, which leads to the notion of complete metric spaces. Similarly, the concept of completeness can be defined for normed spaces.

Definition 6.1.4. Let X be a normed space, and we say the a sequence $\{x_n\}$ in X ***converges to*** $a \in X$ if $\lim_{n\to\infty} \|x_n - a\| = 0$. That is, for every $\epsilon > 0$, there is $N \in \mathbf{N}$ such that $\|x_n - a\| < \epsilon$ for all $n \geq N$. In this case, $\{x_n\}$ is said to be convergent and a is called a limit of the sequence $\{x_n\}$ and we write $\lim_{n\to\infty} x_n = a$.

A sequence (x_n) in X is called a ***Cauchy sequence*** if for every $\epsilon > 0$, there is $N \in \mathbf{N}$ such that $\|x_m - x_n\| < \epsilon$ for all $m, n \geq N$.

Proposition 6.1.1. Every convergent sequence in a normed space X is Cauchy. The converse does not hold generally. This relationship between Cauchy sequences and convergent sequences is the same as in metric spaces.

Proof. Let (x_n) be a convergent sequence with the limit $a \in X$. Then for every $\epsilon > 0$, there is a positive integer N such that $\|x_n - a\| < \epsilon$ for all $n \geq N$. This implies that $\|x_m - x_n\| \leq \|x_n - a\| + \|a - x_m\| < 2\epsilon$ for all $n, m \geq N$. Thus, (x_n) is a Cauchy sequence. □

If, in a given metric space, Cauchy sequences are equivalent to convergent sequences, then the metric space is said to be complete. For normed spaces, such spaces have a special name: Banach spaces.

Definition 6.1.5. A normed space X is said to be a ***Banach space*** if X is complete, that is every Cauchy sequence in X must be convergent. That is, $\lim_{n,m} \|x_n - x_m\| = 0$ implies the existence of $a \in X$ such that $\lim_n \|x_n - a\| = 0$.

Example 6.1.3. $(\ell^p, \|\cdot\|_p)$ and $(\ell^\infty, \|\cdot\|_\infty)$ are Banach spaces.

Proof. For $1 \leq p < \infty$, suppose that (x^k) is a Cauchy sequence in ℓ^p with $x^k = (x_1^k, x_2^k, \dots)$. For each $n \in \mathbf{N}$, $\left|x_n^k - x_n^m\right| \leq \left\|x^k - x^m\right\|_p$, so (x_n^k) is Cauchy in $(\mathbf{K}, \|\cdot\|_1)$. Since $\mathbf{K}$ is complete, there exists $y_n \in \mathbf{K}$ such that $\lim_{k\to\infty} x_n^k = y_n$. Now, we show that $y = (y_n) \in \ell^p$. Since (x^k) is Cauchy, it is bounded. So there exists $R > 0$ such that $\left\|x^k\right\|_p \leq R$ for all $k \in \mathbf{N}$. Therefore,

$$\left(\sum_{n=1}^{j} \left|x_n^k\right|^p\right)^{1/p} \leq \left(\sum_{n=1}^{\infty} \left|x_n^k\right|^p\right)^{1/p} \leq R \text{ for each } j,k \in \mathbf{N}.$$

Let $k \to \infty$ in the finite sum to conclude that $(\sum_{n=1}^{j} |y_n|^p)^{1/p} \leq R$. This is true for all $j \in \mathbf{N}$, so $(\sum_{n=1}^{\infty} |y_n|^p)^{1/p} \leq R$. Hence $y \in \ell^p$ as desired. Now, we show that $x^k \to y$ in $\|\cdot\|_p$. For every $\epsilon > 0$, there exists $N \in \mathbf{N}$ such that $\left\|x^k - x^m\right\|_p < \epsilon$ for all $k, m \geq N$. If $j \in \mathbf{N}$ is fixed, consider

$$\left(\sum_{n=1}^{j} \left|x_n^k - x_n^m\right|^p\right)^{1/p} \leq \left\|x^k - x^m\right\|_p < \epsilon.$$

Letting $m \to \infty$ in the finite sum to obtain $(\sum_{n=1}^{j} \left|x_n^k - y_n\right|^p)^{1/p} \leq \epsilon$. This is true for all $j \in \mathbf{N}$, so $(\sum_{n=1}^{\infty} \left|x_n^k - y_n\right|^p)^{1/p} \leq \epsilon$ for all $k \geq N$. Thus, $\left\|x^k - y\right\|_p \to 0$ as desired. □

Example 6.1.4. The normed space $(C[a,b], \|\cdot\|_\infty)$ is a Banach space. But the normed space $(C[a,b], \|\cdot\|_1)$ is not a Banach space.

Proof. Suppose that f_n is a Cauchy sequence under uniform convergence in $C[a,b]$. For each $x \in [a,b]$, we have

$$|f_n(x) - f_m(x)| \leq \|f_n - f_m\|_\infty \to 0.$$

If follows that for every $x \in [a,b]$, $f_n(x)$ is a Cauchy sequence in the normed space $(\mathbf{R}, \|\cdot\|_1)$ and hence convergent to some limit in $\mathbf{R}$ which we denote by $f(x)$. Let $\epsilon > 0$ and suppose that $\|f_n - f_m\|_\infty < \epsilon$ for all $n, m > N$. Then

$$\begin{aligned}|f(x) - f_m(x)| &= \left|\lim_{n\to\infty} f_n(x) - f_m(x)\right| = \lim_{n\to\infty} |f_n(x) - f_m(x)| \\ &\leq \lim_{n\to\infty} \|f_n - f_m\|_\infty \leq \epsilon,\end{aligned}$$

holds for all $x \in [a,b]$ and all $m > N$. Thus, $\|f - f_m\|_\infty \leq \epsilon$ holds when $m > N$. Since ϵ was arbitrary, we have shown that $\{f_m\}$ converges uniformly to f. We turn to the continuity of f. Let $x_0 \in [a,b]$ and let $\epsilon > 0$. Choose $N \in \mathbf{N}$ such that $\|f_n - f\|_\infty < \epsilon$ holds whenever $n > N$, and then use the continuity of f_N to choose $\delta > 0$ such that $|f_N(x) - f_N(x_0)| < \epsilon/3$ holds if $x \in [a,b]$ and $|x - x_0| < \delta$. It follows that if $x \in [a,b]$ and $|x - x - 0| < \delta$, then

$$\begin{aligned}|f(x) - f(x_0)| &\leq |f(x) - f_N(x)| + |f_N(x) - f_N(x_0)| + |f_N(x_0) - f(x_0)| \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon, \text{ as desired.}\end{aligned}$$

□

Theorem 6.1.7. A normed space $(X, \|\cdot\|)$ is a Banach space iff $\sum_{k=1}^{\infty} x_k$ converges for every sequence $(x_1, x_2, \ldots)$ where $x_k \in X$ such that $\sum_{k=1}^{\infty} \|x_k\| < \infty$.

Proof. First assume that X is a Banach space and (x_k) is a sequence in X such that $\sum_{k=1}^{\infty} \|x_k\| < \infty$. For $\epsilon > 0$, there exists $N \in \mathbf{N}$ such that $\sum_{k=N+1}^{\infty} \|x_k\| < \epsilon$. Let $y_n = x_1 + x_2 + \cdots + x_n$. If $m > n > N$, then

$$\begin{aligned} \|y_m - y_n\| &= \|x_{n+1} + x_{n+2} + \cdots + x_m\| \\ &\le \|x_{n+1}\| + \|x_{n+2}\| + \cdots + \|x_m\| \\ &\le \sum_{k=n+1}^{\infty} \|x_k\| < \epsilon. \end{aligned}$$

Thus, $y_1, y_2, \ldots$ is a Cauchy sequence in X. Since X is a Banach space, we conclude that (y_n) converges to some element of X, which means for $\sum_{k=1}^{\infty} x_k$ converge. To see the converse, assume that $\sum_{k=1}^{\infty} x_k$ converges for every sequence (x_k) in X such that $\sum_{k=1}^{\infty} \|x_k\| < \infty$. Suppose that (y_k) is a Cauchy sequence in X. It suffices to show that a subsequence converges. Since (y_k) is Cauchy, we can find $k_1 < k_2 < \ldots$ such that

$$\left\|y_{k_{n+1}} - y_{k_n}\right\| \le 2^{-n} \text{ for each } n = 1, 2, \ldots.$$

To show that the subsequence, note that $\sum_{n=1}^{\infty} \left\|y_{k_{n+1}} - y_{k_n}\right\| \le 1 < \infty$, by the hypothesis, $\sum_{n=1}^{\infty} (y_{k_{n+1}} - y_{k_n})$ converges. That is $\lim_{n\to\infty} y_{k_{n+1}}$ exists. □

6.2 Linear Operators

Linear operators are a core concept in functional analysis, among which linear functionals are a special and most important class of linear operators. In this preliminary stage, we will study some fundamental concepts.

A Linear Functional

Definition 6.2.1. Let X, Y be vector spaces on $\mathbf{K}$. A function $T : X \to Y$ is called ***linear*** if: $T(x_1 + x_2) = T(x_1) + T(x_2)$ for all $x_1, x_2 \in X$ and $T(cx) = cT(x)$ for all $c \in \mathbf{K}$ and $x \in X$. In functional analysis it is common practice to use the term ***linear operator*** instead of ***linear map***. If Y is the field of $\mathbf{K}$, then we use the term linear functional on X.

Definition 6.2.2. Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces and $T : X \to Y$ is a linear operator. The linear operator is called ***continuous*** if it is continuous in the topological sense. The linear operator T is called ***bounded*** if there exists a constant $M \ge 0$ such that

$$\|T(x)\|_Y \le M \|x\|_X \text{ for all } x \in X. \tag{1}$$

Example 6.2.1. Let $C[a, b]$ be the normed space of continuous functions from

$[a, b]$ to $\mathbf{R}$ with $\|f\|_\infty = \sup_{x\in[a,b]} |f(x)|$. Define $T : C[a, b] \to C[a, b]$ by

$$T(f) = \int_a^b f(t)dt,$$

then $\|Tf\| = (b - a) \sup_{[a,b]} |f| = (b - a) \|f\|$. Thus, T is a bounded linear functional on $C[a, b]$ with a bound $M = b - a$.

Theorem 6.2.1. Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces and let $T : X \to Y$ be a linear operator. The following statements are equivalent:

1. T is bounded.
2. T is continuous.
3. T is continuous at $x = 0$.

Proof. (1) $\Rightarrow$ (2): let T be bounded. Suppose that $a \in X$ and (x_n) is a sequence in X such that $\lim_{n\to\infty} x_n = a$. Then we have

$$\|Tx_n - Ta\|_Y = \|T(x_n - a)\|_Y \le M \|x_n - a\|_X \text{ for some } M \ge 0.$$

Thus, $\lim_{n\to\infty} Tx_n = Ta$. Hence T is continuous. That (2) $\Rightarrow$ (3) follows directly from the definition of continuity. We now prove that (3) $\Rightarrow$ (1), assume that T is continuous at $x = 0$. Then for $\epsilon = 1$, there exists $\delta > 0$ such that

$$\|Tx\|_Y < 1 \text{ whenever } \|x\|_X < \delta.$$

This implies that $\|Tx\|_Y \le 1$ for every $x \in X$ with $\|x\|_X = \delta$. Let $x \in X \setminus 0$. Then $\left\|\delta \|x\|_X^{-1} x\right\|_X = \delta$ and so $\left\|T(\delta \|x\|_X^{-1} x)\right\|_Y \le 1$. Multiply both sides of this last inequality by $\delta^{-1} \|x\|_X$ to obtain the inequality

$$\|Tx\|_Y \le \delta^{-1} \|x\|_X \text{ for all } x \in X, \text{ as desired.}$$

□

Definition 6.2.3. Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces. The set of all linear operators from X to Y is denoted by $L(X, Y)$. And the set of all bounded linear operators from X to Y is denoted by $B(X, Y)$. (It is clear that $B(X, Y)$ is a vector space).

We now want to define a norm on $B(X, Y)$, and it is called ***operator norm***. Define the function $\|\cdot\|$ by

$$\|T\| := \inf \{M > 0 : \|Tx\|_Y \le M \|x\|_X \text{ for all } x \in X\}.$$

It is clear that $\|Tx\|_Y \le \|T\| \|x\|_X$ for all $x \in X$. Note that the set on the right hand side is not empty and thus $\|T\| < \infty$.

Theorem 6.2.2. $(B(X, Y), \|\cdot\|)$ is a normed space.

Proof. It is clear that $\|T\| \geq 0$ and $\|0\| = 0$. If $\|T\| = 0$, then $\|Tx\|_Y \leq 0$ which implies that $T = 0$. Now, let $c \in \mathbf{K}$ and $c \neq 0$. Define $A_1 = \{M > 0 : \|Tx\|_Y \leq M\|x\|_X \text{ for all } x \in X\}$ and $A_2 = \{K > 0 : \|cT(x)\|_Y \leq c\|x\|_X \text{ for all } x \in X\}$. For every $M \in A_1$,

$$\|cT(x)\|_Y = \|T(cx)\|_Y \leq M\|cx\|_X = M|c|\,\|x\|_X \text{ for all } x \in X.$$

Therefore, $M|c| \in A_2$ and $\|cT\| = \inf A_2 \leq M|c|$. This is true for every $M \in A_1$. So $\|cT\| \leq |c| \inf A_1 = |c|\,\|T\|$. Replacing c with $1/c$ and T by cT, we conclude that $\|T\| = \|1/c \cdot cT\| \leq 1/|c|\,\|cT\|$. Hence $\|cT\| \geq |c|\,\|T\|$. Let $S, T \in B(X,Y)$, then for every $x \in X$, we have

$$\begin{aligned}\|(S+T)(x)\|_Y &= \|Sx + Tx\|_Y \leq \|Sx\|_Y + \|Tx\|_Y \\ &\leq \|S\|\,\|x\|_X + \|T\|\,\|x\|_X = (\|S\| + \|T\|)\,\|x\|_X\,.\end{aligned}$$

It follows that $\|S+T\| \leq \|S\| + \|T\|$, as desired. □

Theorem 6.2.3. Let X, Y be normed spaces. Y is a Banach space $\Rightarrow B(X,Y)$ is a Banach space. (even if X is not a Banach space).

Proof. Let (T_n) be a Cauchy sequence in $B(X,Y)$. Then for every $x \in X$, the inequality $\|T_nx - T_mx\|_Y \leq \|T_n - T_m\|\,\|x\|_X$ implies that $(T_nx) \subset Y$ is Cauchy. Since Y is complete, this sequence converges in Y. Hence we may define $T : X \to Y$ by

$$Tx = \lim_{n\to\infty} T_nx.$$

It is clear that T is linear. Furthermore, since (T_n) is Cauchy, there exists $M > 0$ such that $\|T_n\| \leq M$ for all $n \in \mathbf{N}$. Hence $\|Tx\|_Y \leq M\|x\|_X$ for all $x \in X$, so that $T \in B(X,Y)$. Now, we want to prove that $\lim_{n\to\infty} \|T_n \to T\| = 0$. To see this, let $\epsilon > 0$, there exists $N \in \mathbf{N}$ such that $\|T_n - T_m\| < \epsilon$ for all $n, m > N$. Then for every $x \in X$ and $n > N$,

$$\|Tx - T_nx\|_Y = \lim_{m\to\infty} \|T_mx - T_nx\|_Y \leq \lim_{m\to\infty} \|T_m - T_n\|\,\|x\|_X \leq \epsilon\|x\|_X\,.$$

Therefore, $\|T - T_n\| < \epsilon$ for all $n > N$. Thus, $T_n \to T$ in $B(X,Y)$ as desired. □

Theorem 6.2.4. A linear functional Λ on a normed space X is continuous iff $\ker(\Lambda) = f^{-1}(0)$ is a closed subspace of X. In particular, the linear functional is bounded $\Leftrightarrow$ continuous $\Leftrightarrow \ker(\Lambda)$ is closed.

Proof. First we assume that Λ is continuous, then $\ker(\Lambda) = \Lambda^{-1}(0)$ is closed since it is the inverse image of the closed set $\{0\}$. To see the converse, if $\ker(\Lambda) = \Lambda^{-1}(0) = X$, then Λ is the zero, which is continuous. Otherwise, there exists $x \in \ker(\Lambda)$. We may assume that $\Lambda(x) = 1$, if not, we could choose $x/\Lambda(x)$ instead of x. Since $\ker(\Lambda)$ is closed, x must be exterior to $\ker(\Lambda)$. Thus, there exists an open ball $B(x,\delta)$ such that

$$B(x,\delta) \cap \ker(\Lambda) = \emptyset.$$

We want to show that $|\Lambda(\omega)| < 1$ whenever $\|\omega\| < \delta$, which is similar to the proof of theorem 5.1.6. As in that proof, this statement will show that Λ is continuous because it has a bound of $M = \delta^{-1}$. To prove this, we assume that it is not ture. Then there exists $\omega \in B(0,\delta)$ with $|\Lambda(\omega)| \geq 1$. But then $\gamma = -\omega/\Lambda(\omega) \in B(0,\delta)$ as well. Translating by x, we have $x+\gamma \in B(x,\delta)$. But $x+\gamma \in \ker(\Lambda)$. Thus

$$x+\gamma \in B(x,\delta) \cap \ker(\Lambda),$$

which contradicts that $B(x,\delta) \cap \ker(\Lambda) = \emptyset$, as desired. $\square$

Definition 6.2.4. The ***algebraic dual space*** of a normed space X is the Banach space: $X^{\#} := B(X, \mathbf{K})$. That is $X^{\#}$ is the space of all linear functionals on the normed space X. The ***topological dual space*** of a normed space X is the space of all continuous linear functionals is denoted with X^*.

Definition 6.2.5. Let X, Y be two normed spaces. A linear operator $T : X \to Y$ is said to be ***isometric isomorphism*** if T is an isomorphism and $\|Tx\|_Y = \|x\|_X$ for all $x \in X$. In this case, we write $X = Y$.

B Hahn-Banach Theorem

Next, we will focus on the core theorem of this section: the Hahn-Banach extension theorem. This theorem states that every continuous linear functional (which we have already proven to be equivalent to a bounded linear functional) defined on a subspace of a normed space can be extended to the entire space. This theorem was proposed by Hahn and Banach in 1928 (it is worth noting that the content we are learning has finally moved beyond the first decade of the 20th century). However, the proof of this theorem also relies on Zorn's lemma...

Definition 6.2.6. Let X be a vector space with a subspace U. Suppose that Λ is a linear functional on U. A linear functional Γ on all of X is called a ***linear extension*** of Λ if

$$\Gamma(x) = \Lambda(x) \text{ for all } x \in U.$$

Hahn-Banach Theorem for Real. Let X be a normed space over $\mathbf{R}$ and U is a subspace of X, and $\Lambda \in U^{\#}$ we say that Λ is dominated by $\|\cdot\|^*$ on U if $|\Lambda(x)| \leq \|x\|^*$ for all $x \in U$ where $\|\cdot\|^*$ is a seminorm $\|\cdot\|^*$ on X, then Λ has a linear extension Γ to all of X which is dominated by $\|\cdot\|^*$ on all of X.

Proof. Consider the family $\mathscr{A}$ of all subspaces W of X such that $U \subset W \subset X$ and such that Λ has a linear extension Γ_W to W dominated by $\|\cdot\|^*$. Observe that

$\mathscr{A}$ is non-empty. Now, we define the order relation $\preceq$ on $\mathscr{A}$:

$$W_1 \preceq W_2 \text{ if } W_1 \subset W_2 \text{ and } \Gamma_{W_2}(x) = \Gamma_{W_1}(x) \text{ for all } x \in W_1.$$

It is clear that $\mathscr{A}$ is partially ordered by $\preceq$. Now, choose any chain $\mathscr{C}$ of $(\mathscr{A}, \preceq)$. If we can find the maximal element in $\mathscr{C}$ (the element belongs to $\mathscr{A}$), then Zorn,s lemma can be applied. The most straightforward approach is to prove that the union of all elements in the chain $\bigcup_{W\in\mathscr{C}} W$ is the maximal element: the maximality is quite clear, and next we show that it belongs to $\mathscr{A}$:

By Zorn's lemma, there is a maximal element of $\mathscr{A}$. It remains to be shown that this maximal subspace M is X. Assume not, then there exists $x_0 \in X \setminus M$ and consider the set

$$M_{x_0} = span\,\{M, x_0\} = \{x + \alpha x_0 : x \in M, \alpha \in \mathbf{R}\}.$$

It is clear that $M \subsetneq M_{x_0}$, if we can find a linear extension $\Gamma_{M_{x_0}}$ to M_{x_0} dominated by $\|\cdot\|^*$, then it would be strictly larger than M, contradicting the maximality of M. To see this, let $x, y \in M$ and Γ_M be the linear extension dominated by $\|\cdot\|^*$ on M, then

$$\begin{aligned}\Gamma_M(x) - \Gamma_M(y) = \Gamma_M(x - y) \le \|x - y\|^* &= \|(x + x_0) - (y + x_0)\| \\ &\le \|x + x_0\|^* + \|y + x_0\|^*.\end{aligned}$$

And hence $-\|y + x_0\|^* - \Gamma_M(y) \le \|x + x_0\|^* - \Gamma_M(x)$. For each $y \in M$, we take the infimum of the right side and let $d(x_0) = \inf\left\{\|x + x_0\|^* - \Gamma_M(x) : x \in M\right\}$, then $-\|y + x_0\|^* - \Gamma_M(x) \le d(x_0)$. Then we take the supremum of the left side to obtain that $\sup\left\{-\|y + x_0\|^* - \Gamma_M(y) : y \in M\right\} \le d(x_0)$. Thus, for all $y \in M$, we have

$$|\Gamma_M(y) + d(x_0)| \le \|y + x_0\|^*. \tag{$\bigstar$}$$

Now, for every $\omega = x + \alpha x_0 \in M_{x_0}$, define the mapping $\Gamma_{M_{x_0}}$ by $\Gamma_{M_{x_0}}(\omega) = \Gamma_M(x) + \alpha d(x_0)$. It is clear that $\Gamma_{M_{x_0}}$ is a linear functional on M_{x_0} and $\Gamma_{M_{x_0}}$ is a linear extension of Λ to M_{x_0}. To show that $\Gamma_{M_{x_0}}$ is dominated by $\|\cdot\|^*$, let $y = x/\alpha$ for $\alpha \neq 0$. From ($\bigstar$) we obtain that

$$\left|\frac{\Gamma_M(x) + \alpha d(x_0)}{\alpha}\right| \le \frac{1}{|\alpha|}\|x + \alpha x_0\|.$$

It follows that for $\omega = x + \alpha x_0$, we have $\left|\Gamma_{M_{x_0}}(\omega)\right| = |\Gamma_M(x) + \alpha d(x_0)| \le \|x + \alpha x_0\|^* = \|\omega\|^*$, which shows that the extension $\Gamma_{M_{x_0}}$ is dominated by $\|\cdot\|^*$ on the larger M_{x_0}, contradicting the maximality of M. Thus, X is the maximal element, as desired. □

Hahn-Banach Theorem for Complex. Let X be a normed space over $\mathbf{C}$ and U is a subspace of X, and $\Lambda \in U^{\#}$ with $|\Lambda(x)| \le \|x\|^*$ for all $x \in U$ where $\|\cdot\|^*$ is a seminorm $\|\cdot\|^*$ on X, then Λ has a linear extension Γ with $|\Gamma(x)| \le \|x\|^*$ for all $x \in X$.

Proof. Split Λ into its real an imaginary parts $\Lambda = \Lambda_1 + i\Lambda_2$. Then Λ_1, Λ_2 are real linear functional on U, where U is viewed as a vector space over $\mathbf{R}$. It is clear that $|\Lambda_1(x)| \leq |\Lambda(x)| \leq \|x\|^*$ on U, by the real version of the Hahn-Banach theorem, there exists an extension Γ_1 of Λ_1 to X. How about Λ_2 ? Note that

$$\Lambda(ix) = \Lambda_1(ix) + i\Lambda_2(ix) = i\Lambda(x) = i\Lambda_1(x) - \Lambda_2(x).$$

Thus, $\Lambda_1(ix) = -\Lambda_2(x)$ on U. Naturally, we can define Γ by $\Gamma(x) = \Gamma_1(x) - i\Gamma_1(ix)$ for all $x \in X$. It is clear that Γ is a complex linear functional. For every $x \in U$, we have

$$\Gamma(x) = \Gamma_1(x) - i\Gamma_1(ix) = \Lambda_1(x) - i\Lambda_1(ix) = \Lambda_1(x) + i\Lambda_2(x) = \Lambda(x).$$

Therefore, Γ is an extension of Λ to X. It remains to show that Γ is dominated by $\|\cdot\|^*$, to see this, we write $\Gamma(x) = re^{i\theta}$ for $x \in X$. Then $e^{-i\theta}\Gamma(x) = r$ is real and nonnegative, so that

$$|\Gamma(x)| = e^{-i\theta}\Gamma(x) = \Gamma_1(e^{-i\theta}x) \leq \left\|e^{-i\theta}x\right\|^* = \left|e^{-i\theta}\right| \|x\|^* = \|x\|^*.$$

□

Hahn-Banach Theorem for Normed Space. Let X be a normed space with a subspace U. Let $\Lambda \in U^*$ be a continuous linear functional on U. Then there exists a linear extension Γ of Λ that is continuous on all of X with $\|\Gamma\| = \|\Lambda\|$.

Proof. Let $\|x\|^* = \|\Lambda\| \|x\|$. By the Hahn-Banach theorem, $|\Gamma(x)| \leq \|\Lambda\| \|x\|$ for all $x \in X$, which implies that $\|\Gamma\| \leq \|\Lambda\|$. Since $U \subset X$, it is clear that $\|\Gamma\| \geq \|\Lambda\|$, as desired. □

►**Corollary 1.** Let x be a nonzero vector in a normed space X. Then there exists $\Gamma \in X^*$ such that $\Gamma(x) = \|x\|$ and $\|\Gamma\| = 1$. It follows that, if $X \neq \{0\}$, then $X^* \neq \{0\}$.

Proof. Let $U = \{cx : c \in \mathbf{K}\}$ be the span of x. We define the linear functional Λ on U as $\Lambda(cx) = c\|x\|$ for $cx \in U$. Since $x \neq 0$, then $\|\Lambda\| = 1$ on U and $\Lambda(x) = \|x\|$. By Hahn-Banach theorem, there exists $\Gamma \in X^*$ such that $\Gamma(x) = \Lambda(x) = \|x\|$ and $\|\Gamma\| = \|\Lambda\|$ □

►**Corollary 2.** Let X be a normed space with $x \neq y$, then there exists $\Lambda \in V'$ such that $\Lambda(x) \neq \Lambda(y)$.

►**Corollary 3.** Let X be a normed space with $x \in X$. If $\Lambda(x) = 0$ for all $\Lambda \in X^*$, then $x = 0$.

►**Corollary 4.** Let U be a subspace of a normed space X. If $x \in X \setminus U$ and $d(x, U) > 0$, then there exists $\Lambda \in X^*$ such that $\Lambda = 0$ on U and $\Lambda(x) = 1$.

Proof. □

►**Corollary 5.** Let U be a closed subspace of a normed space X. If $x \in X \backslash U$, then there exists $\Lambda \in X^*$ such that $\Lambda(x) = 1$ but $\Lambda(\omega) = 0$ for all $\omega \in U$.

C Separability and Reflexivity

Theorem 6.2.8. If X^* is separable, then X is separable.

Proof. The metric space induced by the normed space X^* is separable, thus, every subspace of X^* is separable, in particular, $B = \{\Lambda \in X^* : \|\Lambda\| = 1\}$ is separable as a subspace and then has a countable dense subset. Suppose that $\{\Lambda_1, \Lambda_2, \dots\}$ is the countable dense subset of B. Since we have

$$\|\Lambda_n\| = \inf \{M : |\Lambda_n(x)| \le M \|x\|_X\} = 1 \text{ for each } n = 1, 2, \dots,$$

there exists $x^n \in X$ with $\|x^n\|_X = 1$ such that $|\Lambda_n(x^n)| > 1/2$. We claim that $U = span \left\{x^1, x^2, \dots\right\}$ is dense in X, which shows that X is separable. Assume that U is not dense in X. Since U is a closed subspace of the normed space X, there exists a nonzero $\Lambda \in X^*$ such that $\Lambda(x^n) = 0$ for all n. WLOG, we assume that $\|\Lambda\| = 1$, then $\Lambda \in B$. Since $\{\Lambda_1, \Lambda_2, \dots\}$ is dense in B, $B(\Lambda, \epsilon) \cap \{\Lambda_1, \Lambda_2, \dots\} \neq \emptyset$, that is, for some n, we have $\|\Lambda_n - \Lambda\| < 1/2$. But this contradicts

$$1/2 < |\Lambda_n(x^n)| = |(\Lambda_n - \Lambda)(x^n)| \le \|\Lambda_n - \Lambda\| \|x^n\|_X = \|\Lambda_n - \Lambda\| .$$

□

Definition 6.2.7. Let X be a normed space. And note that X^* is always a Banach space. The dual of the Banach space X^* is called the ***second dual*** of X and written X^{**}.

Definition 6.2.8. We say that X is ***reflexive*** if X is equivalent to its second dual X^{**}. Since a dual space is always a Banach space, a normed space X which is not a Banach space cannot be reflexive.

6.3 The Baire Category Theorem

A The Baire Category Theorem

Definition 6.3.1. Let X be a topological space, and A is a subset of X, a set A is ***nowhere dense*** in a metric space if $(\overline{A})^c$ is dense in X. It is clear that A is nowhere dense iff $\overline{A}$ is nowhere dense in X.

Definition 6.3.2. A set A in a metric space X is said to be of the ***first category*** in X if it is the union of countable many nowhere dense sets. Otherwise, if A cannot be written as the union of countable many nowhere dense sets, it is called the ***second category***.

Baire's Category Theorem. Let X be a complete metric space.

1. If $G_1, G_2, \ldots$ is a sequence of dense open sets in X, then $\bigcap_{k=1}^{\infty} G_k$ is dense.
2. X is of the second category in X.

Proof. (1). Let U be an arbitrary open ball in X. Since G_1 is dense in X, then $G_1 \cap U \neq \emptyset$. Choose $x_1 \in G_1 \cap U$ and $0 < r_1 < 1$ such that $\overline{B}(x_1, r_1) \subset G_1 \cap U$ (this is possible since G_1 and U are open). Since G_2 is dense in X, $G_2 \cap B(x_1, r_1) \neq \emptyset$, choose $x_2 \in G_2 \cap B(x_1, r_1)$ and $0 < r_2 < 1/2$ such that $\overline{B}(x_2, r_2) \subset G_2 \cap B(x_1, r_1)$. Continue this process to get a nest sequence of closed balls such that $\overline{B}(x_n, r_n) \subset G_n \cap B(x_{n-1}, r_{n-1})$ and $0 < 1/r_n < 1/n$. Note that $x_n \in B(x_m, r_n)$ for all $n > m$. Thus, for every $\epsilon > 0$, there exists $N \in \mathbf{N}$ such that $1/N < \epsilon$ and for all $n, m > N$, $|x_m - x_n| < r_n < \epsilon$. Since X is complete metric space, $x_n \to x \in X$. By the definition of closure. we know that $x \in \overline{B}(x_n, r_n)$. Since our sequence of balls is nested, and $\overline{B}(x_n, r_n) \subset G_n$, we know that $x \in G_n$ for all n. Moreover, since $\overline{B}(x_1, r_1) \subset U$, we know that $x \in U$. Therefore, we have shown that $(\bigcap G_n) \cap U$ contains x and is thus nonempty. Since U is arbitrary, we have shown that $\bigcap G_n$ is dense in X.

(2). Assume that X is of the first category. Then $X = \bigcup_{k=1}^{\infty} A_k$, where A_k is nowhere dense and then each $\overline{A_k}$ is nowhere dense as well. Then $\emptyset = \bigcap_{k=1}^{\infty} (\overline{A_k})^c = \bigcap_{k=1}^{\infty} G_k$ where $G_k = (\overline{A_k})^c$ is open and dense for each k (by the definition of nowhere dense). This contradicts the part (1). Thus, X is of the second category in X. □

Example 6.3.1.

B Applications

Uniform Boundedness Principle. Let $T_1, T_2, \ldots$ be a sequence of continuous linear operators from a Banach space X into a normed space Y. If for every $x \in X$, we have $\sup_k \|T_k x\|_Y < \infty$, then $\sup_k \|T_k\| < \infty$.

Proof. For $n = 1, 2, \ldots$, let $B_n = \{x \in X : \|T_k(x)\|_Y \leq n \text{ for all } k = 1, 2, \ldots.\}$ For every $x \in X$, since $\sup_k \|T_k(x)\| < \infty$, the union $\bigcup_{n=1}^{\infty} B_n = X$. Now, we claim that each B_n is closed: since T_k is continuous and the set $B_{nk} =$

$\{x \in X : \|T_k(x)\|_Y \leq n\}$ is the inverse image of the closed interval $[0, n]$, is closed. Hence the intersection $B_n = \bigcap_{k=1}^{\infty} B_n k$ is also closed. By the Baire Category theorem (X is Banach (complete)), there exists $N \in \mathbf{N}$ such that B_N has non-empty interior (since B_N is closed). Therefore there exists $x_0 \in X$ and $r > 0$ such that $B(x_0, r) \subset B_N$, which implies that $\|T_k(x)\| \leq N$ for all $k = 1, 2, \ldots$ and all $x \in B(x_0, r)$. Let $x \in X$ with $\|x\|_X \leq 1$. Then $x_0 + rx \in B(x_0, r)$, so that $\|T_k(x_0 + rx)\|_Y \leq N$ for all $k = 1, 2, \ldots$. Thus,

$$\|T_k(x)\|_Y = \left\|T_k\left(\frac{x_0 + rx}{r} - \frac{x_0}{r}\right)\right\| \leq \frac{2N}{r} < \infty, \text{as desired.}$$

□

6.4 $\mathscr{L}^p$ and L^p

A Lebesgue Spaces: $1 \leq p < \infty$

Definition 6.4.1. Let (X, Σ, μ) be a measure space and $1 \leq p < \infty$, and let $f : X \to \mathbf{R}$ be a measurable function. A function $f : X \to \overline{\mathbf{R}}$ is called ***p-integrable*** if it is measurable and $|f|^p$ is integrable. The space of p-integrable functions is denoted by

$$\mathscr{L}^p(X, \Sigma, \mu) := \left\{f : X \to \overline{\mathbf{R}} : f \text{ is } p\text{-integrable}\right\}.$$

It is worth noting that in general, there is no inclusion relationship between $\mathscr{L}^p(X, \Sigma, \mu)$ and $\mathscr{L}^q(X, \Sigma, \mu)$ when $1 \leq p < q < \infty$. For example: $f_1(x) = 1/x$, $f_2(x) = 1/\sqrt{x}\chi_{(0,1)}(x) + 1/x^2\chi_{[1,\infty)}(x)$, one can check that $f_1 \in \mathscr{L}^2$, $f_1 \notin \mathscr{L}^1$ and $f_2 \in \mathscr{L}^1$, $f_2 \notin \mathscr{L}^1$.

It is easy to prove that $\mathscr{L}^p$ is a vector space (use the inequality $(a + b)^p \leq 2^p(a^p + b^p)$). Now, for $1 \leq p < \infty$, we define the function $\|\cdot\|_p^*$ on the vector space $\mathscr{L}^p(X, \Sigma, \mu)$:

$$\|f\|_p^* := \left(\int_X |f|^p \, d\mu\right)^{1/p}.$$

Now, we show that this function is indeed a seminorm on the vector space $\mathscr{L}^p$. The first property is easy to verify, and the proof of the second property follows from the integral form of Minkowski's inequality:

Hölder's Inequality for Integrals. Suppose $p, q > 1$ are Hölder conjugates. Then for any $f, g \in \mathscr{L}^p(\mu)$, we have $fg \in \mathscr{L}^1(\mu)$ and the inequality

$$\|fg\|_1^* = \int_X |fg| \, d\mu \leq \left(\int_X |f|^p \, d\mu\right)^{1/p} \left(\int_X |g|^q \, d\mu\right)^{1/q} = \|f\|_p^* \, \|g\|_q^*$$

Proof. If $\|f\|_p^* = 0$ or $\|g\|_q^* = 0$, then $f = 0$ or $g = 0$ μ-a.e. and the result is clear. Otherwise, the proof follows the one for Hölder's Inequality for sums, that is, let $a = |f(x)|^p / \|f\|_p^*$ and $b = |g(x)| / \|g\|_q^*$, then we have

$$\frac{|f(x)g(x)|}{\|f\|_p^* \|g\|_q^*} \leq \frac{|f(x)|^p}{p \cdot (\|f\|_p^*)^p} + \frac{|g(x)|^q}{q \cdot (\|g\|_q^*)^q}.$$

Integrating both sides we obtain that

$$\frac{\int_X |fg|}{\|f\|_p^* \|g\|_q^*} \leq \frac{\int_X |f|^p}{p(\|f\|_p^*)^p} + \frac{\int_X |g|^q}{q(\|g\|_q^*)^q} = \frac{1}{p} + \frac{1}{q} = 1.$$

□

Minkowski's Inequality for Integrals. Suppose $p \geq 1$. Then for any $f, g \in \mathscr{L}^p(\mu)$, we have $f + g \in \mathscr{L}^p(\mu)$ and the inequality

$$\|f + g\|_p^* \leq \|f\|_p^* + \|g\|_p^* .$$

Proof. The proof follows the pattern of Minkowski's inequality for sums. □

Remark 6.4.1. We rarely discuss the case $0 < p < 1$, because in this case, the function $\|\cdot\|_p^*$ defined in such a manner fails to satisfy the triangle inequality. This essentially stems from the restriction on p in Minkowski's inequality, as discussed earlier. An example can illustrate this issue: take $f = \chi_{(0,1/2]}$ and $g = \chi_{(1/2,1)} \in \mathscr{L}^p([0,1], \Sigma, \mu)$, then $\|f\|_p^* = \|g\|_p^* = 2^{-1/p}$ and $\|f + g\|_p = 1$ and

$$\|f\|_p^* + \|g\|_p^* = 2^{-1/p} + 2^{-1/p} > 1 = \|f + g\|_p^* .$$

It can be shown that, in this case, the reverse triangle inequality holds, and it is natural to define a new space — the F-space. However, this is not the main focus of this lecture. Interested readers are encouraged to consult relevant materials.

It is easy to see through Minkowski's inequality that $\|\cdot\|_p^*$ is indeed a seminorm. However, we are more interested in studying norms. The situation here is similar to the examples we encountered earlier: the $(\mathscr{L}^p(X, \Sigma, \mu), \|\cdot\|^*)$ is still not a normed space. The same method using Theorem 5.1.6 can be applied to address this issue. Next, we will provide a detailed explanation of the refinement process:

Let (X, Σ, μ) be a measure space, and let $\mathscr{N}^p(X, \Sigma, \mu)$ be the subset of $\mathscr{L}^p(X, \Sigma, \mu)$ that consists of those functions that belong to $\mathscr{L}^p(X, \Sigma, \mu)$ and satisfies $\|f\|_p^* = 0$, that is, $\mathscr{N}^p(X, \Sigma, \mu) = \{f \in \mathscr{L}^p(X, \Sigma, \mu) : f = 0 \ \mu\text{-a.e.}\}$. It is clear that $\mathscr{N}^p(X, \Sigma, \mu)$ is a subspace of $\mathscr{L}^p(X, \Sigma, \mu)$ (that is $\mathscr{N}^p(X, \Sigma, \mu)$ is a vector space). And define

$$L^p(X, \Sigma, \mu) := \mathscr{L}^p(X, \Sigma, \mu) / \mathscr{N}^p(X, \Sigma, \mu).$$

Then $L^p(X, \Sigma, \mu)$ is a vector space. Recall that this means that $L^p(X, \Sigma, \mu)$ is the family of cosets of $\mathscr{N}^p(X, \Sigma, \mu)$ in $\mathscr{L}^p(X, \Sigma, \mu)$. Those cosets are by definition the equivalence classes induced by the equivalence relation $\sim$, where $f \sim g$ holds iff $f - g \in \mathscr{N}^p(X, \Sigma, \mu)$, that is $f = g$ μ-a.e..

For $f, g \in \mathscr{L}^p(X, \Sigma, \mu)$, It is easy to check that $f\mathscr{N}^p + g\mathscr{N}^p = (f+g)\mathscr{N}^p$ and $c(f\mathscr{N}^p) = (cf)\mathscr{N}^p$ define operations that make $L^p(X, \Sigma, \mu)$ into a vector space. Then we define the function

$$\|f\mathscr{N}^p\|_p := \|f\|_p^* .$$

It is easy to check that $\|\cdot\|_p$ is a norm on $L^p(X, \Sigma, \mu)$, that is $(L^p(X, \Sigma, \mu), \|\cdot\|_p)$ is a normed space. In fact, some authors use the same symbol for $\mathscr{L}^p$ and L^p. We will try to avoid this identification of functions and classes of functions, since it can lead to confusion. However, to simplify notation we will often deal with $\mathscr{L}^p$ when proving theorems about L^p. Now, we can prove that it is a Banach space:

Riesz-Fischer Theorem. For $1 \leq p < \infty$, $(L^p(X, \Sigma, \mu), \|\cdot\|)_p$ is Banach.

Proof. Suppose that (f_k) of $\mathscr{L}^p(X, \Sigma, \mu)$ and satisfy $\sum_{k=1}^{\infty} \|f\|_P^* < \infty$. Define $g : X \to [0, \infty]$ by $g(x) = \left(\sum_{k=1}^{\infty} |f_k(x)|\right)^p$. Minkowski's inequality implies that

$$\left(\int_X \left(\sum_{k=1}^{n} |f_k|\right)^p d\mu\right)^{1/p} = \left\|\sum_{k=1}^{n} |f_k|\right\|_p^* \leq \sum_{k=1}^{n} \|f_k\|_p^* .$$

By the monotone convergence theorem, we obtain

$$\int_X g d\mu = \lim_{n\to\infty} \int_X \left(\sum_{k}^{n} |f_k|\right)^p d\mu \leq \left(\sum_{k=1}^{\infty} \|f_k\|_p^*\right)^p < \infty.$$

Therefore, g is integrable and hence g is finite μ-a.e.. Now we claim that $\sum_{k=1}^{\infty} f_k$ is convergent μ-a.e.. Define the function f on X by

$$f(x) = \begin{cases} \sum_{k=1}^{\infty} f_k(x), & \text{if } g(x) < \infty. \\ 0, & \text{other.} \end{cases}$$

Then f is measurable and satisfies $|f|^p \leq g$, and so it belongs to $\mathscr{L}^p(X, \Sigma, \mu)$. Now define $s_n = \sum_{k=1}^{n} f_k$, then $s_n \in \mathscr{L}^p(X, \Sigma, \mu)$ and we want to prove that $\|s_n - f\|_p^* \to 0$. Since $\lim_{n\to\infty} |s_n - f| = 0$ and $|f - s_n|^p \leq (2g)^p$. Therefore by Dominated convergence theorem,

$$\lim_{n\to\infty} \int_X |f - s_n|^p d\mu = \int_X \lim_{n\to\infty} |s_n - f|^p d\mu = 0.$$

That is $\|f - s_n\|_p^* = 0$. By theorem 5.1.5, $L^p(X, \Sigma, \mu)$ is complete. □

Just like in the case of sequence spaces where $p = \infty$ is allowed, the Lebesgue space can also be defined for $p = \infty$. Discussing the case of $p = \infty$ is complex but crucially important. I prefer not to directly present the definitions of

$\mathscr{L}^\infty, L^\infty$ and $\|\cdot\|_\infty$ here without proper motivation. Therefore, I will address this case later.

►**Corollary 1.** Let (X, Σ, μ) be a measure space and $1 \leq p < \infty$. If $f \in \mathscr{L}^p(\mu)$ and (f_k) is a sequence of $\mathscr{L}^p(\mu)$ such that $\lim_{k\to\infty} \|f_k - f\|_p^* = 0$, then there exists a subsequence (f_{k_m}) such that $f_{k_m} \to f$ μ-a.e..

Proof. □

Theorem 6.4.4. Let (X, Σ, μ) be a measure space and let $1 \leq p < \infty$. Then the simple functions in $L^p(\mu)$ form a dense subspace of $L^p(\mu)$.

Proof. Let $f \in \mathscr{L}^p(X, \Sigma, \mu, \mathbf{R})$, choose increasing sequences $\{\phi_k\}$ and $\{\psi_k\}$ of nonnegative simple functions such that $f^+ = \lim_k \phi_k$ and $f^- = \lim_k \psi_k$, and define the sequence $\eta_k = \phi_k - \psi_k$. Then each η is simple that satisfies $|\eta_k| \leq |f|$ and so belongs to $\mathscr{L}^p(X, \Sigma, \mu, \mathbf{R})$. Since these functions satisfy $|\eta_k(x) - f(x)| \leq 2\,|f(x)|$ and $\lim_k |\eta_k(x) - f(x)|^p = 0$, the dominated convergence theorem implies that $\lim_k \|f - \eta_k\|_p^* = 0$. □

Definition 6.4.2. Let $[a, b]$ be a closed bounded interval. A real valued function f on $[a, b]$ is a ***step function*** if there are real numbers $a_0, \ldots, a_n$ such that $a = a_0 < a_1 < \cdots < a_n = b$ and f is constant on each interval (a_{i-1}, a_i).

Now, we will use $\mathscr{L}^p(a, b)$ and $L^p([a, b])$ for $\mathscr{L}^p([a, b], \mathcal{B}([a, b]), \lambda)$ and $L^p([a, b], \mathcal{B}([a, b]), \lambda)$ where $\mathcal{B}([a, b])$ is the σ-algebra of Borel subsets of $[a, b]$ and λ is the Lebesgue measure.

►**Corollary 1.** Let $[a, b]$ be a closed bounded interval and $1 \leq p < \infty$. Then the subspace of $L^p([a, b])$ determined by the step functions on $[a, b]$ is dense in $L^p([a, b])$.

Proof. It is clear that every step function on $[a, b]$ belongs to $\mathscr{L}^p([a, b])$. Since the Borel simple functions on $[a, b]$ determine a dense subspace, it suffices to show that if f is a Borel simple function and $\epsilon > 0$ is arbitrary, then there exists a step function g such that $\|f - g\|_p^* < \epsilon$. By the definition of simple function, they can write as a linear combination of finite characteristic functions, it suffices to show that for every characteristic function on a Borel susbet A of $[a, b]$, and for every $\epsilon > 0$ there exists a step function g such that $\|\chi_A - g\|_p^* =< \epsilon$. Let χ_A be the characteristic function of a Borel subset $A \in \mathcal{B}([a, b])$, for every $\epsilon > 0$, there is a sequence $\{(a_n, b_n)\}$ of open intervals such that $A \subset \bigcup_{n=1}^\infty (a_n, b_n)$ and $\sum_{n=1}^\infty (b_n - a_n) < \lambda(A) + \epsilon$. And then choose $N \in \mathbf{N}$ such that $\sum_{n=N+1}^\infty (b_n - a_n) < \epsilon$. Let h be the step function of $[a, b] \cap (\bigcup_{n=1}^\infty (a_n, b_n))$ and let g be the characteristic function of $[a, b] \cap \left(\bigcup_{n=1}^N (a_n, b_n)\right)$, then g is a step function and then

$$\begin{aligned}\|\chi_A - g\|_p^* &\le \|\chi_A - h\|_p^* + \|h - g\|_p^* \\ &\le \left(\lambda\left(\bigcup_{n=1}^{\infty}(a_n, b_n) - A\right)\right)^{1/p} + \left(\lambda\left(\bigcup_{n=N+1}^{\infty}(a_n, b)n\right)\right)^{1/p} \\ &< \epsilon^{1/p} + \epsilon^{1/p} = 2\epsilon^{1/p}.\end{aligned}$$

□

►**Corollary 2.** Let $[a, b]$ is a closed bounded interval and $1 \le p < \infty$. Then the subspace of $L^p([a, b])$ determined by the continuous functions on $[a, b]$ is dense in $L^p([a, b])$.

Proof. Since every continuous function is bounded on all closed intervals, all the continuous functions on $[a, b]$ belong to $\mathscr{L}^p([a, b])$. Since the step functions on $[a, b]$ determine a dense subspace, it suffices to prove that for every step function f and every $\epsilon > 0$ there is a continuous function g such that $\|f - g\|_p^* < \epsilon$. Let f be a step function on $[a, b]$ and $M = \sup\{|f(x) : x \in [a, b]|\}$ and $\epsilon > 0$. □

Theorem 6.4.5. Let (X, Σ, μ) be a σ-finite measure space and $1 \le p < \infty$. If $\Sigma = \sigma(\mathscr{C})$ where $\mathscr{C}$ is a countable subfamily of $\mathscr{A}$, then $L^p(X, \Sigma, \mu)$ is separable.

B *Lebesgue Space:* $p = \infty$

Let (X, Σ, μ) be a measure space, $1 < p < \infty$. In this subsection, we study an important theorem: for $1 < p < \infty$ the dual space of $L^p(X, \Sigma, \mu)$ is isometrically isomorphisc to $L^q(X, \Sigma, \mu)$ where p, q are Hölder conjugates. For $g \in \mathscr{L}^q(X, \Sigma, \mu)$ where p, q are Hölder conjugates. For every p-integrable function $f \in \mathscr{L}^p(X, \Sigma, \mu)$, the function fg is integrable by the Hölder Inequality. Then we define

$$\Lambda_g f = \int_X g f \, d\mu, \text{ for } f \in \mathscr{L}^p(X, \Sigma, \mu). \tag{1}$$

It is clear that Λ_g is a linear functional on $\mathscr{L}^p(X, \Sigma, \mu)$. Moreover, $\left\|\Lambda_g f\right\|_p^* \le \|g\|_q^* \|f\|_p^*$ by the Hölder inequality, and hence $\left\|\Lambda_g\right\| \le \|g\|_q^*$ and Λ_g is a bounded linear functional. On the otherhand, the function $f_1 = |g|^{q-2} g$ satisfies

$$\int_X |f_1|^p \, d\mu = \int_X |g|^{(q-1)p} \, d\mu = \int_X |g|^q \, d\mu < \infty,$$

so $f_1 \in \mathscr{L}^p(X,\Sigma,\mu)$. And we obtain that $\left|\Lambda_g f_1\right| \leq \left\|\Lambda_g\right\| \|f_1\|_p^*$, that is

$$\int_X |g|^q \, d\mu \leq \left\|\Lambda_g\right\| \left(\int_X |g|^q \, d\mu\right)^{1/p},$$

which means that $\|g\|_q^* \leq \left\|\Lambda_g\right\|$. Thus, $\left\|\Lambda_g\right\| = \|g\|_q^*$. We conclude that the linear operator $T : \mathscr{L}^q(X,\Sigma,\mu) \to (\mathscr{L}^p(X,\Sigma,\mu))^*$ given by $g \mapsto \Lambda_g$ is norm-preserving. Then we set $\Lambda_g(\langle f\rangle) = \Lambda_g(f)$ and $T(\langle g\rangle) = T(g)$. We refine the linear operator $T : L^q(X,\Sigma,\mu) \to (L^p(X,\Sigma,\mu))^*$ and given by $\langle g\rangle \mapsto \Lambda_g$. It is clear that T is injective, next we show that it is also surjective.

Theorem 6.4.6. Let (X,Σ,μ) be a measure space. Let p,q be Hölder conjugates and $1 < p < \infty$. Then for every $\Gamma \in (L^p)^*$, there exists $\langle g\rangle \in L^q$ such that $\Gamma(\langle f\rangle) = \Lambda_g\langle f\rangle = \int_X fg d\mu$ for all $\langle f\rangle \in L^p(X,\Sigma,\mu)$. That is L^q is isometrically isomorphisc to $(L^p)^*$.

Proof. We suppose that $\Gamma \in (L^p(X,\Sigma,\mu))^*$. First we assume that μ is finite, so that all simple functions belong to $\mathscr{L}^p(X,\Sigma,\mu)$. We define a function ν on the σ-algebra Σ by $\nu(A) = \Gamma(\langle\chi_A\rangle)$. If $\{A_k\}$ is a sequence of disjoint sets in Σ and $A = \bigcup_{k=1}^\infty A_k$, then

$$\left\|\chi_A - \sum_{k=1}^n \chi_{A_k}\right\|_p^* = \left\|\sum_{k=n+1}^\infty \chi_{A_k}\right\|_p^* = \mu\left(\bigcup_{n+1}^\infty A_k\right)^{1/p} \to 0.$$

And since Γ is continuous and linear, this implies that $\Gamma(\langle\chi_A\rangle) = \sum_{k=1}^\infty \Gamma(\langle\chi_{A_k}\rangle)$ and hence that $\nu(A) = \sum_{k=1}^\infty \nu(A_k)$, Thus ν is a finite signed measure. If $\mu(E) = 0$, then $\chi_E = 0$ and so $\nu(E) = 0$, that is $\nu \ll \mu$. By the Radon-Nikodym theorem, there is a measurable function $g \in \mathscr{L}^1(X,\Sigma,\mu)$ such that $\nu(A) = \int_A g d\mu$ for each $A \in \Sigma$. Now, we claim that $g \in \mathscr{L}^q(X,\Sigma,\mu)$ and $\Gamma(\langle f\rangle) = \int_X fg d\mu$ holds for each $f \in \mathscr{L}^p(X,\Sigma,\mu)$:

For each positive integer n let $E_n = \{x \in X : |g(x) \leq n|\}$. Then $g\chi_{E_n}$ is bounded and so belongs to $\mathscr{L}^q(X,\Sigma,\mu)$ (since μ is finite). Define a functional Γ_{E_n} on $L^p(X,\Sigma,\mu)$ by $\Gamma_{E_n}(\langle f\rangle) = \Gamma(\langle f\chi_{E_n}\rangle)$ and consider the relation

$$\Gamma_{E_n}(\langle f\rangle) = \int_X fg\chi_{E_n} d\mu. \tag{2}$$

It f is the characteristic function of a measurable set A, that is if $f = \chi_A$, then both sides of (2) are equal to $\nu((A \cap E_n))$, thus (2) holds if f is the characteristic function of a measurable set and hence if f is a simple function. Since the simple functions determine a dense subspace of $L^p(X,\Sigma,\mu)$, (2) holds for all $\langle f\rangle$ in $L^p(X,\Sigma,\mu)$. It follows from (1) that

$$\left\|g\chi_{E_n}\right\|_q^* = \left\|\Gamma_{E_n}\right\| \leq \|\Gamma\|.$$

Since $q < \infty$, the monotone convergence theorem implies that

$$\int_X |g|^q \, d\mu = \lim_{n\to\infty} \int_X \left|g\chi_{E_n}\right|^q d\mu \leq (\|\Gamma\|)^q.$$

Which implies that $g \in \mathscr{L}^q(X,\Sigma,\mu)$ and $\|g\|_q^* \leq \|\Gamma\|$. Now, we can take limits in (2) and conclude that $F(\langle f\rangle) = \int_X fgd\mu$.

To see the case that μ is not finite. Suppose that $B \in \Sigma$. Let Σ_B be the σ-algebra on B consisting of those subsets of B that belongs to Σ, and let μ_B be the restriction of μ ot Σ_B. If f is a real valued function on B then we will denote by $\tilde{f}$ the function on X that agrees with f on B and vanishes outside B. The formula $\Gamma_B(\langle f\rangle) = \Gamma(\langle \tilde{f}\rangle)$ defines a linear functional Γ_B on $L^p(B,\Sigma_B,\mu_B)$, this functional satisfies $\|\Gamma_B\| \leq \|F\|$.

Now, assume that μ is σ-finite. Let $\{B_k\}$ ba a sequence of disjoint measurable sets such that $X = \bigcup_{k=1}^\infty B_k$ and $\mu(B_k) < \infty$ for each k. From the first part of this proof, there is a function $g_k \in \mathscr{L}^q(B_k,\Sigma_{B_k},\mu_{B_k})$ that represents Γ_{B_k} on $L^p(B_k,\Sigma_{B_k},\mu_{B_k})$ and satisfies $\|g_k\|_q^* \leq \left\|\Gamma_{B_k}\right\|$. Define $g = \sum_{k=1}^\infty g_k\chi_{B_k}$ and define $g^{(n)} = \sum_{k=1}^n g_k\chi_{B_k}$, then $\left\|g^{(n)}\right\|_q^* = \sum_{k=1}^n \|g_k\|_p^* \leq \left\|\Gamma_{\bigcup_{k=1}^n B_k}\right\| \leq \|\Gamma\|$. By the monotone convergence theorem,

Finally, we assume that μ is arbitrary. Let $\mathscr{S}$ be the family of sets in Σ that are σ-finite under μ. Note that if $B \in \mathscr{S}$, then (B,Σ_B,μ_B) is σ-finite, and so by what we proved, there is a function $g \in \mathscr{L}^q(B,\Sigma_B,\mu_B)$ such that

$$\Gamma_B(\langle f\rangle) = \int_X fgd\mu_B \text{ for each } \langle f\rangle \in L^p(B,\Sigma_B,\mu_B).$$

Furthermore, we claim that: if B_1, B_2 are disjoint sets in $\mathscr{S}$, then $\left\|\Gamma_{B_1\cup B_2}\right\|^q = \left\|\Gamma_{B_1}\right\|^q + \left\|\Gamma_{B_2}\right\|^q$: it is clear that $(B_1\cup B_2, \Sigma_{B_1\cup B_2}, \mu_{B_1\cup B_2})$ is σ-finite, then we suppose that $g \in \mathscr{L}^q(B_1\cup B_2, \Sigma_{B_1\cup B_2}, \mu_{B_1\cup B_2})$ such that $\Gamma_{B_1\cap B_2} = \int_{B_1\cup B_2} fgd\mu_{B_1\cup B_2}$ for every $f \in \mathscr{L}^p(B_1 \cup B_2, \Sigma_{B_1\cup B_2}, \mu_{B_1\cup B_2})$ and note that

$$\begin{aligned}\left\|\Gamma_{B_1\cup B_2}\right\|^q &= \int_{B_1\cup B_2} |g|^q \, d\mu_{B_1\cup B_2} \\ &= \int_{B_1} |g|^q \, d\mu_{B_1} + \int_{B_2} |g|^q \, d\mu_{B_2} = \left\|\Gamma_{B_1}\right\|^q + \left\|F_{B_2}\right\|^q.\end{aligned}$$

Now we choose a sequence $\{C_n\}$ of sets in $\mathscr{S}$ such that

$$\lim_{n\to\infty} \left\|\Gamma_{C_n}\right\| = \sup\{\|\Gamma_B\| : B \in \mathscr{S}\}.$$

Let $C = \bigcup_{n=1}^\infty C_n$. Then $C \in \mathscr{S}$ and $\|\Gamma_C\| = \sup|\|\Gamma_B\| : B \in \mathscr{S}|$ and we can choose a functional g_C in $\mathscr{L}^q(C,\Sigma_C,\mu_C)$ such that

$$\Gamma_C(\langle f\rangle) = \int_X fg_Cd\mu_C \text{ for each } \langle f\rangle \in L^p(C,\Sigma_C,\mu_C).$$

Note that if $f \in \mathscr{L}^p(X,\Sigma,\mu)$ and vanishes on C, then $\Gamma(\langle f\rangle) = 0$. It follows that if g is the function on X that agrees with g_C on C and vanishes out C, then $g \in \mathscr{L}^q(X,\Sigma,\mu)$ and

$$\Gamma(\langle f\rangle) = \int_X fgd\mu \text{ for each } \langle f\rangle \in L^p(X,\Sigma,\mu).$$

Hence $\Gamma = \Lambda_g$ and the surjectivity of T is complete. $\square$

Some textbooks directly present the definition of the $\mathscr{L}^{\infty}(X, \Sigma, \mu)$ and its seminorm $\|\cdot\|_{\infty}^{*}$ when introducing it. However, I believe it is possible to develop it more motivationally by considering the limiting relationship between the p-norm and the supremum norm, and attempting to use limits to derive the desired $\mathscr{L}^{\infty}(X, \Sigma, \mu)$ and $\|\cdot\|_{\infty}^{*}$.

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III Differentiation and Radon Measure

7

Radon Measure and Riesz Representation Theorem

7.1 Measures on Locally Compact Spaces

A Locally Hausdorff Space

Theorem 7.1.1. Let X be a LCHS, $K \subset X$ be a compact subset and let $K \subset U$ be open. Then there exists an open set $V \subset X$ such that the closure $\overline{V}$ is compact and $K \subset V \subset \overline{V} \subset U$.

Proof. □

Let X be LCHS and denote $C_c(X) = \{f \in C(X) : \text{supp}(f) \text{ is a compact set}\}$ where $\text{supp}(f) = \overline{\{x \in X : f(x) \neq 0\}}$ is called the support of f.

Theorem 7.1.2. Let X be a LCHS, $K \subset X$ be a compact subset and let $K \subset U$ be open. Then there exists $f \in C_c(X)$ such that $\chi_K(x) \leq f(x) \leq \chi_U(x)$ and is $\text{supp}(f) \subset U$.

Proof. □

Theorem 7.1.3. Let X be a LCHS, $f \in C_c(X)$ and $\text{supp}(f) \subset \bigcup_{i=1}^n U_i$, where U_i is an open set of X. Then there exist $f_i \in C_c(X)$, $1 \leq i \leq n$ such that $f = \sum_{i=1}^n f_i$ and $\text{supp}(f_i) \subset U_i$ for each i. Moreover, if $f \geq 0$, we can choose $f_i \geq 0$ for each i.

Proof. □

Theorem 7.1.4. Let X be a LCHS with a countable basis $\mathscr{B}$, then every open subset of X can be expressed as a union of countable compact sets, so is a F_σ set.

Proof. □

B Radon Measure

Definition 7.1.1. Let $(X, \mathscr{T})$ be a locally compact Hausdorff space and let μ^* be a Borel outer measure induced by a set function μ. Then μ^* is a ***Radon outer measure*** if

1. If K is a compact subset of X, then $\mu^*(K) < \infty$,
2. If $A \in 2^X$, then $\mu^*(A) = \inf\{\mu^*(U) : A \subset U,\ \text{U open}\}$,
3. If $A \in \mathscr{T}$, then $\mu^*(A) = \sup\{\mu^*(K) : A \supset K,\ \text{K compact}\}$.

That is, the Radon outer measure is a regular Borel outer measure on the LCHS that is finite on every compact subset. And we reter to $\mu = \mu^*|_{\Sigma_{\mu^*}}$ as the ***Radon measure*** on X.

Theorem 7.1.5. Let X be a LCHS with a countable basis (A_2) and μ is a Borel measure on X. Then every Borel set is regular with respect to μ, that is the conditons (2) and (3) of definition before hold.

Proof. Since X has a countable basis, every open set of X can be expressed as a union of countable compact sets. In particular, $X = \bigcup_{i=1}^{\infty} K_i$ where K_i is compact. □

C Riesz Representation Theorem

Riesz Representation Theorem. Let X be a LCHS and if I is a positive linear functional on $C_c(X)$, then there exists a unique Radon measure μ on X such that

$$I(f) = \int_X f\,d\mu \text{ for every } f \in C_c(X).$$

7.2 Sigend Measure and Function — Act II

8

Differentiation

8.1 Covering Theorems

Before formally delving into the study of differentiation theory, we will first introduce two covering theorems as a preliminary section (which will be used later). Of course, the content is both highly important and quite fascinating. In fact, this could be considered part of geometric measure theory. However, I have chosen to approach differentiation theory through covering theorems because I believe this method avoids introducing too many unnecessary new definitions (indeed, many textbooks handle it this way).

A Vitali Covering Theorem for Lebesgue Measure

A nondegenerate closed ball means $\overline{B}_r(x) = \{y \in \mathbf{R}^n : |y - x| \le r\}$ for some $x \in \mathbf{R}^n$ and $r > 0$. We will use $\overline{B}$ to denote a nondegenerate closed ball and $\mathscr{B}$ is the family of all nondegenerate closed balls whose diameters are bounded uniformly (there is a number M such that $diam(\overline{B}) \le M$ for all $\overline{B} \in \mathscr{B}$).

Lemma 8.1.1. Let $\mathscr{B}$ be the family of nondegenerate closed balls whose diameters are bounded uniformly. Then there exists a countable subfamliy $\mathscr{V}$ of $\mathscr{B}$ consisting of disjoint balls such that

$$\bigcup_{\overline{B} \in \mathscr{B}} \overline{B} \subset \bigcup_{\overline{B} \in \mathscr{V}} 5\overline{B}$$

where $5\overline{B}_r(x) := \overline{B}_{5r}(x)$.

Proof. □

Vitali Covering Theorem. Let A be a subset of $\mathbf{R}^n$ and let $\mathscr{B}$ be a family of nondegenerate closed balls whose diameters are bounded uniformly such that covering A in the sense: for every $x \in A$, there exists a sequence of balls centered at x with diameters going to 0 from $\mathscr{B}$, that is

$$\inf\left\{diam(\overline{B}) : x \in \overline{B} \in \mathscr{B}\right\} = 0 \text{ for } x \in A.$$

Then for every open set G containing A, there is a family $\mathscr{V}$ of disjoint balls of $\mathscr{B}$ such that

$$\bigcup_{\overline{B} \in \mathscr{V}} \overline{B} \subset G \text{ and } \mu_n^*\left(A \setminus \bigcup_{\overline{B} \in \mathscr{V}} \overline{B}\right) = 0$$

Proof. ***Case 1***: we assume that $\mu_n^*(A) < \infty$. For every open set G containing A, there exists an open set $G_1 \subset G$ containing A such that

$$\mu_n(G_1) \le \mu_n^*(A) + \epsilon_0 \mu_n^*(A),$$

where $1 + 2\epsilon_0 - 1/5^n = \theta \in (0, 1)$. Let

$$\mathscr{B}_1 = \left\{ \overline{B}(x) : x \in A, \overline{B}(x) \in \mathscr{B}, \overline{B}(x) \subset G_1 \right\}.$$

By Vitali covering theorem, there is a subfamliy of pairwise disjoint balls $\mathscr{V}_1 = \left\{ \overline{B}_{1,k} \right\} \subset \mathscr{B}_1$ that satisfies

$$A \subset \bigcup_{\overline{B} \in \mathscr{B}_1} \overline{B} \subset \bigcup_{\overline{B} \in \mathscr{V}_1} 5\overline{B} = \bigcup_k 5\overline{B}_{1,k}.$$

Then we have

$$\begin{aligned}
\mu_n^* \left(A \setminus \bigcup_k \overline{B}_{1,k} \right) &\le \mu_n^* \left(G_1 \setminus \bigcup_k \overline{B}_{1,k} \right) \\
&\le \mu_n(G_1) - \sum_k \mu_n(\overline{B}_{1,k}) \\
&= \mu_n(G_1) - \frac{1}{5^n} \sum_k \mu \left(5\overline{B}_{1,k} \right) \\
&\le (1 + \epsilon_0)\mu_n^*(A) - \frac{1}{5^n} \sum_k \mu_n \left(5\overline{B}_{1,k} \right) \\
&\le (1 + \epsilon_0 - \frac{1}{5^n})\mu_n^*(A) < \theta \mu_n^*(A).
\end{aligned}$$

Next, let $A_1 = A \setminus \bigcup_k \overline{B}_{1,k}$, there exists an open set $G_2 \supset A_1$ such that

$$\mu_n(G_2) \le \mu_n^*(A_1) + \epsilon_0 \mu_n^*(A_1).$$

Similarly, we construct a new family of balls

$$\mathscr{B}_2 = \left\{ \overline{B}(x) : x \in A_1, \overline{B}(x) \in \mathscr{B}, \overline{B}(x) \cap \overline{B}_{1,k} = \emptyset, \overline{B}(x) \subset G_2 \right\}.$$

By the covering condition, for every $x \in A_1$, there exists an arbitrarily small ball containing x, which does not intersect with $\bigcup_k \overline{B}_{1,k}$ as $\bigcup_k \overline{B}_{1,k}$ is closed and $x \notin \bigcup_k \overline{B}_k$. So, $A_1 \subset \bigcup_{\overline{B} \in \mathscr{B}_2} \overline{B}$. Repeating the same argument for A_1, G_2, we obtain that

$$\mu_n^* \left(A_1 \setminus \bigcup_k \overline{B}_{2,k} \right) \le \theta \mu_n^*(A_1) < \theta^2 \mu_n^*(A).$$

Repeating this argument, we will obtain a countable, pairwise disjoint balls from $\mathscr{B}$ such that

$$\mu_n^* \left(A \setminus \bigcup_k \overline{B}_k \right) \le \liminf_{n \to \infty} \theta^n \nu^*(A) = 0.$$

Case 2: consider the unbounded set A, let

$$R_n = \{ n < |x| < n + 1 \}, A_0 = A \cap B_1(0), A_n = A \cap R_n, \ n \ge 1$$

Apply the previous result to every $A_n, n \ge 0$. Note that $\bigcup_{n=0}^{\infty} \{ x : |x| = n \} \cap A$ is a null set. □

Both the preceding lemma and this theorem rely heavily on metric properties in their proofs, specifically the five-fold covering relationship between balls. Additionally, the theorem depends on the scaling of the Lebesgue measure relative to the measures of balls to control the complement sets.

B Besicovitch Covering Theorem for Radon Measure

The Vitali Covering Theorem only applies to a small part of measures similar to the Lebesgue measure. The reader should note that in the proof, this is because the Lebesgue measure can control the relationship between $\mathcal{L}^n(\overline{B})$ and $\mathcal{L}^n(5\overline{B})$. Therefore, the Vitali covering theorem cannot be applied to any Borel measure on $(\mathbf{R}^n, \mathcal{B}(\mathbf{R}^n))$ that does not satisfy this requirement. Therefore, this requirement is too restrictive, we extend the theorem to Radon measures on $(\mathbf{R}^n, \mathcal{B}(\mathbf{R}^n))$.

Recall that a topological space is said to be Hausdorff if it has the property that for every pair of distinct points $x, y \in X$, there are open sets U, V with $x \in U, y \in V$ and $U \cap V = \emptyset$. Most of the rest of the discussion this chapter takes place in locally compact Hausdorff space, a space X is said to be locally compact if for each $x \in X$, there is a neighborhood U_x of x such that the closure $\overline{U_x}$ is compact.

Lemma 8.1.2. Let $\mathscr{B}$ be a family of nondegenerate closed balls whose diameters are uniformly bounded. And let A be the set consisting of the centers of these balls. There exists finite subfamilies $\mathscr{B}_1, \mathscr{B}_2, \ldots, \mathscr{B}_N$ of $\mathscr{B}$ in which each $\mathscr{B}_j$ is composed of countably many disjoint balls such that

$$A \subset \bigcup_{j=1}^{N} \bigcup_{\overline{B} \in \mathscr{B}_j} \overline{B}.$$

*Proof. **Case 1***: we assume that A is bounded. First we choose a ball $\overline{B}_1 = \overline{B}_{r_1}(a_1)$ such that

$$r_1 \geq \frac{3}{4} \sup \left\{ r : \overline{B}_r(x) \in \mathscr{B} \right\}.$$

Now, we choose balls $B_j, j \geq 2$ as follow sense: if we have chosen $\overline{B}_1, \ldots, \overline{B}_{j-1}$, let $A_j = A \setminus \bigcup_{i=1}^{j-1} B_i$, if $A_j = \emptyset$, then we stop choose and let $J = j-1$; if $A_j \neq \emptyset$, choose $\overline{B}_i = \overline{B}_{r_j}(a_j)$ such that $a_j \in A_j$ and

$$r_j \geq \frac{3}{4} \sup \left\{ r : \overline{B}_r(a) \in \mathscr{B}, a \in A_j \right\}.$$

If the process could be forever, then we let $J = \infty$. Now, we obtain a family $\left\{ \overline{B}_{r_i}(a_i) \right\}_{i=1}^{J}$. We claim that they have the following properties

1. If $i < j$, then $r_j \leq \frac{4}{3} r_i$;

2. If $i \neq j$, then $\overline{B}_{\frac{r_i}{3}}(a_i) \cap \overline{B}_{\frac{r_j}{3}}(a_j) = \emptyset$;

3. $\lim_i r_i = 0$;

4. $A \subset \bigcup_{i=1}^{J} \overline{B}_{r_i}(a_i)$.

For (1), for $i < j$, $a_j \in A_i$, thus

$$r_i \geq \frac{3}{4} \sup\left\{r : \overline{B}_r(a) \in \mathscr{B}, a \in A_i\right\} \geq \frac{3}{4} r_j.$$

For (2), let $j > i$, then $a_j \notin \overline{B}_{r_i}(a_i)$, and hence

$$\left|a_i - a_j\right| \geq r_i = \frac{r_i}{3} + \frac{2r_i}{3} \geq \frac{r_i}{3} + \frac{2}{3}\frac{3}{4} r_j > \frac{r_i}{3} + \frac{r_j}{3}.$$

For (3), A is bounded and $\left\{\overline{B}_{\frac{r_i}{3}}(a_i)\right\}$ are disjoint, and hence $r_i \to 0$ as $i \to \infty$. For (4), if $J < \infty$, this is trival. If $J = +\infty$, let $a \in A \setminus \bigcup_{i=1}^{\infty} \overline{B}_i$, then there exists a closed ball $\overline{B}_r(a) \in \mathscr{B}$ and $r_i \geq \frac{3}{4} r > 0$, which contradicting (3). $\square$

Besicovitch Covering Theorem. Let μ be a Radon measure on $(\mathbf{R}^n, \mathcal{B}(\mathbf{R}^n))$. And let $\mathcal{B}$ be a family of nondegenerate closed balls with uniformly bounded diameters that covering A in the sense: for every $x \in A$, there exists a sequence of balls centered at x with diameters going to 0 from $\mathscr{B}$. For every open set G containing A, there exists a countable subfamliy $\mathscr{V}$ such that

$$\bigcup_{\overline{B} \in \mathscr{V}} \overline{B} \subset G \text{ and } \mu\left(A \setminus \bigcup_{\overline{B} \in \mathscr{V}} \overline{B}\right) = 0.$$

Proof. The proof process is almost identical to that of the Vitali theorem. For every open set G containing A, there exists an open set $G_1 \subset G$ containing A such that

$$\mu(G_1) \leq \mu(A) + \epsilon_0 \mu(A),$$

where $1 + 2\epsilon_0 - 1/N_1 = \theta \in (0,1)$. Let

$$\mathscr{B}_1 = \left\{\overline{B}(x) : x \in A, \overline{B}(x) \in \mathscr{B}, \overline{B}(x) \subset G_1\right\}.$$

Then A consisting of the centers of all balls in $\mathscr{B}_1$. By lemma 5.2.2, there exist subfamilies $\mathscr{V}_{1,1}, \mathscr{V}_{1,2}, \ldots, \mathscr{V}_{1,N_1}$ of $\mathscr{B}_1$ in which each $\mathscr{V}_{1,j}$ is composed of countable many disjoint balls such that

$$A \subset \bigcup_{j=1}^{N_1} \bigcup_{\overline{B} \in \mathscr{V}_{1,j}} \overline{B}.$$

Then we have

$$\mu(A) \leq \sum_{j=1}^{N_1} \mu\left(\bigcup_{\overline{B} \in \mathscr{V}_{1,j}} \overline{B}\right).$$

It follows that there exists an integer j between 1 and N_1 such that

$$\mu\left(A \cap \bigcup_{\overline{B}\in\mathscr{V}_{1,j}}\right) \geq \frac{1}{N_1}\mu(A).$$

Then we have

$$\begin{aligned}\mu\left(A \setminus \bigcup_{\overline{B}\in\mathscr{V}_{1,j}} \overline{B}\right) &= \mu(A) - \mu\left(A \cap \bigcup_{\overline{B}\in\mathscr{V}_{1,j}} \overline{B}\right)\\ &\leq \mu(G_1) - \frac{1}{N_1}\mu\,(A)\\ &\leq (1+\epsilon_0)\mu(A) - \frac{1}{N_1}\mu(A)\\ &= (1+\epsilon_0 - \frac{1}{N_1})\mu(A) < \theta\mu(A).\end{aligned}$$

Next, let $A_2 = A \setminus \bigcup_{\overline{B}\in\mathscr{V}_{1,j}} \overline{B}$, then there exists an open set $G_2 \supset A_1$ such that

$$\mu(G_2) \leq \mu(A_2) + \epsilon_0\mu(A_2),$$

where $1 + 2\epsilon_0 - 1/N_2 = \theta$. Similarly, we construct a new family of balls

$$\mathscr{B}_2 = \left\{\overline{B}(x) : x \in A_2, \overline{B}(x) \in \mathscr{B}, \overline{B}(x) \in G_2\right\}.$$

It is clear that A_2 is the set consisting of the centers of all ball from $\mathscr{B}_2$. Repeating the same argument for A_2, G_2, we obtain that

$$\mu\left(A \setminus \bigcup_{\overline{B}\in\mathscr{V}_{2,j_2}} \overline{B}\right) \leq \theta\mu(A_2) < \theta^2\mu(A).$$

Repeating this argument, we will obtain a countable, pairwise disjoint balls from $\mathscr{B}$ such that

$$\mu\left(A \setminus \bigcup_{\overline{B}\in\mathscr{V}\overline{B}}\right) \leq \liminf_{n\to\infty} \theta^n\mu(A) = 0.$$

□

8.2 *Differentiation of Measures*

In Section 1, when introducing the RN theorem, we have already presented the fundamental motivation for learning this section: to transform certain special measurable functions into the form of measures or signed measures. Consequently, we will shift our study from the perspective of differentiating functions to the perspective of differentiating measures. Let us first understand the differentiation of measures, which is a direct application of the covering theorem:

A Differentiation of Measures

Definition 8.2.1. Let μ, ν be Radon measures on $(\mathbf{R}^n, \mathcal{B}(\mathbf{R}^n))$. The ***upper and lower derivatives of*** ν ***respect to*** μ at a point $x \in \mathbf{R}^n$ are defined by

$$\overline{D}_\mu \nu(x) = \begin{cases} \lim_{r\downarrow 0} \sup \left\{ \frac{\nu(\overline{B}_r(x))}{\mu(\overline{B}_r(x))} \right\}, & \mu(\overline{B}_r(x)) > 0, \forall r > 0 \\ \infty, & \mu(\overline{B}_r(x)) = 0, \exists r = 0. \end{cases}$$

and

$$\underline{D}_\mu \nu(x) = \begin{cases} \lim_{r\downarrow 0} \inf \left\{ \frac{\nu(\overline{B}_r(x))}{\mu(\overline{B}_r(x))} \right\}, & \mu(\overline{B}_r(x)) > 0, \forall r > 0 \\ \infty, & \mu(\overline{B}_r(x)) = 0, \exists r = 0. \end{cases}$$

At the point x where the limit exists we define the ***derivative*** of ν by

$$D_\mu \nu(x) := \overline{D}_\mu \nu(x) = \underline{D}_\mu \nu(x).$$

Lemma 8.2.1. Let μ, ν be Radon measures on $(\mathbf{R}^n, \mathcal{B}(\mathbf{R}^n))$. For every set A and we fix $\alpha \in (0, \infty)$,

1. $A \subset \left[\overline{D}_\mu \nu \geq \alpha\right]$ implies that $\nu(A) \geq \alpha\mu(A)$;
2. $A \subset \left[\underline{D}_\mu \nu \leq \alpha\right]$ implies that $\nu(A) \leq \alpha\mu(A)$;

Proof. Assume that A is bounded. For every $\epsilon > 0$, we choose an open set G containing A. Let

$$\mathscr{B} = \left\{ \overline{B}(x) : x \in A, \overline{B}(x) \subset G, \nu(\overline{B}(x)) \leq (\alpha + \epsilon)\mu(\overline{B}(x)) \right\}.$$

Then $\inf \left\{ diam(\overline{B}(x)) : \overline{B}(x) \in \mathscr{B} \right\} = 0$ for every $x \in A$. By theorem 5.3.4, there exists a countable family $\mathscr{V}$ of disjoint balls of $\mathscr{B}$ such that

$$\nu\left(A \setminus \bigcup_{\overline{B} \in \mathscr{V}} \overline{B} \right) = 0.$$

It follows that

$$\begin{aligned} \nu(A) &\leq \nu\left(A \setminus \bigcup_{\overline{B} \in \mathscr{V}} \overline{B} \right) + \nu\left(\bigcup_{\overline{B} \in \mathscr{V}} \overline{B} \right) \\ &= \nu\left(\bigcup_{\overline{B} \in \mathscr{V}} \overline{B} \right) \\ &= \sum_{\overline{B} \in \mathscr{B}} \nu(\overline{B}) \\ &\leq (\alpha + \epsilon) \sum_{\overline{B} \in \mathscr{V}} \mu(\overline{B}) = (\alpha + \epsilon)\mu(G). \end{aligned}$$

This holds for every open set $G \supset A$. Taking infimum over all G, by the regularity property ot μ we get

$$\mu(A) = \inf\{\mu(G) : A \subset G, G \text{ open}\}.$$

Therefore, $\nu(A) \leq (\alpha + \epsilon)\mu(A)$ for every $\epsilon > 0$, as desired. When A is unbounded, apply the proof to $A \cap B(0, n)$ and then let n go to ∞. □

Theorem 8.2.1. Let μ, ν be Radon measures on $(\mathbf{R}^n, \mathcal{B}(\mathbf{R}^n))$. Then

1. $[\overline{D}_\mu \nu = \infty]$ is a ν-null set.
2. $[\underline{D}_\mu \nu < \overline{D}_\mu \nu < \infty]$ is a ν-null set.

Proof. For each $k \geq 1$, $[\overline{D}_\mu \nu = \infty] \subset [\overline{D}_\mu \nu \geq k]$. By lemma 5.4.1,

$$\mu\left([\overline{D}_\mu \nu = \infty]\right) \leq \frac{1}{k}\nu\left([\overline{D}_\mu \nu = \infty]\right).$$

Taking $k \to \infty$, we obtain that $\mu\left([\overline{D}_\mu \nu = \infty]\right) = 0$. Next, for $0 < \alpha < \beta$, let $A(\alpha, \beta) = [\underline{D}_\mu \nu < \alpha < \beta < \overline{D}_\mu \nu < \infty]$. By lemma 5.4.1,

$$\nu(A(\alpha, \beta)) \leq \alpha\mu(A(\alpha, \beta)), \quad \beta\mu(A(\alpha, \beta)) \leq \nu(A(\alpha, \beta)).$$

It follows that $\mu(A(\alpha, \beta)) = 0$. Moreover,

$$[\underline{D}_\mu \nu < \overline{D}_\mu \nu < \infty] = \bigcup_{\alpha, \beta \in \mathbf{Q}} A(\alpha, \beta).$$

We conclude that $\mu\left([\underline{D}_\mu \nu < \overline{D}_\mu \nu < \infty]\right) = 0$. □

This theorem shows that $\overline{D}_\mu \nu = \underline{D}_\mu \nu < \infty$ except on a μ-measurable zero set N where $N = I \cup R$, $I = \left[\overline{D}_\mu \nu(x) < \infty\right]$ and $R = \left[\underline{D}_\mu \nu(x) < \overline{D}_\mu \nu(x) < \infty\right]$. To make the derivative well defined everywhere, we set $D_\mu \nu(x) = \overline{D}_\mu \nu(x)$ when $x \in \mathbf{R}^n \setminus N$ and equals ∞ when $x \in N$.

Proposition 8.2.1. Let μ, ν be Radon measures on $(\mathbf{R}^n, \mathcal{B}(\mathbf{R}^n))$. Then $D_\mu \nu$ is Borel measurable.

Proof. We claim that $x \mapsto \mu(\overline{B}(x, r))$ is upper semicontinuous. Let $x_n \to x$, for every $\epsilon > 0$, $\overline{B}(x_n, r)$ is contained in $\overline{B}(x, r + \epsilon)$ for all large n. It follows that $\limsup_{n\to\infty} \mu(\overline{B}(x_n, r)) \leq \mu(\overline{B}(x, r + \epsilon))$. Now, letting $\epsilon \to 0$, by the continuity of measures,

$$\limsup_{n\to\infty} \mu(\overline{B}(x_n, r)) \leq \mu(\overline{B}(x, r)).$$

Which means that $x \mapsto \mu(\overline{B}(x, r))$ is upper semicontinuous. Similarily, we can show that $x \mapsto \nu(\overline{B}(x, r))$ is upper semicontinuous. For $x \in \mathbf{R}^n \setminus N$, noting that

$\mu(\overline{B}(x,r)) > 0$ for all r, we can fix a sequence $r_k \to 0$ to get

$$D_\mu \nu(x) = \lim_{r\to 0} \frac{\nu(\overline{B}(x,r))}{\mu(\overline{B}(x,r))} = \lim_{r_k\to 0} \frac{\nu(\overline{B}(x,r_k))}{\mu(\overline{B}(x,r_k))} < \infty.$$

Since N is a μ-measurable null set where $D_\mu \nu = \infty$, we conclude that $D_\mu \nu$ is measurable. $\square$

Radon-Nikodym Theorem. Let μ, ν be Radon measures on $(\mathbf{R}^n, \mathcal{B}(\mathbf{R}^n))$. If $\nu \ll \mu$, then $D_\mu \nu = d\nu/d\mu$ (since $D_\mu \nu$ is Borel measurable) that is

$$\nu(A) = \int_A D_\mu \nu d\mu \text{ for every } A \in \mathcal{B}(\mathbf{R}^n).$$

Proof. Assume that A is bounded. We claim that A is μ-measurable implies that A is ν-measurable. Let A be μ-measurable, then there exists a Borel set $B \supset A$ with $\mu(B \setminus A) = 0$. Thus, $\nu(B \setminus A) = 0$ and so $A = B \setminus (B \setminus A)$ is ν-measurable. Let

$$\begin{aligned} Z &= \left[D_\mu \nu = 0\right], \\ I &= \left[D_\mu \nu = \infty\right], \\ R &= \left[\underline{D}_\mu \nu < \overline{D}_\mu \nu < \infty\right]. \end{aligned}$$

Then Z, I, R are μ-measurable and $\mu(I) = \mu(R) = 0$. $\mu(Z)$ may not vanish, but $\nu(Z) = 0$ by lemma 5.4.1. For a μ-measurable set A, let

$$A_m = [t^m < D_\mu \nu \le t^{m+1}], m \in \mathbf{Z}.$$

Then A_m is μ-measurable and so also ν-measurable. Moreover, we have $A \setminus \bigcup_{-\infty}^{\infty} A_m \subset Z \cup I \cup R$. and hence

$$\nu\left(A \setminus \bigcup_{-\infty}^{\infty} A_m\right) = 0.$$

Then we have

$$\begin{aligned} \nu(A) = \sum_{-\infty}^{\infty} \nu(A_m) &\le \sum_{-\infty}^{\infty} t^{m+1}\mu(A_m) \text{ (by lemma 5.4.1)} \\ &= t\sum_{-\infty}^{\infty} t^m \mu(A_m) \le t \sum_{-\infty}^{\infty} \int_{A_m} D_\mu \nu d\mu \\ &\le t \int_A D_\mu \nu d\mu. \end{aligned}$$

Similarly,

$$\begin{aligned}\nu(A) = \sum_{-\infty}^{\infty} \nu(A_m) \geq \sum_{-\infty}^{\infty} t^m \mu(A_m) \text{ (by lemma 5.4.1)} \\ = t^{-1} \sum_{-\infty}^{\infty} t^{m+1} \mu(A_m) \geq t^{-1} \sum_{-\infty}^{\infty} \int_{A_m} D_\mu \nu d\mu \\ \leq t^{-1} \int_A D_\mu \nu d\mu.\end{aligned}$$

Letting $t \to 1$, we obtain $\nu(A) = \int_A D_\mu \nu d\mu$. Finally, when A is unbounded, applying the above result to $A \cap \overline{B}_n(0)$ and let $n \to \infty$. □

►**Corollary 1. Lebesgue-Radon-Nikodym Theorem.** Let μ, ν be Radon measures on $(\mathbf{R}^n, \mathcal{B}(\mathbf{R}^n))$. Then there exists a Borel set D such that $\mu(B^c) = 0$ and $\nu = \nu_{ac} + \nu_s$ where $\nu_{ac} \ll \mu, \nu_s \perp \mu$ and

$$\nu_{ac} = \nu\lfloor_D, \quad \nu_s = \nu\lfloor_{D^c}.$$

Moreover, $D_\mu \nu_s = 0$ μ-a.e. so that $D_\mu \nu = D_\mu \nu_{ac}$ μ-a.e.. and consequently

$$\nu(A) = \int_A D_\mu \nu d\mu + \nu_s(A) \text{ for every } A \in \mathcal{B}(\mathbf{R}^n).$$

Proof. Let $\mathcal{F} = \{A \subset \mathbf{R}^n : \mu(A^c) = 0, A \text{ is Borel}\}$. We claim that there exists an $D \in \mathcal{F}$ such that $\nu(D) \leq \nu(A)$ for all $A \in \mathcal{F}$. Let $A_k \in \mathcal{F}$ satisfy

$$\nu(A_k) \leq \inf_{A \in \mathcal{F}} \nu(A) + 1/k \text{ and } D = \bigcap_{k=1}^{\infty} A_k.$$

It is clear that $D \in \mathcal{F}$ and $\nu(D) \leq \nu(A)$ for all $A \in \mathcal{F}$. Now, define $\nu_{ac} = \nu\lfloor_D$ and $\nu_s = \nu\lfloor_{D^c}$. Since $\nu_s(D) = \nu(D \cap D^c) = \nu(\emptyset) = 0$ and $\mu(D^c) = 0$, $\nu_s \perp \mu$ has been proved. To show that $\nu_{ac} \ll \mu$, note that $E = (E \cap D) \cup (E \setminus D)$. By the definition of ν_{ac}, $\nu_{ac}(E \setminus D) = \nu(D \cap (E \setminus D)) = 0$, it suffices to assume that $E \subset D$ and check that $\mu(E) = 0$ implies that $\nu_{ac}(E) = 0$. Suppose on the contrary there is some $E_1 \subset D$ such that $\mu(E_1) = 0$ but $\nu_{ac}(E_1) > 0$. Find a Borel set E_2 such that $E_1 \subset E_2$ such that $\mu(E_1) = \mu(E_2) = 0$ (regularity) and $\nu_{ac}(E_1) = \nu_{ac}(E_2) > 0$. The set $D \setminus E_2 \in \mathcal{F}$ but

$$\nu_{ac}(D \setminus E_2) = \nu_{ac}(D) - \nu_{ac}(E_2) < \nu_{ac}(D) = \inf_{A \in \mathcal{F}} \nu(A),$$

contradiction holds. Hence $\nu_{ac} \ll \mu$. To show that $D_\mu \nu_s = 0$ μ-a.e., let $E = [D_\mu \nu_s \geq \alpha]$ where $\alpha > 0$ is fixed. By lemma 5.4.1, we have $\nu_s(E \cap D) \geq \alpha \mu(E \cap D)$. Since $\nu_s(D) = 0$, we have $\mu(E \cap D) = 0$. But then $\mu(E) = \mu(E \cap D) + \mu(E \cap D^c) = \mu(\emptyset) = 0$, that is $D_\mu \nu_s = 0$ μ-a.e.. □

B Lebesgue Differentiation Theorem

Definition 8.2.2. A measurable function $f : \mathbf{R}^n \to \mathbf{R}$ on $(\mathbf{R}^n, \mathcal{B}(\mathbf{R}^n), \mu)$ where μ is a Radon measure is called ***locally integrable*** if

$$\int_A |f| \, d\mu < \infty \text{ for every bounded set } A \in \mathcal{L}(\mathbf{R}^n).$$

We denote the space of locally integrable by $L^1_{loc}(\mu)$.

Theorem 8.2.3. Let μ be a Borel measure on $(\mathbf{R}^n, \mathcal{B}(\mathbf{R}^n))$ that is finite on compact sets. If $f \in L^1_{loc}$, then we have

$$\lim_{r \downarrow 0} \frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} f d\mu \xlongequal{\mu-a.e.} f(x) \text{ for } x \in \mathbf{R}^n.$$

Proof. Define $\nu^+(A) = \int_A f^+ d\mu$, $\nu^-(A) = \int_A f^- d\mu$ for $A \in \mathcal{B}(\mathbf{R}^n)$. Then ν^+, ν^- are Borel measure on $(\mathbf{R}^n, \mathcal{B}(\mathbf{R}^n))$ that finite on compact sets and $\nu^+ \ll \mu$, $\nu^- \ll \mu$. By theorem 5.4.2,

$$\begin{aligned} \nu^+(A) &= \int_A D_\mu \nu^+ d\mu = \int_A f^+ d\mu; \\ \nu^-(A) &= \int_A D_\mu \nu^- d\mu = \int_A f^- d\mu. \end{aligned}$$

Thus, $D_\mu \nu^+ = f^+$ μ-a.e. and $D_\mu \nu^- = f^-$ μ-a.e.. Therefore,

$$\begin{aligned} \lim_{r \downarrow 0} \frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} f d\mu &= \lim_{r \downarrow 0} \frac{1}{\mu(\overline{B}_r(x))} \left(\nu^+(\overline{B}_r(x)) - \nu^-(\overline{B}_r(x)) \right) \\ &= D_\mu \nu^+(x) - D_\mu \nu^-(x) \\ &= f^+(x) - f^-(x) = f(x) \ \mu\text{-a.e. point } x. \end{aligned}$$

□

►**Corollary 1.** Let μ be a Borel measure on $(\mathbf{R}^n, \mathcal{B}(\mathbf{R}^n))$ that is finite on compact sets and let $f \in L^1_{loc}$. Then

$$\lim_{r \downarrow 0} \frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} |f - f(x)| \, d\mu = 0 \text{ for } \mu\text{-a.e. } x \in \mathbf{R}^n. \qquad (\bigstar)$$

Proof. Let $\{r_j\}$ be a sequence of rational numbers. Apply theorem 5.4.3 to each $|f - r_j|$ to get

$$\lim_{r \downarrow 0} \frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} |f - r_j| \, d\mu = |f(x) - r_j| \text{ for } x \notin N_j,$$

where N_j is a μ-null set. Letting $N = \bigcup_{j=1}^{\infty} N_j$, then $\mu(N) = 0$ and $x \notin N$. For every $\epsilon > 0$ and $x \notin N$, choose r_j such that $\left|f(x) - r_j\right| < \frac{\epsilon}{2}$, we have

$$\begin{aligned}\frac{1}{\mu\overline{B}_r(x)} \int_{\overline{B}_r(x)} |f - f(x)| \, d\mu &\leq \frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} \left|f - r_j\right| d\mu \\ &\quad + \frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} \left|f(x) - r_j\right| d\mu \\ &= \frac{1}{\mu(\overline{B}_r(x))} \int_{\overline{B}_r(x)} \left|f - r_j\right| d\mu + \left|f(x) - r_j\right|.\end{aligned}$$

Thus,

$$\lim_{r \downarrow 0} \frac{1}{\mu\overline{B}_r(x)} \int_{\overline{B}_r(x)} |f - f(x)| \, d\mu \leq 2\left|f(x) - r_j\right| < \epsilon.$$

$\square$

A point x such that ($\bigstar$) holds is called a ***Lebesgue point*** of f with respect to μ. And the set of all Lebesgue points of f with respect to μ forms the ***Lebesgue set***.

►**Corollary 2.** Let μ be a Borel measure on $(\mathbf{R}^n, \mathcal{B}(\mathbf{R}^n))$ that is finite on compact sets and let $f \in L^1_{loc}$. Then

$$\lim_{\overline{B} \downarrow \{x\}} \frac{1}{\mu(\overline{B})} \int_{\overline{B}} |f - f(x)| \, d\mu = 0 \text{ for } \mu\text{-a.e. } x \in \mathbf{R}^n.$$

where the limit is taken over all closed balls $\overline{B}$ containing x (not necessarily centered at x).

Proof. For every Lebesgue point $x \in \mathbf{R}^n$, choose a sequence of closed balls $\{\overline{B}_j\}$ such that

$$x \in \overline{B}_j \text{ and } \overline{B}_j \subset \overline{B}_{r_j}(x)$$

where $r_j = diam(\overline{B}_j)$ and consequently,

$$\frac{1}{\mu(\overline{B}_j)} \int_{\overline{B}_j} |f - f(x)| \, d\mu \leq \frac{2^n}{\mu(\overline{B}_j)} \int_{\overline{B}_{r_j}(x)} |f - f(x)| \, d\mu$$

The right-hand side goes to zero. $\square$

Now we consider taking average over other sets. Let $\{E_j\}$ be a sequence of μ-measurable sets. We call it ***shrinks regularly*** to x if there is $\overline{B}_{r_j}(x)$ such that

$$\alpha\mu(E_j) \geq \mu(\overline{B}_{r_j}(x)), \;\; E_j \subset \overline{B}_{r_j}(x),$$

for some $\alpha > 0$.

►**Corollary 3.** Let $\{E_j\}$ shrink regularly to x, a Lebesgue point of $f \in L^1_{loc}$. Then

$$\lim_{j\to\infty} \frac{1}{\mu(E_j)} \int_{E_j} f d\mu = f(x).$$

Proof. □

Differentiability of Monotone Functions. If a function $F : \mathbf{R} \to \mathbf{R}$ is monotone increasing, then $F'(x)$ exists for almost every $x \in \mathbf{R}$. In particular, every monotone function on $[a, b]$ is differentiable almost everywhere.

Proof. First we assume F is bounded, increasing and right continuous, and that it vanishes at $-\infty$. Let μ be the finite Borel measure such that $F(x) = \mu((-\infty, x])$. By Lebesgue Decomposition theorem $\mu = \mu_{ac} + \mu_s$ where $\mu_{ac} \ll \mu_1, \mu_s \perp \mu_1$. By the corollary of theorem 5.3.3, there exists a set A_1 with $\mu_1(A_1^c) = 0$ such that

$$D_{\mu_1}\mu_s = 0, D_{\mu_1}\mu_{ac} = D_{\mu_1}\mu \text{ on } A_1.$$

and

$$\frac{\mu((x, x+\delta])}{\mu_1((x, x+\delta])} = \frac{1}{\mu_1((x, x+\delta])} \int_{(x,x+\delta]} D_{\mu_1}\mu_{ac} d\mu_1 + \frac{\mu_s((x, x+\delta])}{\mu_1((x, x+\delta])}.$$

By corollary 3 of theorem 5.3.3,

$$\lim_{\delta\to 0} \frac{\mu_{ac}((x, x+\delta])}{\delta} = \frac{1}{\delta} \lim_{\delta\to 0} \int_{(x,x+\delta]} D_{\mu_1}\mu_{ac} d\mu_1 = D_{\mu_1}\mu_{ac}(x),$$

for x in a set A_2 with $\mu_1(A_2^c) = 0$. Moreover,

$$\limsup_{\delta\to 0} \frac{\mu_s((x, x+\delta])}{\delta} \le 2 \limsup_{\delta\to 0} \frac{\mu_s(\overline{B}_\delta(x))}{\mu_1(\overline{B}_\delta(x))} = 0,$$

for $x \in A_1$. Therefore, for $x \in A_1 \cap A_2$,

$$\lim_{\delta\to 0} \frac{\mu((x, x+\delta])}{\mu_1((x, x+\delta])} = D_{\mu_1}\mu_{ac}(x) = D_{\mu_1}\mu(x).$$

Since $\mu((x, x+\delta]) = F(x+\delta) - F(x)$, we conclude that

$$\frac{d^+F}{dx}(x) = D_{\mu_1}\mu(x) \text{ for } x \in A_1 \cap A_2.$$

Similarily, we can show that

$$\frac{d^-F}{dx}(x) = D_{\mu_1}\mu(x) \text{ for } x \in B_1 \cap B_2.$$

Now, remove the requirement that F be right continuous. Define $G : \mathbf{R} \to \mathbf{R}$ by $G(x) = F(x+)$. Then G is right continuous and so, G is differentiable almost everywhere. Note that F and G are continuous at the same points and they agree at each point at which they are continuous; moreover, if $F(x_0) = G(x_0)$, then

$$\frac{G(x-) - G(x_0)}{x - x_0} \le \frac{F(x) - F(x_0)}{x - x_0} \le \frac{G(x) - G(x_0)}{x - x_0}.$$

Hence if G is differentiable at x_0, then F is differentiable at x_0, and $F'(x_0) = G'(x_0)$. The almost everywhere differentiability of F follows. □

►**Corollary 1.** If $f : [a,b] \to \mathbf{R}$ is monotone increasing on $[a,b]$, then $f' \in L^1[a,b]$, $f' \geq 0$ a.e. and

$$0 \leq \int_{[a,b]} f' d\nu \leq f(b) - f(a).$$

Proof. □

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9

Product Measure

If f is a continuous function in a rectangle $[a,b]\times[c,d]$, then the integral can be computed by two one-dimensional integrals:

$$\iint_{[a,b]\times[c,d]} f(x,y)\,dx\,dy = \int_a^b \left(\int_c^d f(x,y)\,dy\right) dx.$$

This has been known since the time of Cauchy. Here one says that the integral in the plane has been computed by "repeated integration". Certainly, this is the most familiar method of computing integrals in higher dimensions. Indeed, apart from numerical methods, we have almost no other way to obtain the value of such integrals.

To beyond continuous functions requires some caution. For example, the function $f(x,y) = (x^2-y^2)/(x^2+y^2)^2$ in the unit square $[0,1]\times[0,1]$ has just the one discontinuity at the origin. But

$$\int_0^1 \left(\int_0^1 f(x,y)\,dx\right) dy = -\frac{\pi}{4}$$
$$\int_0^1 \left(\int_0^1 f(x,y)\,dy\right) dx = +\frac{\pi}{4}.$$

Clearly, there are some technicalities need to addressed in any such study. The idea was carried out first by Lebesgue and then by two Italian mathematicians, Guido Fubini and Leonida Tonelli. Most of the work was done by Lebesgue. Even so, the name of Fubini is now most firmly attached to the theory in most accounts. The ideas underlying the formula above for the repeated integral are simple. The measure in the plane of the rectangle $[a,b]\times[c,d]$ is

$$\lambda_2([a,b]\times[c,d]) = (b-a)(d-c).$$

Indeed, it is this trivial fact that allows the two-dimensional Riemann integral for continuous functions to be reduced to one-dimensional Riemann integrals. This fact extends general rectangles by the formula

$$\lambda_2(A\times B) = \lambda_1(A)\lambda_1(B),$$

again using λ_2 as two-dimensional Lebesgue measure and λ_1 as one-dimensional Lebesgue measure. We can place these ideas in general measure setting.

9.1 Product Measure

A Product Measurable Space

Definition 9.1.1. Let μ^* be an outer measure on a set X and ν^* be an outer measure onn a set Y. And μ, ν are the Caratheodory measures induced by μ^*, ν^* respectively. Then we define the ***product outer measure*** $\mu^* \times \nu^*$ on each subset $S \subset X \times Y$ as

$$(\mu^* \times \nu^*)(S) = \inf \left\{ \sum_{n=1}^{\infty} \mu(A_n)\nu(B_n) : S \subset \bigcup_{n=1}^{\infty} A_n \times B_n \right\},$$

where A_n are μ^*-measurable and B_n are ν^*-measurable. One should prove that $\mu^* \times \nu^*$ definitely an outer measure. And it called the ***product outer measure*** with respect to μ^* and ν^*.

Definition 9.1.2. Let $(X_1, \Sigma_1), (X_2, \Sigma_2), \ldots, (X_n, \Sigma_n)$ be n measurable spaces. The family of all ***measurable rectangles***

$$\mathscr{S} = \left\{ \mathop{\times}_{i=1}^{n} A_i : A_i \in \Sigma_i, i = 1, 2, \ldots, n \right\}.$$

is a semi algebra (one should prove this). And the σ-algebra $\sigma(\mathscr{S})$ is called the ***product σ-algebra***, denoted with $\times_{i=1}^n \Sigma_i$, that is $\sigma(\mathscr{S}) = \times_{i=1}^n \Sigma_i$. And we refer to $\left(\times_{i=1}^n X_i, \times_{i=1}^n \Sigma_i\right)$ as the ***product measurable space*** of $(X_1, \Sigma_1), \ldots, (X_n, \Sigma_n)$.

Definition 9.1.3. Let $\{(X_\gamma, \Sigma_\gamma), \gamma \in \Gamma\}$ be a family of measurable spaces. And let $\pi_i : \times_{\gamma\in\Gamma} X_\gamma \to X_\gamma$ be the coordinate mappings (the projective mapping is formed by $\pi_I^J : \times_J X_\gamma \to \times_I X_\gamma$, where $\Gamma \supset J \supset I \neq \emptyset$). We refer to

$$\mathop{\times}_{\gamma\in\Gamma} \Sigma_\gamma = \sigma\left(\bigcup_{\gamma\in\Gamma} \pi_\gamma^{-1}(\Sigma_\gamma) \right)$$

as the product σ-algebra of $\Sigma_\gamma, \gamma \in \Gamma$. And we refer to $\left(\times_{\gamma\in\Gamma} X_\gamma, \times_{\gamma\in\Gamma} \Sigma_\gamma\right)$ as the product measurable space of $(X_\gamma, \Sigma_\gamma)$.

Lemma 9.1.1. Let $I \neq \emptyset$ and $I \subset \Gamma$. Then $\pi_I = \pi_I^\Gamma$ is a $\left(\times_\Gamma \Sigma_\gamma, \times_I \Sigma_\gamma\right)$-measurable mapping.

Proof. It is clear that $\pi_\gamma = \pi_\gamma^I \circ \pi_I^\Gamma$. Note that π_γ is $(\times_\Gamma \Sigma_\gamma, \Sigma_\gamma)$-measurable, and so $\pi_\gamma^I \circ \pi_I^\Gamma$ is $(\times_\Gamma \Sigma_\gamma, \Sigma_\gamma)$-measurable. Then $\pi_I^{-1}(\pi_\gamma^{I^{-1}}(\Sigma_\gamma)) \subset \times_\Gamma \Sigma_\gamma$ and hence $\pi_I^{-1}\left(\bigcup_{\gamma\in I} \pi_\gamma^{I^{-1}}(\Sigma_\gamma)\right) \subset \times_\Gamma \Sigma_\gamma$. Since $\sigma\left(\bigcup_{\gamma\in I} \pi_\gamma^{I^{-1}}(\Sigma_\gamma)\right) = \times_I \Sigma_\gamma$, π_I is measurable. $\square$

Theorem 9.1.1. Let $\mathcal{P}_0$ be the family of all nonempty finite subsets of Γ.

1. the family of all measurable rectangle sets:
$$\mathscr{S} = \bigcup_{I\in\mathcal{P}_0} \left\{\pi_I^{-1}\left(\bigtimes_{\gamma\in I} A_\gamma : A_\gamma \in \Sigma_\gamma, \gamma \in I\right)\right\},$$
is a semi-algebra on $\times_{\gamma\in\Gamma} X_\gamma$ and $\sigma(\mathscr{S}) = \times_{\gamma\in\Gamma} \Sigma_\gamma$.

2. the family of all measurable cylinder sets:
$$\mathscr{A} = \bigcup_{I\in\mathcal{P}_0} \left\{\pi_I^{-1}\left(\bigtimes_{\gamma\in I} \Sigma_\gamma\right)\right\},$$
is an algebra on $\times_{\gamma\in\Gamma} X_\gamma$ and $\sigma(\mathscr{A}) = \times_{\gamma\in\Gamma} \Sigma_\gamma$.

Proof. For (1), we first show that $\mathscr{S}$ is a semi-algebra.

- For every $I \in \mathcal{P}_0$, let $A_\gamma = \emptyset$ for all $\gamma \in I$, then $\pi_I^{-1}(\times_I A_\gamma) = \pi_I^{-1}(\emptyset) = \emptyset$, that means $\emptyset \in \mathscr{S}$;
- for every $I \in \mathcal{P}_0$, let $A_\gamma = X_\gamma$ for all $\gamma \in I$, then $\pi_I^{-1}(\times_I A_\gamma) = \pi_I^{-1}(\times_I X_\gamma) = \times_\Gamma X_\gamma$, that means $\times_\Gamma X_\gamma \in \mathscr{S}$;
- let $I_1, I_2 \in \mathcal{P}$ and let $\pi_{I_1}^{-1}(\times_{I_1} A_\gamma), \pi_{I_2}^{-1}(\times_{I_2} B_\gamma) \in \mathscr{S}$, where $\times_{I_1} A_\gamma \in \times_{I_1}\Sigma_\gamma$ and $\times_{I_2} B_\gamma \in \times_{I_2}\Sigma_\gamma$. Let $I = I_1 \cup I_2 \in \mathcal{P}$, and then we suppose that $A_\gamma = X_\gamma, \gamma \in I \setminus I_1$ and $B_\gamma = X_\gamma, \gamma \in I \setminus I_2$, then
$$\begin{aligned}\pi_{I_1}^{-1}\left(\bigtimes_{\gamma\in I_1} A_\gamma\right) \cap \pi_{I_2}^{-1}\left(\bigtimes_{\gamma\in I_2} B_\gamma\right) &= \pi_I^{-1}\left(\bigtimes_{\gamma\in I} A_\gamma\right) \cap \pi_I^{-1}\left(\bigtimes_{\gamma\in I} B_\gamma\right)\\ &= \pi_I^{-1}\left(\bigtimes_{\gamma\in I} A_\gamma \cap \bigtimes_{\gamma\in I} B_\gamma\right)\\ &= \pi_I^{-1}\left(\bigtimes_{\gamma\in I} (A_\gamma \cap B_\gamma)\right),\end{aligned}$$
that means $\mathscr{S}$ is closed in intersections.
- let $I_1, I_2 \in \mathcal{P}_0$ and let $\pi_{I_1}^{-1}(\times_{I_1} A_\gamma), \pi_{I_2}^{-1}(\times_{I_2} B_\gamma) \in \mathscr{S}$ and $\pi_{I_1}^{-1}(\times_{I_1} A_\gamma) \subset \pi_{I_2}^{-1}(\times_{I_2} B_\gamma) \in \mathscr{S}$. Let $I = I_1 \cup I_2 \in \mathcal{P}_0$, and then suppose that $A_\gamma = X_\gamma, \gamma \in I \setminus I_1$ and $B_\gamma = X_\gamma, \gamma \in I \setminus I_2$, then
$$\begin{aligned}\pi_{I_2}^{-1}\left(\bigtimes_{\gamma\in I_2} B_\gamma\right) \setminus \pi_{I_1}^{-1}\left(\bigtimes_{\gamma\in I_1} A_\gamma\right) &= \pi_I^{-1}\left(\bigtimes_{\gamma\in I} B_\gamma\right) \setminus \pi_I^{-1}\left(\bigtimes_{\gamma\in I} A_\gamma\right)\\ &= \pi_I^{-1}\left(\bigtimes_{\gamma\in I} B_\gamma \setminus \bigtimes_{\gamma\in I} A_\gamma\right).\end{aligned}$$
Since $\times_I A_\gamma$ and $\times_I B_\gamma \in \times_I\Sigma_\gamma$ and $\times_I\Sigma_\gamma$ is a σ-algebra, $\times_I B_\gamma \setminus \times_I A_\gamma$ can be written as a finite union of disjoint elements formed $\times_I C_\gamma \in \times_I\Sigma_\gamma$.

For the last statement of (1), for every $\pi_I^{-1}(\times_I A_\gamma) \in \mathscr{S}$, we have $\pi_I^{-1}(\times_I A_\gamma) = \cap_{\gamma\in I}\pi_\gamma^{-1}(A_\gamma) \in \times_\Gamma\Sigma_\gamma$, that means $\sigma(\mathscr{S}) \subset \times_\Gamma\Sigma_\gamma$. To see the converse, for every $A_\gamma \in \Sigma_\gamma$, and $\pi_\gamma^{-1}(A_\gamma) \in \mathscr{S}$, that is $\pi_\gamma^{-1}(\Sigma_\gamma) \subset \mathscr{S}$. It follows that $\bigcup_{\gamma\in\Gamma} \pi_\gamma^{-1}(\Sigma_\gamma) \subset \sigma(\mathscr{S})$. So $\times_\Gamma\Sigma_\gamma \subset \sigma(\mathscr{S})$. $\square$

Lemma 9.1.2. Let $X_1, X_2, \ldots, X_n$ be topological spaces. Then $\times_{i=1}^n \mathcal{B}(X_i) \subset \mathcal{B}(\times_{i=1}^n X_i)$. In particular, if $X_1, X_2, \ldots, X_n$ are A_2 spaces, then $\times_{i=1}^n \mathcal{B}(X_i) = \mathcal{B}(\times_{i=1}^n X_i)$. Moreover, These two conclusions can be generalized to countably many topological spaces $X_1, X_2, \ldots$.

Proof. We only prove the countable case; the finite case can be proved similarly. It is clear that the coordinate mappings $\pi_n : \times_{n=1}^\infty X_n \to X_n$ is continuous, and so is measurable. Therefore, if $A_n \in \mathcal{B}(X_n)$, then

$$\pi_n^{-1}(A_n) \in \mathcal{B}\left(\mathop{\times}_{n=1}^{\infty} X_n\right).$$

It follows that $\pi_1^{-1}(A_1) \cap \pi_2^{-1}(A_2) \cap \cdots \in \mathcal{B}(\times_{n=1}^\infty X_n)$ since $\mathcal{B}(\times_{n=1}^\infty X_n)$ is a σ-algebra. So, $\times_{n=1}^\infty \mathcal{B}(X_n) \subset \mathcal{B}(\times_{n=1}^\infty X_n)$. Now, assume that $X_1, X_2, \ldots$ are A_2 spaces and choose the countable bases $\mathscr{B}_1, \mathscr{B}_2, \ldots$. We claim that

$$\begin{aligned}\mathscr{B} = &\{B_1 \times X_2 \times X_3 \times \cdots : B_1 \in \mathscr{B}_1\} \cup \\ &\{B_1 \times B_2 \times X_3 \times \cdots : B_1 \in \mathscr{B}_1, B_2 \in \mathscr{B}_2\} \cup \ldots.\end{aligned}$$

is a countable base of $\times_{n=1}^\infty X_n$ (left as an exercise). Therefore, every open set in $\times_{n=1}^\infty X_n$ can be written as a union from $\mathscr{B}$. Moreover, since every set in $\mathscr{B}$ comes from $\times_{n=1}^\infty \mathcal{B}(X_n)$, every open set in $\times_{n=1}^\infty X_n$ belongs to $\times_{n=1}^\infty \mathcal{B}(X_n)$, and hence $\mathcal{B}(\times_{n=1}^\infty X_n) \subset \times_{n=1}^\infty \mathcal{B}(X_n)$. □

►**Corollary 1.** $\times_{i=1}^n \mathcal{B}(\mathbf{R}) = \mathcal{B}(\times_{i=1}^n \mathbf{R}) = \mathcal{B}(\mathbf{R}^n)$.

B Product Measure

Now, suppose that $(X_1, \Sigma_1, \mu_1), (X_2, \Sigma_2, \mu_2), \ldots, (X_n, \Sigma_n, \mu_n)$ are σ-finite measure spaces, we will formally define the "product" measure on the product measurable space of them. Recalling that in n-dimensional Eucliden space, the volume of a rectangle equals the product of the lengths of its edges, we can analogously define the "measure" of a measurable rectangle $\times_{i=1}^n A_i \in \mathscr{S}$ as follows:

$$\mu\left(\mathop{\times}_{i=1}^{n} A_i\right) = \prod_{i=1}^{n} \mu_i(A_i).$$

Then we expect this formula holds for all $\times_{i=1}^n \Sigma_i$-measurable set. It is natural to consider directly applying Carathéodory-Hahn theorem to define the product measure. However, we first need to vertify that the set function μ defined on the semi-algebra $\mathscr{S}$ is at least a finitely additive measure (charge). This is not straightforward, indeed, proving it is rather complex. Therefore, we adopt an alternative approach for constructing the product measure: the method of sections.

Definition 9.1.4. Let $(X,\mathscr{A})$, $(Y,\mathscr{B})$ be two σ-finite measure space. And let $E \subset X \times Y$. For every $x \in X$, $y \in Y$, we define the ***sections*** E_x and E^y by

$$E_x = \{y \in Y : (x,y) \in E\} \text{ and } E^y = \{x \in X : (x,y) \in E\}.$$

If f is a function on $X \times Y$, then the ***sections*** of f is defined by

$$f_x(\cdot) = f(x,\cdot) \text{ and } f^y(\cdot) = f(\cdot,y).$$

Lemma 9.1.3. Let $(X,\mathscr{A})$ and $(Y,\mathscr{B})$ be measurable spaces.

1. If $E \subset X \times Y$ that belongs to $\mathscr{A} \times \mathscr{B}$, then each section E_x belongs to $\mathscr{B}$ and each section E^y belongs to $\mathscr{A}$.
2. If f is an extended real valued $\mathscr{A} \times \mathscr{B}$-measurable function on $X \times Y$, then each section f_x is $\mathscr{B}$-measurable and each section f^y is $\mathscr{A}$-measurable.

Proof. For (1), let $x \in X$ and let $\mathscr{F}$ be the collection of all subsets E of $X \times Y$ such that $E_x \in \mathscr{B}$, that is $\mathscr{F} = \{E \subset X \times Y : E_x \in \mathscr{B}\}$. It is clear that the family of measurable rectangles $\mathscr{S}$ is contained the $\mathscr{F}$. In particular, $X \times Y \in \mathscr{F}$. Furthermore, one should check that $(E^c)_x = (E_x)^c$ and $(\bigcup_{n=1}^\infty E_n)_x = \bigcup_{n=1}^\infty((E_n)_x)$, which implies that $\mathscr{F}$ is closed under complementation and uner the formation of countable unions, so that $\mathscr{F}$ is a σ-algebra on $X \times Y$. It follows that $\sigma(\mathscr{S}) = \mathscr{A} \times \mathscr{B} \subset \mathscr{F}$ and hence that $E_x \in \mathscr{B}$ whenever $E \in \mathscr{A} \times \mathscr{B}$. Similarily, $E^y \in \mathscr{A}$ whenever $E \in \mathscr{A} \times \mathscr{B}$.

For (2), let $B \in \mathcal{B}(\mathbf{R})$, then $f^{-1}(B) = \{(x,y) \in X \times Y : f(x,y) \in B\} \in \mathscr{A} \times \mathscr{B}$ by the assumption that f is measurable. For every $x \in X$,

$$\begin{aligned} f_x^{-1}(B) &= \{y \in Y : f_x(y) \in B\} = \{y \in Y : f(x,y) \in B\} \\ &= \left\{y \in Y : (x,y) \in f^{-1}(B)\right\} = (f^{-1}(B))_x. \end{aligned}$$

Then by part (1), we conclude that $f_x^{-1}(B) \in \mathscr{B}$ that means f_x is $\mathscr{B}$-measurable. Similarily, f^y is $\mathscr{A}$-measurable. □

Lemma 9.1.4. Let $(X,\mathscr{A},\mu)$ and $(Y,\mathscr{B},\nu)$ be σ-finite measure spaces. If $E \in \mathscr{A} \times \mathscr{B}$, then the function $x \mapsto \nu(E_x)$ is $\mathscr{A}$-measurable and the function $y \mapsto \mu(E^y)$ is $\mathscr{B}$-measurable.

Proof. First, we assume that ν is finite. Let $\mathscr{F}$ be the family of sets E in $\mathscr{A} \times \mathscr{B}$ such that the function $x \mapsto \nu(E_x)$ is $\mathscr{A}$-measurable, that is

$$\mathscr{F} = \{E \in \mathscr{A} \times \mathscr{B} : x \mapsto \nu(E_x) \text{ is } \mathscr{A}\text{-measurable}\}.$$

If $A \in \mathscr{A}$ and $B \in \mathscr{B}$, then $\nu((A \times B)_x) = \nu(B)\chi_A(x)$ and hence the family of measurable rectangles $\mathscr{S} \subset \mathscr{F}$. In particular, $X \times Y \in \mathscr{F}$. Now, we claim that $\mathscr{F}$ is a λ system: if $E, F \in \mathscr{F}$ with $E \subset F$, then $\nu((F \setminus E)_x) = \nu(F_x) - \nu(E_x)$; if $\{E_n\}$ is an increasing sequence of sets in $\mathscr{F}$, then $\nu((\bigcup_{n=1}^\infty E_n)_x) = \lim_n \nu((E_n)_x)$. Since $\mathscr{S}$ is a π system, $\sigma(\mathscr{S}) = \mathscr{A} \times \mathscr{B} \subset \mathscr{F}$ by Dynkin's

Lemma. Now, we assume that ν is σ-finite and let $Y = \bigcup_{n=1}^{\infty} Y_n$ where $Y_n \cap Y_m = \emptyset$ and $Y_n \in \mathscr{B}$ with $\nu(Y_n) < \infty$ for each n. Let

$$\nu_n(B) = \nu(B \cap Y_n) \text{ for every } B \in \mathscr{B}.$$

Then ν_n is finite and $\nu = \sum_{n=1}^{\infty} \nu_n$. If follows that

$$\nu(E_x) = \sum_{n=1}^{\infty} \nu_n(E_x) \text{ for every } E \in \mathscr{A} \times \mathscr{B}.$$

By what we have proved that for each n, the function $x \mapsto \nu_n(E_x)$ is $\mathscr{A}$-measurable, $x \mapsto \nu(E_x)$ is a $\mathscr{A}$-measurable. Similarly, $y \mapsto \mu(E^y)$ is $\mathscr{B}$-measurable. □

Theorem 9.1.2. Let $(X, \mathscr{A}, \mu)$ and $(X, \mathscr{B}, \nu)$ be two σ-finite measure spaces. Then there is a unique measure on $\mathscr{A} \times \mathscr{B}$, denoted with $\mu \times \nu$, such that

$$(\mu \times \nu)(A \times B) = \mu(A)\nu(B) \text{ for } A \in \mathscr{A}, B \in \mathscr{B}.$$

Furthermore, for every $E \in \mathscr{A} \times \mathscr{B}$, the measure $(\mu \times \nu)(E)$ is given by

$$(\mu \times \nu)(E) = \int_X \nu(E_x) d\mu(x) = \int_Y \mu(E^y) d\nu(y).$$

Proof. Since we have shown the measurability of $x \mapsto \nu(E_x)$ and $y \mapsto \mu(E^y)$, we can define the functions μ_1 and μ_2 on $\mathscr{A} \times \mathscr{B}$ by

$$\mu_1(E) = \int_X \nu(E_x) d\mu(x), E \in \mathscr{A} \times \mathscr{B},$$
$$\mu_2(E) = \int_Y \mu(E^y) d\nu(y), E \in \mathscr{A} \times \mathscr{B}.$$

We claim that μ_1, μ_2 are measures on $\mathscr{A} \times \mathscr{B}$: it is clear that $\mu_1(\emptyset) = \mu_2(\emptyset) = 0$; let $\{E_n\}$ be a sequence of disjoint sets in $\mathscr{A} \times \mathscr{B}$ and $E = \bigcup_n E_n$, then $\{(E_n)_x\}$ is a sequence of disjoint sets in $\mathscr{B}$ such that $E_x = \bigcup_n ((E_n)_x)$ and hence such that $\nu(E_x) = \sum_{n=1}^{\infty} \nu((E_n)_x)$, by the corollary of Levi's montone theorem, we obtain

$$\mu_1(E) = \int_X \nu(E_x) d\mu(x) = \sum_{n=1}^{\infty} \int_X \nu((E_n)_x) d\mu(x) = \sum_{n=1}^{\infty} \mu_1(E_n),$$

and so μ_1 is σ-additive. Similarily, μ_2 is σ-additive. Since $\mu((A \times B)^y) = \mu(A)\chi_B(y)$ and $\nu((A \times B)_x) = \nu(B)\chi_A(x)$,

$$\mu_1(A \times B) = \int_X \nu(B)\chi_A(x) d\mu(x) = \nu(B) \int_A d\mu(x) = \nu(B)\mu(A),$$
$$\mu_2(A \times B) = \int_Y \mu(A)\chi_B(y) d\nu(y) = \mu(A) \int_B d\nu(y) = \mu(A)\nu(B).$$

Now, μ_1 is a measure on the semi-algebra $\mathscr{S}$ and μ_1 is σ-finite, applying the Caratheodory-Hahn Theorem, there is a unique measure on $\mathscr{A} \times \mathscr{B}$ that extends μ_1, denoted with $\mu \times \nu$, that is $\mu_1 = \mu \times \nu = \mu_2$. □

Now, we call the measure $\mu \times \nu$ the ***product measure*** of μ and ν. And the measure space $(X \times Y, \mathscr{A} \times \mathscr{B}, \mu \times \nu)$ is called the ***product measure space*** of $(X, \mathscr{A}, \mu)$ and $(Y, \mathscr{B}, \nu)$. Moreover, it is clear that

$$\begin{aligned}(\mu \times \nu)(E) &= \int_X \left(\int_Y \chi_E(x, y) d\nu(y) \right) d\mu(x) \\ &= \int_Y \left(\int_X \chi_E(x, y) d\mu(x) \right) d\nu(y).\end{aligned}$$

Example 9.1.1. Let λ_1 be the Lebesgue measure on the Borel subsets $\mathcal{B}(\mathbf{R})$ of $\mathbf{R}$, and let λ_2 be the Lebesgue measure on the Borel subsets $\mathcal{B}(\mathbf{R}^2) = \mathcal{B}(\mathbf{R}) \times \mathcal{B}(\mathbf{R})$. There exists a unique measure $(\lambda_1 \times \lambda_1)$ on $\mathcal{B}(\mathbf{R}^2)$ such that

$$(\lambda_1 \times \lambda_1)((a, b] \times (c, d]) = (b - a)(d - c).$$

Where $\mathscr{S} = \{(a, b] \times (c, d]\}$ is a semiring (so is a π system) and $\mathcal{B}(\mathbf{R}^2) = \sigma(\mathscr{S})$. Thus, $\lambda_2 = \lambda_1 \times \lambda_1$ by theorem 1.2.4.

►**Corollary 1.** Let $(X_i, \Sigma_i, \mu_i), i = 1, 2, \ldots, n$ be σ-finite measure spaces. Then there is a unique measure on $\times_{i=1}^n \Sigma_i$, denoted with $\times_{i=1}^n \mu_i$, such that

$$\left(\mathop{\times}_{i=1}^{n} \mu_i \right) \left(\mathop{\times}_{i=1}^{n} A_i \right) = \prod_{i=1}^{n} \mu_i(A_i) \text{ for } A_i \in \Sigma_i.$$

Furthermore, for every $E \in \times_{i=1}^n \Sigma_i$, the measure $\left(\times_{i=1}^n \mu_i \right)(E)$ is given by

$$\left(\mathop{\times}_{i=1}^{n} \mu_i \right)(E) = \int_{X_{k_1}} \left(\cdots \left(\int_{X_{k_i}} \chi_E(x_1, x_2, \ldots, x_n) d\mu_{k_i}(x_{k_i}) \right) \cdots \right) d\mu_{k_n}(x_{k_n}).$$

9.2 *Fubini's Theorem*

A *Tonelli's Theorem and Fubini's Theorem*

Tonelli's theorem applies to non-negative measurable functions, while Fubini's theorem applies to general measurable and integrable functions -—- just as we followed a similar progression when first learning integration: first consider the non-negative case, then the general case. Interestingly, Tonelli's theorem originated in 1909, whereas Fubini's theorem dates back to 1907. In fact, the original version of Fubini's theorem was stated for measurable sets (which can be translated into the case of characteristic functions) rather than for general integrable functions.

Tonelli's Theorem. Let $(X, \mathscr{A}, \mu)$ and $(Y, \mathscr{B}, \nu)$ be σ-finite measure spaces, and let $f : X \times Y \to [0, \infty]$ be a $(\mathscr{A} \times \mathscr{B})$-measurable function. Then

1. $x \mapsto \displaystyle\int_Y f_x(y)d\nu(y)$ is $\mathscr{A}$-measurable, $y \mapsto \int_X f^y d\mu(x)$ is $\mathscr{B}$-measurable.

2. $$\begin{aligned}\int_{X\times Y} fd(\mu\times\nu) &= \int_X\left(\int_Y f_x d\nu(y)\right)d\mu(x) = \int_Y\left(\int_X f^y d\mu(x)\right)d\nu(y)\\ &= \int_X\left(\int_Y f(x,y)d\nu(y)\right)d\mu(x)\\ &= \int_Y\left(\int_X f(x,y)d\mu(x)\right)d\nu(y).\end{aligned}$$

Note that f_x and f^y are nonnegative and measurable, thus the expression $\int_Y f_x d\nu$ is defined for each x in X and the expression $\int_X f^y d\mu$ is defined for each y in Y.

Proof. Let $f = \chi_E$ where $E \in \mathscr{A}\times\mathscr{B}$. Then $f_x = \chi_{E_x}$ and $f^y = \chi_{E^y}$. And so the relations $\int_Y f_x d\nu = \nu(E_x)$ and $\int_X f^y d\mu = \mu(E^y)$ hold for each x and y. By lemma 6.1.4, (1) holds if f is a characteristic function. By theorem 6.1.2, (2) holds if f is a characteristic function. Therefore, (1) and (2) hold if f is a simple function. By the Levi's Theorem, (1) and (2) hold for arbitrary nonnegative $\mathscr{A}\times\mathscr{B}$-measurable function. □

►**Corollary 1.** Let $(X,\mathscr{A},\mu)$ and $(Y,\mathscr{B},\nu)$ be σ-finite measure spaces, and let $E \in \mathscr{A}\times\mathscr{B}$ be a null set with respect to $\mu\times\nu$. Then E_x is a ν-null set for $x \in X$ μ-a.e; and E^y is a μ-null set for $y \in Y$ ν-a.e..

Proof. Applying the Tonelli theorem to χ_E, we have

$$\begin{aligned}0 &= \int_X\left(\int_Y \chi_E(x,y)d\nu(y)\right)d\mu(x) = \int_Y\left(\int_X \chi(x,y)d\mu(x)\right)d\nu(y)\\ &= \int_X \nu(E_x)d\mu(x) = \int_Y \mu(E^y)d\nu(y).\end{aligned}$$

Thus, E_x is a ν-null for $x \in X$ μ-a.e. and E^y is a μ-null set for $y \in Y$ ν-a.e.. □

►**Corollary 2. (Tonelli's Theorem for Complete Measure).** Let $(X,\mathscr{A},\mu)$ and $\underline{(Y,\mathscr{B},\nu)}$ be complete σ-finite measure spaces, and let $f : X\times Y \to [0,\infty]$ be a $\overline{\mathscr{A}\times\mathscr{B}}$-measurable function. Then

1. For $x \in X$ μ-a.e., the section f_x is $\mathscr{B}$-measurable and the function $x \mapsto \int_Y f_x(y)d\nu(y)$ is $\mathscr{A}$-measurable.

2. For $y \in Y$ ν-a.e., the section f^y is $\mathscr{A}$-measurable and the function $y \mapsto \int_X f^y(x)d\mu(x)$ is $\mathscr{B}$-measurable.

3.
$$\int_{X\times Y} f d(\overline{\mu\times\nu}) = \int_X\left(\int_Y f_x d\nu(y)\right) d\mu(x) = \int_Y\left(\int_X f^y d\mu(x)\right) d\nu(y)$$
$$= \int_X\left(\int_Y f(x,y) d\nu(y)\right) d\mu(x)$$
$$= \int_Y\left(\int_X f(x,y) d\mu(x)\right) d\nu(y).$$

Fubini's Theorem. Let $(X,\mathscr{A},\mu)$ and $(Y,\mathscr{B},\nu)$ be complete σ-finite measure spaces, and let $f : X\times Y \to [-\infty,\infty]$ be $\overline{\mathscr{A}\times\mathscr{B}}$-measurable function and $\overline{\mu\times\nu}$-integrable. Then

1. f_x is ν-integrable μ-a.e., and f^y is μ-integrable ν-a.e..
2. the functions I_f and J_f defined by
$$I_f(x) = \begin{cases} \int_Y f_x d\nu, & \text{if } f_x \text{ is } \nu\text{-integrable.}\\ 0, & \text{other.}\end{cases}$$
$$J_f(y) = \begin{cases} \int_X f^y d\mu, & \text{if } f^y \text{ is } \mu\text{-integrable.}\\ 0, & \text{other.}\end{cases}$$
belongs to $L^1(X,\mathscr{A},\mu,\mathbf{R})$ and $L^1(Y,\mathscr{B},\nu,\mathbf{R})$, respectively. And
$$\int_{X\times Y} f d(\overline{\mu\times\nu}) = \int_X I_f(x) d\mu(x) = \int_Y J_f(y) d\nu(y).$$

Proof. Let f^+, f^- be the positive and negative parts of f. Then $f_x, (f^+)_x, (f^-)_x$ are $\mathscr{B}$-measurable and the Tonelli's theorem implies that the functions $x \mapsto \int_Y (f^+)_x d\nu$ and $x \mapsto \int_Y (f^-)_x d\nu$ are $\mathscr{A}$-measurable and μ-integrable (since $|f|$ is $\mu\times\nu$-integrable). And hence that they are finite μ-a.e.. So f_x is ν-integrable. Let
$$N = \left\{x\in X : \int_Y (f^+)_x(y) d\nu(y) = \infty \text{ or } \int_Y (f^-)(y) d\nu(y) = \infty\right\}.$$
Then $N \in \mathscr{A}$ since $\{\infty\}\in\mathcal{B}(\mathbf{R}^n)$. And
$$I_f(x) = \begin{cases} \int_Y (f^+)_x(y) d\nu(y) - \int_Y (f^-)(y) d\nu(y), & x\notin N\\ 0, & x\in N.\end{cases}$$
and hence $I_f \in L^1(X,\mathscr{A},\mu,\mathbf{R})$. By Tonelli's theorem, we have
$$\int_{X\times Y} f d(\mu\times\nu) = \int_{X\times Y} f^+ d(\mu\times\nu) - \int_{X\times Y} f^- d(\mu\times\nu)$$
$$= \int_X\left(\int_Y (f^+)_x(y) d\nu(y)\right) d\mu(x) - \int_X\left(\int_Y (f^-)_x(y) d\nu(y)\right) d\mu(x)$$
$$= \int_X I_f(x) d\mu(x).$$
Similar arguments apply to the functions f^y and J_f, and so the proof is complete. $\square$

Remark 9.2.1. If "$f : X \times Y \to [-\infty, \infty]$ is $\mu \times \nu$-integrable" is changed to "the integral of $f : X \times Y \to [-\infty, \infty]$ with respect to $\mu \times \nu$ exists", then the word "integrable" in conclusions should be replaced by "the integral with respect to μ or ν exists."

B Applications

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